Introduction

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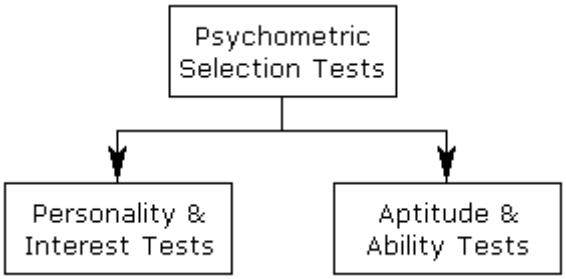
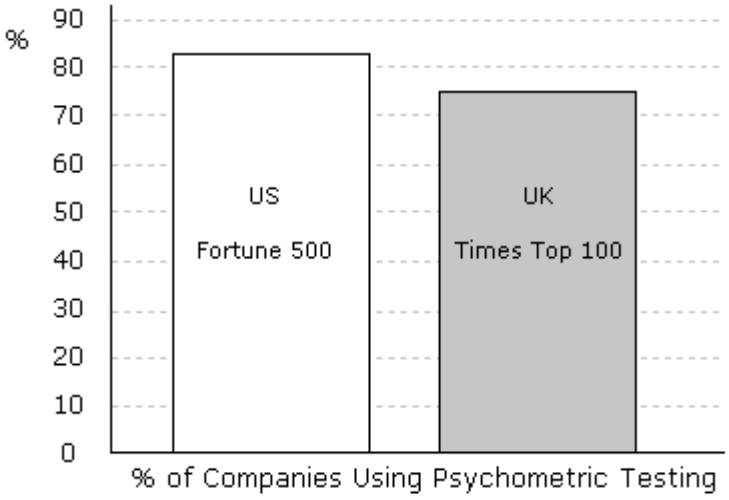
Mechanical Aptitude Tests

An Introduction to Psychometric Tests

You are most likely to encounter psychometric testing as part of the recruitment or selection process. Tests of this sort are devised by occupational psychologists and their aim is to provide employers with a reliable method of selecting the most suitable job applicants or candidates for promotion.



Psychometric tests aim to measure aspects of your personality or your mental ability, as illustrated in the diagram above. Personality questionnaires seek to measure aspects of your personality, whereas aptitude and ability tests aim to measure your intellectual and reasoning abilities.



Source: "Psychometric, Personality and Aptitude Testing" Ramada Consulting (2005)

The graph above shows the use of psychometric testing is slightly higher in America than in the UK and that these types of test are used extensively.

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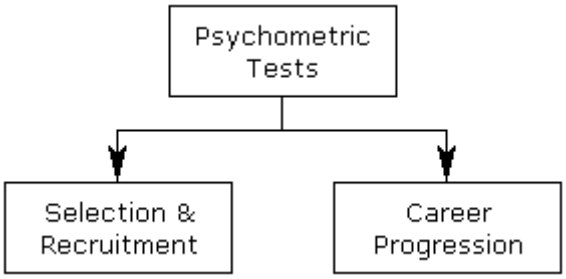
Psychometric tests can help prospective employers in number of ways. As an indicator of your personality, preferences and abilities they can find the best match of individual to occupation and working environment. As a recruitment and selection tool, these tests can be applied in a straightforward way at the early stages of selection to screen-out candidates who are likely to be unsuitable for the job.

You are very likely to be asked to take a psychometric test if you work in any of the following:

- IT Companies
- Large Corporations
- Armed Forces
- Local Authorities
- Civil Service
- Emergency Services

If you are applying for a job or seeking in promotion in one of these type of organisations then familiarity with these tests will give you a major advantage over candidates who are not familiar with them.

Human Resources departments may also use psychometric tests to help individuals develop their careers and understanding how these tests are used can help you achieve your career goals.



Because the results of psychometric tests are used to influence such important personnel decisions it is vital that the tests themselves produce accurate results based on standardized methods and statistical principles.

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A psychometric test should be:

Objective	The score must not affected by the testers' beliefs or values.
Standardised	It must be administered under controlled conditions.
Reliable	It must minimize and quantify any intrinsic errors.
Predictive	It must make an accurate prediction of performance.
Non-Discriminatory	It must not disadvantage any group on the basis of gender, culture, ethnicity, etc.

Why Test Use is Increasing

The single most frequently given reason for the increases in testing is:

The need to have a selection process that can withstand legal challenges.

The increased use of psychometric tests by organisations can be seen in part as a defensive strategy, which has been introduced and standardized in response to regulation and legislation. Organisations may need to demonstrate the fair treatment of all candidates during the selection process. Another factor, which must not be ignored, is the ease with which these tests can now be delivered online. This offers organizations a substantially decreased time-scale and cost for the whole selection exercise.

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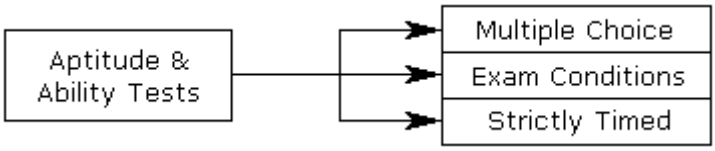
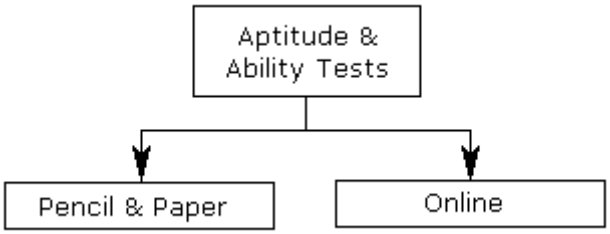
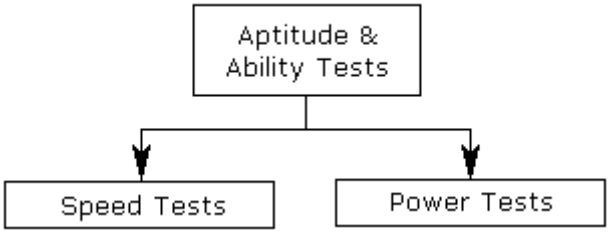
Aptitude and Ability Tests



Aptitude and ability tests are designed to assess your logical reasoning or thinking performance. They consist of multiple choice questions and are administered under exam conditions. They are strictly timed and a typical test might allow 30 minutes for 30 or so questions.



You may be asked to answer the questions either on paper or online. The advantages of online testing include immediate availability of results and the fact that the test can be taken at employment agency premises or even at home. This makes online testing particularly suitable for initial screening as it is obviously very cost-effective.



Aptitude and ability tests can be classified as speed tests or power tests. In a speed test the scope of the questions is limited and the methods you need to use to answer them is clear from the form of the question. Taken individually, the questions appear relatively straightforward. These tests are concerned with how many questions you can answer correctly in the allotted time. A power test on the other hand will present a smaller number of more complex questions. The methods you need to use to answer these questions are not obvious, and working out how to answer the question is the difficult part. Once you have determined this, arriving at the correct answer is usually relatively straightforward.

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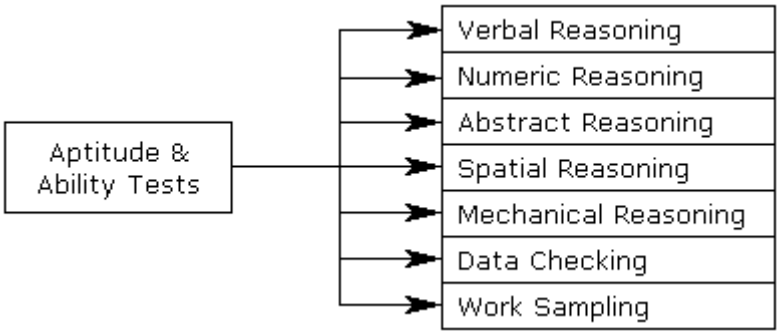

There are at least 5000 aptitude and ability tests on the market. Some of them contain only one type of question (for example, verbal ability, numeric reasoning ability etc) while others are made up of different types of question.

Types of Question
You can expect to find all of the following types of question in a typical mechanical aptitude test.

Mechanical Reasoning - Designed to assess your knowledge of physical and mechanical principles. For example, pulleys, levers, simple electrical circuits etc. Mechanical reasoning tests are used to select for a wide range of technical occupations.

Spatial Ability - Measures your ability to manipulate shapes in two dimensions or to visualize three-dimensional objects presented as two-dimensional pictures. These questions often appear in technical aptitude tests where many jobs require good spatial skills.

Numeric Ability - Includes basic arithmetic, number sequences and simple mathematics. These questions appear in most technical aptitude tests because employers usually want some indication of your ability to use numbers even if this is not a major part of the job.

Abstract Reasoning - Measures your ability to identify the underlying logic of a pattern and then determine the solution. Because abstract reasoning ability is believed to be the best indicator of fluid intelligence and your ability to learn new things quickly these questions appear in most technical aptitude tests.

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How to Improve Your Test Scores

You may hear people say that you can't improve your scores in psychometric tests. This is simply untrue. Everyone, ***if they practice***, can improve their test scores. The biggest gains are achieved quite quickly and result from becoming familiar with the types of question and from getting 'into the groove' of answering them.

Research suggests that the amount of improvement you can expect will depend on three areas.

1. **Educational Background**

   The longer that you have been out of the educational system and the less formal your educational background, the more likely you are to benefit from practice. Both of these factors suggest that familiarity with any type of examination process, both formal and timed, will give you an advantage.

2. **Personal Interests**

   Personal interests also play a part. For example if you are someone who habitually takes things apart and fixes them then your mechanical abilities may be highly developed. However, if you don't do these things on a regular basis then you will need some practice to refresh these dormant skills.

3. **Quality of Practice Material**

   If you are unfamiliar with the types of test questions then you will waste valuable time trying to determine what exactly the questions are asking you to do. This unfamiliarity also causes you to worry about whether you have understood the question correctly and this also wastes mental energy, which you could otherwise spend on getting the correct answer. By increasing your familiarity with the style and types of questions you will improve your scores.

The first of these factors is beyond your control, the second may be worth addressing in the longer term if you feel that increasing your facility with English would benefit your career. However, solving crossword puzzles is not going to make much difference to your psychometric test results in the short term. This leaves you with the '*quality of the practice material*' as the best way to improve your score.

Firstly, the material itself needs to match as closely as possible the tests that you expect to take. The questions in this book are based on the question types used by the most popular test providers in the industry and are updated regularly to reflect the latest trends.

Secondly, you should practice the material in the most realistic way possible. Find somewhere where you will not be disturbed and go through each paper without interruption and try to stick to the time limit. Do not have anything with you that are not allowed on the day of the test (dictionary, thesaurus or calculator) and switch off your mobile phone.

The tests are generally about twenty minutes long. If you don't have an uninterrupted twenty minutes for a practice paper, then try to complete the first half of the questions in ten minutes and treat the second half as another ten minute paper. Concentrate one hundred percent for the duration of the test as this keeps the practice as realistic as possible.

Small Improvements Make a BIG Difference

*'Can you improve your score **enough** to make a difference in the selection process?*	The answer is **'YES'**.

To understand why this is true, you need to consider three things.

I. The number of candidates sitting the test.
II. The educational background of candidates
III. How test scores are compared.

The most important concept to understand is that of the '***percentile score***'. This is the score most often used by organizations when comparing your score with that of other candidates. It has the advantage of being easily understood and percentiles are very widely used when reporting test results to managers.

To calculate your percentile score, your actual score is converted to a number indicating the percentage of the test group who scored below you.

For example,

SCORE	MEANS THAT
60th percentile	your score is the **same as or higher** than the scores of **60%** of those who took the test.
85th percentile	your score is the **same as or higher** than the scores of **85%** of those who took the test

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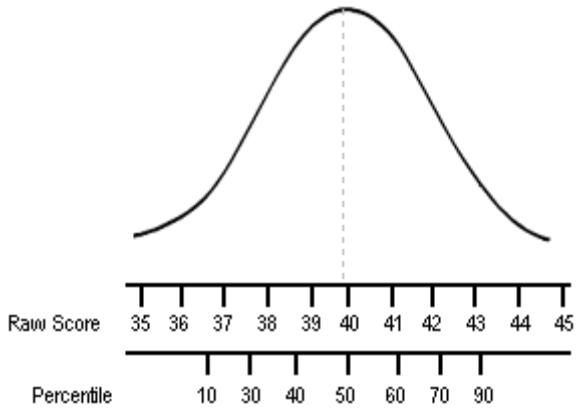
The aptitude and ability tests used in selection have relatively few questions and the people sitting a particular test tend to be from a similar group in terms of their education and background. As a result, the scores tend to cluster quite tightly around the average. This combination of relatively few questions, clustering and the use of percentiles has important implications for you as a job candidate. This means:

A small improvement in your actual score will result in a

big improvement to your percentile score.

To illustrate this point, consider a typical test consisting of 50 questions. Most of the candidates, who are a fairly similar group in terms of their educational background and achievements, will score around 40 (raw score on the diagram). It is very unlikely that any of them will score less than 35 or more than 45.



This means the difference between the 35th (38/50) and the 70th (42/50) percentile ***is only 4 marks out of the possible 50.***

Although an experienced statistician would never use percentiles on this type of data; nine times out of ten this is exactly what organisations do. So therefore, as the previous example shows, a few extra marks can take you from the 35th to the 70th percentile.

Those '4 marks' can be the difference to your chances of success and it's all attributable to ***your preparation****.*

That is why preparing for these tests is so worthwhile. Even a small improvement of two or three marks can make you appear a far superior candidate. It is extremely important that you find effective ways to motivate yourself to practice, and the next section gives you some guidelines.

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Motivating Yourself to Succeed

Your personal experience has probably taught you that someone who is motivated can achieve far more than someone who is not, even though their natural abilities may be the same. Whilst everyone learns this, it does not make it any easier to become motivated, particularly if the task is not obviously enjoyable or interesting. There are literally thousands of publications dedicated to the subject of motivation. Entire books have been written about it and high-profile careers have been carved out by people who claim to know its secret.

If you are already someone who has embraced a particular motivational philosophy then you may want to go straight to the next section. Use what you have and approach the job selection process with the determination to succeed and no doubt you will achieve your potential.

If you find it difficult to buy into any self motivational philosophy then you will probably need some help to push yourself to achieve your potential. Those who have achieved success, business professionals and motivational gurus, have done so by setting themselves a 'SMART' goal.

S pecific	The goal says what you want to achieve. *"I want to achieve my full potential in verbal ability tests"* *"With practice I want to increase my score in the tests."*
M easurable	Goals need to be measurable so that you know when you have succeeded. *"I want to increase my test score each time I practice."* *"I want to increase the number of questions I complete with each practice."*
A chievable	Goals need to be challenging but *realistic.* *"I must allocate three 1-hour sessions each week to practice."* *"I will complete half a test every day before my interview."*
R elevant	The goal has to *mean something to you; an emotional tie*. *"Once I get this new job I can afford my holiday."* *"My increase in salary will enable me to buy that house."*
T ime-bound	All goals must be time bound. *"From the 2nd till the 18th I can do practice papers."* *"From today for 10-days I will practice verbal ability tests."*

Motivation is much easier to achieve and maintain when the objective is clear and you can see that every minute of the time you are spending is taking you nearer to that goal; not always easy in life. But, the nature of verbal ability tests means that they are ideal for use with 'SMART' as the test goals are easy to define and you can measure your progress.

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To develop your techniques, you MUST complete sufficient practice papers to identify any weak areas that you have and allow sufficient time to take remedial action. It is important to remember that the difference in scores between those candidates who are selected to go forward to the interview, and those who are not, is likely to be quite small. An improvement of just a few percent could put you in the first group, rather than the latter.

> *It's your career that is at stake here and* ***30-60 minutes practice a day for 2 weeks*** *is a small investment to make when you consider the potential payoff.*

It is important that you view these tests in a positive way, as something that you can excel at. Remember that employers see test results as indicating potential and good results will encourage them to view you in a positive way. If you see the test as an obstacle it will be much harder for you to motivate yourself. It is vital that you focus on gaining a higher score, which will increase your ability to eclipse a candidate with better qualifications or more experience.

The Practice Environment

Once you've defined your goal, you then need to consider the practicalities of achieving within your time frame. You may find another acronym 'TAP' will help you to identify the most effective way for you to practice in the time available to you.

T ime slot	Identify the best time for you and your schedule.
A tmosphere	Create the right atmosphere, one that matches the 'real' test situation as best you can. Avoid ALL interruptions and turn off the mobile phone!
P lace	Find the most ideal situation for you to practice in so that you can give the test you full concentration for that time slot.

In order to develop your techniques and complete sufficient practice papers to identify any weak areas you will need to spend between 30-60 minutes each day. You will then be able to take remedial action to address your weak areas.

If you think that this could be a problem then you are not alone. Very few people feel that they have a 'spare' hour a day just waiting to be filled. You will need to take positive action to schedule this task. The conditions in which you practice will need to be as near to the actual test conditions as possible. It is vital that the environment is free of distractions and interruptions.

Some ideas that you may find useful are:

- Practice in your lunch hour,
- Practice at a local library,
- Practice in an empty office or conference room,
- Stay after work and do it at your desk before going home,
- Set the alarm an hour early and do it first thing.

If you miss a day or two the temptation is to try to make up for lost time by doing a long session. You should avoid this at all costs, *you can only concentrate fully for about an hour* and you will find that if you try to concentrate longer than that your performance will decline. This is doubly frustrating because you will feel as though you are getting worse instead of better and the more that you 'stick at it' the worse it will get. This will destroy your motivation – so don't do it.

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Using the Practice Tests in this Book

This book has been designed so that it is easier for you to practice the areas where you feel that you are weakest, giving you the greatest benefit in the shortest time. So each test in the book contains more questions of each type than you would get in a real test. It also ensures that you receive maximum exposure to as wide a variety of questions styles and types as possible.

The best strategy is to attempt one of each type of question. It is vital that you complete the whole test before you look at the answers to maximise the benefit to you and help you achieve your SMART goal.

- Mechanical Reasoning
- Fault Diagnosis
- Spatial Ability
- Numerical Reasoning – Computation
- Numerical Reasoning - Estimation
- Numerical Reasoning – Critical Reasoning
- Abstract Reasoning

After this exercise you will have a much clearer idea of the type of questions that you are most likely to benefit from practicing. This is a personal decision and you should go with your feelings.

It's important to remember that a real test will contain a mixture of question types and that **most tests allocate one mark to each correct answer -** there is **no** differential marking. For example, this means that you will get one mark for understanding the meaning of a word or for spelling it correctly and you will get one mark for untangling the meaning of a complex sentence and answering a question about it.

Do not linger on questions that you find difficult - complete all those you can answer easily and then go back and fill in the gaps. This will also highlight the style of questions you find more difficult.

Managing Stress

The job selection process will always involve an element of stressfulness. This is mostly due to confronting a situation over which you do not have total control. For example, you cannot predict or influence the personality or behaviour of the interviewer or know in advance which questions you will be asked. However you can, and probably have already, prepared for the most likely questions. You can also make educated guesses as to which areas of your resume the interviewer will concentrate on.

With regard to the psychometric test component of the selection process, your preparation should be far more straightforward. If you haven't taken this type of test for a long time this will increase the degree of stress and nervousness you experience. This is mostly due to a simply fear of the unknown, as well as, a feeling that you will 'let yourself down' and that the test will not be a fair reflection of your strengths and abilities. You may experience physical symptoms such as a lack of ability to get to sleep and psychological symptoms such as loss of concentration and mild depression. You must act immediately to tackle this stress before its effects become more corrosive as the test date approaches.

You will hear a lot of advice for coping with the symptoms of stress and anxiety, including: relaxation, exercise and visualization. While all of these things can help, ***the most effective solution is to take direct action and spend your time practicing these tests*** in the most systematic and efficient way possible.

Mechanical Reasoning

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Introduction to Mechanical Reasoning Tests

Mechanical reasoning tests measure your knowledge of straightforward mechanical and physical concepts. They do not measure your underlying mechanical aptitude in the same way that abstract reasoning questions measure your underlying intellectual ability. For example, you could sit an abstract reasoning test without having seen one before and still get a reasonable score. The same is not true of mechanical reasoning where your score will depend significantly on your knowledge of:

- Levers
- Pulleys
- Gears
- Springs
- Simple Electrical Circuits
- Tools
- Shop Arithmetic

You may have come across: levers, pulleys, gears, springs and simple circuits in elementary science and the questions on these topics are fairly straightforward. If elementary science classes seem like a long time ago then you may need to refresh your memory before attempting these questions.

If you are taking a mechanical reasoning test as part of the selection process for the emergency services or the military then the questions you can expect will tend to concentrate on principles rather than on making calculations. For example, you may be shown 3 diagrams of a lever and asked which one is the most efficient. If however, you are taking a test for a craft or technical job then you may be expected to calculate the actual force required to move a particular lever. In this case, knowing the principle is not enough, you need to know the formula.

This section also contains fault diagnosis questions which are used to select personnel for technical roles where they need to be able to find and repair faults in electronic control systems. The questions in fault diagnosis tests appear rather abstract and require pure logic to solve. This is because when a fault develops in an electronically controlled system, there is often no physical clue as to the cause. In the absence of such a clue, for example an obviously burned-out component, the only way to diagnose the problem is by a process of logical elimination.

This type of test is used extensively to select technical and maintenance personnel as well as to select for artificer (technical) roles within the armed forces where the ability to approach problems logically in order to find the cause of the fault is increasingly important.

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Overview of Mechanical Reasoning Tests

These tests are designed to assess your knowledge of physical and mechanical principles. Questions are in the form of a question and a diagram and you will need to determine which mechanical principle is being illustrated. No specialist knowledge is required to answer these questions, only an understanding of basic mechanical and physical principles.

Mechanical reasoning tests are used to select for a wide range of jobs including the military (Armed Services Vocational Aptitude Battery), police forces, fire services, as well as many engineering occupations. These tests are usually of 20-30 minutes duration and contain 20-30 questions which concentrate on the following:

- Levers
- Pulleys
- Gears
- Springs
- Gravity
- Electricity

Levers
A lever consists of a bar which pivots at a fixed point known as the fulcrum. In the example shown the fulcrum is at the center of the lever. This lever provides no mechanical advantage and the force needed to lift the weight is equal to the weight itself.

However, if you want to lift a weight that is heavier than the force applied you can move the fulcrum closer to the weight to be lifted. This affects the force required in the following way:

$$w \times d1 = f \times d2$$

Where:

W = weight	d1 = distance from fulcrum to weight
f = force needed	d2 = distance from fulcrum to point where force is applied

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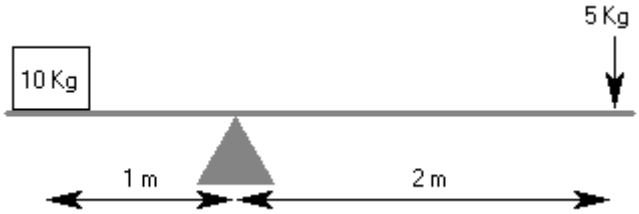
In this example the fulcrum has been moved towards the weight so that the weight is 1 meter from the fulcrum. This means that the force can now be applied 2 meters from the fulcrum.



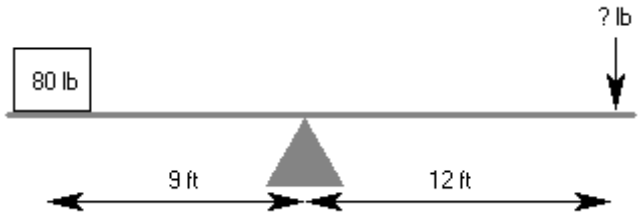
If you needed to calculate the force needed to lift the weight then you can rearrange the formula.

w x d1 = f x d2 can be rearranged to **f = (w x d1)/d2**

f = (10 x 1)/2 (10/2 is the same as 5/1, the force required is 5 Kg)

Example Questions

1. How much force is required to lift the weight?



A) 40lbs B) 50lbs C) 60lbs D)70lbs

Answer

1. C - 60lbs is needed to lift the weight. It can be calculated like this:

f = (w x d1)/d2
f = (80 x 9)/12
f = (720)/12
f = 60 lbs

In practice, levers are used to reduce the force needed to move an object, in other words to make the task easier. However, in mechanical aptitude questions it is possible that you will see questions where the fulcrum has been placed closer to the force than the weight. This will mean that a force greater than the weight will be required to lift it.

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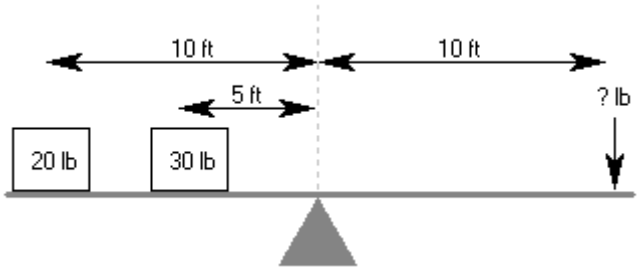
You may see more complex questions involving levers, for example, there may be more than one weight. In this case you need to work out the force required to lift each weight independently and then add them together to get the total force required.

2. How much force is required to lift the weights?



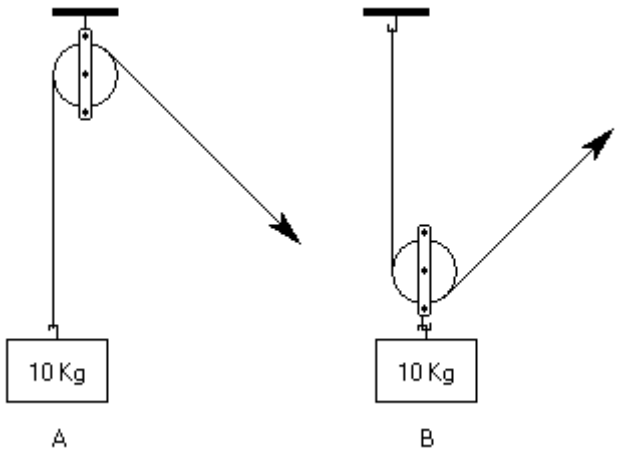
A) 25lbs B) 35lbs C) 40lbs D)45lbs

Answer
2. B - 35lbs is needed to lift the weight. It can be calculated like this:

f = (w1 x d1) + (w1a x d1a)/d2
f = (20 x 10) + (30 x 5)/10
f = (200 + 150)/10
f = 35 lbs

Pulleys
The pulleys used in this type of question consist of a grooved wheel and a block which holds it. A rope runs in the groove around the wheel and one end will be attached to either: a weight, a fixed object like the ceiling or to another pulley. For the purposes of these questions you can ignore the effect of friction.

Single Pulley
3. Which weight requires the least force to move?



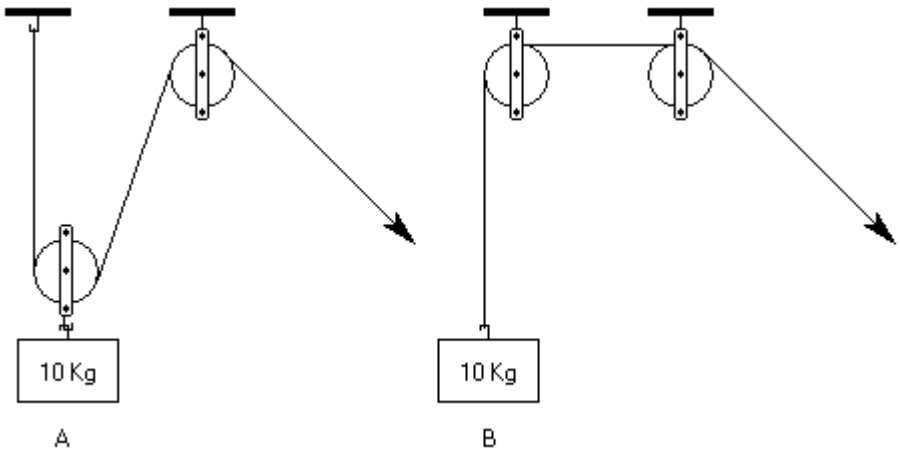
A) A B) B C) Both require the same force

Answer
3. B – Weight B requires a force equal to 5 Kg whereas A requires a force equal to 10 Kg.

Single pulley questions are relatively straightforward. If the pulley is fixed, then the force required is equal to the weight. If the pulley moves with the weight then the force is equal to half of the weight. Another way of thinking about this is to divide the weight by the number of sections of rope supporting it to obtain the force needed to lift it. In A there is only one section of rope supporting the weight, so 10/1 = 10 Kg required to lift the weight. In B there are two sections of rope supporting the weight, so 10/2 = 5 Kg required to lift it.

Double Pulleys
There are two possible ways that two pulleys can be used. Either one pulley can be attached to the weight or neither of them can be.

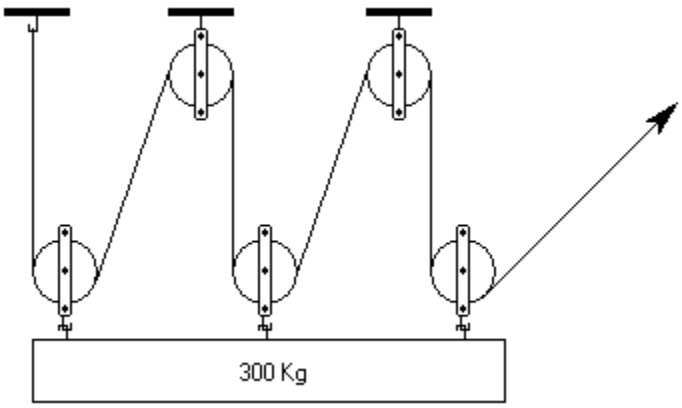
4. Which weight requires the least force to move?



A) A B) B C) Both require the same force

Answer
4. A – Weight A requires a force equal to 5 Kg whereas weight B requires a force equal to 10 Kg. Remember to divide the weight by the number of sections of rope supporting it to get the force needed to lift the weight.

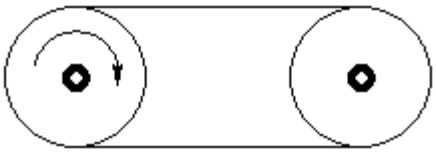
Using More Than Two Pulleys
5. How much force is required to move the weight?



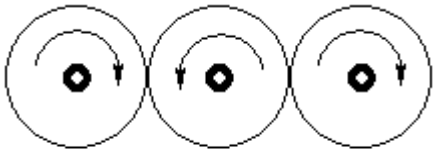
A) 100 Kg B) 150 Kg C) 50 Kg D) 60 Kg

Answer
5. C – The weight is 300 Kg and there are 6 sections of rope supporting it. Divide 300 by 6 to get 50 Kg. In all cases, just divide the weight by the number of sections of rope supporting it to get the force needed to lift the weight.

Gears
A gear is a toothed wheel or cylinder that meshes with another toothed component to transmit motion or to change speed or direction. Gears are attached to a rotating shaft turned by an external force, which is not usually illustrated in these types of question. Two gears may be connected by touching each other directly or by means of a chain or belt. If gears are connected by a chain or belt then they move in the same direction.



Gears Connected by Belt



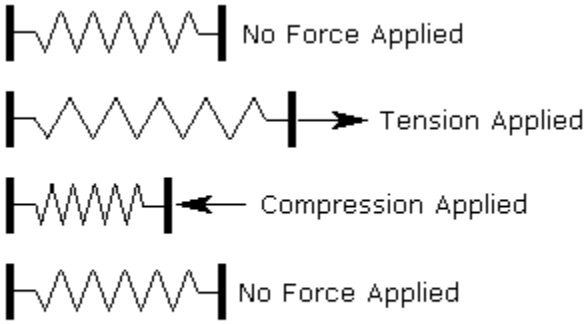
Gears Meshed

If the gears are touching (meshed) then adjacent gears move in opposite directions. In this example the first and third gear will turn in the same direction. When there are an odd number of meshed gears then the last gear will always turn in the same direction as the first one.

Meshed gears with an equal number of teeth will turn at the same speed. If they have an unequal number of teeth then the gear with the fewest teeth will turn faster. To work out how fast one is turning with respect to the other you need to count the teeth.

Mechanical Aptitude Tests

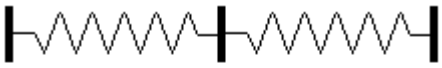
Springs
A spring is piece of wire or metal that can be extended or compressed by an external force but which then returns to its original length when that force is no longer applied.

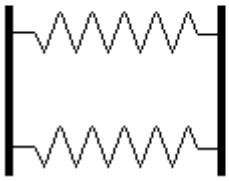


Spring Under Tension & Compression

There are many different types of spring including, spiral coil, leaf springs and torsion springs. Springs are used in many applications including clocks, vehicle suspensions etc. In the type of questions that you will be asked in mechanical aptitude tests, you can assume that springs behave in a linear way. That is, doubling the force applied will stretch or compress the spring twice as much.

Springs in Series & Parallel
If more than one spring is used then they can be arranged in one of two ways, either in series or in parallel.

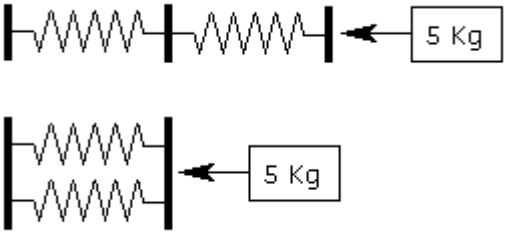

Springs in Series


Springs in Parallel

When springs are arranged in series, each spring is subjected to the force applied. When the springs are arranged in parallel the force is divided equally between the springs.

Mechanical Aptitude Tests

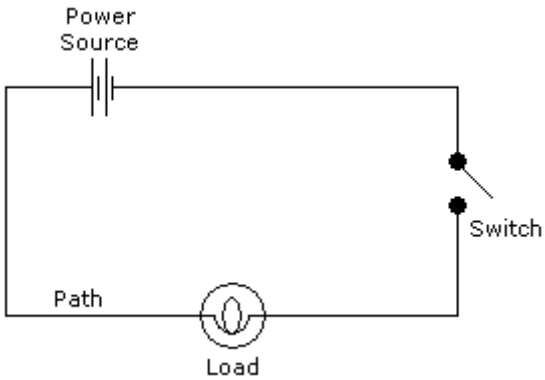
Example Question
6. A force of 5 Kg compresses the springs in series by 10cm.What will be the total distance that the springs in parallel are compressed?



A) 10 cms B) 2.5 cms C) 5 cms D) 7.5 cms

Answer
6. C – The total force will be divided equally between the 2 springs in parallel. Since the force is divided in half, the distance moved will also be halved. The springs in series were compressed by10 cms, therefore the springs in parallel will be compressed by 5 cms.

Electricity
Questions on electricity usually take the form of simple circuit diagrams.

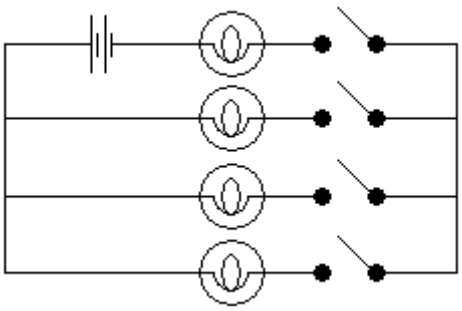


These diagrams are usually restricted to showing the power source, switches, loads (typically bulbs), and the path of the wiring. To answer these questions you need a basic understanding of how electricity flows around a circuit.

Mechanical Aptitude Tests

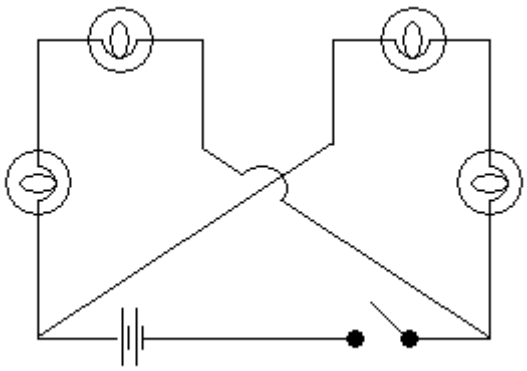
Example Question

7. How many switches need to be closed to light up one bulb?



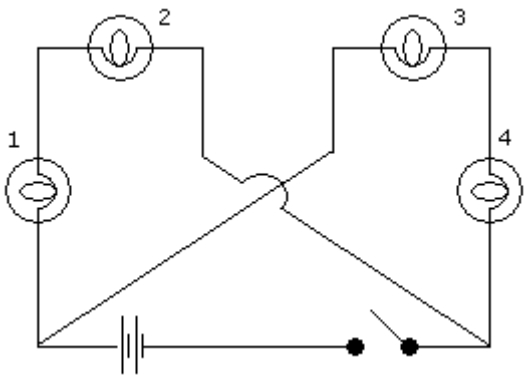
A) 1 B) 2 C) 3 D) 4

8. How many bulbs will light up when the switch is closed?



A) 1 B) 2 C) 3 D) 4

9. If bulb 1 is removed, how many bulbs will light up when the switch is closed?



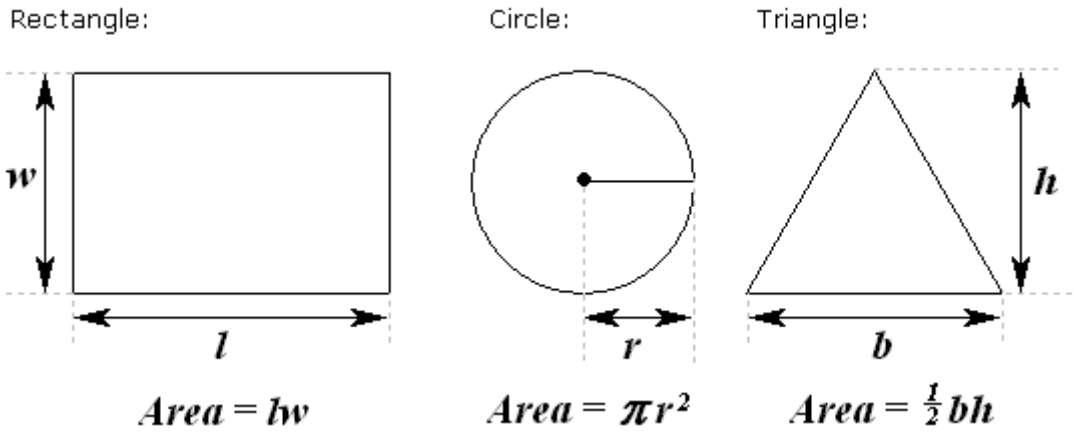
A) 1 B) 2 C) 3 D) 0

Mechanical Aptitude Tests

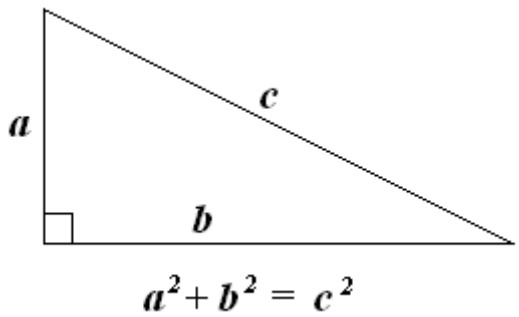
Answers
7. B – Two switches need to be closed to complete a circuit.
8. D – All 4 bulbs will light up.
9. B – Only bulbs 3 and 4 will light up.

Other Things You May Need to Know
Some of the questions you come across may require knowledge of basic mathematics, fractions, decimals, ratios, percentages, and averages. These basic mathematical skills are an important aspect of mechanical ability and making simple calculations on the basis of the information provided and your knowledge of mechanical principles is part of these tests.

Some questions may also require you to know some simple formulas. For example, the areas of shapes like rectangles, circles and triangles,



and some simple geometry like Pythagoras Theorem.



Mechanical Aptitude Tests

Mechanical Reasoning Questions

Mechanical reasoning questions vary widely in difficulty from test to test. For example, if you are taking a mechanical reasoning test as part of the selection process for the emergency services or the military then the questions will tend to concentrate on principles rather than on making calculations. For example, you may be shown three diagrams of a lever and asked which one is the most efficient.

If however, you are taking a test for a craft or technical job then you may be expected to calculate the actual force required to move a particular lever. In this case, knowing the principle is not enough, you need to know the formula.

You may also be asked some questions about tools and how they are used. These questions are again straightforward and if you have spent significant time fixing or making things, they won't present any problems. However, mechanical goods of all types are relatively cheaper, less prone to breakdown and often less repairable than they were 20 years ago. This means that many people under 30 years of age don't have much practical experience of fixing things or of watching someone else do so. If you feel that this applies to you, then you will need to make some effort to improve your knowledge of everyday tools. You can do this by getting hold of a catalog for a tool hire company and simply reading through it – boring but effective.

If you are taking a test as part of the selection for a craft or apprenticeship job you should also expect some shop arithmetic questions. These questions approximate the type of reasoning and maths that are needed to estimate materials costs etc. You will need to know the formulae for calculating the areas and volumes of simple shapes, as well as familiarity with imperial units including: inches, feet and yards. For example, you need to know that there are nine square feet in a square yard, something which surprises many people brought up using only metric units!

One final point, many of the questions in mechanical reasoning tests are 'industry' specific. For example, tests used by the fire service tend to frame the questions in terms of fire-fighting whereas tests used to select for an aircraft maintenance job would tend to frame the questions in aviation industry terms. It doesn't matter if the questions you practice on aren't specific to the industry you are applying for. It is the substance of the question that is important – the scenario is incidental.

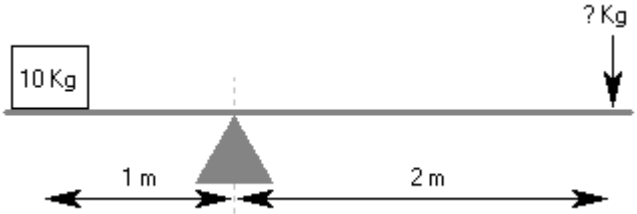
Mechanical Reasoning		Operatives	Supervisory	Management
Craft & Technical		Y	Y	Y
Clerical & Administrative				
Police, Fire, Military etc.	Y			
Management Trainee				
Graduate & Professional				

These sample question papers each contain 40 questions and have a suggested time limit of 30 minutes each. The questions are presented in Letter/A4 format for easy printing and self-marking.

Mechanical Aptitude Tests

Test 1: 20 Questions
Answer as many questions as you can in 15 minutes. Circle the letter below the question which corresponds to the correct answer. You are advised to use a calculator.

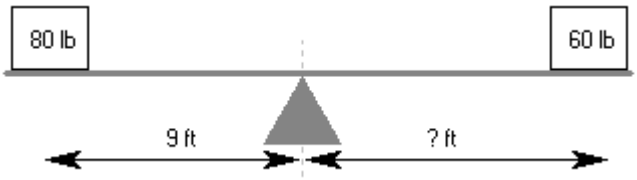
1) How much weight is required to balance the lever?



A	B	C	D	E
15Kg	5Kg	10Kg	7.5Kg	20Kg

A B C D E

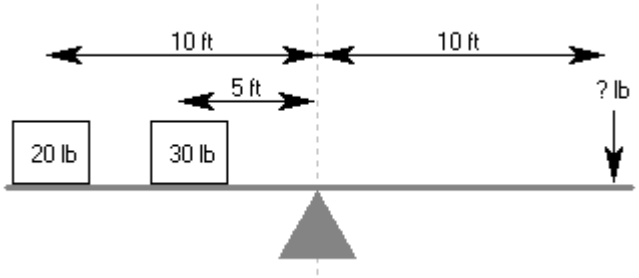
2) How far from the fulcrum does the 60 lb weight need to be to balance the lever?



A	B	C	D	E
9 ft	7 ft	14 ft	12 ft	10 ft

A B C D E

3) How much weight is required to balance the lever?

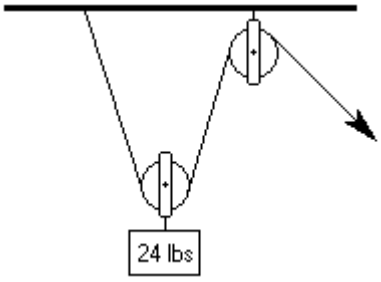


A	B	C	D	E
30 lbs	25 lbs	28 lbs	40 lbs	35 lbs

A B C D E

Mechanical Aptitude Tests

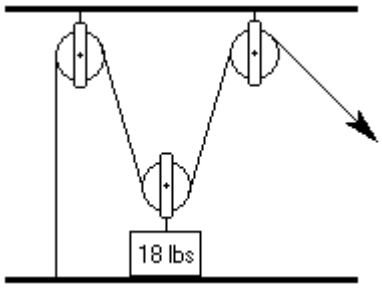
4) Approximately how much force is needed to lift the weight?



A	B	C	D	E
24 lbs	10 lbs	48 lbs	12 lbs	18 lbs

A B C D E

5) Approximately how much force is needed to lift the weight?



A	B	C	D	E
36 lbs	10 lbs	18 lbs	9 lbs	14 lbs

A B C D E

6) If gear X turns clockwise at a constant speed of 10 rpm. How does gear Y turn?

A	B	C	D	E
anti c/w 10 rpm	c/w 10 rpm	c/w 5 rpm	anti c/w 5 rpm	c/w 20 rpm

A B C D E

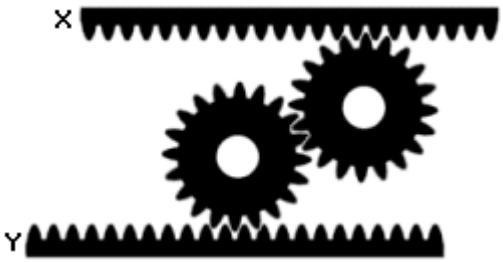
Mechanical Aptitude Tests

7) If gear X turns clockwise at a constant speed of 10 rpm. How does gear Y turn?

A	B	C	D	E
anti c/w 10 rpm	c/w 10 rpm	c/w 5 rpm	anti c/w 5 rpm	c/w 20 rpm

A B C D E

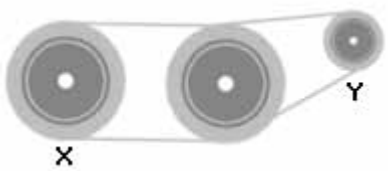
8) If bar Y moves left a constant speed. How does bar X move?



A	B	C	D	E
Left, Faster	Left, Same	Left, Slower	Right, Same	Right, Slower

A B C D E

9) If drive wheel X rotates clockwise at a speed of 10 rpm. How does wheel Y turn?

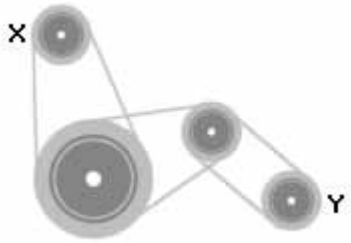


A	B	C	D	E
anti c/w faster	c/w slower	c/w faster	anti c/w slower	anti c/w same

A B C D E

Mechanical Aptitude Tests

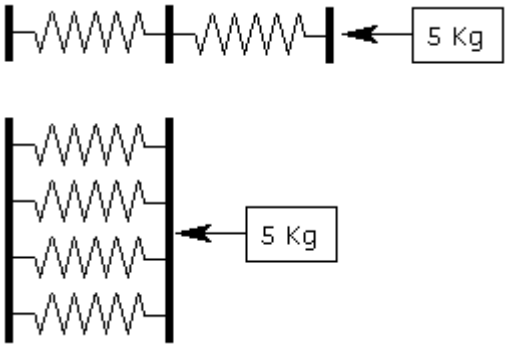
10) If drive wheel X rotates clockwise at a speed of 10 rpm. How does wheel Y turn?



A	B	C	D	E
anti c/w faster	c/w slower	c/w faster	anti c/w slower	c/w same

A B C D E

11) A force of 5 Kg compresses the springs in series 10cm. What will be the total distance that the springs in parallel are compressed?

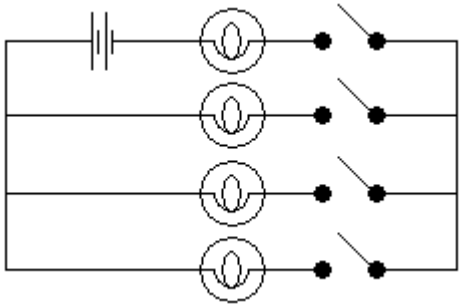


A	B	C	D	E
2.5 cms	5 cms	7.5 cms	10 cms	15 cms

A B C D E

Mechanical Aptitude Tests

12) In the circuit shown, how many switches need to be closed to light up one bulb?



A	B	C	D	E
None	One	Two	Three	Four

A B C D E

13) Which is the most suitable tool for breaking up concrete?

A	B	C	D	E
None	1	2	3	4

A B C D E

14) Which is the most suitable tool for assembling a friction fit wooden frame?

A	B	C	D	E
None	1	2	3	4

A B C D E

Mechanical Aptitude Tests

15) Which tool or combination of tools would be most useful for general woodwork?

A	B	C	D	E
4 & 2	3, 5 & 7	2, 4 & 6	4 & 7	3 & 6

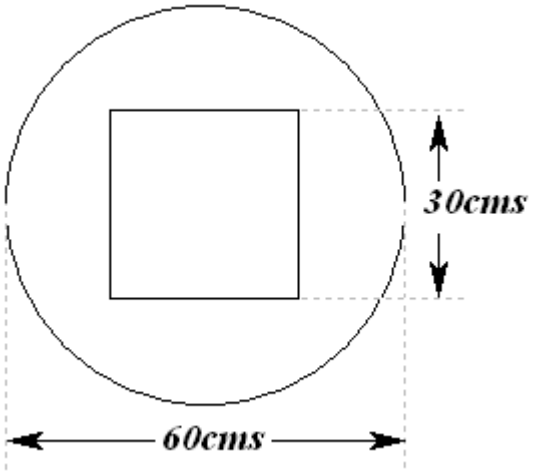
A B C D E

16) Which tool or combination of tools would be most useful for repairing a broken radio?

A	B	C	D	E
1 & 8	3, 5 & 7	8	1 & 9	3 & 6

A B C D E

Mechanical Aptitude Tests



The sketch above shows a component which is stamped out of sheet steel. The square in the center is discarded. These components are stamped out of a continuous steel coil with a width of 70 cms. The stamping process requires a gap of 25mm between each component. The steel coil is supplied in lengths of 25 meters costing $200.

17) What is the approximate area of the remaining shape in square centimetres?

A	B	C	D	E
1938	1855	1926	1880	1760

A B C D E

18) What is the approximate percentage of steel wasted including the center square?

A	B	C	D	E
56%	50%	62%	48%	52%

A B C D E

19) Assuming minimal wastage, how many components can be produced from each 25 meter coil?

A	B	C	D	E
38	40	36	42	37

A B C D E

20) What is the approximate cost of a component if the scrap is sold at 50% of cost?

A	B	C	D	E
$3.60	$3.15	$3.55	$5.00	$4.85

A B C D E

End of Mechanical Reasoning - Test 1

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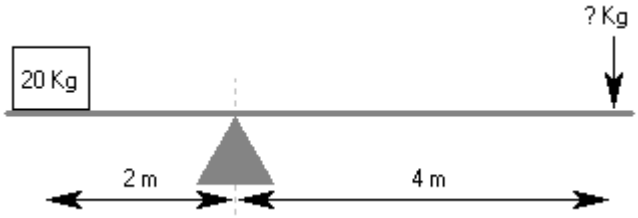
Mechanical Aptitude Tests

Mechanical Reasoning Test 2: 20 Questions
Answer as many questions as you can in 15 minutes. Circle the letter below the question which corresponds to the correct answer. You are advised to use a calculator.

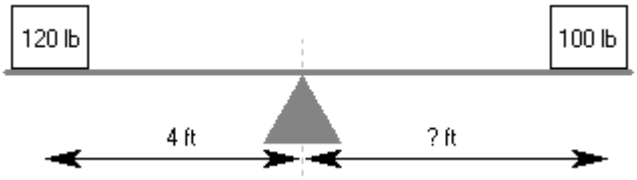
1) How much weight is required to balance the lever?



A	B	C	D	E
15Kg	5Kg	10Kg	7.5Kg	20Kg

A B C D E

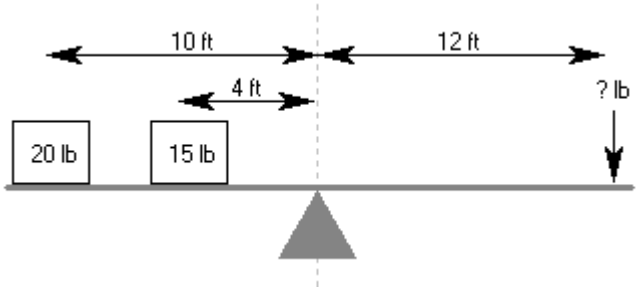
2) How far from the fulcrum does the 100 lb weight need to be to just tip the lever?



A	B	C	D	E
4 ft 8 inches	4 ft 6 inches	5 ft	4 ft 10 inches	4 ft

A B C D E

3) How much weight is required to just tip the lever?

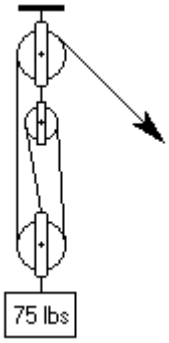


A	B	C	D	E
22 lbs	25 lbs	28 lbs	40 lbs	35 lbs

A B C D E

Mechanical Aptitude Tests

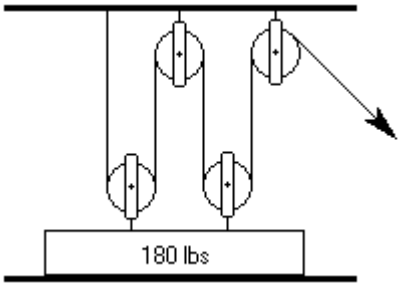
4) Approximately how much force is needed to lift the weight?



A	B	C	D	E
75 lbs	35.5 lbs	25 lbs	50 lbs	15 lbs

A B C D E

5) Approximately how much force is needed to lift the weight?



A	B	C	D	E
30 lbs	45 lbs	60 lbs	90 lbs	120 lbs

A B C D E

6) If gear X turns clockwise at a constant speed of 10 rpm. How does gear Y turn?

A	B	C	D	E
anti c/w 10 rpm	c/w 10 rpm	c/w 20 rpm	anti c/w 5 rpm	anti c/w 20 rpm

A B C D E

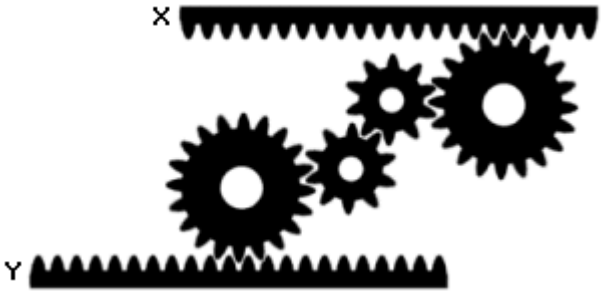
Mechanical Aptitude Tests

7) If gear X turns clockwise at a constant speed of 10 rpm. How does gear Y turn?

A	B	C	D	E
anti c/w 10 rpm	c/w 10 rpm	c/w 5 rpm	anti c/w 5 rpm	c/w 20 rpm

A B C D E

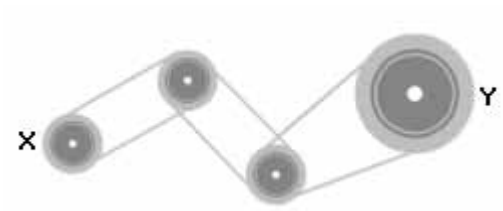
8) If bar Y moves left a constant speed. How does bar X move?



A	B	C	D	E
Left, Faster	Right, Same	Left, Slower	Left, Same	Right, Slower

A B C D E

9) If drive wheel X rotates clockwise at a speed of 10 rpm. How does wheel Y turn?

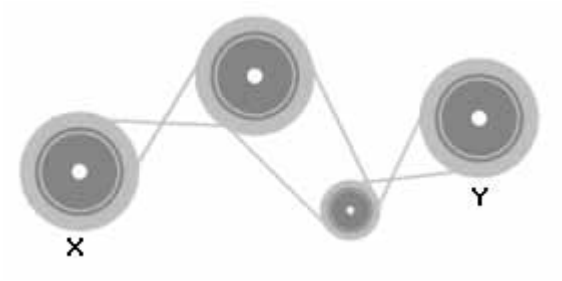


A	B	C	D	E
anti c/w faster	c/w slower	c/w faster	anti c/w slower	anti c/w same

A B C D E

Mechanical Aptitude Tests

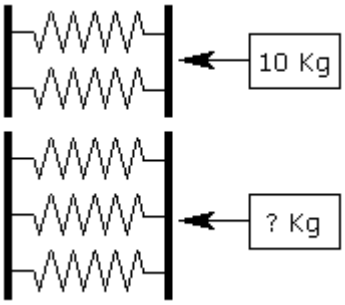
10) If drive wheel X rotates clockwise at a speed of 10 rpm. How does wheel Y turn?



A	B	C	D	E
anti c/w faster	c/w slower	c/w faster	anti c/w slower	c/w same

A B C D E

11) A force of 10 Kg compresses the two springs in parallel 10cm. How much force is required to compresses three springs in parallel 10cm?

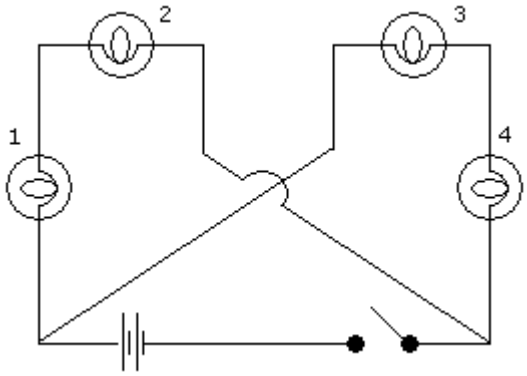


A	B	C	D	E
5 Kg	10 Kg	7.5 Kg	12 Kg	15 Kg

A B C D E

Mechanical Aptitude Tests

12) If bulb 1 is removed, how many bulbs will light up when the switch is closed?



A	B	C	D	E
None	One	Two	Three	Four

A B C D E

13) Which is the most suitable tool for general carpentry?

A	B	C	D	E
None	1	2	3	4

A B C D E

14) Which is the most suitable tool for general metalwork?

A	B	C	D	E
None	1	2	3	4

A B C D E

Mechanical Aptitude Tests

15) Which tool or combination of tools would be most useful for fitting an entertainment system to a vehicle?

A	B	C	D	E
1 & 9	6	8	2 & 8	9

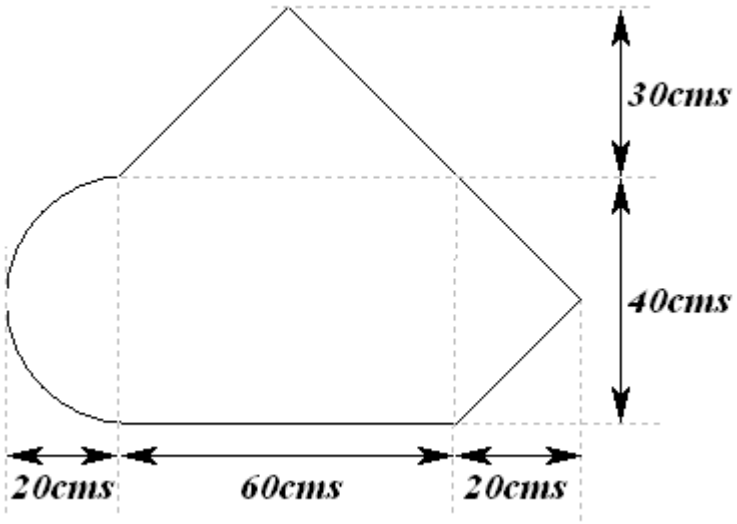
A B C D E

16) Which tool or combination of tools would be most useful for constructing a mild steel frame?

A	B	C	D	E
3 & 4	9	1 & 9	2 & 8	6

A B C D E

Mechanical Aptitude Tests



The sketch above shows a component which is stamped out of sheet steel. These components are stamped out of a continuous steel coil with a width of 75 cms. The stamping process requires a gap of 25mm between each component. The steel coil is supplied in lengths of 30 meters costing $200.

17) What is the approximate area of the component in square centimetres?

A	B	C	D	E
4688	4470	4562	4860	4328

A B C D E

18) What is the approximate percentage of steel wasted?

A	B	C	D	E
42%	35%	44%	37%	39%

A B C D E

19) How many components can be produced from each 30 meter coil?

A	B	C	D	E
30	29	32	37	34

A B C D E

20) What is the approximate cost of a component if the scrap is sold at 50% of cost?

A	B	C	D	E
$4.40	$5.80	$5.66	$5.40	$6.66

A B C D E

End of Mechanical Reasoning - Test 2

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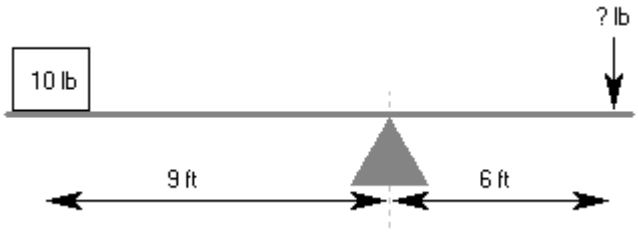
Mechanical Aptitude Tests

Mechanical Reasoning Test 3: 20 Questions
Answer as many questions as you can in 15 minutes. Circle the letter below the question which corresponds to the correct answer. You are advised to use a calculator.

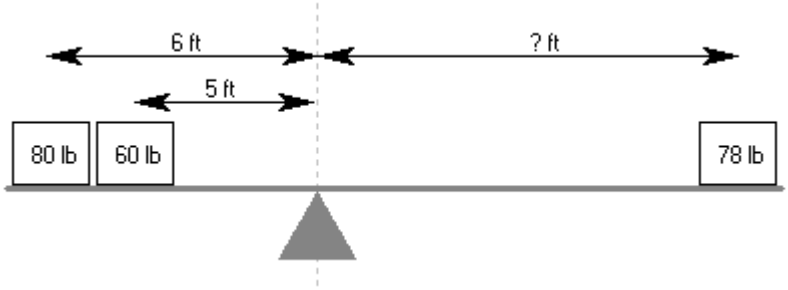
1) How much weight is required to balance the lever?



A	B	C	D	E
15 lbs	12 lbs	18 lbs	16 lbs	20 lbs

A B C D E

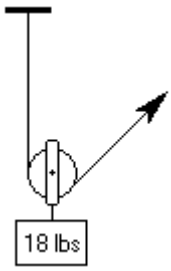
2) How far from the fulcrum does the 78 lb weight need to be to balance the lever?



A	B	C	D	E
6 ft	11 ft	10 ft	8ft	12 ft

A B C D E

3) Approximately how much force is needed to lift the weight?

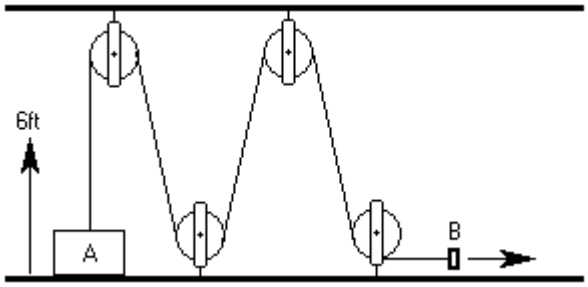


A	B	C	D	E
9 lbs	18 lbs	6 lbs	24 lbs	10 lbs

A B C D E

Mechanical Aptitude Tests

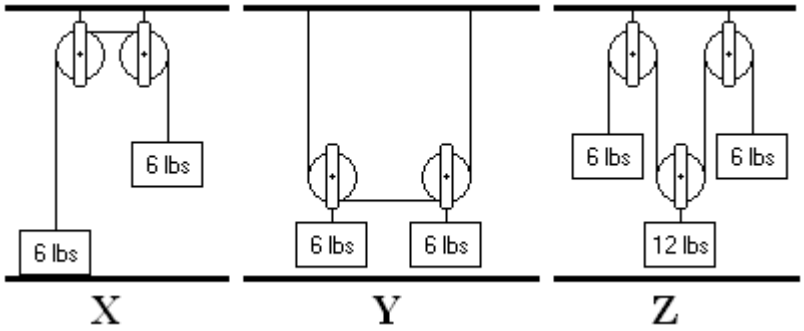
4) How far does B need to move to lift the weight 6 feet?



A	B	C	D	E
3 ft	6 ft	12 ft	8 ft	18 ft

A B C D E

5) Which of the pulley systems shown are stable?



A	B	C	D	E
X only	Y only	Z only	X & Z	Y & Z

A B C D E

6) If gear X turns clockwise at a constant speed of 10 rpm. How does gear Y turn?

A	B	C	D	E
anti c/w 10 rpm	c/w 10 rpm	c/w 5 rpm	anti c/w 5 rpm	c/w 20 rpm

A B C D E

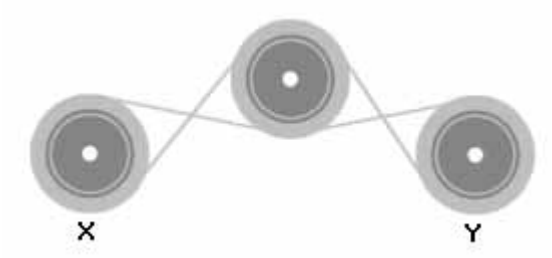
Mechanical Aptitude Tests

7) If gear X turns clockwise at a constant speed of 10 rpm. How does gear Y turn?

A	B	C	D	E
anti c/w 10 rpm	c/w 10 rpm	c/w 5 rpm	anti c/w 5 rpm	c/w 20 rpm

A B C D E

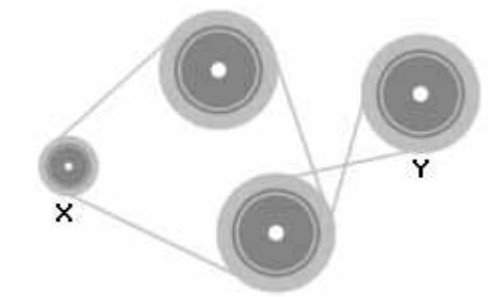
8) If drive wheel X rotates clockwise at a speed of 10 rpm. How does wheel Y turn?



A	B	C	D	E
anti c/w faster	c/w slower	c/w faster	anti c/w slower	c/w same

A B C D E

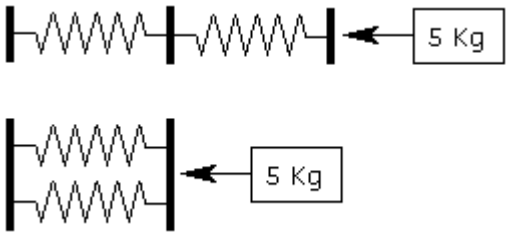
9) If drive wheel X rotates clockwise at a speed of 10 rpm. How does wheel Y turn?



A	B	C	D	E
anti c/w faster	c/w slower	c/w faster	anti c/w slower	c/w same

A B C D E

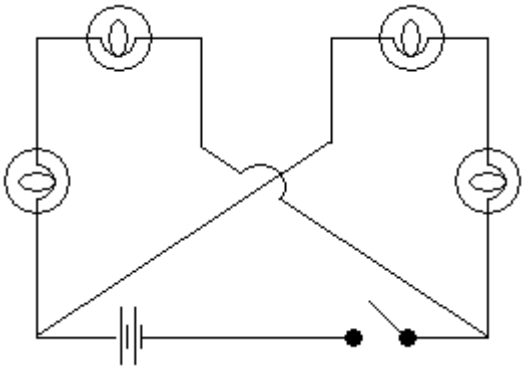
10) A force of 5 Kg compresses the springs in series 10cm. What will be the total distance that the springs in parallel are compressed?



A	B	C	D	E
10 cms	5 cms	2.5 cms	7.5 cms	15 cms

A B C D E

11) In the circuit shown, how many bulbs will light when the switch is closed?



A	B	C	D	E
None	One	Two	Three	Four

A B C D E

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Mechanical Aptitude Tests

12) Which tool or combination of tools would be most useful for working with sheet glass?

A	B	C	D	E
4 & 2	6	9	4	3 & 6

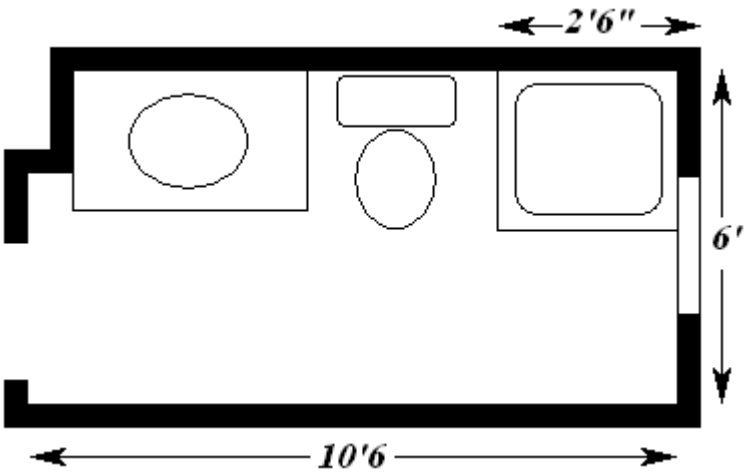
A B C D E

13) Which tool or combination of tools would be most useful for auto body repair work?

A	B	C	D	E
1 & 8	3, 5 & 7	8	1 & 9	3 & 6

A B C D E

Mechanical Aptitude Tests



The sketch shows the floor plan of a bathroom. The shower tray is 2'6" square and is fixed to the floor. The toilet and washbasin are both wall mounted.

14) Allowing for 15% wastage, approximately how many square yards of floor tiles should be ordered?

A	B	C	D	E
7.25	6.25	9.25	5.50	8.50

A B C D E

15) The floor tiles measure 6" square and can be laid at a rate of 30 per hour. Approximately how long will it take to tile the floor?

A	B	C	D	E
14 hrs	4 hrs	12 hrs	8 hrs	10 hrs

A B C D E

16) The bathroom is 8' high, the window measures 2' square and the door measures 7' x 2'6". How many square yards is the remaining wall area of the room?

A	B	C	D	E
32	22	24	25	27

A B C D E

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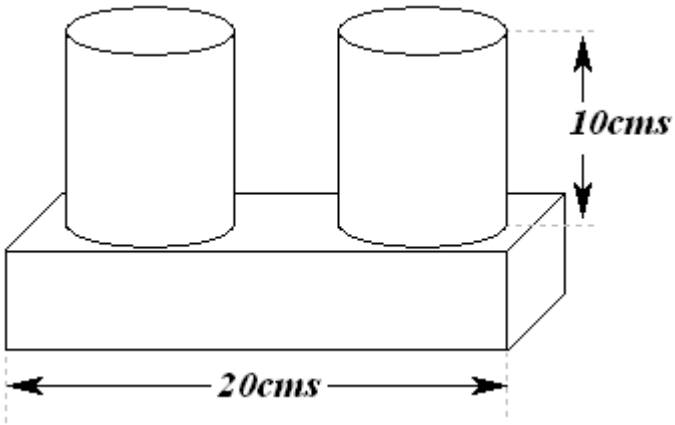
Mechanical Aptitude Tests



The sketch shows a component made from 5cm square bar and 5cm diameter rod. The density of steel is 8g per cubic centimetre. For shipping purposes the components are packed into individual boxes before being packed into shipping crates measuring approximately 0.25m x 0.3m x 0.4m. Shipping crates are packed on pallets to a maximum weight of 800 Kg.

17) What is the approximate total volume of the component in cubic centimetres?

A	B	C	D	E
655	788	967	422	892

A B C D E

18) What is the approximate weight of the component?

A	B	C	D	E
72.4 Kg	7.14 Kg	7.34 Kg	14.4 Kg	5.14 Kg

A B C D E

19) How many boxed components can be fitted into a shipping crate?

A	B	C	D	E
48	24	22	20	18

A B C D E

20) How many shipping crates can be fitted onto a palette?

A	B	C	D	E
5	4	7	6	12

A B C D E

End of Mechanical Reasoning - Test 3

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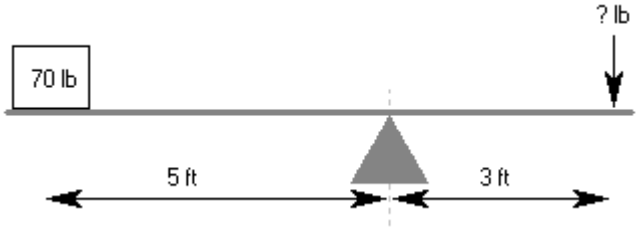
Mechanical Aptitude Tests

Mechanical Reasoning Test 4: 20 Questions
Answer as many questions as you can in 15 minutes. Circle the letter below the question which corresponds to the correct answer. You are advised to use a calculator.

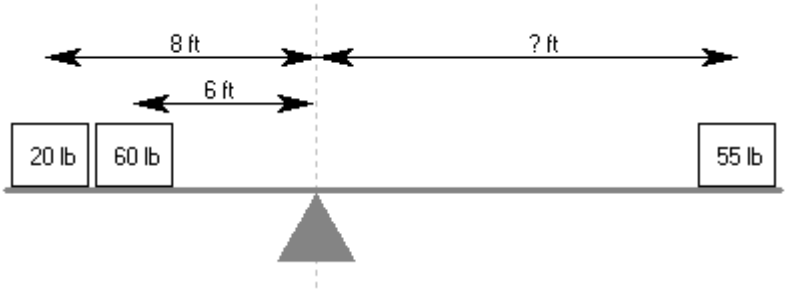
1) How much weight is required to just tip the lever?



A	B	C	D	E
115 lbs	112 lbs	118 lbs	116 lbs	117 lbs

A B C D E

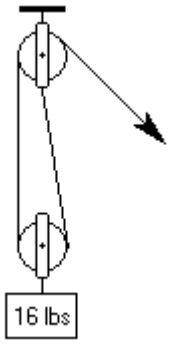
2) How far from the fulcrum does the 55 lb weight need to be to just tip the lever?



A	B	C	D	E
6 ft	9 ft 6 inches	10 ft 6 inches	8ft 6 inches	10 ft

A B C D E

3) Approximately how much force is needed to lift the weight?

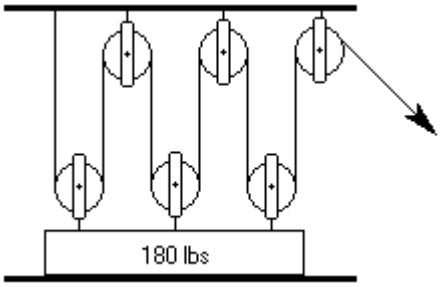


A	B	C	D	E
9 lbs	8 lbs	6 lbs	4 lbs	16 lbs

A B C D E

Mechanical Aptitude Tests

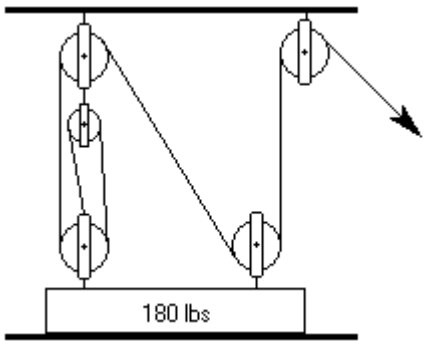
4) Approximately how much force is needed to lift the weight?



A	B	C	D	E
15 lbs	30 lbs	45 lbs	60 lbs	90 lbs

A B C D E

5) Approximately how much force is needed to lift the weight?



A	B	C	D	E
30 lbs	36 lbs	45 lbs	60 lbs	90 lbs

A B C D E

6) If gear X turns clockwise at a constant speed of 10 rpm. How does gear Y turn?

A	B	C	D	E
anti c/w 10 rpm	c/w 10 rpm	c/w 5 rpm	anti c/w 5 rpm	c/w 20 rpm

A B C D E

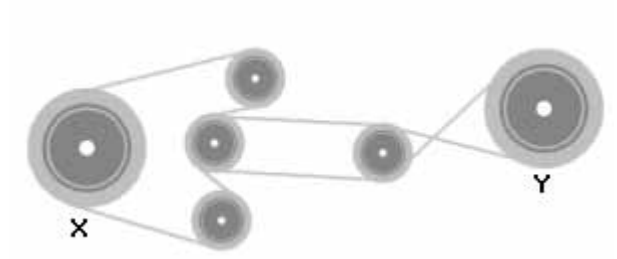
Mechanical Aptitude Tests

7) If gear X turns clockwise at a constant speed of 10 rpm. How does gear Y turn?

A	B	C	D	E
anti c/w 10 rpm	c/w 10 rpm	c/w 5 rpm	anti c/w 5 rpm	anti c/w 20 rpm

A B C D E

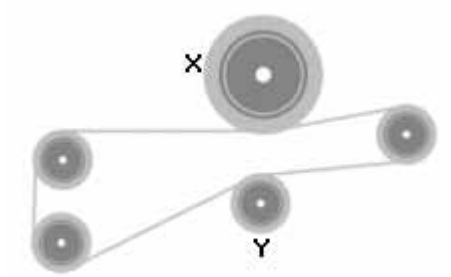
8) If drive wheel X rotates clockwise at a speed of 10 rpm. How does wheel Y turn?



A	B	C	D	E
anti c/w faster	c/w slower	c/w faster	anti c/w slower	c/w same

A B C D E

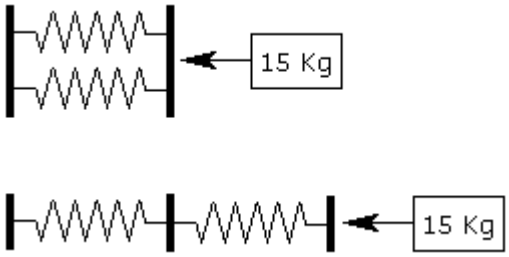
9) If drive wheel X rotates clockwise at a speed of 10 rpm. How does wheel Y turn?



A	B	C	D	E
anti c/w faster	c/w slower	c/w faster	anti c/w slower	c/w same

A B C D E

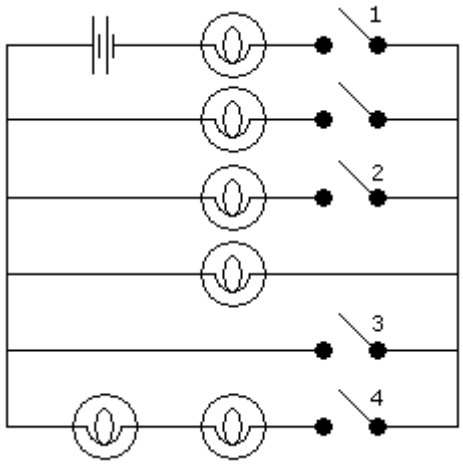
10) A force of 15 Kg compresses the parallel in series 10cm. What will be the total distance that the springs in series are compressed?



A	B	C	D	E
10 cms	5 cms	20 cms	7.5 cms	15 cms

A B C D E

11) How many bulbs will light when switches 1, 2, 3 and 4 are closed?



A	B	C	D	E
None	One	Two	Three	Four

A B C D E

Mechanical Aptitude Tests

12) Which tool or combination of tools would be most useful for masonry work?

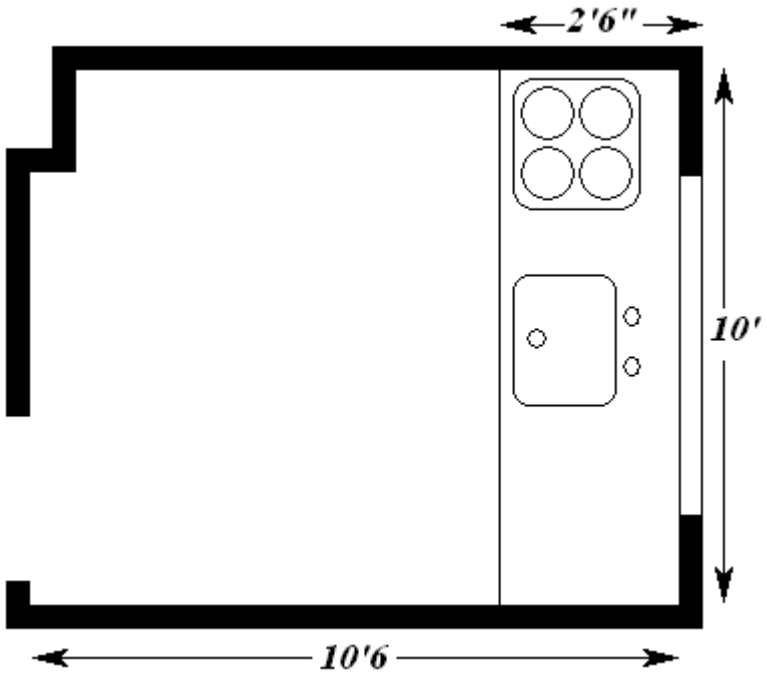
A	B	C	D	E
3	6	4	7	2

A B C D E

13) Which tool or combination of tools would be most useful for fitting a wooden door?

A	B	C	D	E
3,5 & 7	1 & 9	2, 3 & 4	4, 6 & 7	4 & 6

A B C D E



The sketch shows the floor plan of a kitchen. The kitchen units and worktop project 2' 6" from the wall at a height of 36". The window is 7 feet wide and 4 feet high – it is flush with the level of the worktop. The ceiling is 8 feet high. The specification requires 6" x 6" decorative tiles to be fitted above the worktop on three sides to a height of 24".

14) Allowing for 15% wastage, approximately how many tiles should be ordered?

A	B	C	D	E
82	74	64	70	80

A B C D E

15) The door measures 7' x 2'6". Calculate the remaining wall area in square feet (i.e. the area that has not been tiled)

A	B	C	D	E
268	144	306	221	180

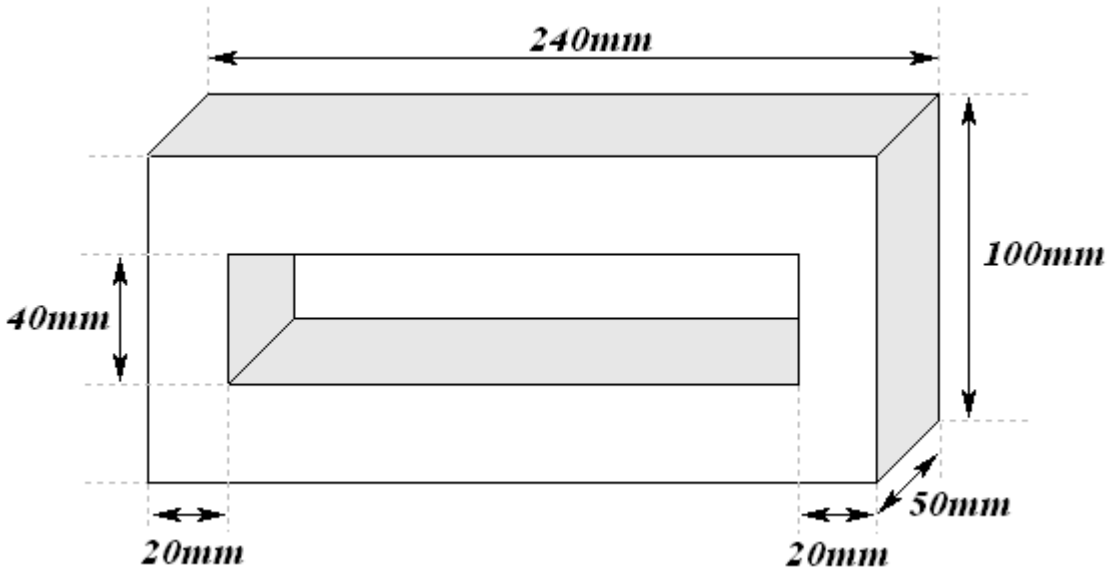
A B C D E

16) The walls and the ceiling are to be painted. How many square yards of paint will be required?

A	B	C	D	E
24	36	30	42	26

A B C D E

Mechanical Aptitude Tests



The sketch shows a component made from titanium. The density of titanium is 4.5g per cubic cm. For shipping purposes the components are packed into individual boxes before being packed into shipping crates measuring 0.24m x 0.3m x 0.4m. Shipping crates are packed on pallets to a maximum weight of 800 Kg.

17) What is the approximate total volume of the component in cubic centimetres?

A	B	C	D	E
800	750	700	680	775

A B C D E

18) What is the approximate weight of the component?

A	B	C	D	E
3.8Kg	4.2Kg	3.6Kg	38Kg	17Kg

A B C D E

19) How many components can be fitted into a shipping crate?

A	B	C	D	E
22	26	18	24	20

A B C D E

20) How many shipping crates can be fitted onto a palette?

A	B	C	D	E
6	9	8	11	7

A B C D E

End of Mechanical Reasoning - Test 4

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Mechanical Aptitude Tests

Answers to Mechanical Reasoning Tests 1-4

Question	Test 1	Test 2	Test 3	Test 4
1)	**B**	**C**	**A**	**E**
2)	**D**	**D**	**C**	**B**
3)	**E**	**A**	**A**	**B**
4)	**D**	**C**	**B**	**B**
5)	**D**	**B**	**D**	**B**
6)	**B**	**C**	**A**	**A**
7)	**D**	**B**	**A**	**E**
8)	**B**	**D**	**E**	**E**
9)	**C**	**B**	**D**	**C**
10)	**E**	**E**	**B**	**C**
11)	**A**	**E**	**E**	**E**
12)	**C**	**C**	**B**	**C**
13)	**D**	**B**	**C**	**A**
14)	**C**	**E**	**A**	**B**
15)	**B**	**A**	**D**	**D**
16)	**D**	**D**	**E**	**B**
17)	**C**	**E**	**E**	**A**
18)	**A**	**C**	**B**	**C**
19)	**B**	**B**	**D**	**D**
20)	**C**	**D**	**A**	**B**

Mechanical Aptitude Tests

Fault Diagnosis Questions

Fault Diagnosis tests are used to select technical personnel who need to be able to find and repair faults in electronic and mechanical systems. As modern equipment of all types becomes more dependent on electronic control systems (and arguably more complex) the ability to approach problems logically in order to find the cause of the fault is increasingly important.

Fault Diagnosis		Operatives	Supervisory	Management
Craft & Technical		Y	Y	Y
Clerical & Administrative				
Police, Fire, Military etc.	M	M - Ususally restricted to technical roles in military		
Management Trainee				
Graduate & Professional				

This type of test is used extensively to select technical and maintenance personnel as well as to select for artificer (technical) roles within the armed forces. For example, aircraft technician.

This type of test usually forms part of a test battery in which verbal and numerical reasoning also feature. No specialised knowledge is required to answer these fault diagnosis questions (unlike mechanical reasoning).

Only when you are happy that you understand how these questions work you should attempt the sample paper. Remember that the object is to work systematically through the questions and avoid mistakes.

Mechanical Aptitude Tests

Example Question 1

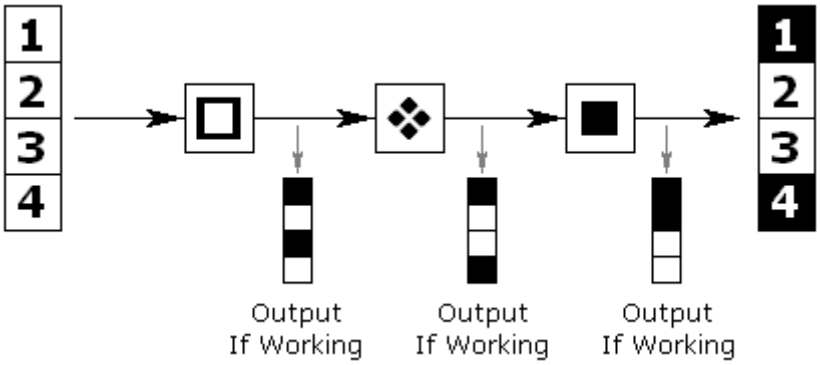
There are two components to this question.

1. A grid which describes the function of a series of switches.
2. A flowchart showing: input – switches – output.

Each of the switches acts to alter the input before the next switch in the series.

Switch	Function	Code
◆	Inverts 1 & 2 (on=off, off=on)	**A** - Indicates Fault
□	Inverts 1 & 3 (on=off, off=on)	**B** - Indicates Fault
■	Inverts 2 & 4 (on=off, off=on)	**C** - Indicates Fault
❖	Inverts 3 & 4 (on=off, off=on)	**D** - Indicates Fault
	N = On N = Off	**E** - Indicates No Faults

This grid shows four switches and their effect on four numbered inputs which may be either on or off. The first switch (diamond symbol) inverts inputs one and two. That is, if the input is on, it is turned off and if it is off it is turned on. The other switches act in a similar way as detailed in the grid. If a switch is not working then it has no effect on the input which passes through unchanged.



In this example flowchart, all of the inputs (numbered 1, 2, 3, 4) are on.
When these inputs pass through the first switch 1 & 3 are inverted (i.e. switched off).
These modified inputs (1=off, 2=on, 3=off, 4=on) then pass through the second switch.
The second switch inverts inputs 3 & 4, which gives (1=off, 2=on, 3=on, 4=off).
These modified inputs (1=off, 2=on, 3=on, 4=off) then pass through the third switch.
The third switch should invert inputs 2 & 4, giving (1=off, 2=off, 3=on, 4=on).

However the third switch is not working so the output from switch two is not changed.
The answer to this question is therefore ‘C’ – because switch type ‘C’ is at fault.

Mechanical Aptitude Tests

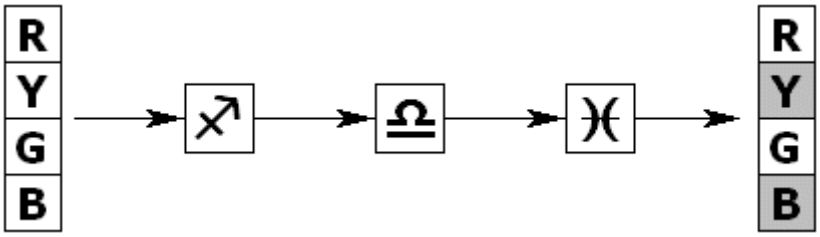
Example Question 2

There are two components to this question.

1. A grid which describes the function of a series of switches.
2. A flowchart showing: input – switches – output.

Switch	Function	Fault	Code
♐	Allows all colours to pass	Allows only red to pass	A
♎	Allows only red & yellow to pass	Allows only green & blue to pass	B
♓	Turns on red & green	Turns on yellow & blue	C
⌘	Toggles all colours	Toggles red & blue	D
No Faults			E
R = Red	Y = Yellow	G = Green	B = Blue	= On	= Off

This grid shows four switches and their effect on four coloured inputs which may be either on or off. For example, the first switch (arrow symbol) allows all colours to pass when working but when at fault allows only red to pass.



In this example, all of the inputs (Red, Yellow, Green, Blue) are on.

When these inputs pass through the first switch, all colours are allowed to pass if the switch is working and only red is allowed to pass if the switch is at fault.

This modified input passes through the second switch which allows only red and yellow to pass if the switch is working and only green and blue to pass if the switch is at fault..

This modified input passes through the third switch which turns on red and green if the switch is working and turns on yellow and blue if the switch is at fault.

Once again, you need to compare the input and output and determine which of the switches is at fault. In this case, it is switch 'A' (the first switch) which allows all colours to pass when working but allows only red to pass when at fault.

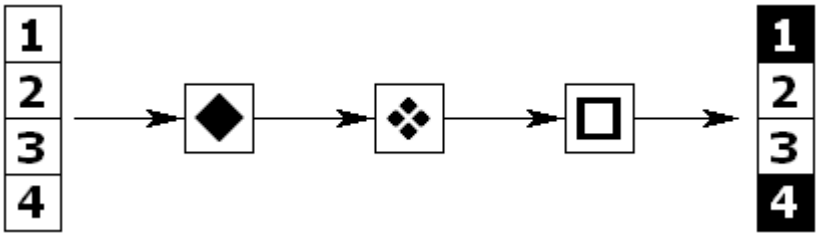
Note that the fourth switch (not used in this question) 'toggles' the lights. That is, it switches a light off if it is on and vice versa.

Mechanical Aptitude Tests

Test 1 - 12 Questions
Answer as many questions as you can in 10 minutes. Circle the letter on the right which corresponds to the correct answer.

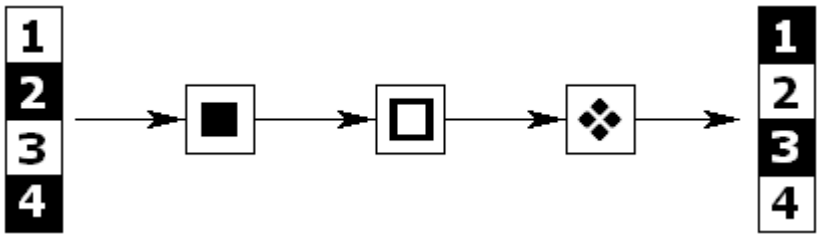
Switch	Function	Code
◆	Inverts 1 & 2 (on=off, off=on)	**A** - Indicates Fault
□	Inverts 1 & 3 (on=off, off=on)	**B** - Indicates Fault
■	Inverts 2 & 4 (on=off, off=on)	**C** - Indicates Fault
❖	Inverts 3 & 4 (on=off, off=on)	**D** - Indicates Fault
N = On	N = Off	**E** - Indicates No Faults

1) Which switch is showing a fault?



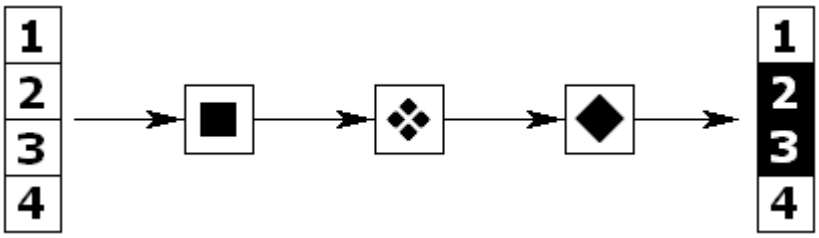
A B C D E

2) Which switch is showing a fault?



A B C D E

3) Which switch is showing a fault?



A B C D E

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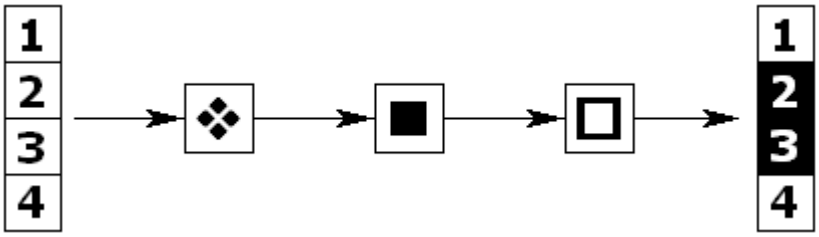
Mechanical Aptitude Tests

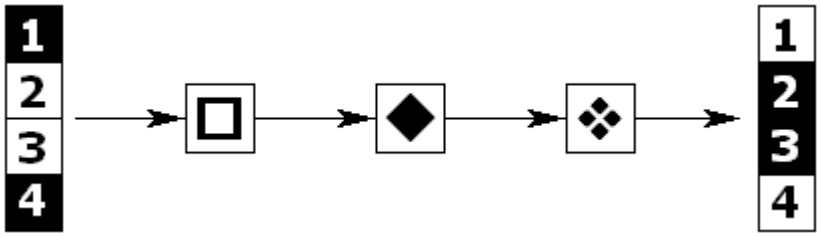
Switch	Function	Code
◆	Inverts 1 & 2 (on=off, off=on)	**A** - Indicates Fault
□	Inverts 1 & 3 (on=off, off=on)	**B** - Indicates Fault
■	Inverts 2 & 4 (on=off, off=on)	**C** - Indicates Fault
❖	Inverts 3 & 4 (on=off, off=on)	**D** - Indicates Fault
N = On	N = Off	**E** - Indicates No Faults

4) Which switch is showing a fault?



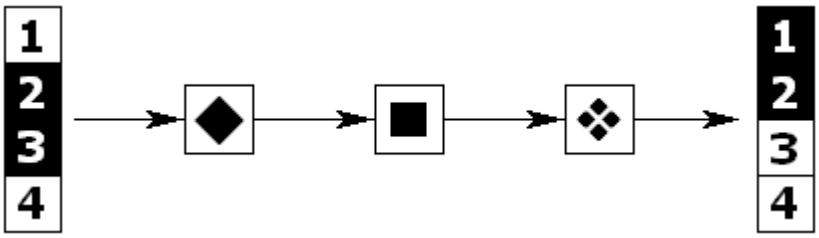
A B C D E

5) Which switch is showing a fault?



A B C D E

6) Which switch is showing a fault?



A B C D E

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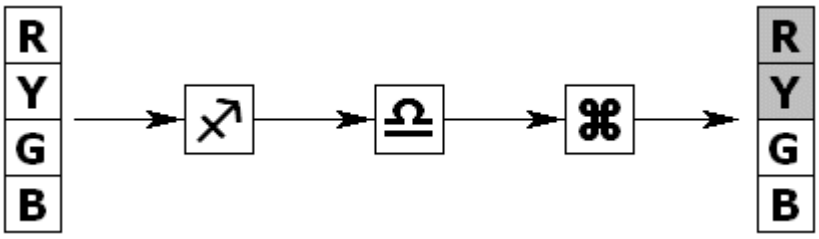
Mechanical Aptitude Tests

Switch	Function	Fault	Code
♐	Allows all colours to pass	Allows only red to pass	A
♎	Allows only red & yellow to pass	Allows only green & blue to pass	B
♓	Turns on red & green	Turns on yellow & blue	C
⌘	Toggles all colours	Toggles red & blue	D
No Faults			E

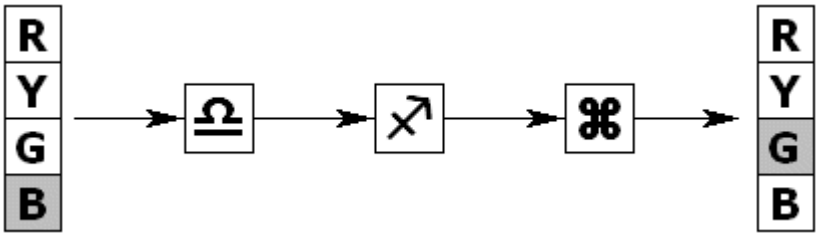
R = Red Y = Yellow G = Green B = Blue = On = Off

7) Which switch is showing a fault?



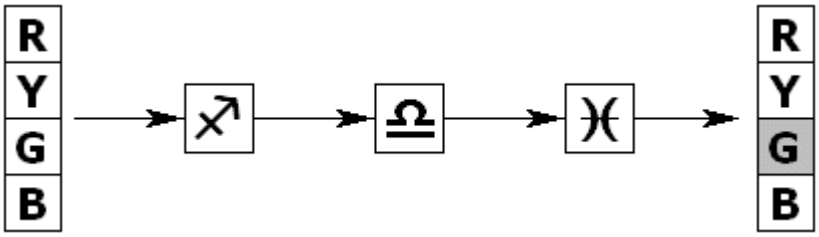
A B C D E

8) Which switch is showing a fault?



A B C D E

9) Which switch is showing a fault?



A B C D E

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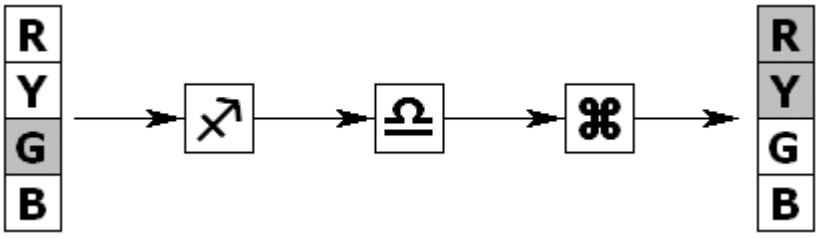
Mechanical Aptitude Tests

Switch	Function	Fault	Code
♐	Allows all colours to pass	Allows only red to pass	A
♎	Allows only red & yellow to pass	Allows only green & blue to pass	B
♓	Turns on red & green	Turns on yellow & blue	C
⌘	Toggles all colours	Toggles red & blue	D
No Faults			E

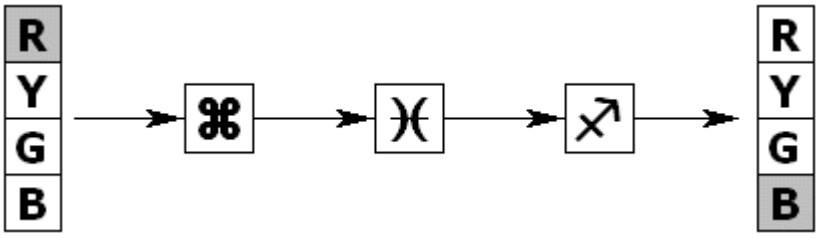
R = Red Y = Yellow G = Green B = Blue = On = Off

10) Which switch is showing a fault?



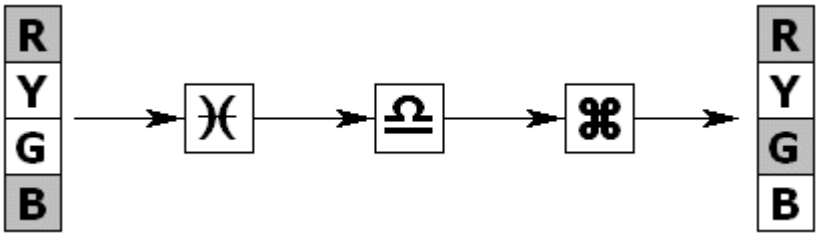
A B C D E

11) Which switch is showing a fault?



A B C D E

12) Which switch is showing a fault?



A B C D E

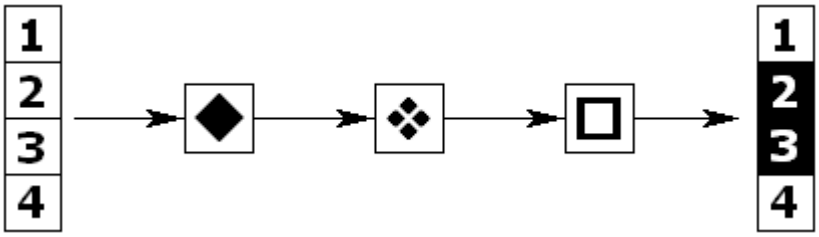
End of Fault Diagnosis - Test 1

Mechanical Aptitude Tests

Test 2 - 12 Questions
Answer as many questions as you can in 10 minutes. Circle the letter on the right which corresponds to the correct answer.

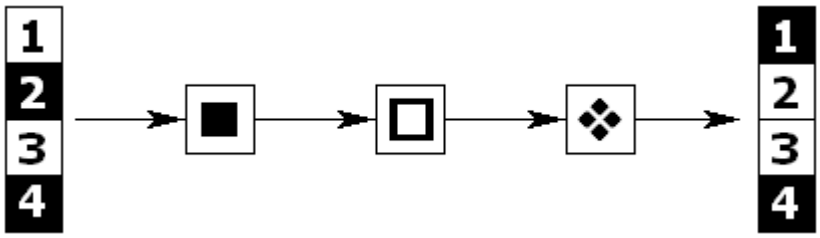
Switch	Function	Code
◆	Inverts 1 & 2 (on=off, off=on)	**A** - Indicates Fault
□	Inverts 1 & 3 (on=off, off=on)	**B** - Indicates Fault
■	Inverts 2 & 4 (on=off, off=on)	**C** - Indicates Fault
❖	Inverts 3 & 4 (on=off, off=on)	**D** - Indicates Fault
N = On	N = Off	**E** - Indicates No Faults

1) Which switch is showing a fault?



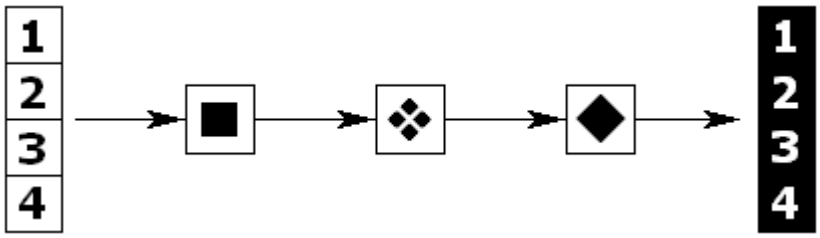
A B C D E

2) Which switch is showing a fault?



A B C D E

3) Which switch is showing a fault?

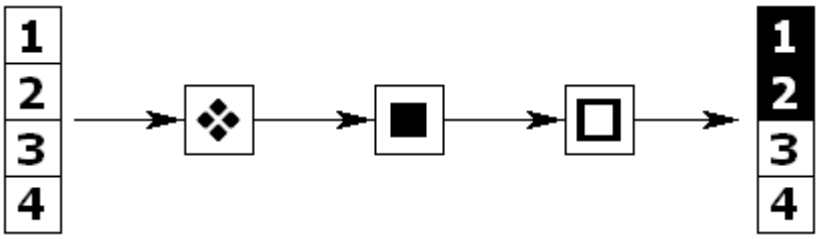


A B C D E

Mechanical Aptitude Tests

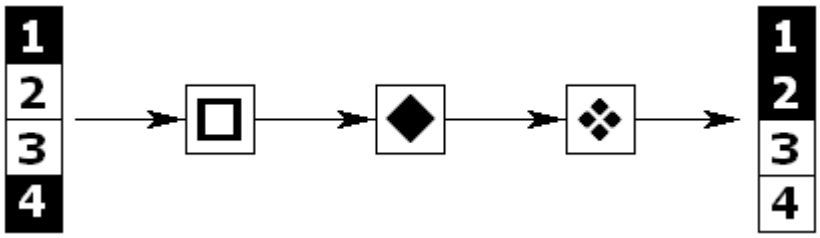
Switch	Function	Code
◆	Inverts 1 & 2 (on=off, off=on)	**A** - Indicates Fault
□	Inverts 1 & 3 (on=off, off=on)	**B** - Indicates Fault
■	Inverts 2 & 4 (on=off, off=on)	**C** - Indicates Fault
❖	Inverts 3 & 4 (on=off, off=on)	**D** - Indicates Fault
N = On	N = Off	**E** - Indicates No Faults

4) Which switch is showing a fault?



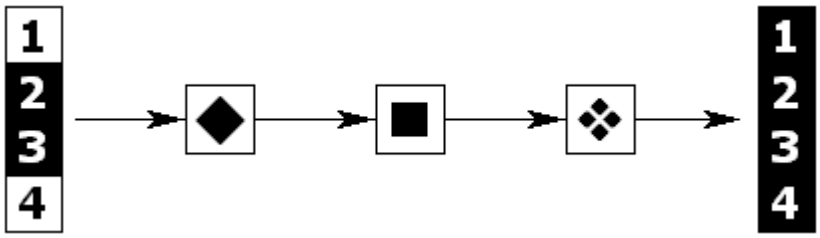
A B C D E

5) Which switch is showing a fault?



A B C D E

6) Which switch is showing a fault?



A B C D E

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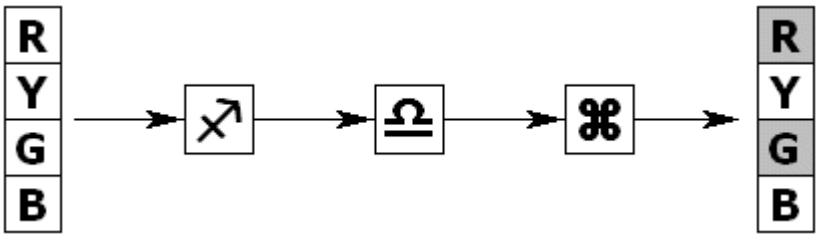
Mechanical Aptitude Tests

Switch	Function	Fault	Code
♐	Allows all colours to pass	Allows only red to pass	A
♎	Allows only red & yellow to pass	Allows only green & blue to pass	B
♓	Turns on red & green	Turns on yellow & blue	C
⌘	Toggles all colours	Toggles red & blue	D
No Faults			E

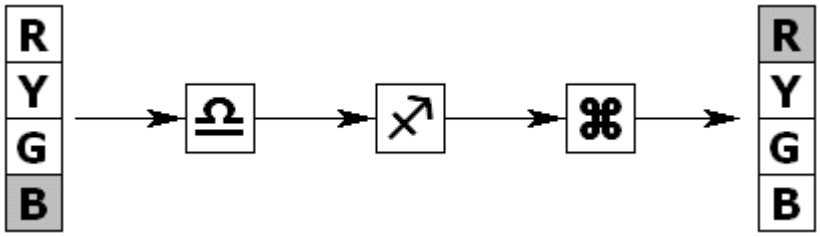
R = Red Y = Yellow G = Green B = Blue = On = Off

7) Which switch is showing a fault?



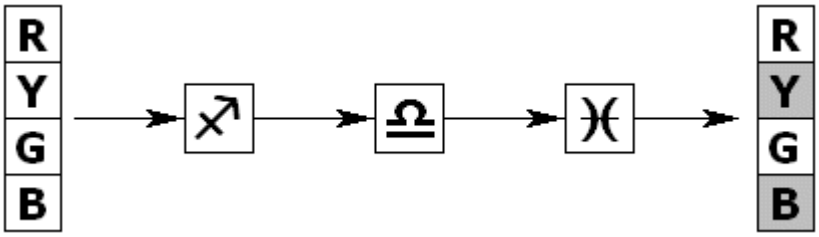
A B C D E

8) Which switch is showing a fault?



A B C D E

9) Which switch is showing a fault?



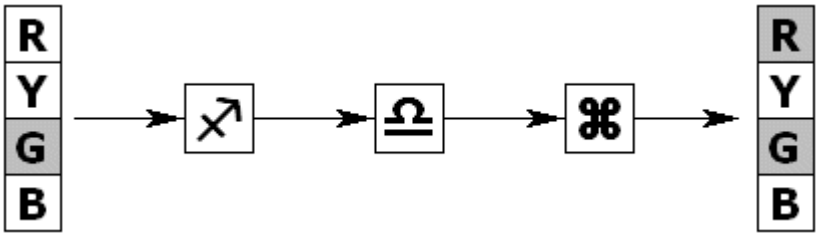
A B C D E

Mechanical Aptitude Tests

Switch	Function	Fault	Code
♐	Allows all colours to pass	Allows only red to pass	A
♎	Allows only red & yellow to pass	Allows only green & blue to pass	B
♓	Turns on red & green	Turns on yellow & blue	C
⌘	Toggles all colours	Toggles red & blue	D
No Faults			E

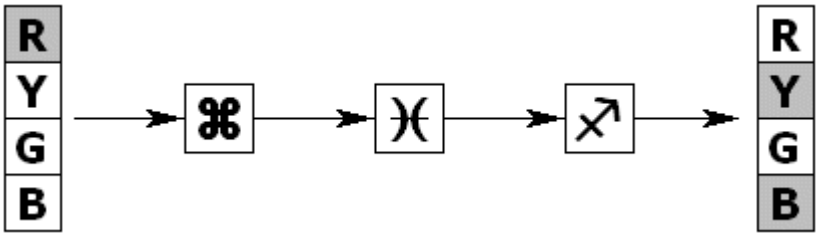
R = Red Y = Yellow G = Green B = Blue = On = Off

10) Which switch is showing a fault?



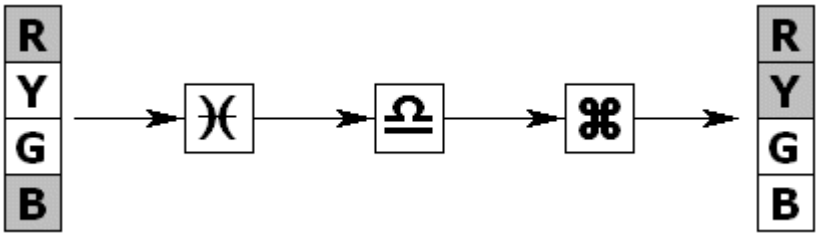
A B C D E

11) Which switch is showing a fault?



A B C D E

12) Which switch is showing a fault?



A B C D E

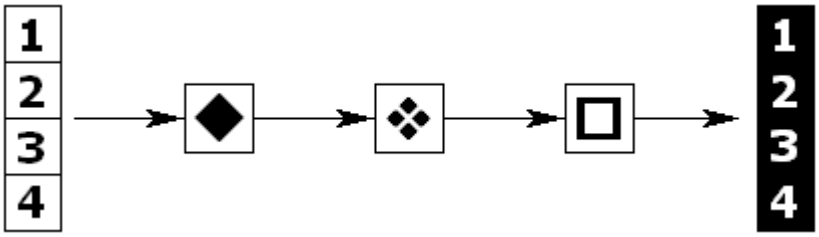
End of Fault Diagnosis - Test 2

Mechanical Aptitude Tests

Test 3 - 12 Questions
Answer as many questions as you can in 10 minutes. Circle the letter on the right which corresponds to the correct answer.

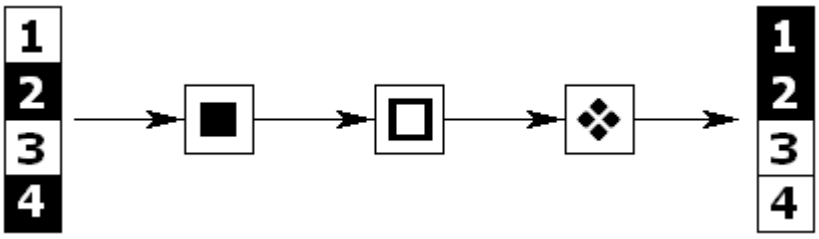
Switch	Function	Code
◆	Inverts 1 & 2 (on=off, off=on)	**A** - Indicates Fault
□	Inverts 1 & 3 (on=off, off=on)	**B** - Indicates Fault
■	Inverts 2 & 4 (on=off, off=on)	**C** - Indicates Fault
❖	Inverts 3 & 4 (on=off, off=on)	**D** - Indicates Fault
N = On	N = Off	**E** - Indicates No Faults

1) Which switch is showing a fault?



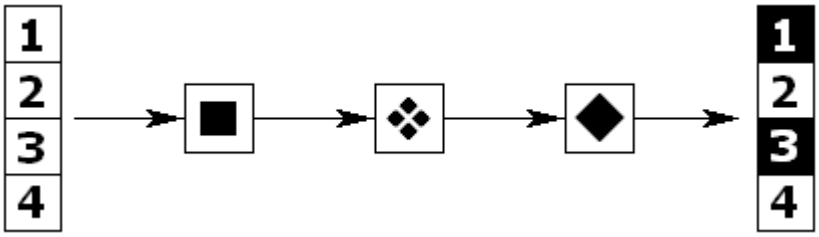
A B C D E

2) Which switch is showing a fault?



A B C D E

3) Which switch is showing a fault?

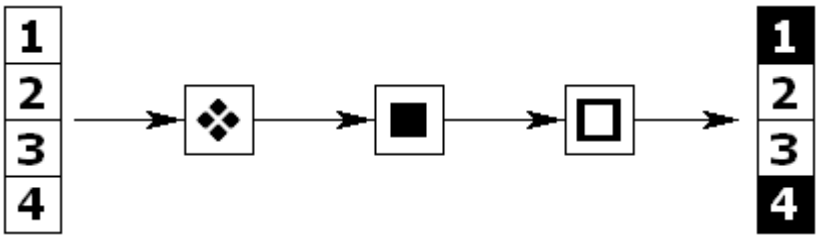


A B C D E

Mechanical Aptitude Tests

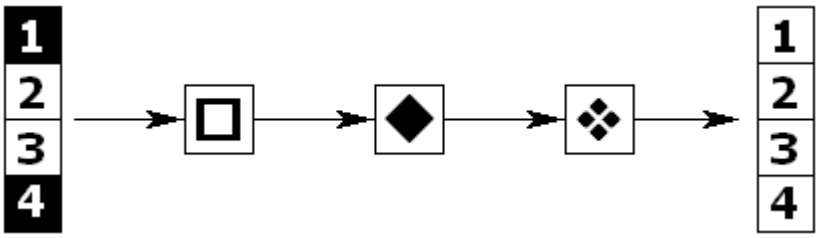
Switch	Function	Code
◆	Inverts 1 & 2 (on=off, off=on)	**A** - Indicates Fault
□	Inverts 1 & 3 (on=off, off=on)	**B** - Indicates Fault
■	Inverts 2 & 4 (on=off, off=on)	**C** - Indicates Fault
❖	Inverts 3 & 4 (on=off, off=on)	**D** - Indicates Fault
N = On	N = Off	**E** - Indicates No Faults

4) Which switch is showing a fault?



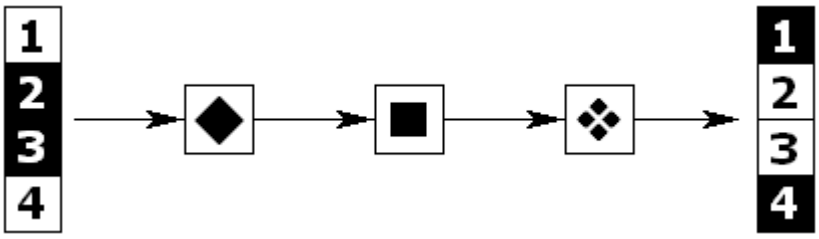
A B C D E

5) Which switch is showing a fault?



A B C D E

6) Which switch is showing a fault?



A B C D E

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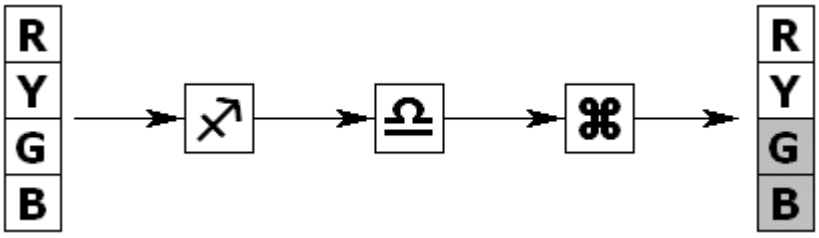
Mechanical Aptitude Tests

Switch	Function	Fault	Code
♐	Allows all colours to pass	Allows only red to pass	A
♎	Allows only red & yellow to pass	Allows only green & blue to pass	B
♓	Turns on red & green	Turns on yellow & blue	C
⌘	Toggles all colours	Toggles red & blue	D
No Faults			E

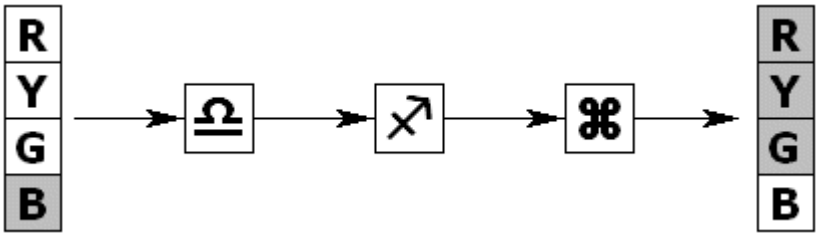
R = Red Y = Yellow G = Green B = Blue = On = Off

7) Which switch is showing a fault?



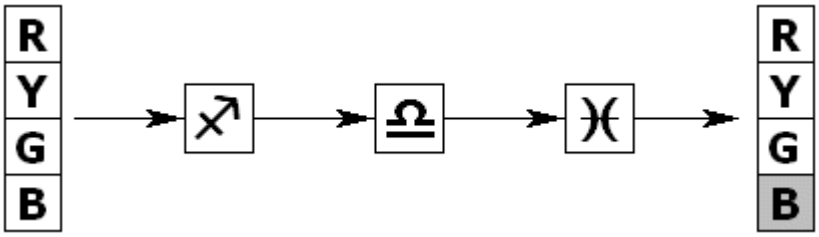
A B C D E

8) Which switch is showing a fault?



A B C D E

9) Which switch is showing a fault?



A B C D E

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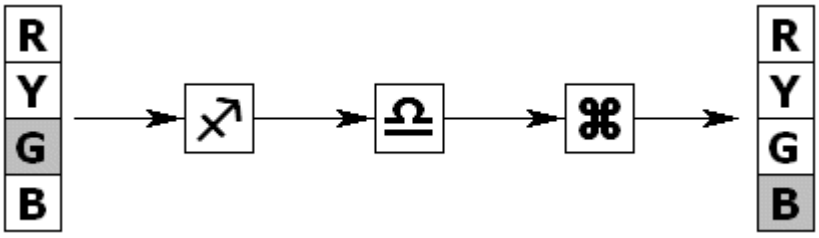
Mechanical Aptitude Tests

Switch	Function	Fault	Code
♐	Allows all colours to pass	Allows only red to pass	A
♎	Allows only red & yellow to pass	Allows only green & blue to pass	B
♓	Turns on red & green	Turns on yellow & blue	C
⌘	Toggles all colours	Toggles red & blue	D
No Faults			E

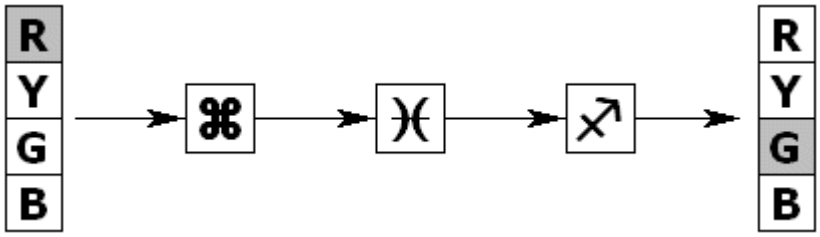
R = Red Y = Yellow G = Green B = Blue = On = Off

10) Which switch is showing a fault?



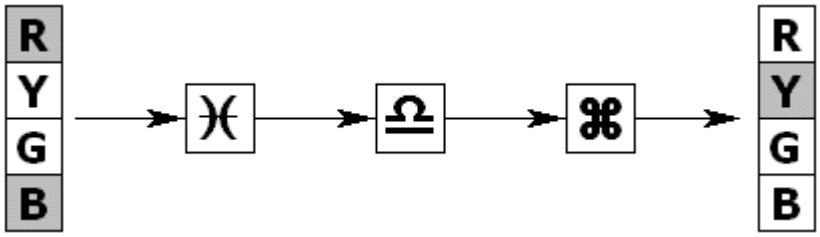
A B C D E

11) Which switch is showing a fault?



A B C D E

12) Which switch is showing a fault?



A B C D E

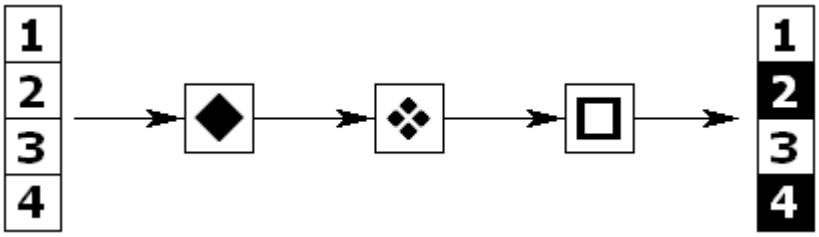
End of Fault Diagnosis - Test 3

Mechanical Aptitude Tests

Test 4 - 12 Questions
Answer as many questions as you can in 10 minutes. Circle the letter on the right which corresponds to the correct answer.

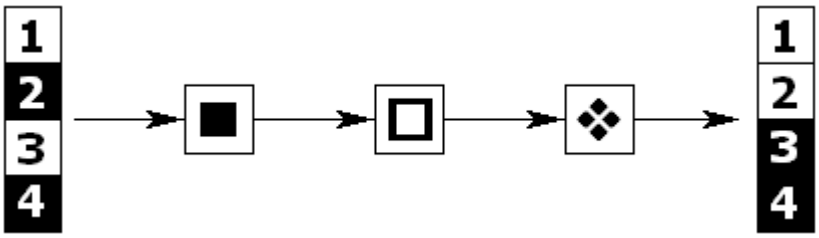
Switch	Function	Code
◆	Inverts 1 & 2 (on=off, off=on)	**A** - Indicates Fault
□	Inverts 1 & 3 (on=off, off=on)	**B** - Indicates Fault
■	Inverts 2 & 4 (on=off, off=on)	**C** - Indicates Fault
❖	Inverts 3 & 4 (on=off, off=on)	**D** - Indicates Fault
N = On	N = Off	**E** - Indicates No Faults

1) Which switch is showing a fault?



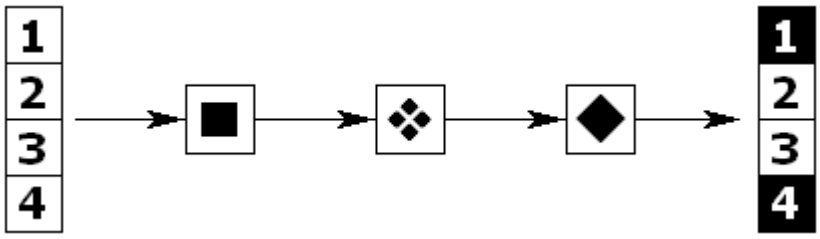
A B C D E

2) Which switch is showing a fault?



A B C D E

3) Which switch is showing a fault?



A B C D E

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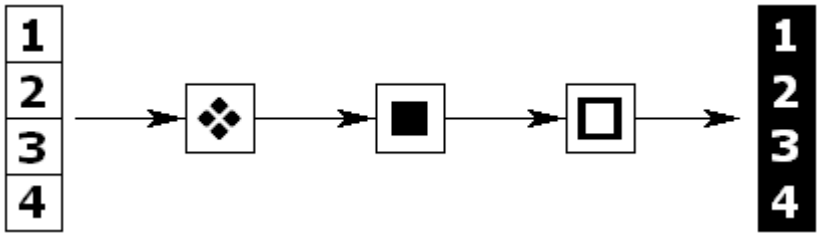
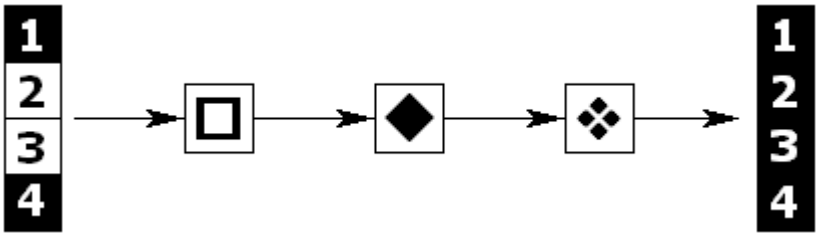
Switch	Function	Code
◆	Inverts 1 & 2 (on=off, off=on)	**A** - Indicates Fault
□	Inverts 1 & 3 (on=off, off=on)	**B** - Indicates Fault
■	Inverts 2 & 4 (on=off, off=on)	**C** - Indicates Fault
❖	Inverts 3 & 4 (on=off, off=on)	**D** - Indicates Fault
N = On	N = Off	**E** - Indicates No Faults

4) Which switch is showing a fault?



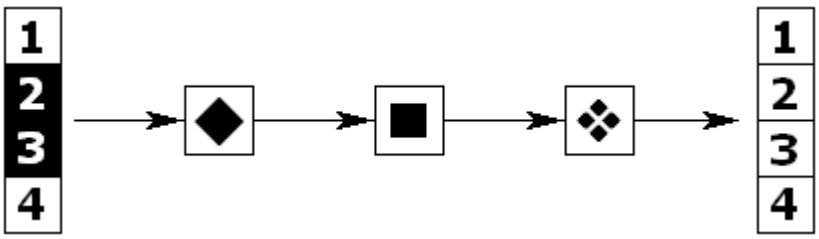
A B C D E

5) Which switch is showing a fault?



A B C D E

6) Which switch is showing a fault?



A B C D E

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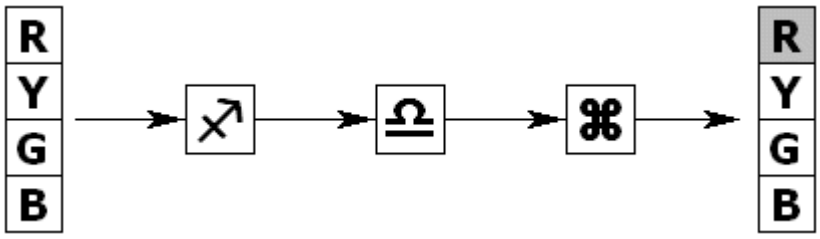
Switch	Function	Fault	Code
♐	Allows all colours to pass	Allows only red to pass	A
♎	Allows only red & yellow to pass	Allows only green & blue to pass	B
♓	Turns on red & green	Turns on yellow & blue	C
⌘	Toggles all colours	Toggles red & blue	D
No Faults			E

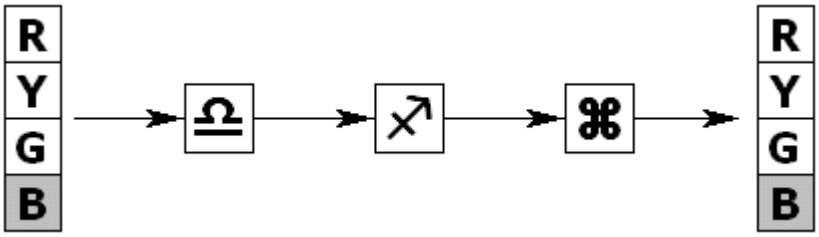
R = Red Y = Yellow G = Green B = Blue = On = Off

7) Which switch is showing a fault?



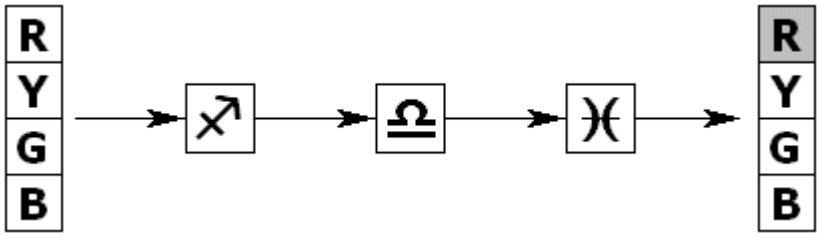
A B C D E

8) Which switch is showing a fault?



A B C D E

9) Which switch is showing a fault?



A B C D E

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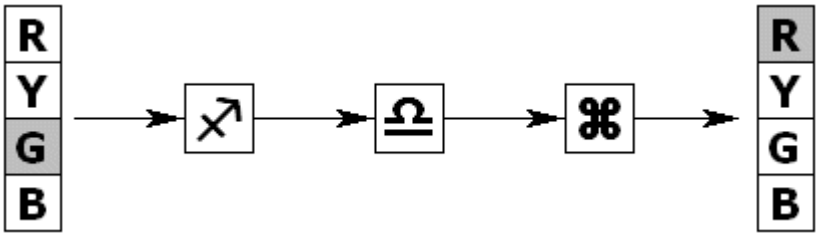
Switch	Function	Fault	Code
♐	Allows all colours to pass	Allows only red to pass	A
♎	Allows only red & yellow to pass	Allows only green & blue to pass	B
♓	Turns on red & green	Turns on yellow & blue	C
⌘	Toggles all colours	Toggles red & blue	D
No Faults			E

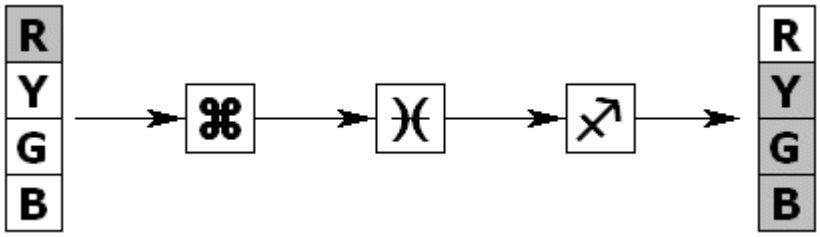
R = Red Y = Yellow G = Green B = Blue = On = Off

10) Which switch is showing a fault?



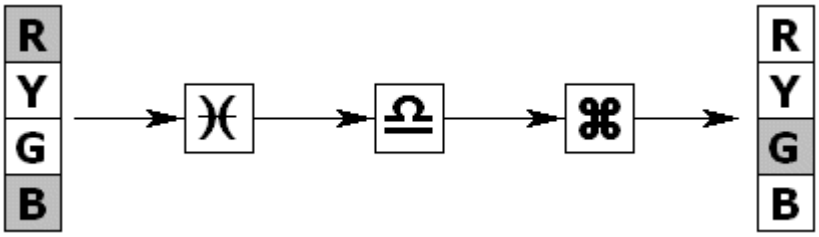
A B C D E

11) Which switch is showing a fault?



A B C D E

12) Which switch is showing a fault?



A B C D E

End of Fault Diagnosis - Test 4

Mechanical Aptitude Tests

Answers

	Test 1	Test 2	Test 3	Test 4
1)	**A**	**D**	**B**	**E**
2)	**D**	**E**	**C**	**B**
3)	**A**	**C**	**E**	**D**
4)	**B**	**E**	**C**	**D**
5)	**B**	**E**	**A**	**D**
6)	**E**	**D**	**C**	**A**
7)	**E**	**D**	**B**	**A**
8)	**B**	**D**	**E**	**A**
9)	**C**	**A**	**E**	**B**
10)	**E**	**D**	**B**	**A**
11)	**D**	**E**	**C**	**A**
12)	**D**	**E**	**C**	**B**

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Spatial Ability

Author : Paul Newton
Version: 2.3

Mechanical Aptitude Tests

The Importance of Spatial Ability Tests

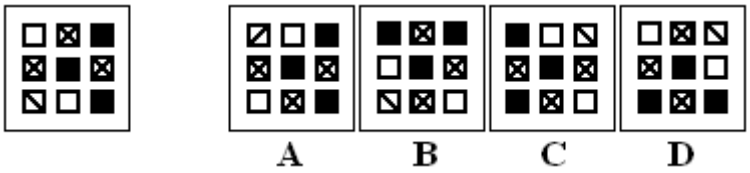
Spatial ability is required in production, technical and design jobs where plans and drawings are used, for example; engineering, architecture, surveying and design. It is also important in some branches of science where the ability to envisage the interactions of 3 dimensional components is essential.

These tests bear a superficial resemblance to abstract reasoning tests, as both types of question contain series of pictorial figures rather than words or numbers. However, spatial ability does not involve analysis and reasoning, it is purely a test of mental manipulation. Remember, you are not looking for the logical relationship between figures - what you are trying to do is form mental images and visualize movement or change between them.

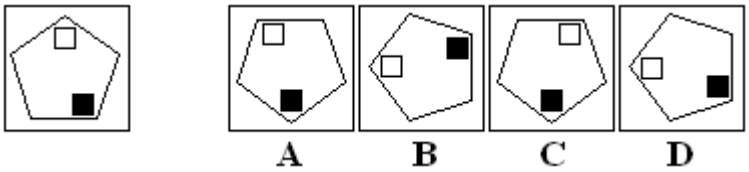
Even if you are applying for a job which does not involve spatial manipulation, you may still find it useful to work through some of these practice questions, purely as a mental exercise.

Example Spatial Ability Test Questions

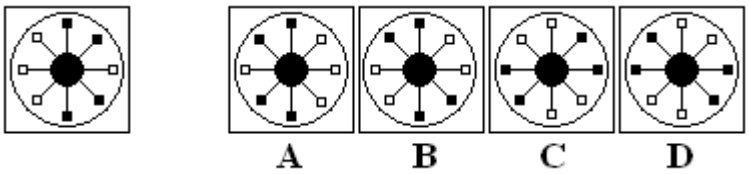
1) Which figure is identical to the first?



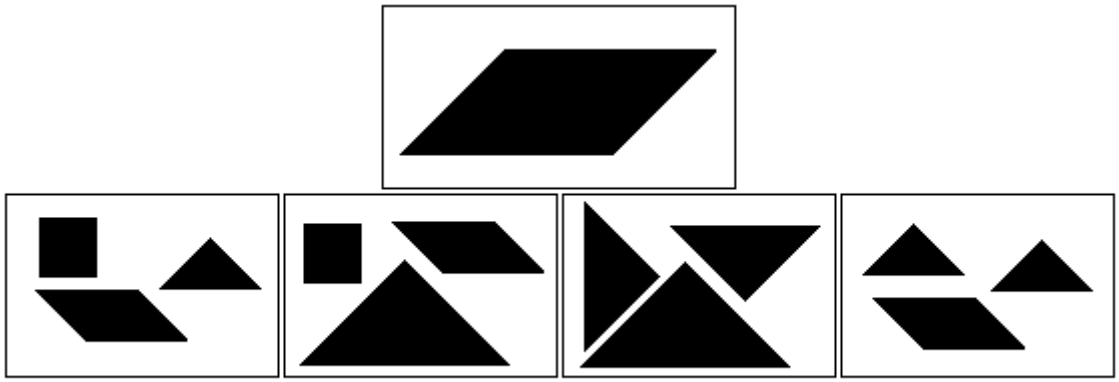
2) Which figure is identical to the first?



3) Which figure is identical to the first?

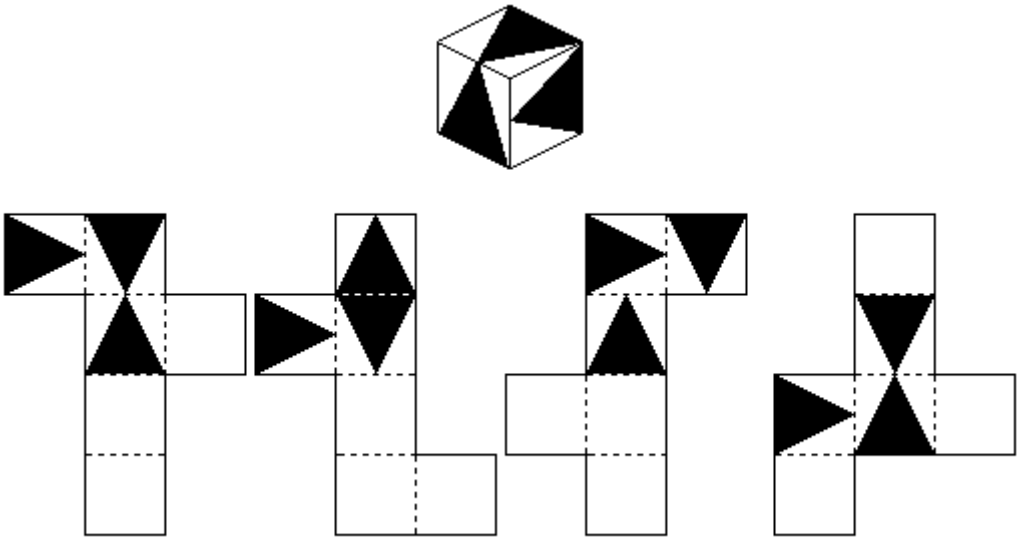


4) Which group of shapes can be assembled to make the shape shown?



A B C D

5) Which pattern can be folded to make the cube shown?



A B C D

The answers to these questions are given at the bottom of the page.
Spatial ability questions often involve the visual assembly and the disassembly of objects that have been rotated, which are viewed from different angles or objects that have different markings on their surfaces.

Generally speaking, if the questions involve the manipulation of 2 dimensional objects then they are probably fairly straightforward, but you will be challenged to answer them all in the time you are given. If the questions involve the manipulation of 3 dimensional objects then many people find them extremely difficult. This is one skill which can be significantly improved with practice.

Answers to Spatial Ability Questions

1) C 2) B 3) A 4) C 5) A

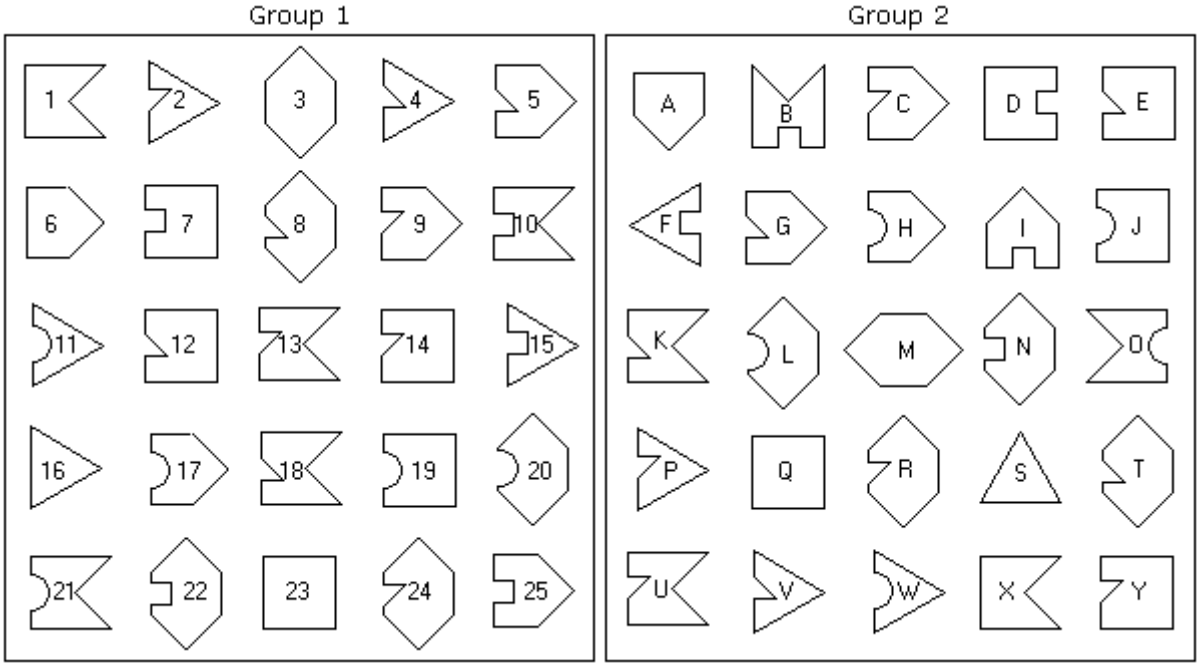
Mechanical Aptitude Tests

Overview of Spatial Ability Tests

Shape Matching – Two Dimensional
Spatial ability questions may test either speed or power. Generally speaking, if the questions involve the manipulation of 2 dimensional objects then they are probably fairly straightforward, but you will be challenged to answer them all in the time you are given.

Example Question
In this example, you are asked to look at two groups of simple, flat objects and find pairs that are exactly the same size and shape. Each group has 25 small drawings of these 2-dimensional objects. The objects in the first group are labelled with numbers and are in numerical order. The objects in the second group are labelled with letters in alphabetical order. Each drawing in the first group is exactly the same as a drawing in the second group. The objects in the second group have been moved and some have been rotated.

1. Which shape in Group 2 corresponds to the shape in Group 1?



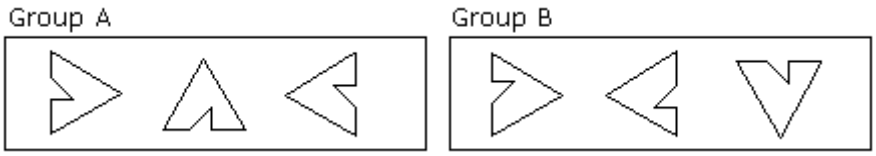
Mechanical Aptitude Tests

Answers

1. X	2. P	3. M	4. V
5. G	6. A	7. D	8. T
9. C	10. B	11. W	12. E
13. U	14. Y	15. F	16. S
17. H	18. K	19. J	20. L
21. O	22. N	23. Q	24. R
25. I			

This is very much a speed test as you will be given far more of these questions than you can comfortably answer in the given time. These questions use a large number of shapes that are presented close together. Some people find this very distracting and find it easier to work through the shapes in the second group systematically rather than trying to look at the whole group at once.

In some questions of this type there may not be a one-to-one match and some of the shapes in the first group may not appear in the second. The way that the question is worded will make this clear. If this is the case then you should be especially careful to look out for reflections in the second group. These are often put in by the test designers to trap the unsuspecting. For example:

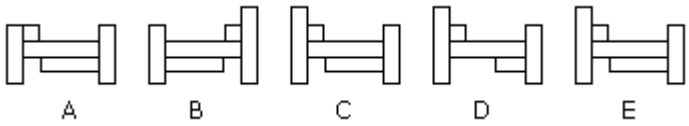


The shapes in Group A are rotations of each other. The shapes in Group B are also rotations of each other. However none of the shapes in group A are rotations of those in Group B.

Mechanical Aptitude Tests

Visual Comparison – Two Dimensional
In these types of question you will be presented with a number of objects only two of which are identical. Once again these are speed questions and you will need to work quickly and attempt to answer as many as possible in the given time

Example Question
26. Which two pictures are identical?



Answer
26. C and E are the only two pictures which are identical

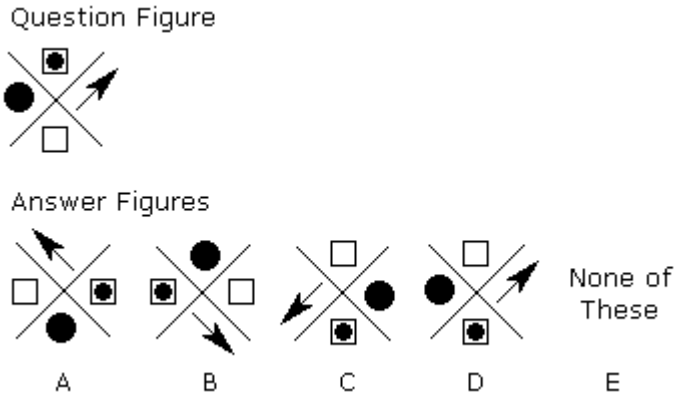
The best strategy for these types of question is to begin with the shape on the left and work through the shapes to the right of it systematically looking for an exact match. If there isn't one then move on to the second shape and repeat the process. It can be quite difficult to discipline yourself to adopt this systematic approach to these types of question, as you may think that it is quicker just to look at all of the shapes until the answer' jumps out' at you. The problem with this is that if the answer doesn't 'jump out' fairly quickly then panic sets in and you usually resort to the systematic approach anyway.

When you are confronted with a large number of similar types of question on a single page, the questions other than the one you are trying to answer can be distracting. It is much easier and less mentally tiring to cover the other questions and concentrate only on the two shapes that you are comparing.

Mechanical Aptitude Tests

Group Rotation – Two dimensional
This is slightly more complex than the rotations in the two-dimensional shape matching question.

27. Which of the Answer Figures is a rotation of the Question Figure?



Answer
27. C

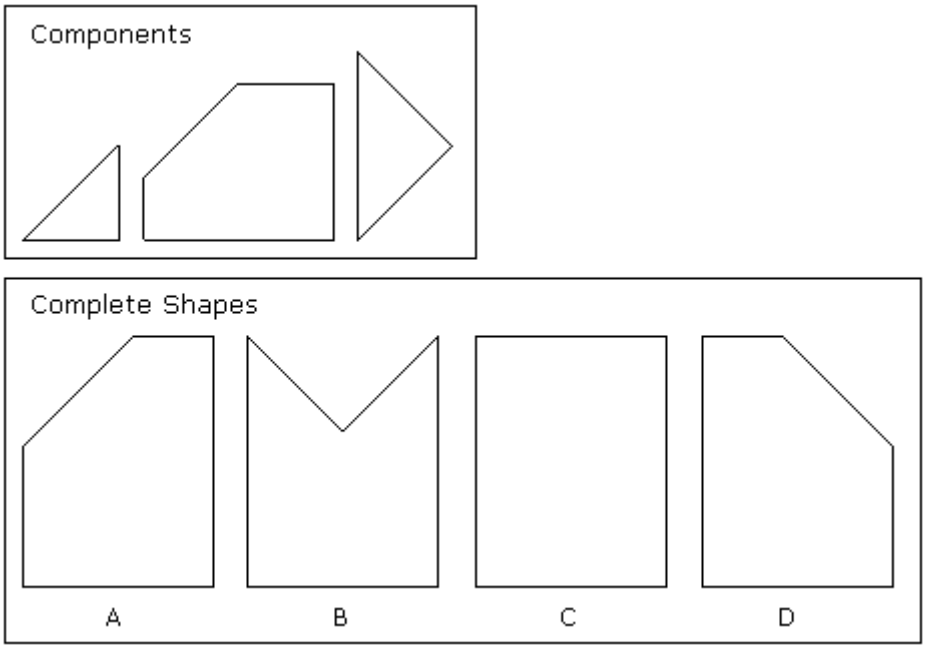
Once again, you need to be careful that you don't identify reflections. The best strategy is to choose the most asymmetrical shape in the group – in this case the arrow. Then determine the shapes 'clockwise' and 'anticlockwise' and opposite. Thinking in these terms is more logical than 'right', 'left' 'above' or 'below' as 'clockwise' etc are constant even when the figures are rotated.

In the example above, the white square is clockwise from the arrow. This means that A, B and D cannot be rotations of the Question Figure. This leaves only C as a possibility which can quickly be checked element by element.

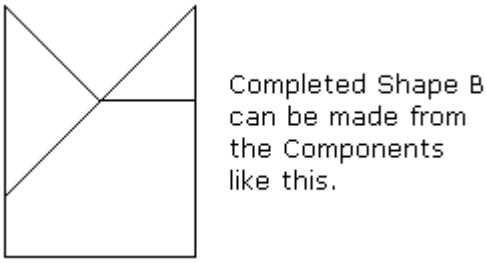
Mechanical Aptitude Tests

Combining Two–Dimensional Shapes
These questions show you a series of 2-dimensional shapes. One of the shapes has been 'cut up' into pieces. The questions presents you with the pieces and you are asked to work out which of the shapes has been 'cut up'.

Example Question
28. Which of the Complete Shapes can be made from the Components shown?



Answer
28. B – is the only shape that can be made from the components shown.



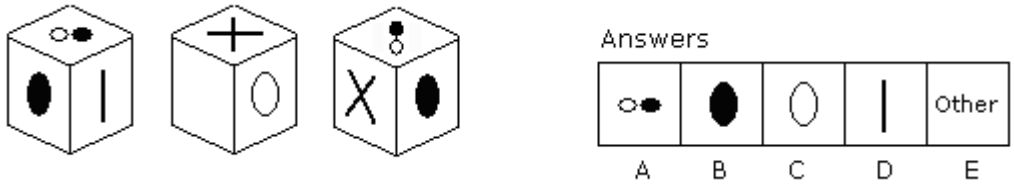
The best strategy for answering these questions is to look at the Complete Shapes and see if there are any distinct features that would make it impossible to construct such a shape from the components. In the example above, this is not very obvious but sometimes there are one or two shapes that can be immediately discounted on the basis of size alone.

One thing to remember is that if the Complete Shapes don't have any bits 'sticking out' (they usually don't) then the components must fit together so that sides of the same length are together. This reduces the number of combinations considerably.

Mechanical Aptitude Tests

Cube Views in 3-Dimensions
These questions show you several (usually 3) views of a 3-dimensional cube with unique symbols or markings on each face and then asks you a question about it. For example, which symbol is on the opposite face?

Example Question
29. Three views of the same cube are shown above. Which symbol is opposite the X?



Answer
29. D

Some people seem to have a natural talent for imagining objects in three dimensions and find these questions straightforward. However, if you're not one of them and you find thinking in three dimensions difficult, there are other ways to get the answer.

In the question above for example, you can simply use a process of elimination. If you can see a symbol on the same illustration as the 'X' then it cannot be opposite. The second and third cubes eliminate A, B and C. This leaves only D and 'other' as possibilities. D has edges shared with A and B which would be consistent with the third cube illustrated. Therefore D is correct.

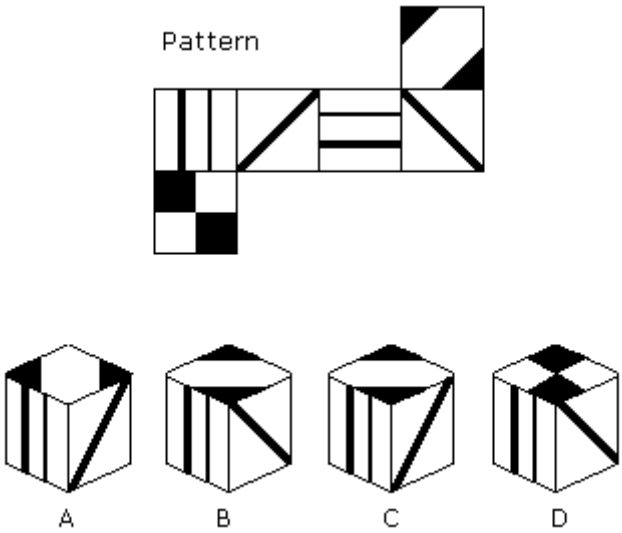
Although it is not usually specified in the instructions, it is almost always true that in these questions each symbol is used only once. This means that even in cases where elimination is not possible, it is sometimes quite easy to see the solution without mentally manipulating the cube too much.

In the example above, you can simply compare the first and third illustrations. The third illustration shows a 90 degree anti-clockwise rotation (looking at the cube from above) of the first illustration. Therefore D must be opposite the 'X'.

Mechanical Aptitude Tests

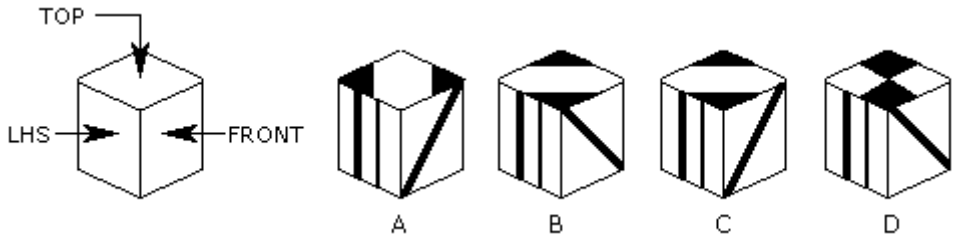
Cubes in 2 and 3 Dimensions
These questions show a flat (2-dimensional) pattern which can be folded to make a cube and a number of 3-dimensional cubes (usually 4). The pattern and the cubes have symbols or marking on each face. You need to look at the pattern and decide which of the cubes, if any, could be made from it.

Example Questions
30. Which of the cubes shown could be made from the pattern?



Answer
30. A - The key to these questions is to remember that only three faces of the cube can be shown in the illustration, this means that you need only to consider the relationship between the three visible elements on each cube and see if the same relationship exists in the pattern.

The best strategy for this type of question is to call one face of the cube the 'front' and then to name the other faces of the cube in relation to it. For example:

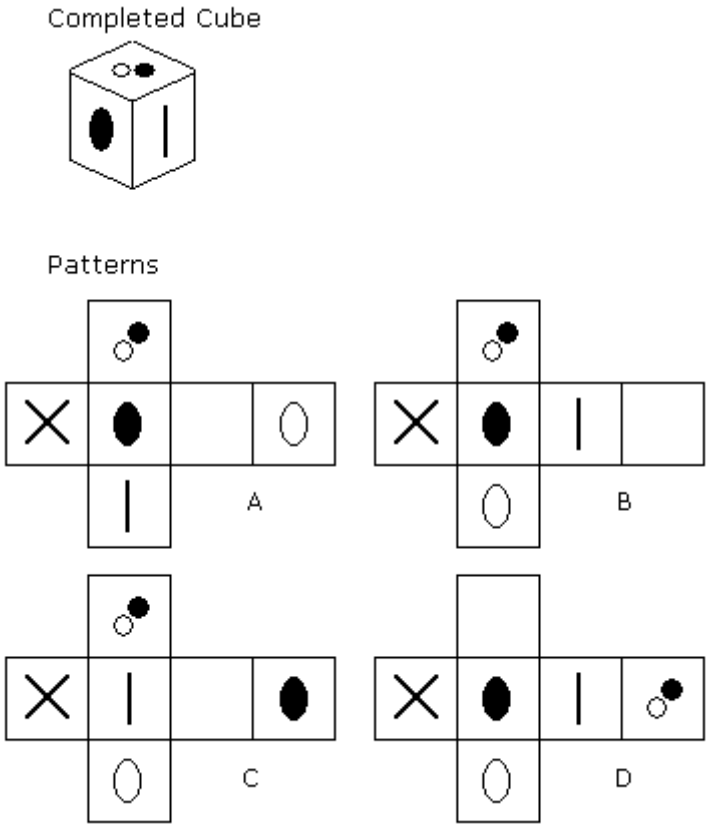


This is obviously an arbitrary decision as you could look at a cube from any side. However, thinking of the problem this way makes it much easier to 'see' the relationship between the faces of the cube. You can then: look at the front of the cube, find that face on the pattern, use the pattern to identify the 'top' and eliminate any options that do not match, use the pattern to identify the other face that touches the 'front' (in this example LHS) and eliminate any options that do not match.

Mechanical Aptitude Tests

Another type of question poses the problem the other way around. Here you have a single 3-dimensional cube and a number of 2-dimensional patterns, only one of which when folded, will make the cube.

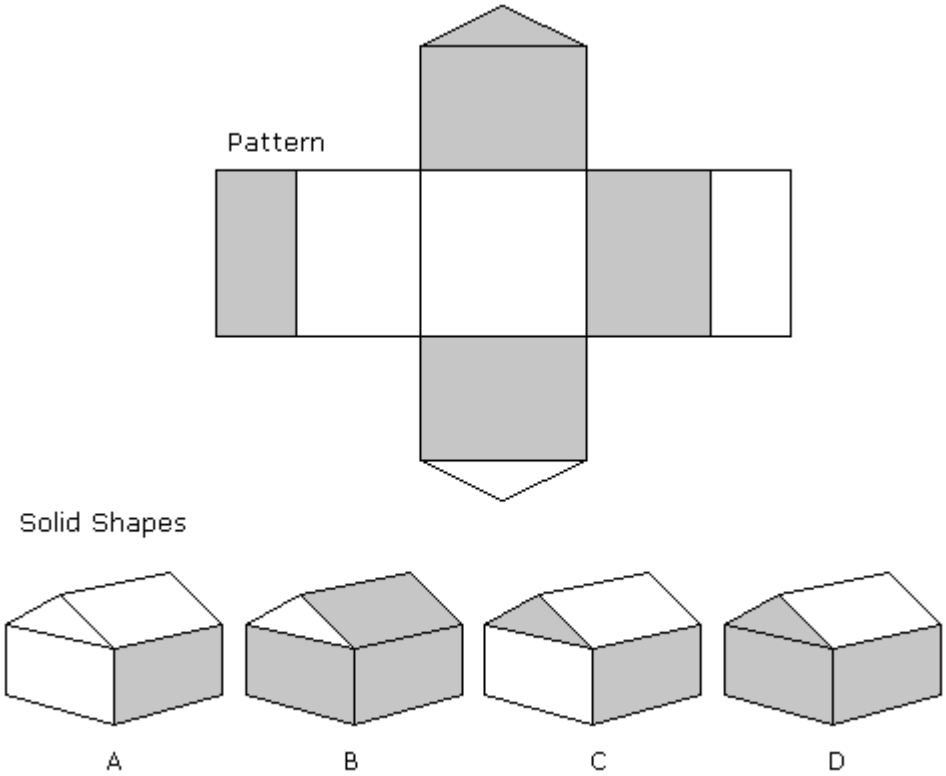
31. Which of the patterns when folded will make the cube shown?



Answer
31. B - The same strategy can be used to solve these questions. Remember don't be intimidated by these problems even if imagining things in three dimensions does not come easily to you. The problem can always be reduced to the relationship between three elements, which you can then try to locate in the answer figures.

Other Solids in 2 and 3 Dimensions
These questions are similar to the cube questions above, but rather than cubes they use other solid shapes which may be irregular. In some respects, these questions are easier than the cube questions as there are more relationships to work with. In other words, each face of the solid shape has a shape of its own rather than just being square.

Example Question
32. Which of the solid shapes shown could be made from the pattern?



Answer
32. D is correct. Shape A can be eliminated because is shows an un-shaded face below a triangular face, both of these faces (below the triangular face) are shaded on the pattern. Shape C can be eliminated for the same reason. Shape B can be eliminated as it has a shaded roof above the shaded side, which does not appear on the pattern.

Once again, you can reduce these problems to the relationship between the visible elements on the 3-dimensional picture. This makes things easier because even though the solid shape may have more faces than a cube, it is unusual for more than four faces to be shown. This means that you need to consider the relationship between the four visible faces, paying particular attention to shading or other patterns on them.

Mechanical Aptitude Tests

Spatial Ability Questions

Spatial ability is required in technical and design jobs where drawings and plans are used, for example; architecture, surveying, engineering, and design. It is also important in some branches of science and technology where 3 dimensional components are interacting.

Spatial ability questions are not routinely used in graduate and management level tests unless the job specifically requires good spatial skills. Do not be misled by the fact that some of these questions look similar to abstract reasoning questions. Spatial ability questions are concerned only with your ability to mentally manipulate shapes, not to identify patterns and make logical deductions.

Spatial Ability		Operatives	Supervisory	Management
Craft & Technical		Y	Y	Y
Clerical & Administrative				
Police, Fire, Military etc.	Y	0 - Occasionally used in selection.		
Management Trainee	0			
Graduate & Professional	0			

Remember, employers use spatial ability questions where the job involves: drawings, plans or the manipulation of shapes. You are not likely to be asked to mentally manipulate three dimensional shapes unless the job requires it.

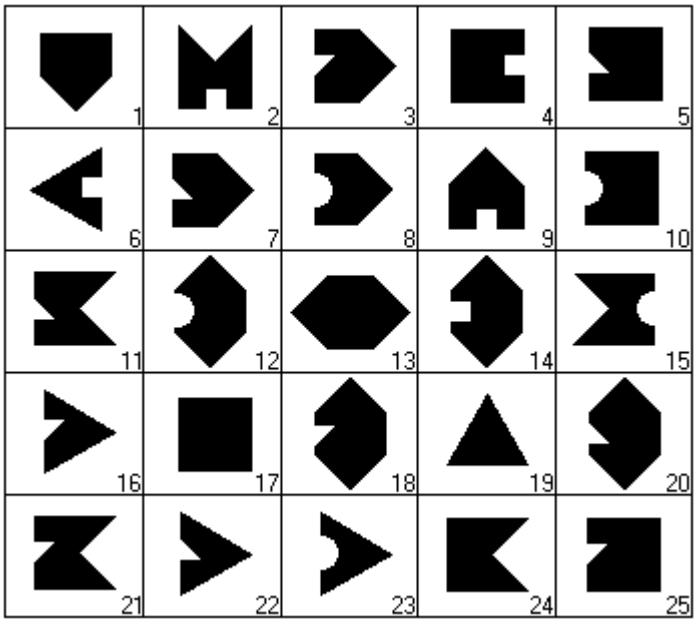
These sample question papers each contain 25 questions and have a suggested time limit of 20 minutes each. The questions are presented in Letter/A4 format for easy printing and self-marking.

Mechanical Aptitude Tests

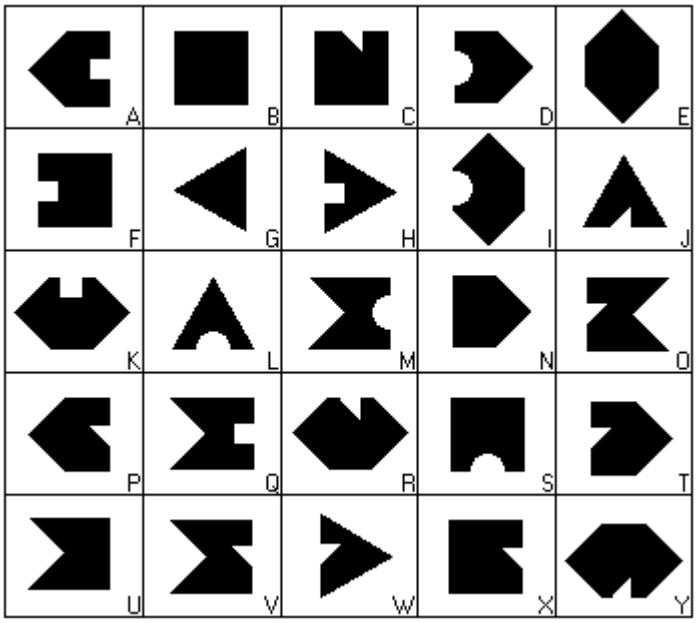
Spatial Ability Test 1: 45 Questions
Answer as many questions as you can in 20 minutes.

The shapes in Group 1 and Group 2 are identical, although some of them may be rotated. Which shape in Group 2 corresponds to the shapes (1 to 25) in Group 1?

Group 1



Group 2



1)	2)	3)	4)	5)
6)	7)	8)	9)	10)
11)	12)	13)	14)	15)
16)	17)	18)	19)	20)
21)	22)	23)	24)	25)

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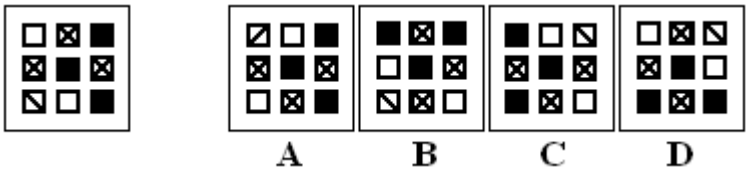
Mechanical Aptitude Tests

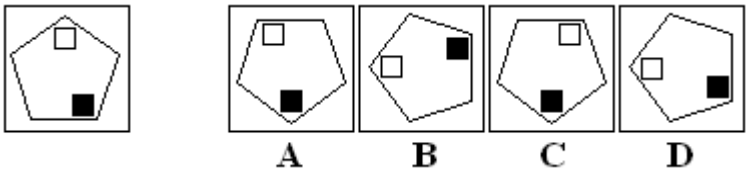
In the figures shown below, one of the shapes (A-D) is identical to the first figure but has been rotated.

26) Which figure is identical to the first?



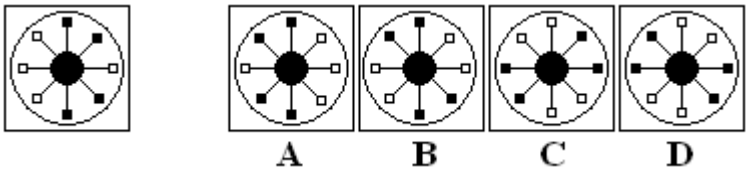
A B C D

27) Which figure is identical to the first?



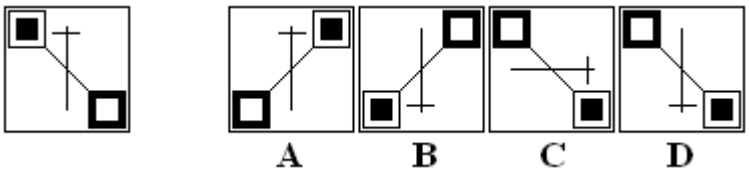
A B C D

28) Which figure is identical to the first?



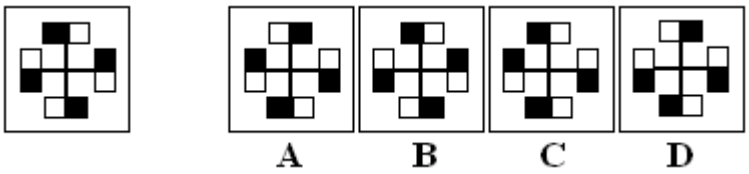
A B C D

29) Which figure is identical to the first?



A B C D

30) Which figure is identical to the first?



A B C D

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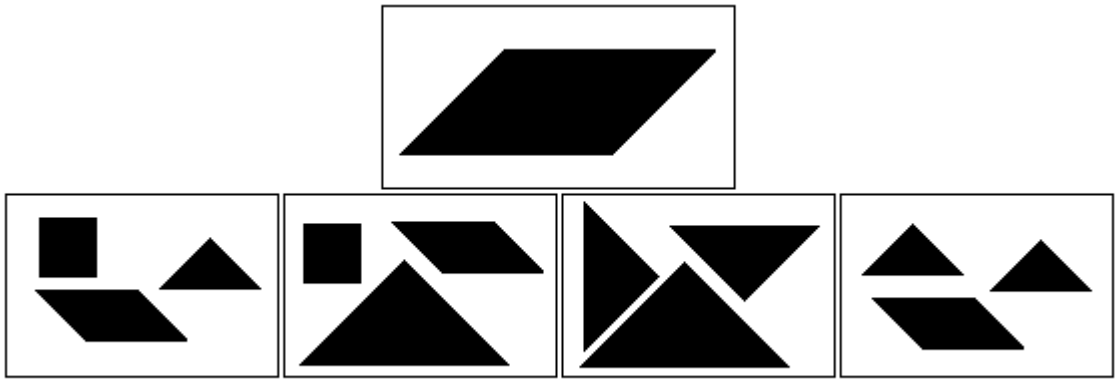
Mechanical Aptitude Tests

31) Which group of shapes can be assembled to make the shape shown?



A B C D

A B C D

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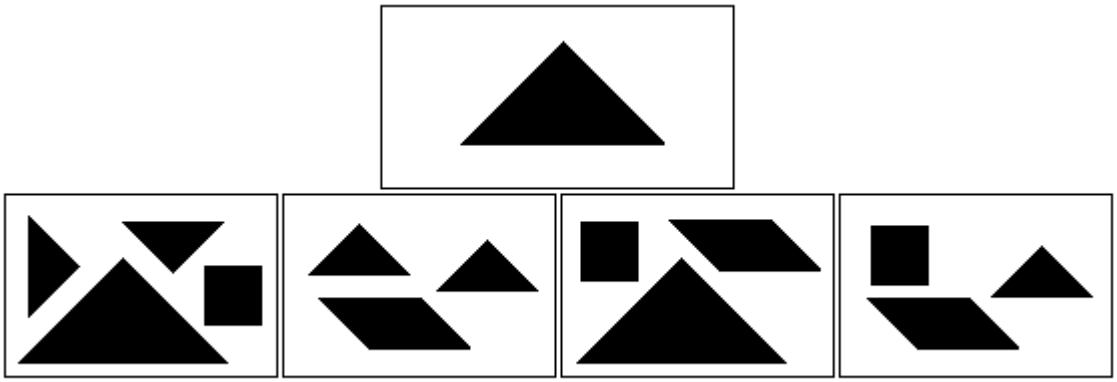

A B C D

A B C D

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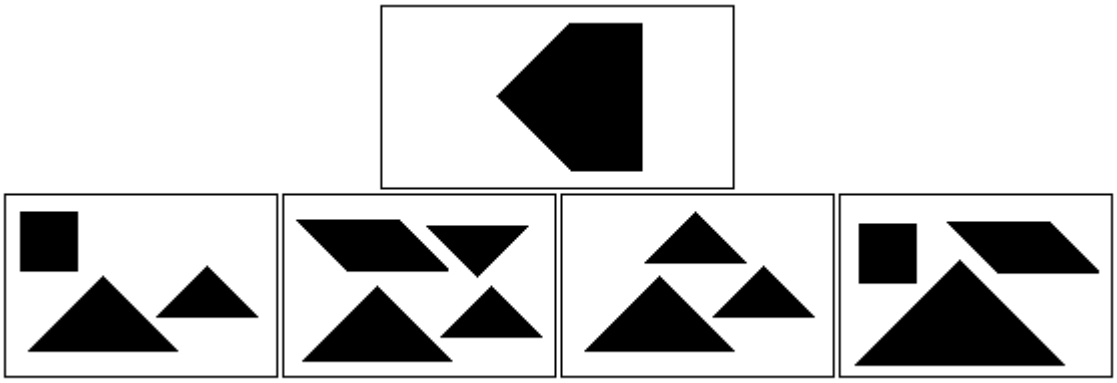
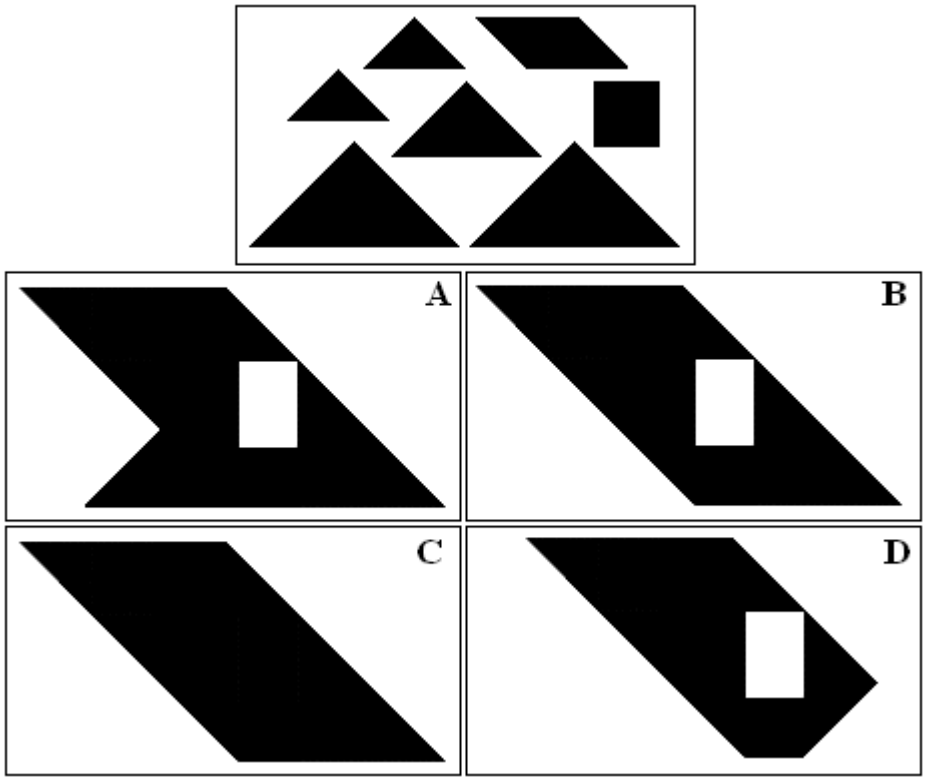


A B C D

A B C D

Mechanical Aptitude Tests

34) Which shape can be assembled using all of the individual shapes shown?



A B C D

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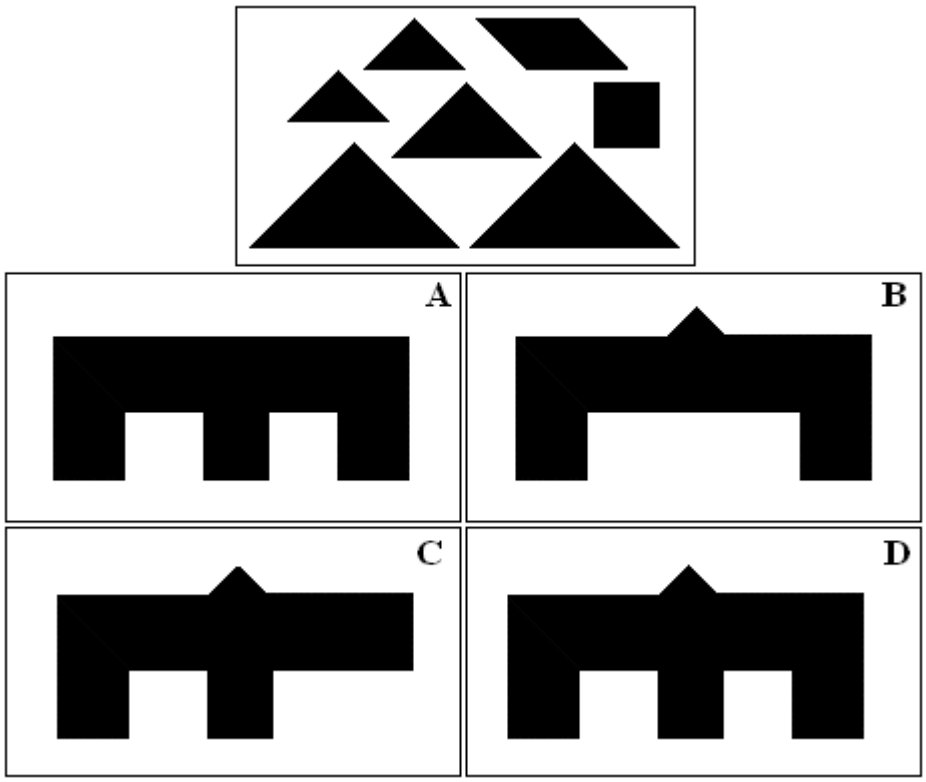
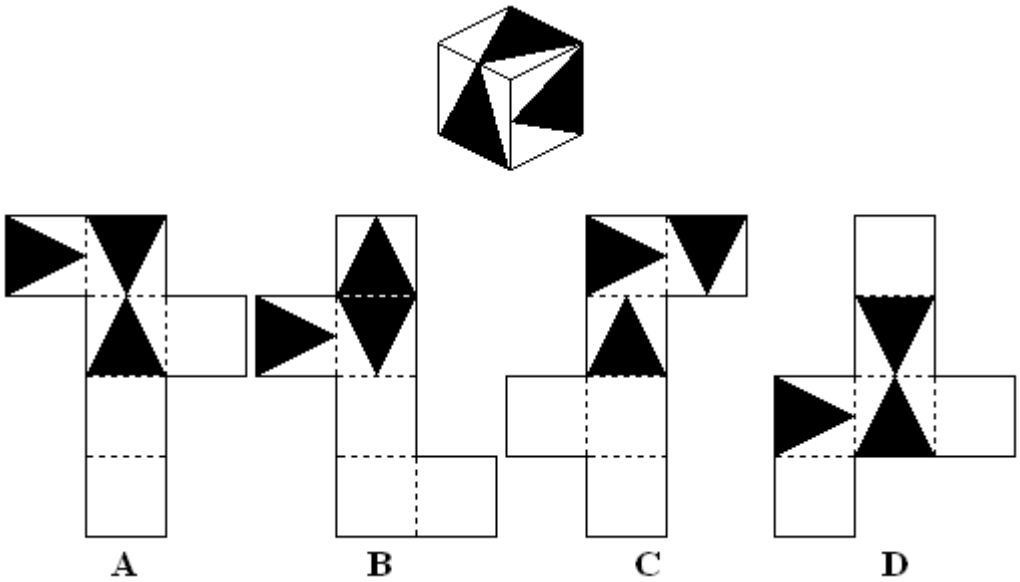


A B C D

Mechanical Aptitude Tests

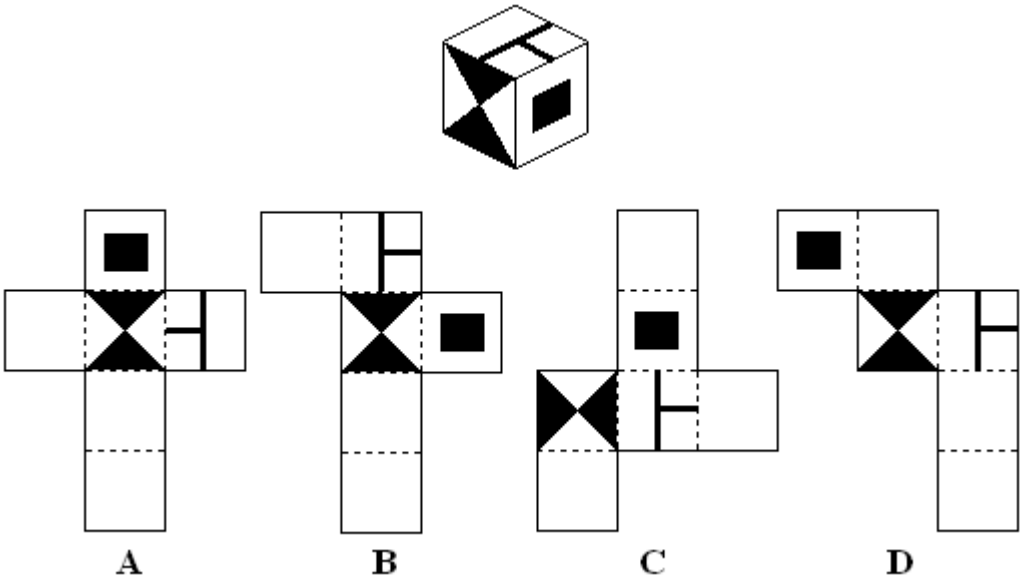
Which pattern can be folded to make the cube shown?

36)



A B C D

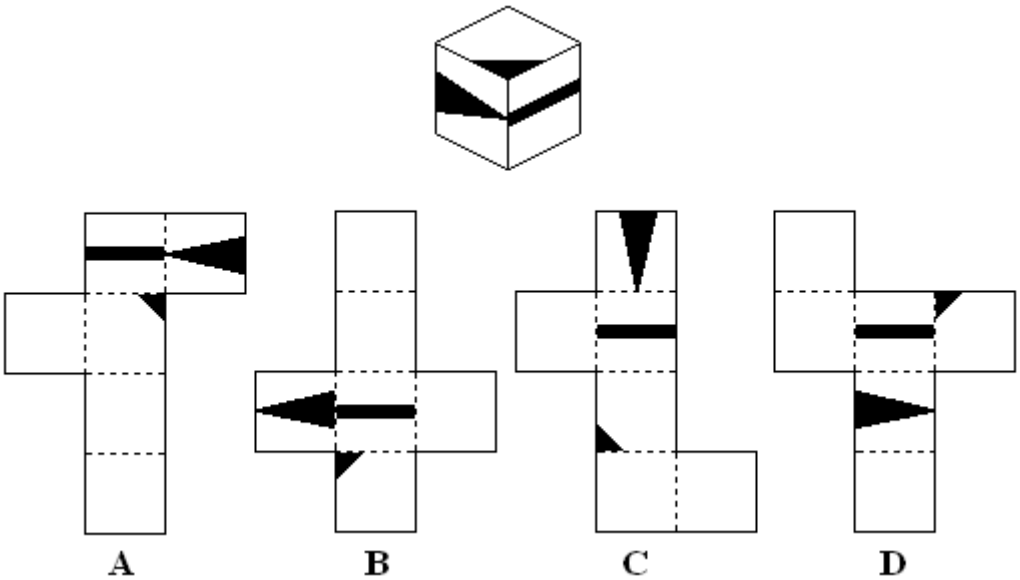
37)



A B C D

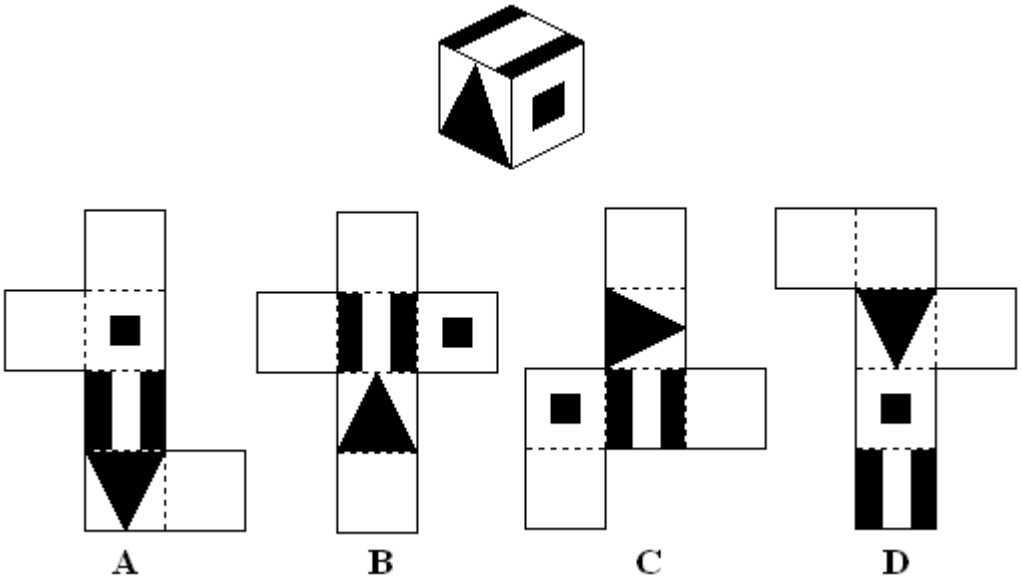
Mechanical Aptitude Tests

38)



A B C D

39)

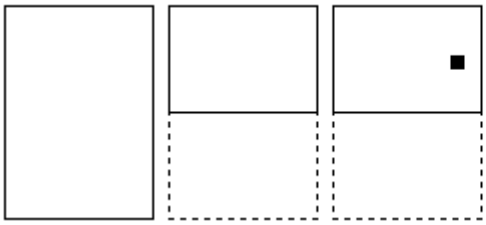


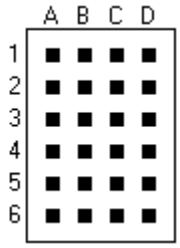
A B C D

Mechanical Aptitude Tests

The drawings show a sheet of paper which has been folded. The dashed lines indicate the whole sheet, each drawing represents a single fold. The black square shows where a hole was punched. Where do the holes appear when the sheet is unfolded?

40)

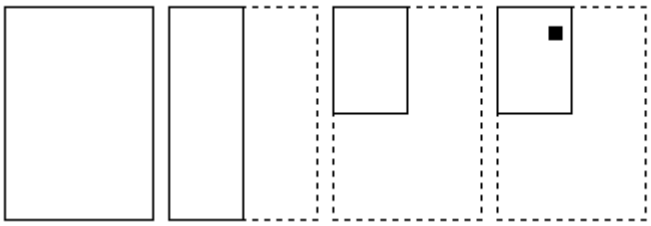


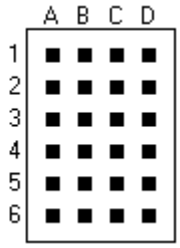


A	B	C	D
2C,5C	2D,5D	3D,3D	2C,2D

A B C D

41)

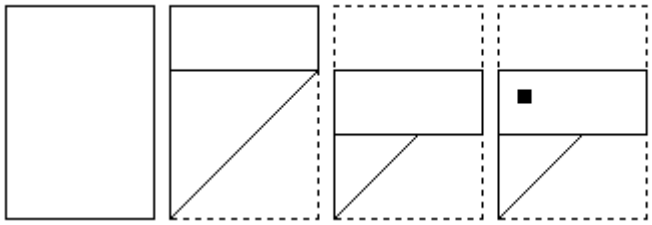


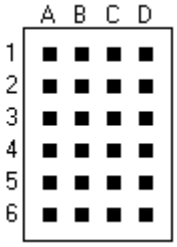


A	B	C	D
1B,1C,5B,5C	2B,2C,5B,5C	1B,2C,6B,6C	1B,1C,6B,6C

A B C D

42)

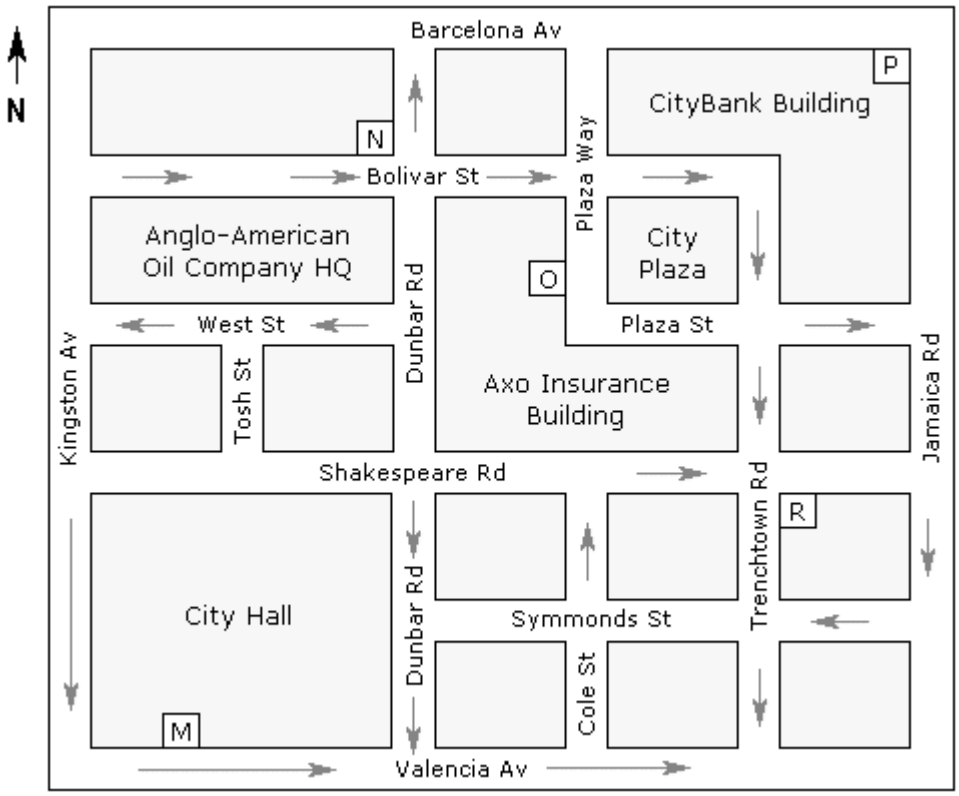




A	B	C	D
3A,2A,6D	3A,5A,6D	3A,5A,3D	3A,2A

A B C D

Mechanical Aptitude Tests



43) Officer Perez is in Tosh St with City Hall to her right. What direction is she facing?

A	B	C	D
North	South	East	West

44) She turns and walks to the junction with West St. She then turns right and walks to the next junction before turning left. Where is location 'O' in relation to her position?

A	B	C	D
North	South	East	West

45) Officer Martinez starts from location 'M' and proceeds as follows: left onto Valencia Av - heading East, second left - heading North, second right - heading East, second left - heading North. He proceeds North for two blocks. What is his location?

A	B	C	D
N	O	R	P

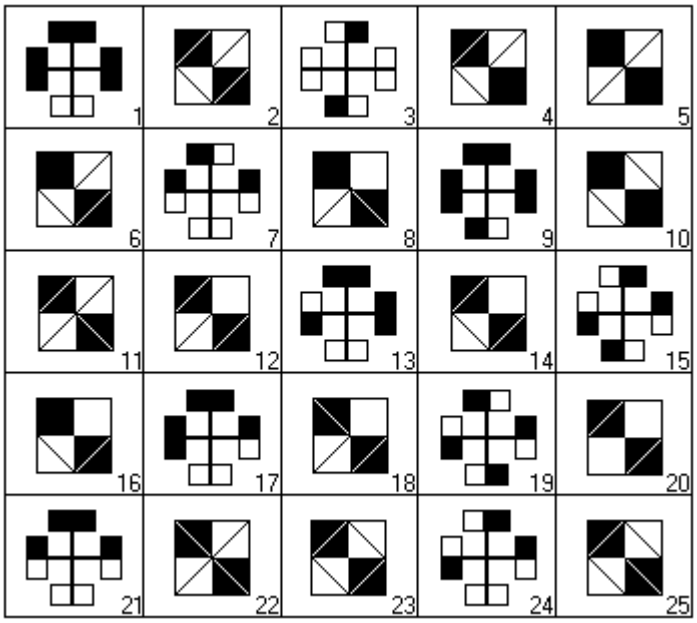
End of Spatial Ability - Test 1

Mechanical Aptitude Tests

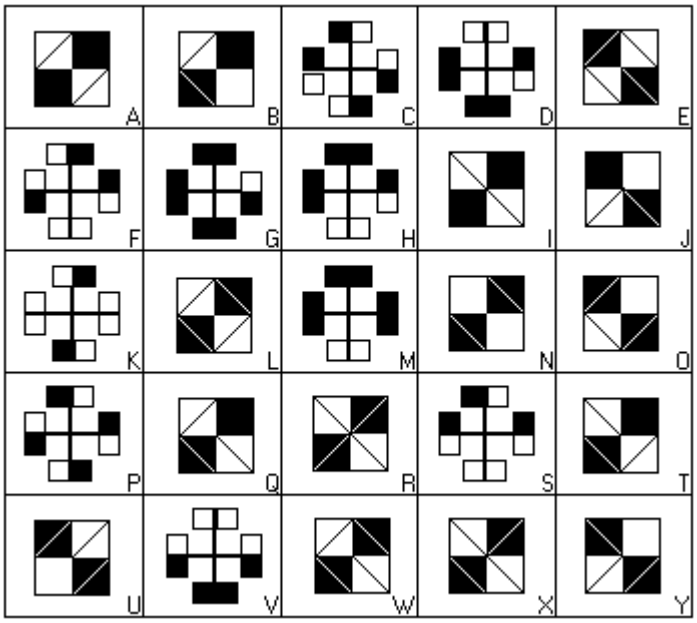
Spatial Ability Test 2: 45 Questions
Answer as many questions as you can in 20 minutes.

The shapes in Group 1 and Group 2 are identical, although some of them may be rotated. Which shape in Group 2 corresponds to the shapes (1 to 25) in Group 1?

Group 1



Group 2



1)	2)	3)	4)	5)
6)	7)	8)	9)	10)
11)	12)	13)	14)	15)
16)	17)	18)	19)	20)
21)	22)	23)	24)	25)

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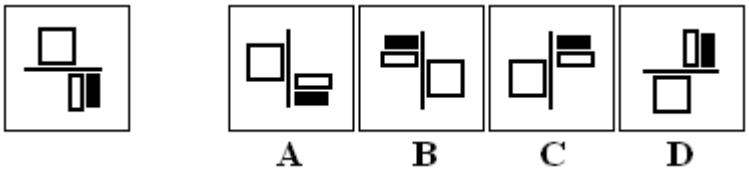
Mechanical Aptitude Tests

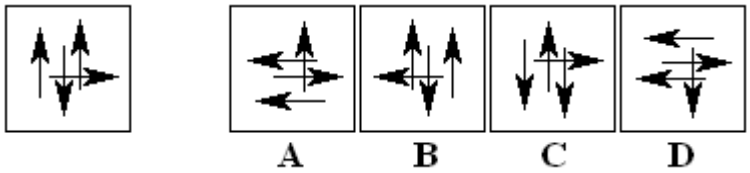
In the figures shown below, one of the shapes (A-D) is identical to the first figure but has been rotated.

26) Which figure is identical to the first?



A B C D

27) Which figure is identical to the first?



A B C D

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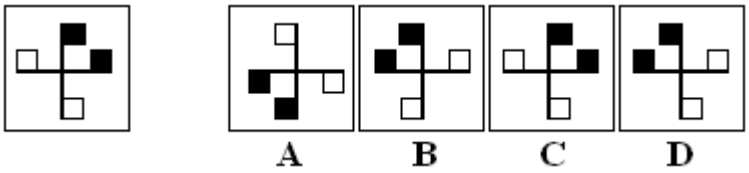

A B C D

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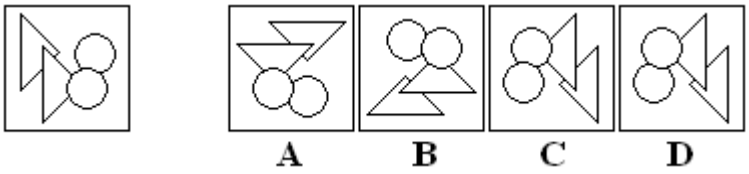

A B C D

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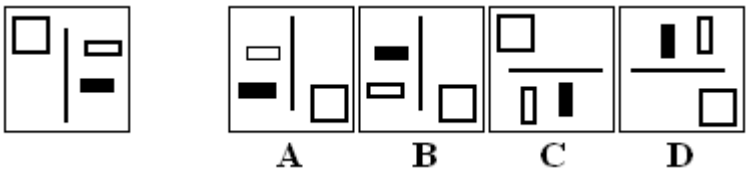

A B C D

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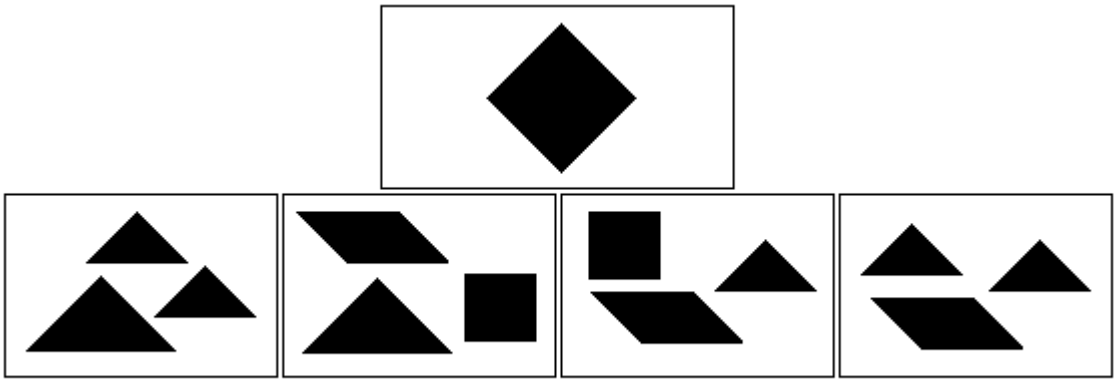
Mechanical Aptitude Tests

31) Which group of shapes can be assembled to make the shape shown?



A B C D

A B C D

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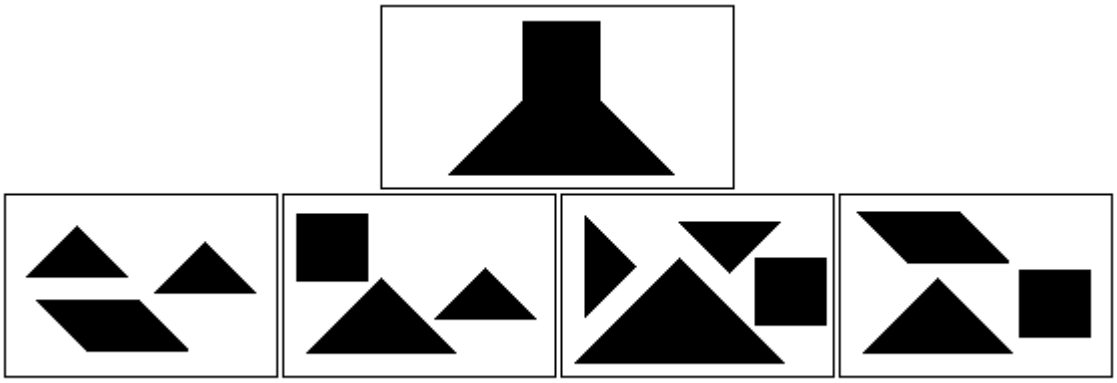

A B C D

A B C D

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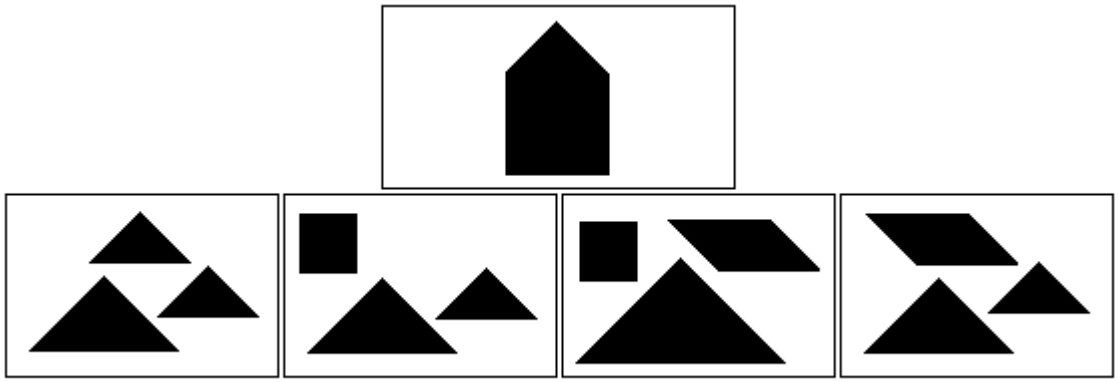
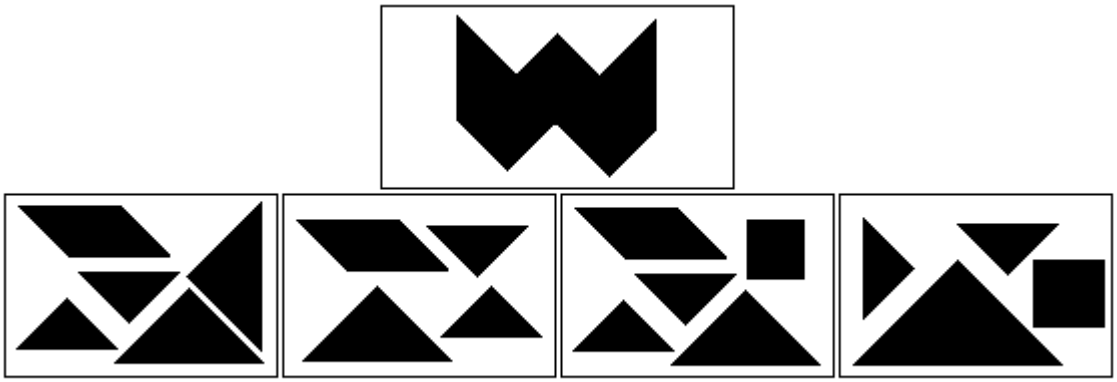


A B C D

A B C D

Mechanical Aptitude Tests

34) Which shape can be assembled using all of the individual shapes shown?



A B C D

A B C D

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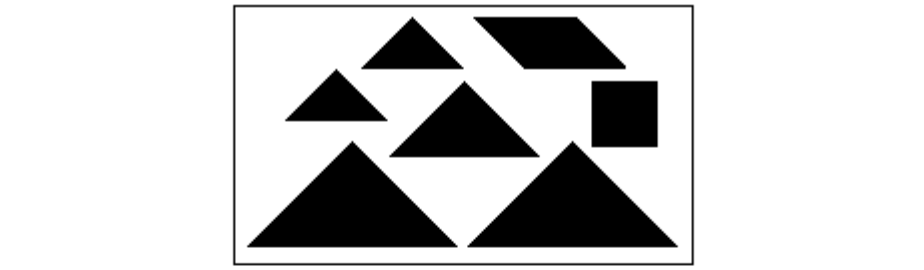
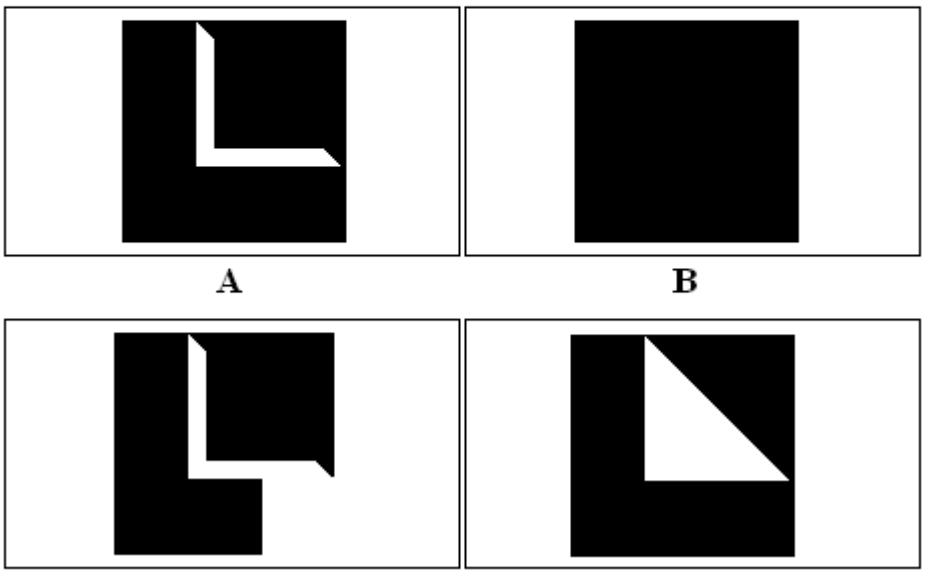




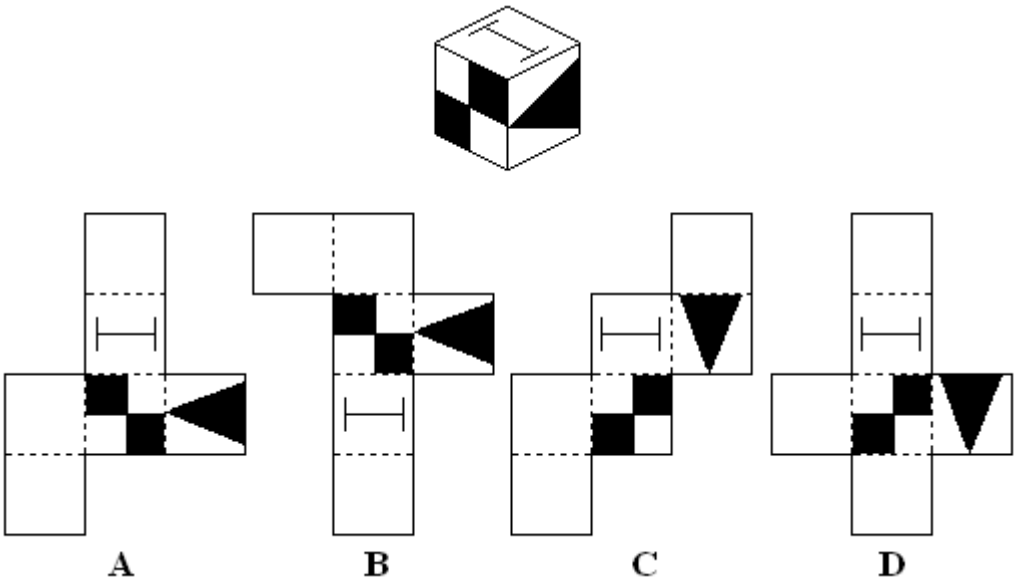
A B C D

A B C D

Mechanical Aptitude Tests

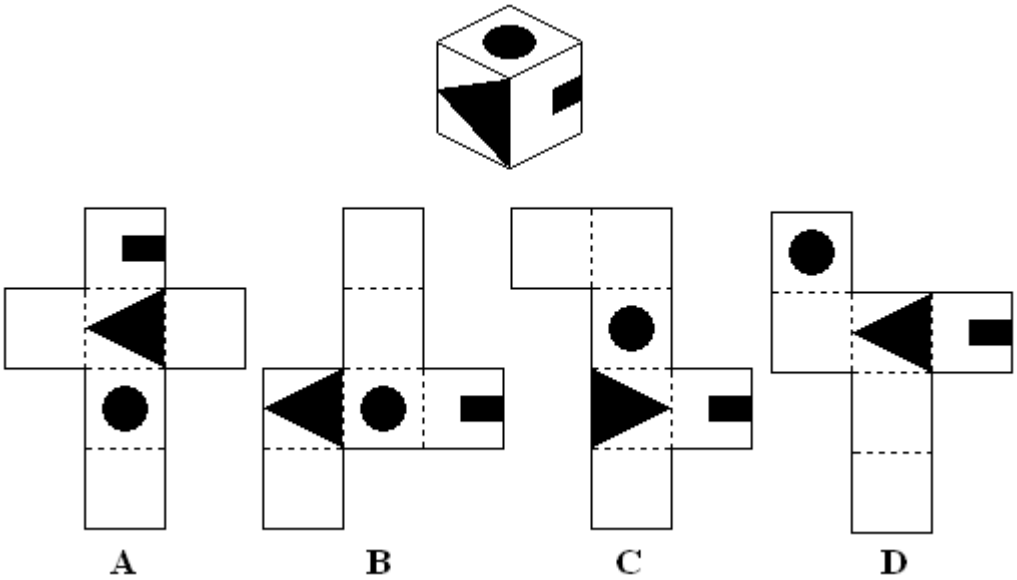
Which pattern can be folded to make the cube shown?

36)



A B C D

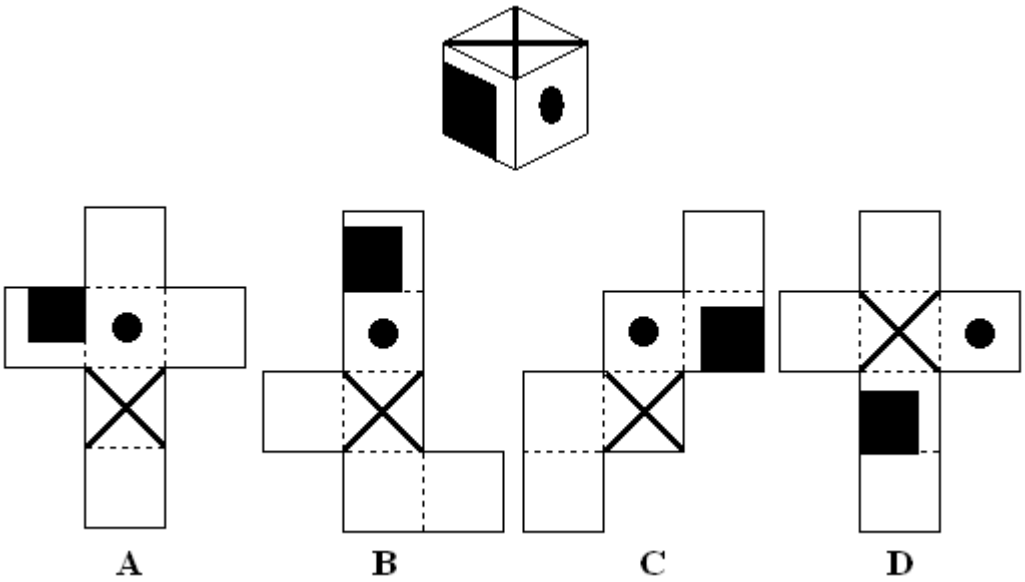
37)



A B C D

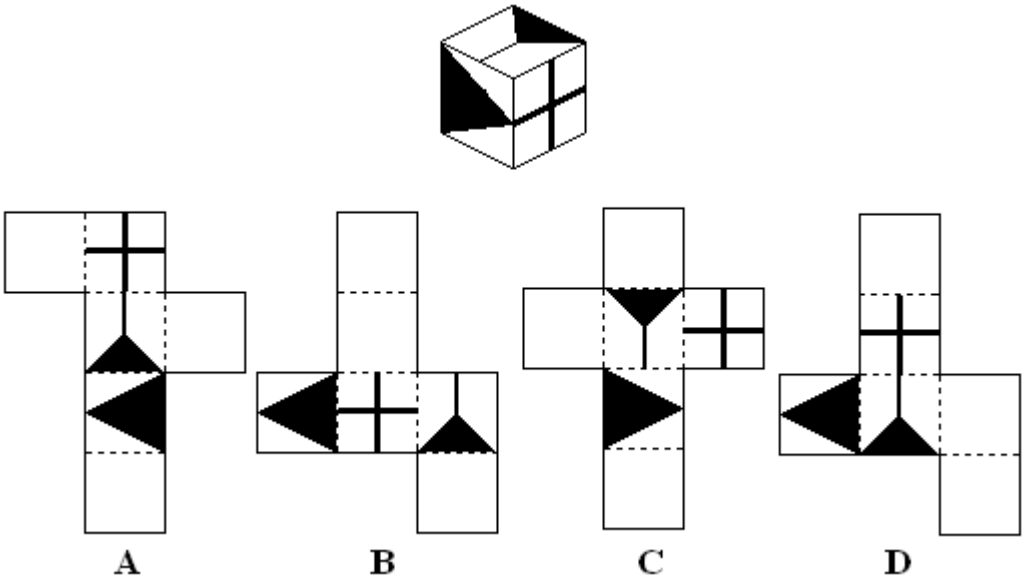
Mechanical Aptitude Tests

38)



A B C D

39)

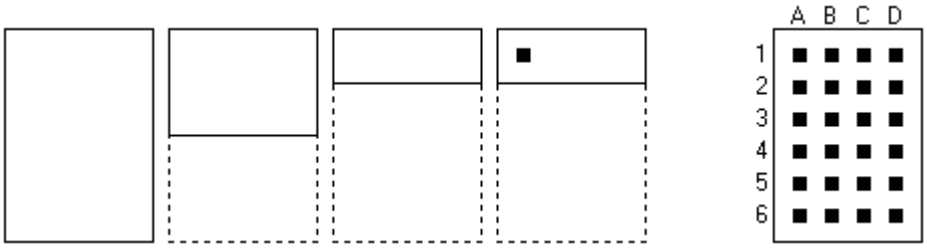


A B C D

Mechanical Aptitude Tests

The drawings show a sheet of paper which has been folded. The dashed lines indicate the whole sheet, each drawing represents a single fold. The black square shows where a hole was punched. Where do the holes appear when the sheet is unfolded?

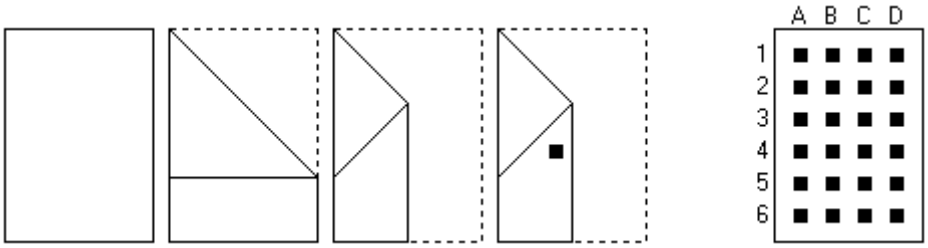
40)



A	B	C	D
1A,3B,4A,6B	1A,3A,4A,6A	1A,2A,4A,6A	1A,2A,4A,5A

A B C D

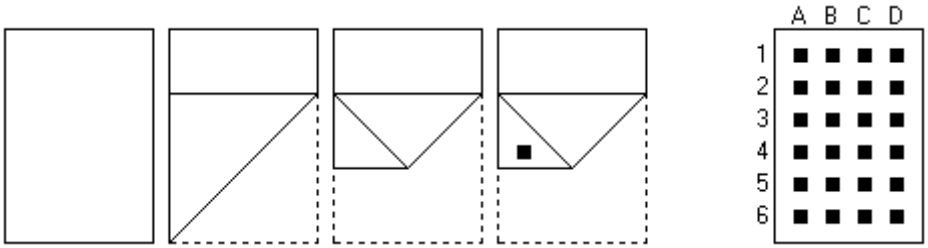
41)



A	B	C	D
4B,4C,3D,4D	4B,4C,2D,2D	4B,4D,2D,3D	4B,4C,2D,3D

A B C D

42)



A	B	C	D
4A,5A,6B,6C	4A,5B,6B,6C	4A,5A,6C,6D	4A,5B,6C,6D

A B C D

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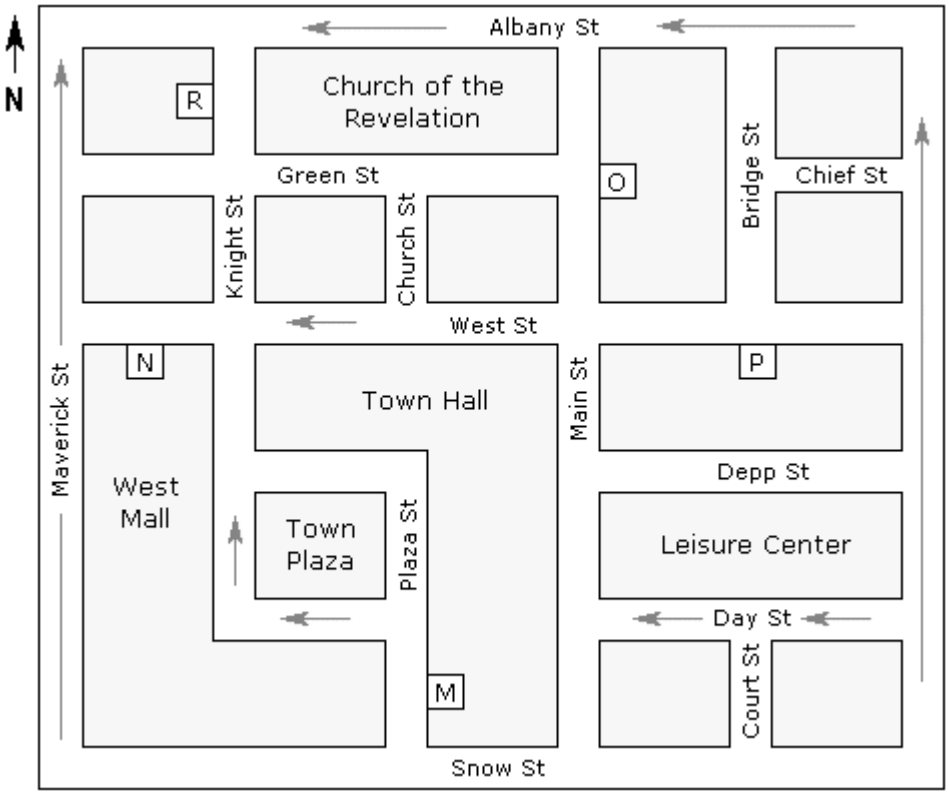


43) Officer Wu is in Green St and can see the Town Hall to her right. What direction is location 'N' in relation to her position?

A	B	C	D
South	South East	South West	North East

44) She turns and walks to the junction with Main St. She turns right and proceeds four blocks before turning right and then taking the next right. Which location is nearest to her current position?

A	B	C	D
P	N	O	M

45) Officer Jones starts from location 'R' and proceeds as follows: right onto Knight St - heading South, first left - heading East, second right - heading South, second left - heading East. He proceeds East for one block. Where is location 'P' in relation to his current position?

A	B	C	D
North	North East	North West	East

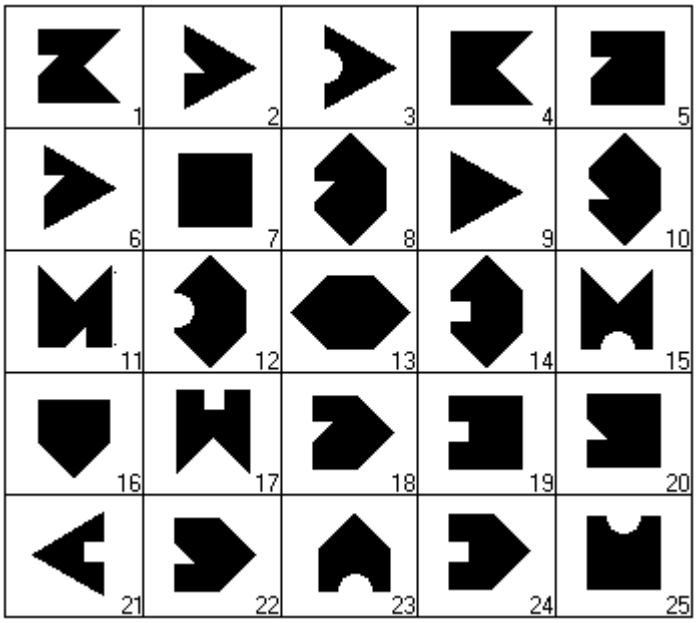
End of Spatial Ability - Test 2

Mechanical Aptitude Tests

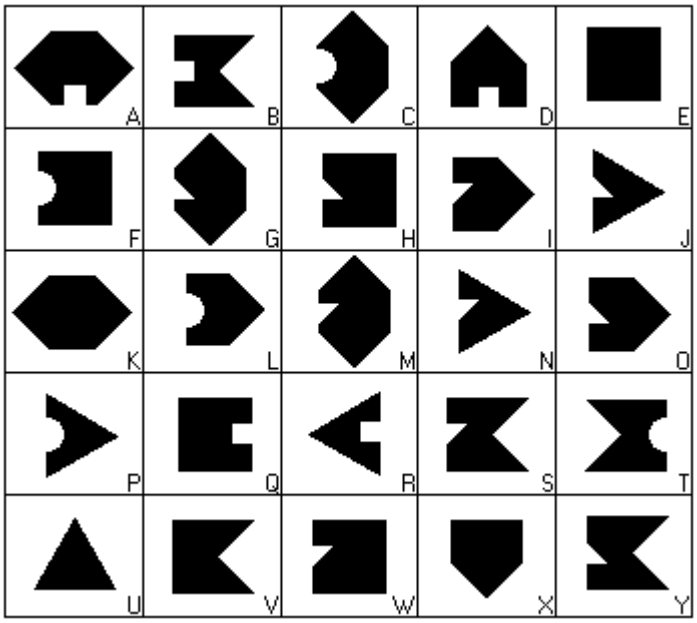
Spatial Ability Test 3: 45 Questions
Answer as many questions as you can in 20 minutes.

The shapes in Group 1 and Group 2 are identical, although some of them may be rotated. Which shape in Group 2 corresponds to the shapes (1 to 25) in Group 1?

Group 1



Group 2



1)	2)	3)	4)	5)
6)	7)	8)	9)	10)
11)	12)	13)	14)	15)
16)	17)	18)	19)	20)
21)	22)	23)	24)	25)

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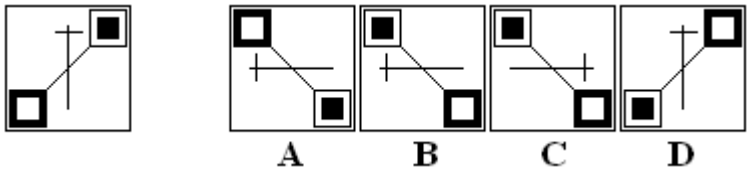
Mechanical Aptitude Tests

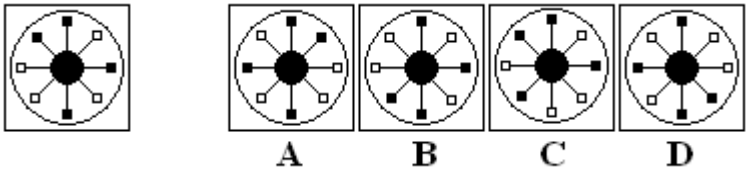
In the figures shown below, one of the shapes (A-D) is identical to the first figure but has been rotated.

26) Which figure is identical to the first?



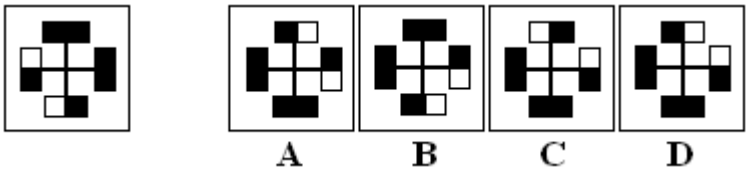
A B C D

27) Which figure is identical to the first?



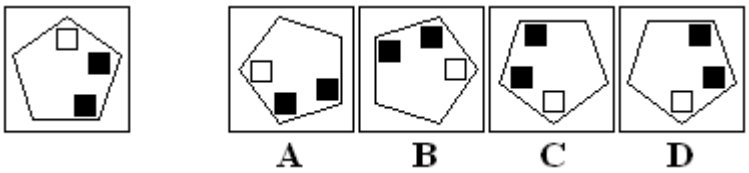
A B C D

28) Which figure is identical to the first?



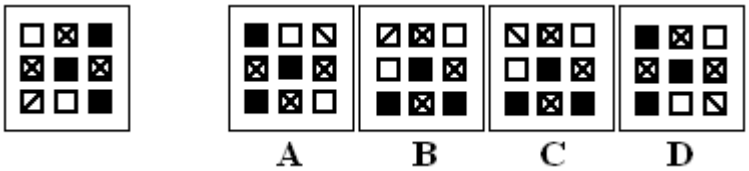
A B C D

29) Which figure is identical to the first?



A B C D

30) Which figure is identical to the first?



A B C D

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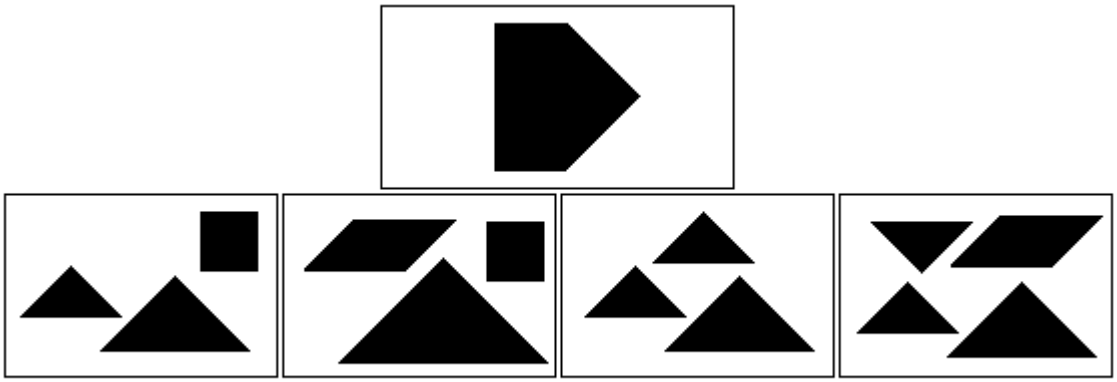
Mechanical Aptitude Tests

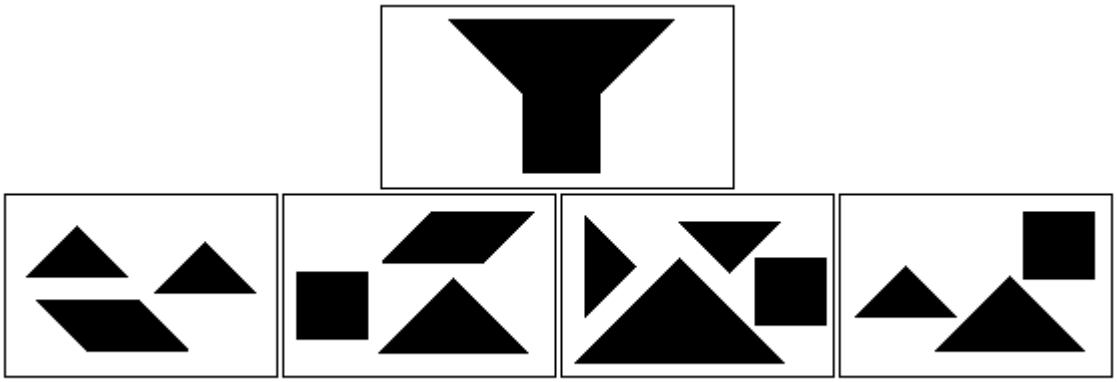
31) Which group of shapes can be assembled to make the shape shown?



A B C D

A B C D

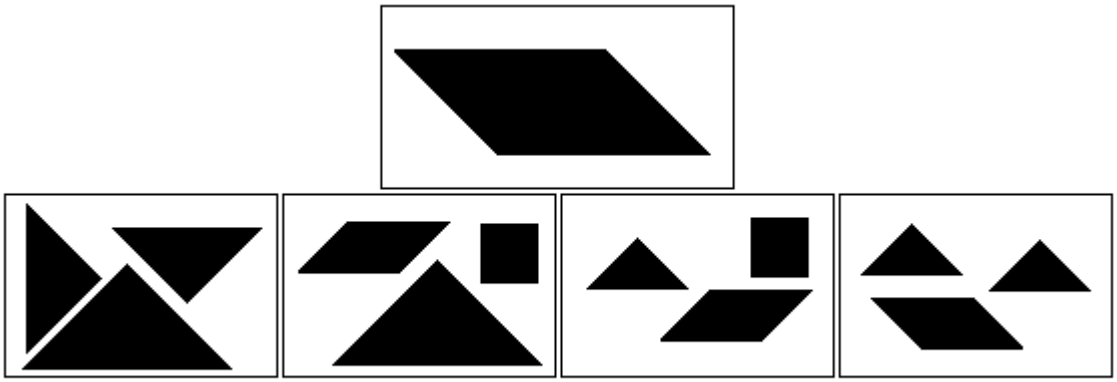
32) Which group of shapes can be assembled to make the shape shown?



A B C D

A B C D

33) Which group of shapes can be assembled to make the shape shown?

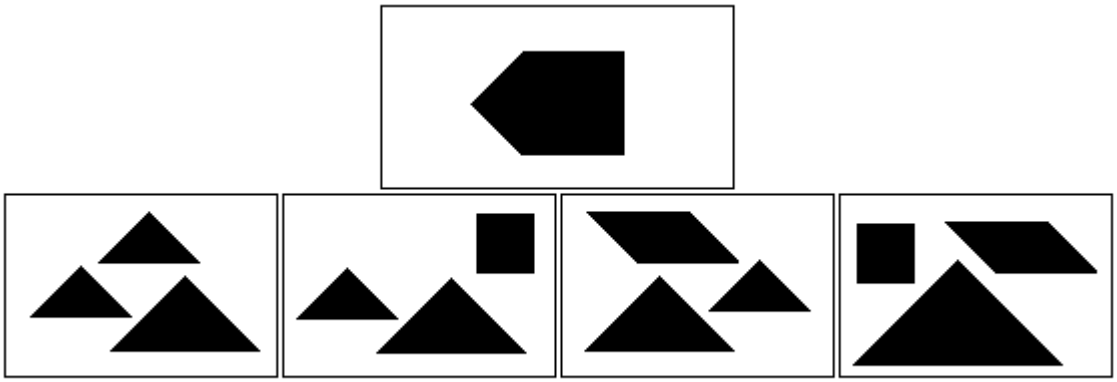


A B C D

A B C D

Mechanical Aptitude Tests

34) Which shape can be assembled using all of the individual shapes shown?



A B C D

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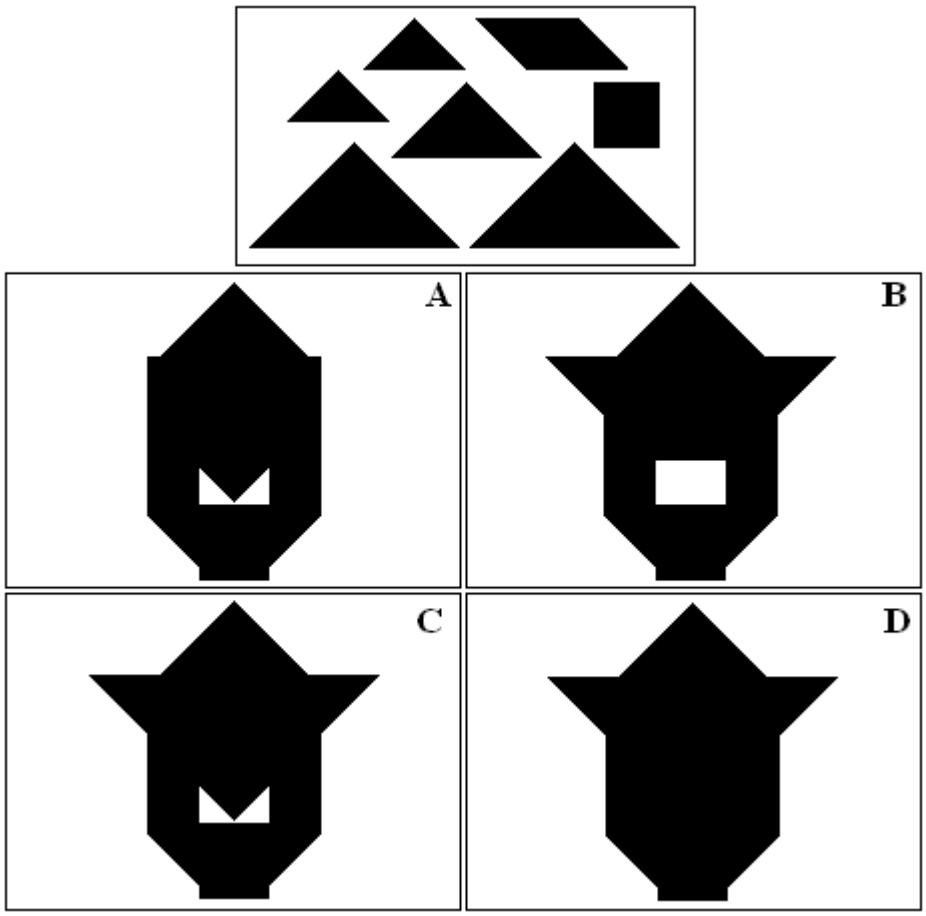
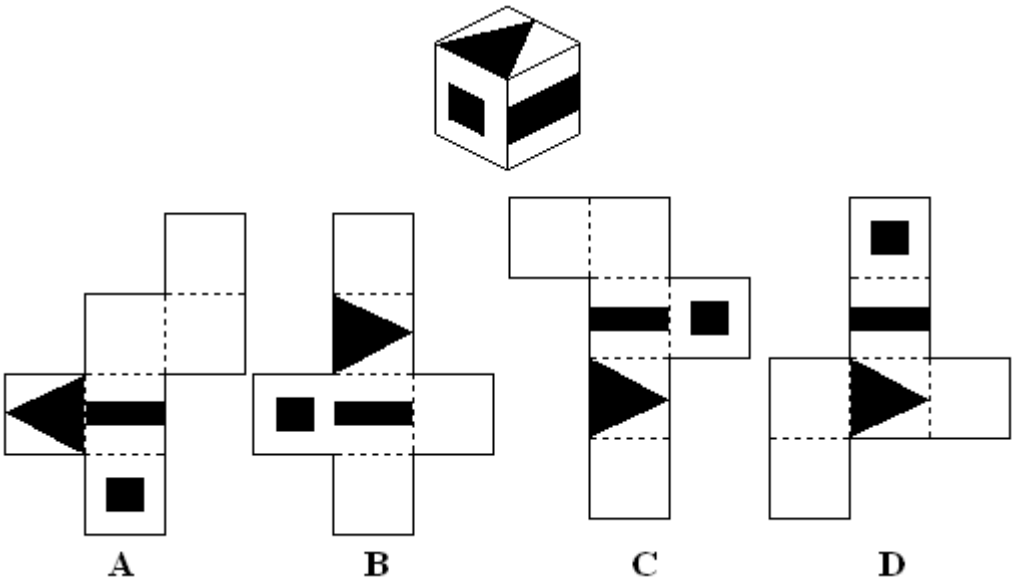


A B C D

Mechanical Aptitude Tests

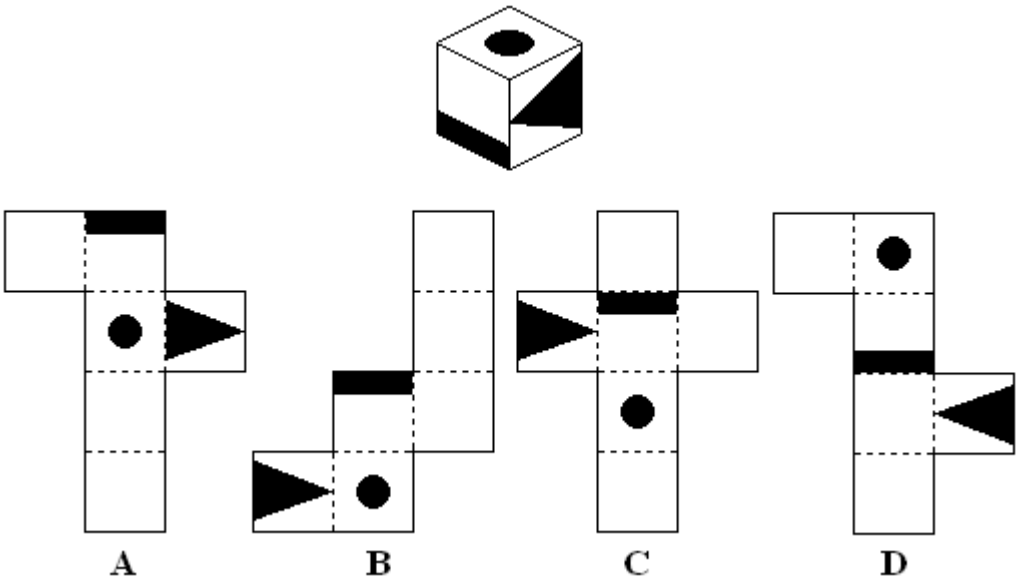
Which pattern can be folded to make the cube shown?

36)



A B C D

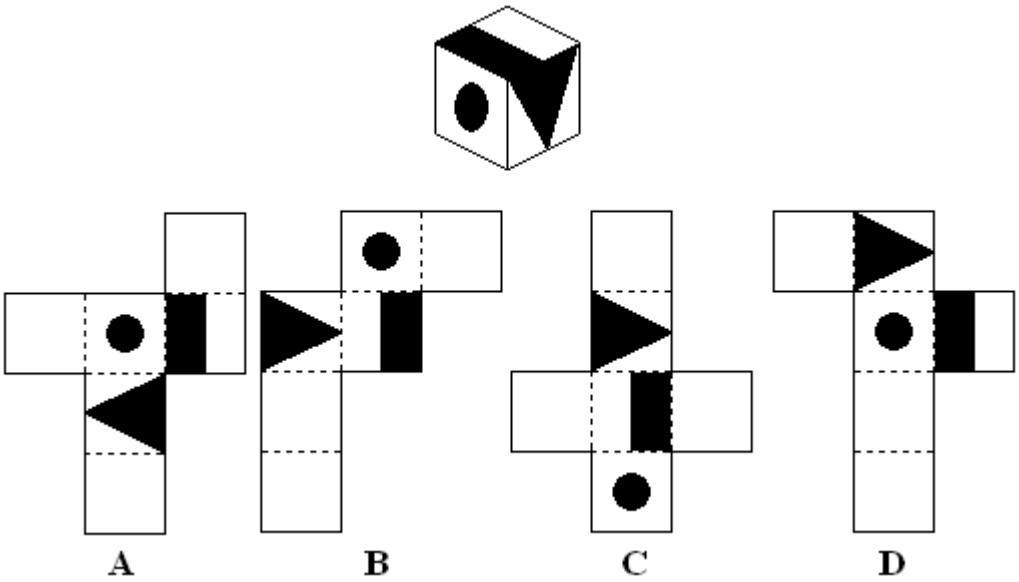
37)



A B C D

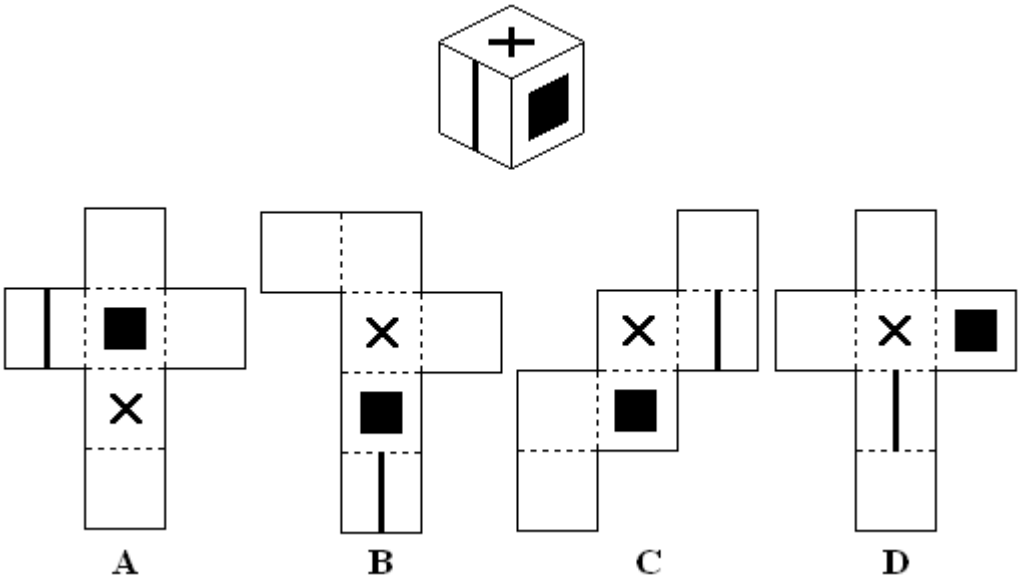
Mechanical Aptitude Tests

38)



A B C D

39)



A B C D

Mechanical Aptitude Tests

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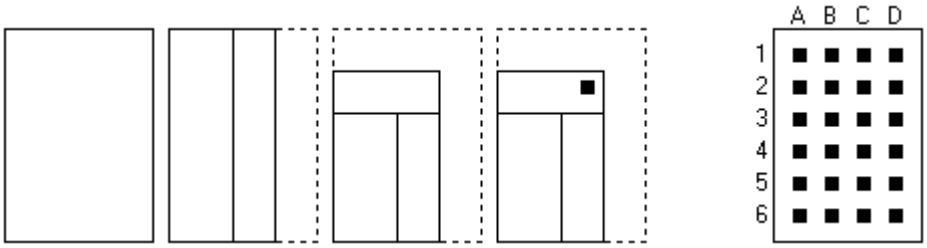
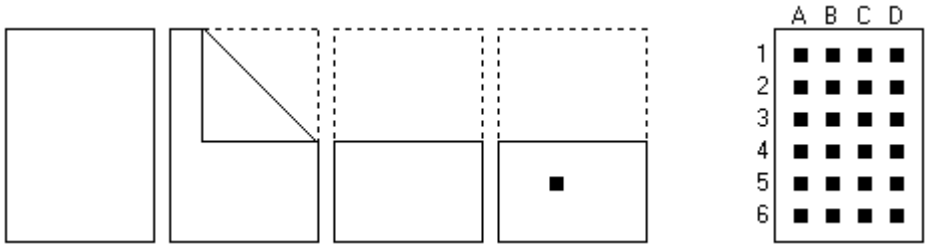
40)



A	B	C	D
1C,2C	1B,1D,2B,2D	1C,1E,2C,2E	1C,1D,2C,2D

A B C D

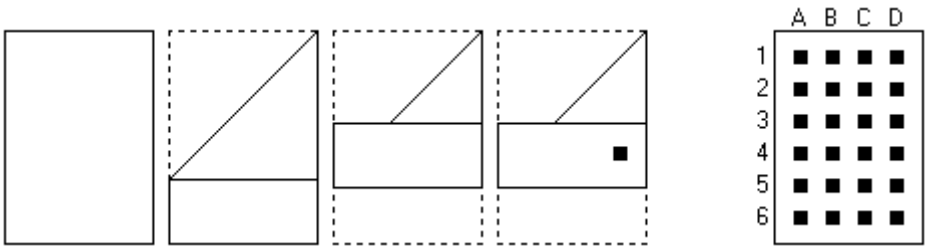
41)



A	B	C	D
1B,2B,5B	1C,2B,5B	1C,2B,5C	1C,2C,5D

A B C D

42)



A	B	C	D
1A,4D,6D	2A,4D,6D	3A,4D,5D	2A,4D,5D

A B C D

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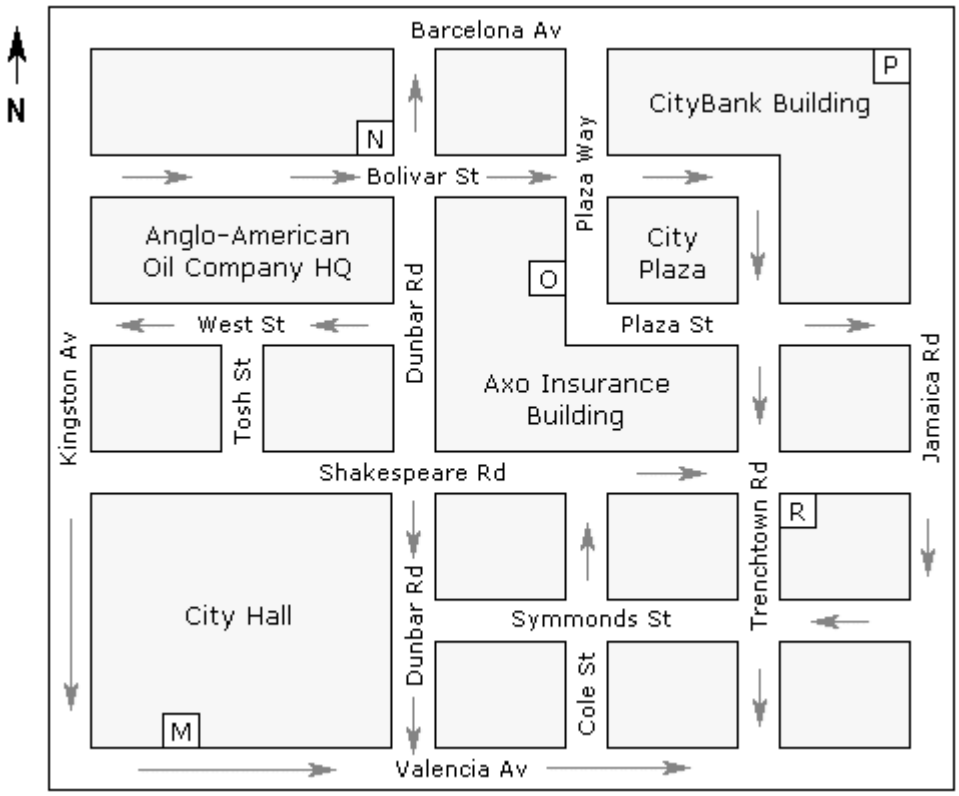
Mechanical Aptitude Tests



43) Patrolman Smith is facing the Anglo-American Oil Company HQ with the Axo Insurance building behind him, which direction is he facing?

A	B	C	D
North	South	East	West

44) He turns and walks to the junction with Shakespeare Rd. He then turns left and walks two blocks before turning right. Where is location 'R' in relation to his position?

A	B	C	D
North	South	East	West

45) Officer McKay starts from location 'O' and proceeds as follows: left onto plaza way - heading North, second left - heading west, second left - heading south, first left - heading east. She proceeds for one block. What is her location?

A	B	C	D
	N		

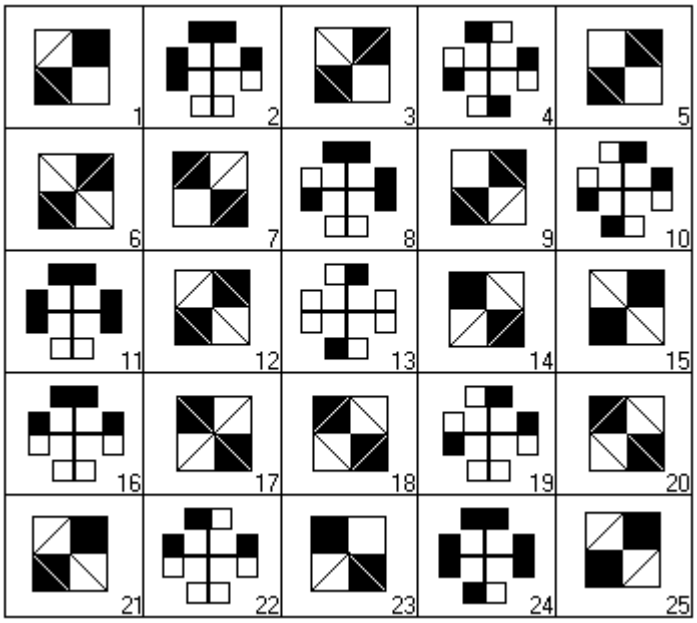
End of Spatial Ability - Test 3

Mechanical Aptitude Tests

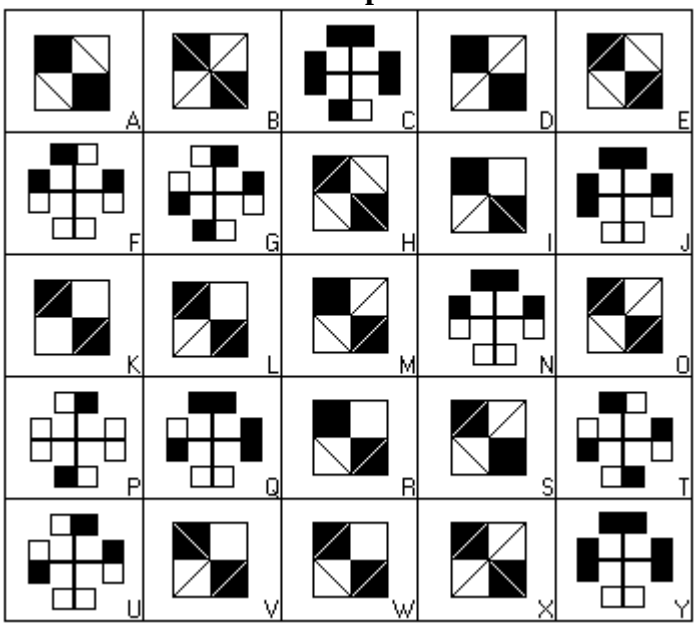
Spatial Ability Test 4: 45 Questions
Answer as many questions as you can in 20 minutes.

The shapes in Group 1 and Group 2 are identical, although some of them may be rotated. Which shape in Group 2 corresponds to the shapes (1 to 25) in Group 1?

Group 1



Group 2



1)	2)	3)	4)	5)
6)	7)	8)	9)	10)
11)	12)	13)	14)	15)
16)	17)	18)	19)	20)
21)	22)	23)	24)	25)

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Mechanical Aptitude Tests

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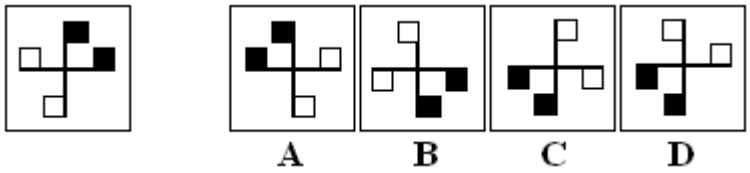
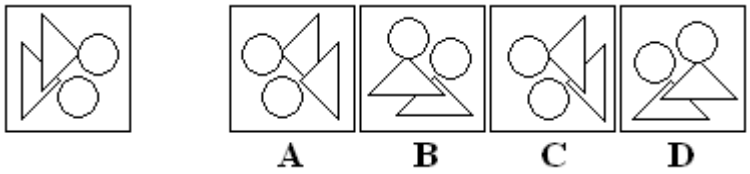
26) Which figure is identical to the first?



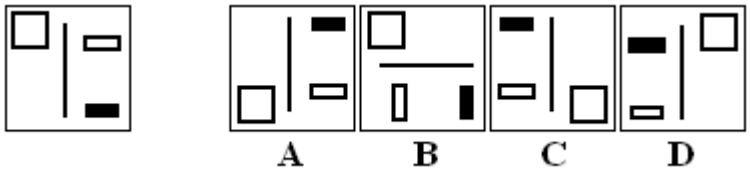
A B C D

27) Which figure is identical to the first?



A B C D

28) Which figure is identical to the first?



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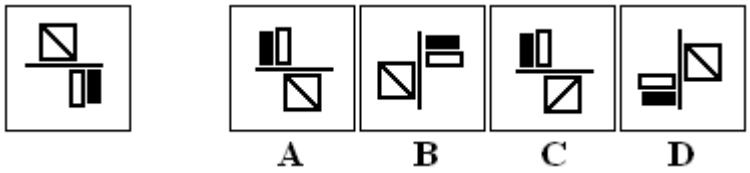

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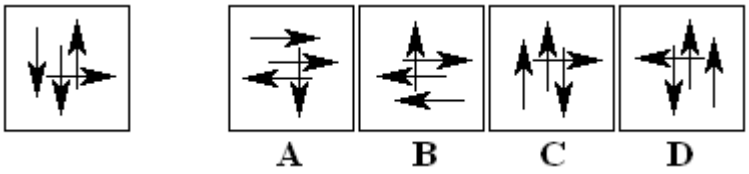

A B C D

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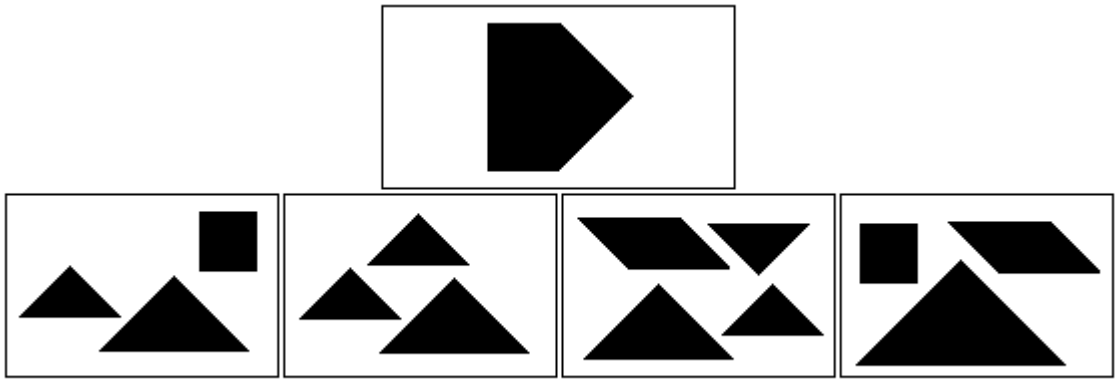
Mechanical Aptitude Tests

31) Which group of shapes can be assembled to make the shape shown?



A B C D

A B C D

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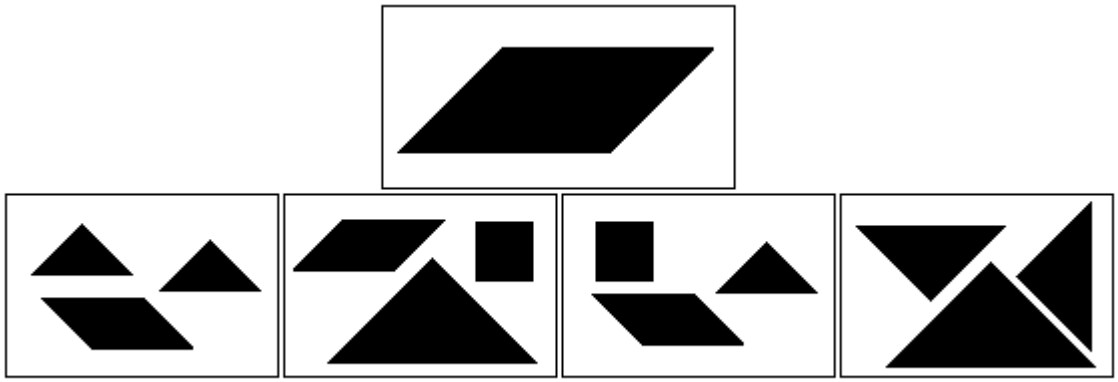

A B C D

A B C D

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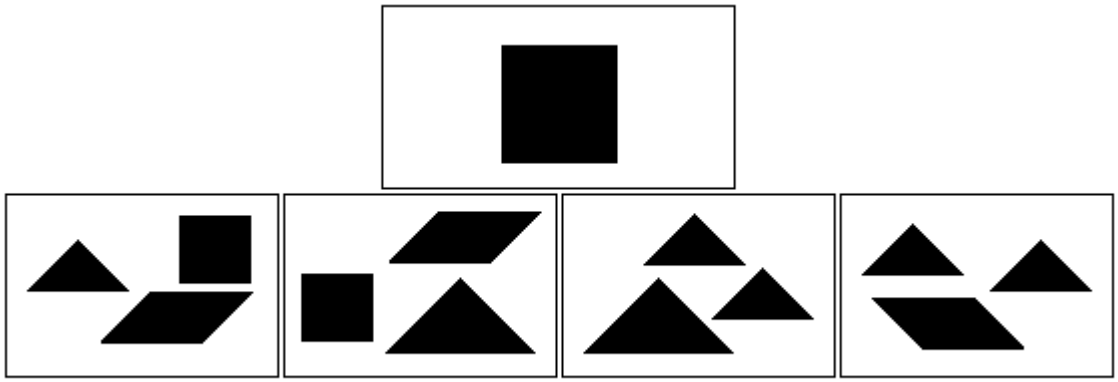
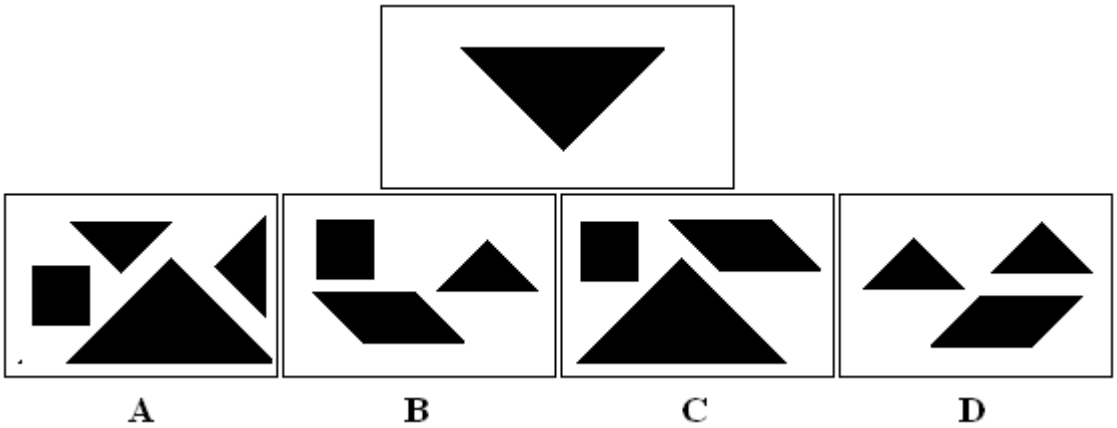


A B C D

A B C D

Mechanical Aptitude Tests

34) Which shape can be assembled using all of the individual shapes shown?



A B C D

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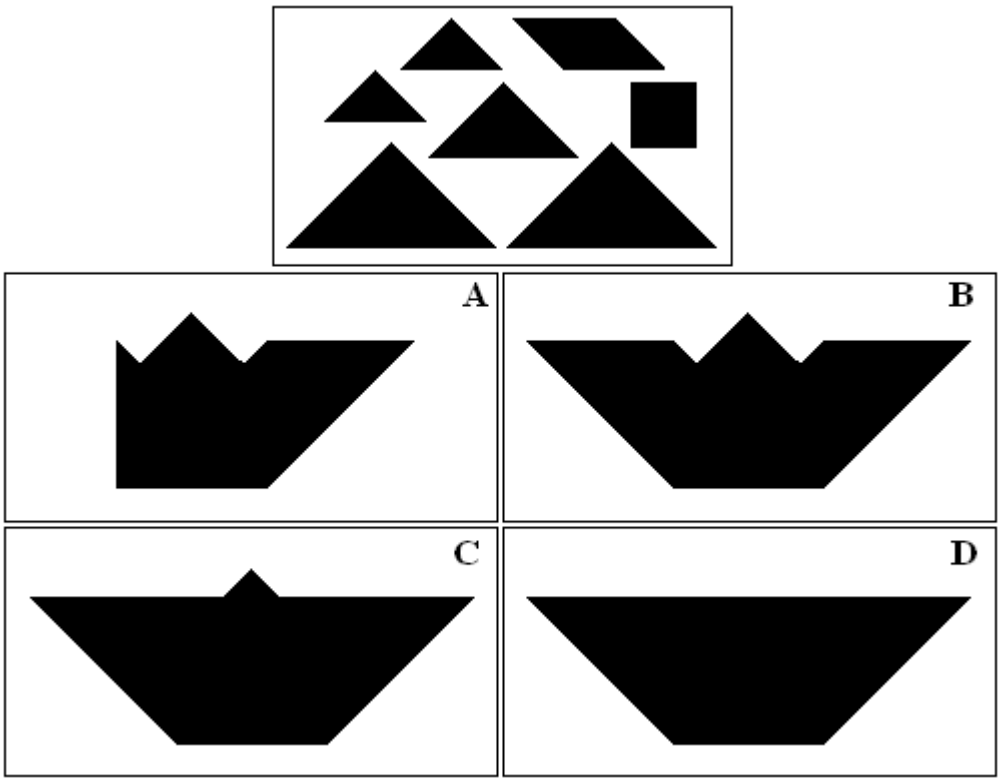
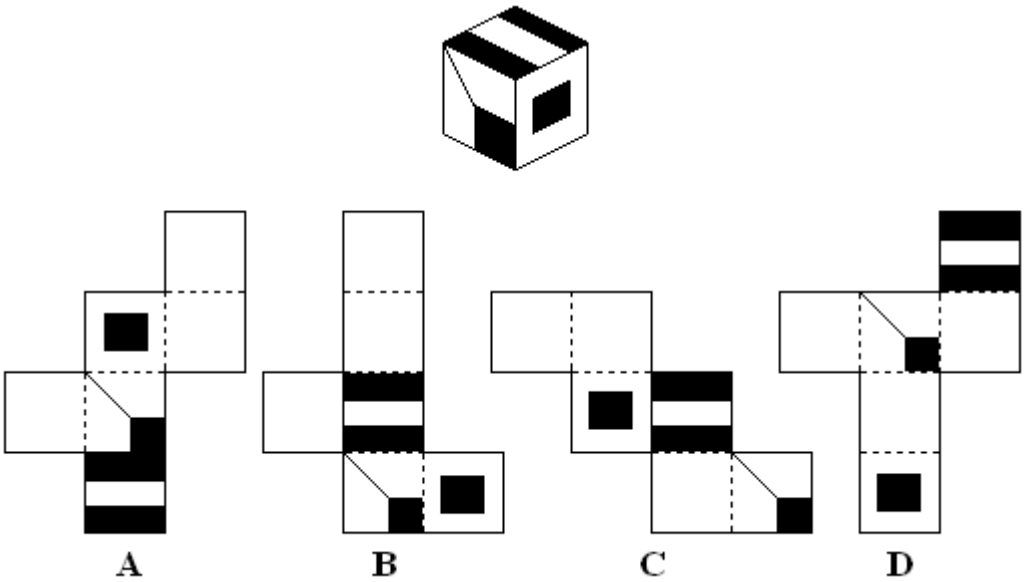


A B C D

Mechanical Aptitude Tests

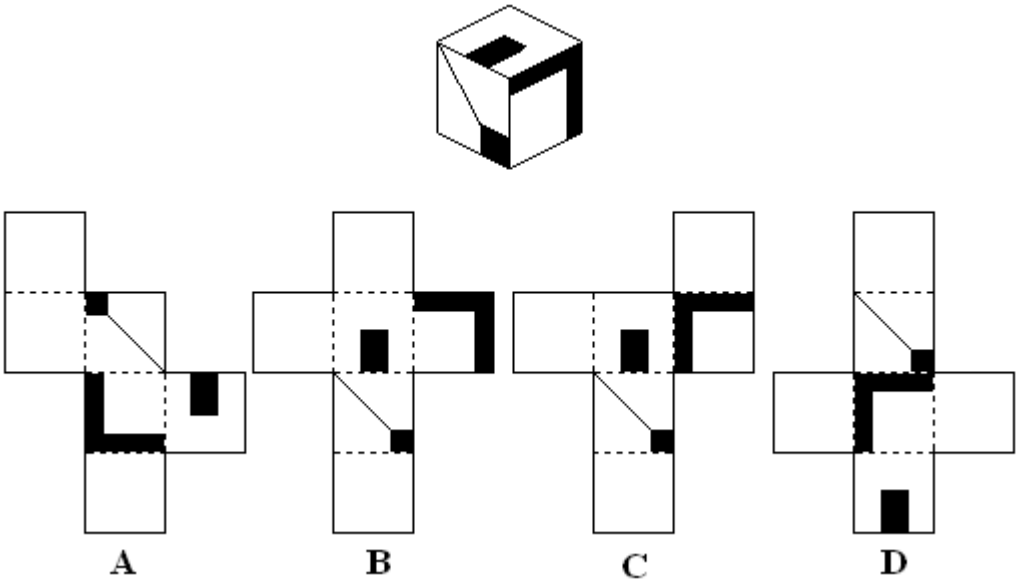
Which pattern can be folded to make the cube shown?

36)



A B C D

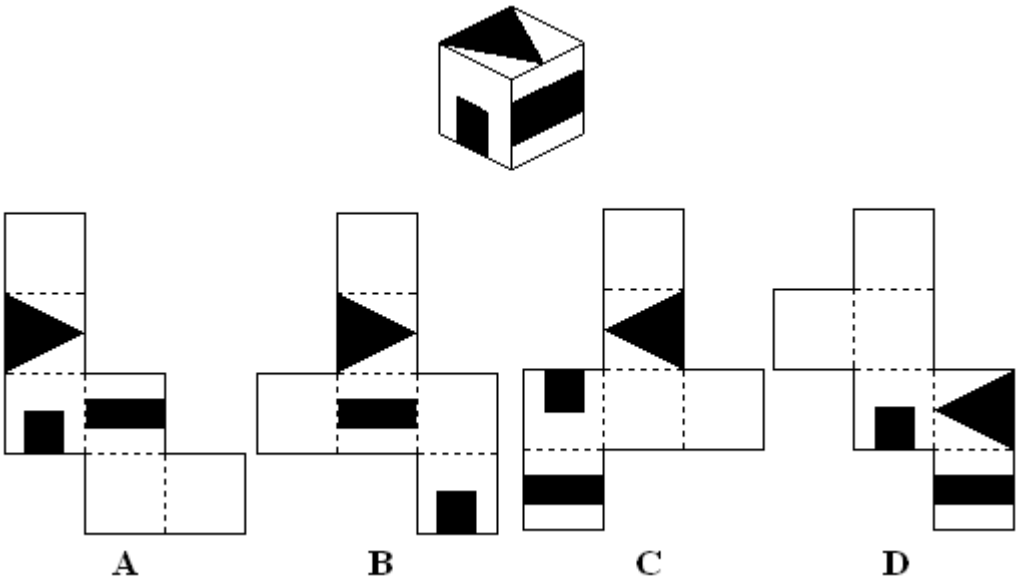
37)



A B C D

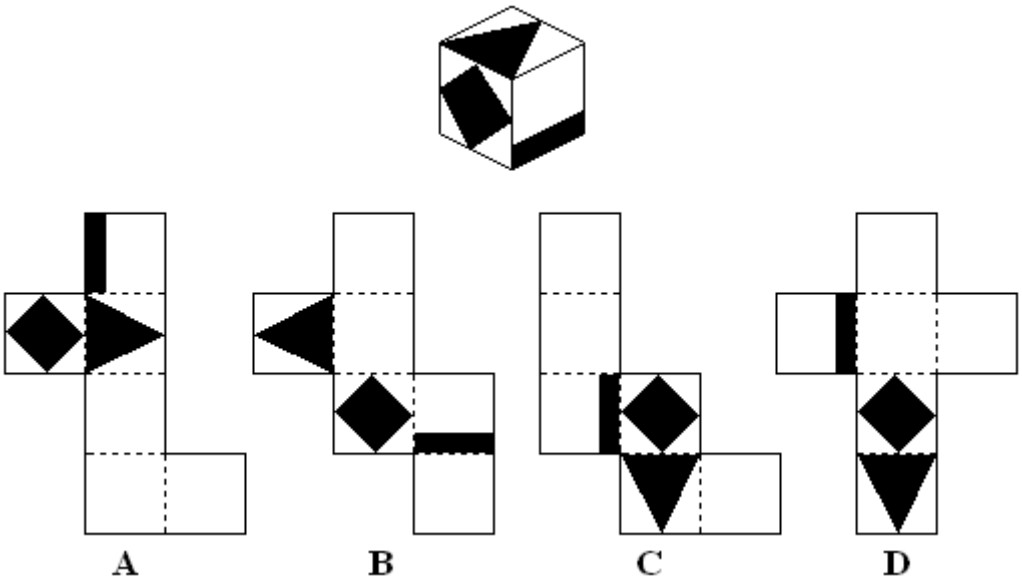
Mechanical Aptitude Tests

38)



A B C D

39)



A B C D

Mechanical Aptitude Tests

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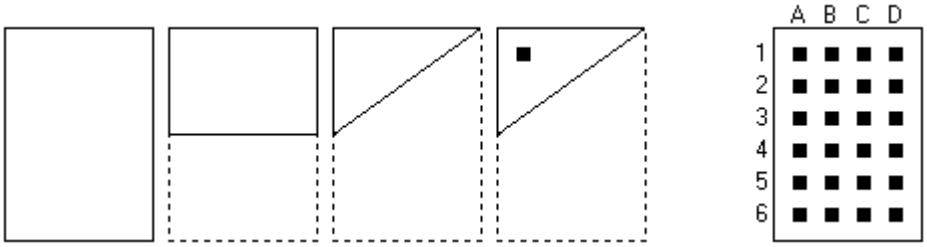
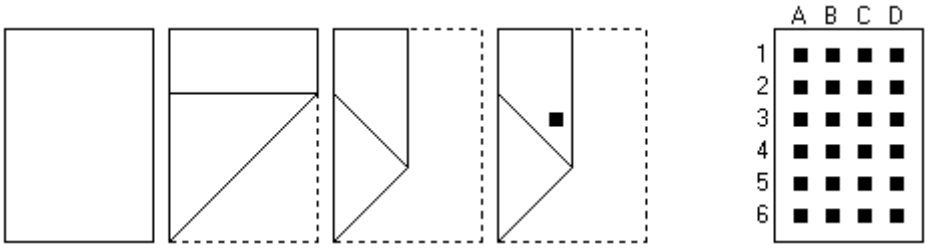
40)



A	B	C	D
1A,4C,4D,6A	1A,4C,4E,6A	1A,3C,4D,6B	1A,3C,4D,6A

A B C D

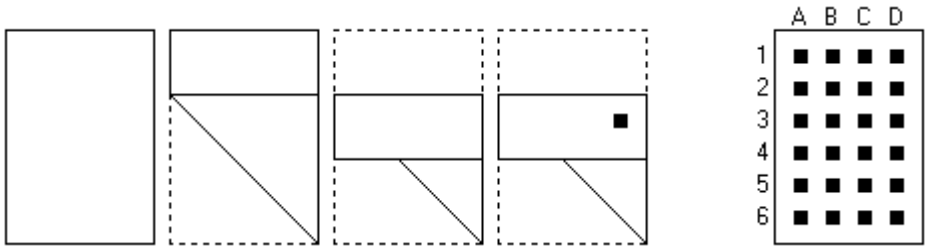
41)



A	B	C	D
3B,3D,5D	3B,3C,5C	3B,3C,5D	3B,3D,5E

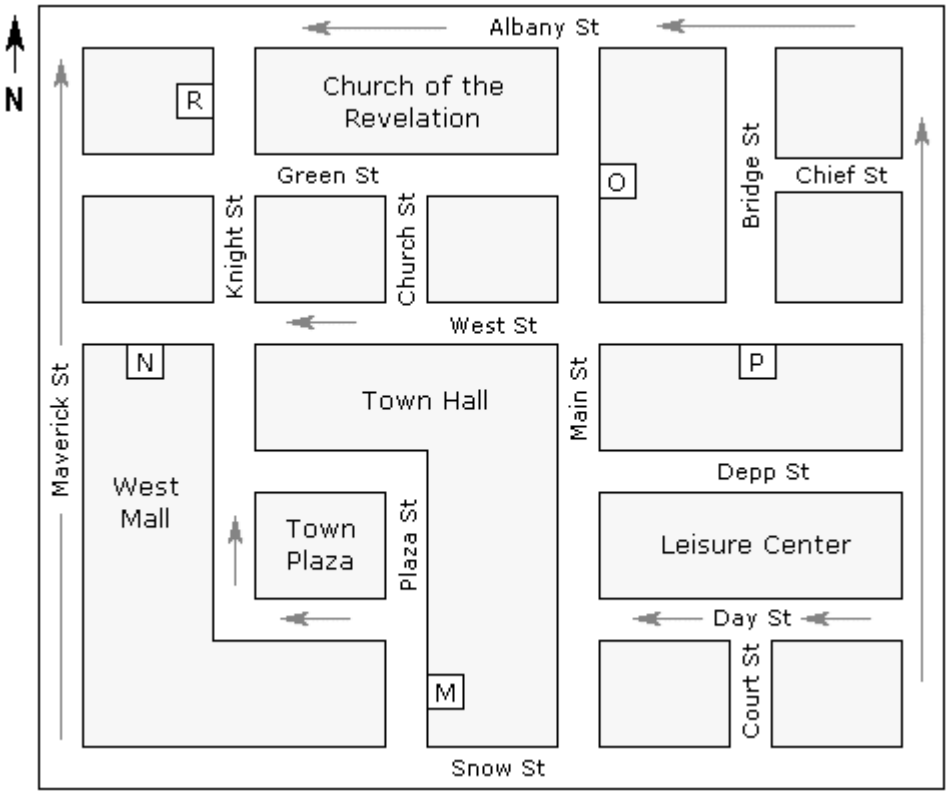
A B C D

42)



A	B	C	D
2D,3D,6A	1D,2D,6A	2D,4D,6A	2D,4D,5A

A B C D



43) Officer Wilkinson is in Depp St and can see the Town Hall to her right. What direction is she facing?

A	B	C	D
North	South	East	West

44) She turns and walks to the junction with Main St. She turns left and proceeds two blocks before turning right, then taking the next right, and walking half a block. Which location is nearest to her current position?

A	B	C	D
M	N	R	P

45) Officer Garcia starts from location 'N' and proceeds as follows: right onto West St - heading East, fourth left - heading North, first right - heading East, first right - heading South, third right – heading West. He proceeds West for one block. Where is location 'P' in relation to his current position?

A	B	C	D
North	South East	North East	North West

End of Spatial Ability - Test 4

Mechanical Aptitude Tests

Answers to Spatial Ability Tests 1-4

Question	Test 1	Test 2	Test 3	Test 4
1)	**N**	**M**	**S**	**R**
2)	**Q**	**W**	**J**	**J**
3)	**T**	**K**	**P**	**V**
4)	**F**	**T**	**V**	**T**
5)	**X**	**I**	**W**	**K**
6)	**H**	**Q**	**N**	**X**
7)	**P**	**S**	**E**	**L**
8)	**D**	**J**	**M**	**Q**
9)	**A**	**G**	**U**	**W**
10)	**S**	**A**	**G**	**G**
11)	**V**	**X**	**Y**	**Y**
12)	**I**	**U**	**C**	**O**
13)	**E**	**D**	**K**	**P**
14)	**K**	**O**	**A**	**S**
15)	**M**	**C**	**T**	**D**
16)	**W**	**B**	**X**	**N**
17)	**B**	**H**	**B**	**B**
18)	**R**	**Y**	**I**	**E**
19)	**G**	**P**	**Q**	**U**
20)	**Y**	**N**	**H**	**H**
21)	**O**	**V**	**R**	**M**
22)	**J**	**R**	**O**	**F**
23)	**L**	**L**	**L**	**I**
24)	**U**	**F**	**D**	**C**
25)	**C**	**E**	**F**	**A**
26)	**C**	**C**	**B**	**C**
27)	**B**	**A**	**D**	**B**
28)	**A**	**D**	**A**	**C**
29)	**D**	**C**	**C**	**A**
30)	**B**	**B**	**C**	**D**
31)	**C**	**A**	**D**	**C**
32)	**B**	**D**	**B**	**D**
33)	**B**	**D**	**A**	**C**
34)	**B**	**C**	**C**	**D**
35)	**D**	**A**	**C**	**B**
36)	**A**	**C**	**B**	**B**
37)	**B**	**D**	**C**	**C**
38)	**A**	**D**	**A**	**A**
39)	**B**	**C**	**D**	**D**
40)	**B**	**B**	**D**	**D**
41)	**D**	**D**	**B**	**C**
42)	**A**	**A**	**A**	**A**
43)	**C**	**C**	**D**	**B**
44)	**C**	**D**	**C**	**A**
45)	**D**	**A**	**B**	**A**

Numerical Reasoning

Author : Paul Newton
Version: 2.3

Mechanical Aptitude Tests

The Importance of Numerical Ability Tests

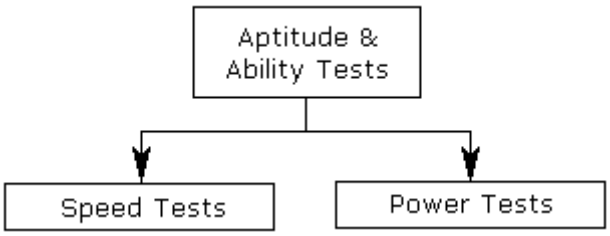
Numeric ability questions appear in most technical aptitude tests because employers usually want some indication of your ability to use numbers even if this is not a major part of the job. The test may include basic arithmetic, number sequences and simple mathematics.

You may also be asked numerical reasoning questions which are designed to test your reasoning ability rather than your ability to do calculations. They invariably include some number series questions where you need to work out which number or numbers are missing from the series and may also include questions where a mathematical problem is posed in words and your task is to apply the necessary logic to find the solution.

What do they Test?

The term 'numerical reasoning test' is often used interchangeably with 'numerical ability test' and 'numerical aptitude test'. There is no widely accepted definition of the difference between numerical ability and numerical aptitude and as far as psychometric tests are concerned the two terms are interchangeable. However the same does not apply to the term 'numerical reasoning' which does have a specific meaning. The term 'numerical reasoning test' should really only be used to describe a specific type of numerical test.

This is best explained as follows: there are basically two types of numerical questions that appear in psychometric tests. Speed questions are so easy that with unlimited time most people taking the test could answer them all successfully. However, the time allowed to complete the test is so short that even the most able person is not expected to finish. This means that the result depends on the number of correct answers made in the relatively short time allowed.



In contrast, a power test contains questions that vary in difficulty and no one is expected to get all of the answers correct even with unlimited time. In practice, a definite but ample time is set for power tests.

Even though there are literally hundreds of numerical ability tests available to employers, the questions used in these tests can be classified into four basic types: Numerical computation, numerical estimation, numerical reasoning and data interpretation. The last group, data interpretation, are restricted to management and graduate level tests.

Numerical Computation
These questions involve the basic principles of arithmetic including: addition, subtraction, multiplication, division, percentages, ratios, fractions and decimals. To score well on these questions you will need to be able to make quick and accurate calculations without using a calculator.

Numerical Estimation
Numerical estimation is key in many technical jobs where you need to quickly and accurately make estimates of material quantities etc. These questions require you to make quick estimates of the answers to straightforward numerical questions. You do not have time to actually calculate these answers because this will take up too much time and prevent you from answering enough questions to get a good score.

Numerical Reasoning
These questions test your reasoning ability rather than your ability to do calculations. In other words, you need to work out how to get the answer rather than simply being told what calculations to apply. They invariably include some number series questions where you need to work out which number or numbers are missing from the series. They also include text based questions where a mathematical problem is posed in words and your task is to apply the necessary logic to find the answer.

Mechanical Aptitude Tests

Numerical Computation Questions

Numerical computation questions involve the basic principles of arithmetic like addition, subtraction, multiplication and division. They also use mathematical terms and methods such as percentages, ratios, fractions and decimals. To score well on numerical computation questions you will need to make quick and accurate calculations.

Numerical Computation		Operatives	Supervisory	Management
Craft & Technical		Y	Y	Y
Clerical & Administrative		Y	Y	Y
Police, Fire, Military etc.	Y			
Management Trainee	Y			
Graduate & Professional	Y			

This type of test can be categorized as a speed test and is used to determine your basic numeracy. Obviously you will not be allowed to use a calculator.

If you are very rusty with arithmetic, try re-learning the times tables up to 12 and practice multiplication, division and percentage calculations. Practice can improve your test scores for all types of aptitude tests, so try as many examples as you can.

The speed at which you can answer these questions is the critical measure, as most people could achieve a very high score given unlimited time in which to answer. You can therefore expect 25-35 questions in 20-30 minutes.

This section contains 4 computation practice papers, answers and some information to refresh your memory about basic maths.

Mechanical Aptitude Tests

Numerical Computation Test 1: 30 Questions

Instructions: Answer as many questions as you can in 10 minutes. Circle the letter on the right which corresponds to the correct answer. Do not use a calculator.

1) 17 + 47 = 7 + ?

A	B	C	D	E
55	57	65	67	35

A B C D E

2) 33 + 18 = 29 + ?

A	B	C	D	E
21	24	32	22	37

A B C D E

3) 56 + 81 = 44 + ?

A	B	C	D	E
93	90	89	91	95

A B C D E

4) 44 – ? = 15

A	B	C	D	E
26	29	28	39	30

A B C D E

5) 87 – 35 = ?

A	B	C	D	E
53	42	51	41	52

A B C D E

6) 54 – 32 = 25 – ?

A	B	C	D	E
3	2	12	14	22

A B C D E

7) 7 × 8 = ?

A	B	C	D	E
49	56	64	54	52

A B C D E

8) 5 × ? = 45

A	B	C	D	E
5	6	7	8	9

A B C D E

9) 17 × 3 = ?

A	B	C	D	E
47	49	51	53	54

A B C D E

10) 140 ÷ 35 = ?

A	B	C	D	E
3	3.5	4	4.5	5

A B C D E

11) 28 ÷ ? = 7

A	B	C	D	E
3	3.5	4	4.5	5

A B C D E

12) 150 ÷ 100 = ?

A	B	C	D	E
1.3	1.5	1.7	15	0.75

A B C D E

13) $\frac{3}{5} \times ? = \frac{2}{5}$

A	B	C	D	E
$\frac{3}{5}$	$\frac{2}{3}$	$\frac{2}{5}$	$\frac{1}{5}$	$\frac{1}{4}$

A B C D E

14) $\frac{1}{4} + ? = \frac{3}{4}$

A	B	C	D	E
$\frac{1}{4}$	$\frac{1}{5}$	$\frac{3}{5}$	$\frac{1}{2}$	$\frac{2}{3}$

A B C D E

15) $2\frac{3}{5} - \frac{4}{5} = ?$

A	B	C	D	E
$1\frac{1}{4}$	$1\frac{1}{5}$	$1\frac{3}{5}$	$1\frac{1}{2}$	$1\frac{4}{5}$

A B C D E

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Mechanical Aptitude Tests

Circle Answer

16) $9\frac{7}{8} - 3\frac{1}{2} = ?$

A	B	C	D	E
$6\frac{3}{8}$	$6\frac{2}{3}$	$7\frac{3}{8}$	$5\frac{3}{8}$	$6\frac{1}{4}$

A B C D E

17) 60% of 120 = ?

A	B	C	D	E
65	70	62	72	54

A B C D E

18) 75% of 400 = ?

A	B	C	D	E
320	300	375	310	250

A B C D E

19) 22% of 200 = ?

A	B	C	D	E
42	44	40	88	46

A B C D E

20) 45% of 500 = ?

A	B	C	D	E
210	225	205	240	230

A B C D E

21) 33.6 + 8.7 = ?

A	B	C	D	E
42.3	43.3	42.5	43.7	38.7

A B C D E

22) 56.9 – 7.4 = ?

A	B	C	D	E
48.3	47.9	45.9	49.3	49.5

A B C D E

23) 0.7 × 0.5 = ?

A	B	C	D	E
0.33	0.35	0.75	1.40	3.50

A B C D E

24) 1.8 × 1.5 = ?

A	B	C	D	E
2.5	2.0	2.4	2.6	2.7

A B C D E

25) 12.8 × ? = 3.2

A	B	C	D	E
0.20	0.25	0.30	0.33	0.40

A B C D E

26) If one ream of paper costs $3.95 how much would 4 reams cost?

A	B	C	D	E
$15.75	$15.70	$15.72	$15.80	$15.77

A B C D E

27) If John starts work at 8:45 am and finishes at 5:15 pm. He has 90 minutes of breaks. How many hours does he work in 5 days?

A	B	C	D	E
38	39	35	40	32

A B C D E

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Circle Answer

28) A restaurant bill is made up of the following: $12.50 for starters, $28.55 for main courses and $8.95 for deserts, plus a 15% service charge. How much is the bill?

A	B	C	D	E
$56.50	$57.50	$57.00	$59.50	$60.50

A B C D E

29) A team of eight lumberjacks cut an average of 15,000 cubic feet of timber in a week. How many cubic feet will four lumberjacks cut in four weeks?

A	B	C	D	E
30,000	25,000	32,000	16,000	28,000

A B C D E

30) A discount of 15% is offered on an item which previously cost $1.80. What is the discounted price?

A	B	C	D	E
$1.53	$1.40	$1.55	$1.60	$1.52

A B C D E

End of Numerical Computation Test 1

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Mechanical Aptitude Tests

Numerical Computation Test 2: 30 Questions

Instructions: Answer as many questions as you can in 10 minutes. Circle the letter on the right which corresponds to the correct answer. Do not use a calculator.

1) 19 + 36 = 8 + ?

A	B	C	D	E
49	44	37	47	46

A B C D E

2) 37 + 17 = 9 + ?

A	B	C	D	E
45	47	54	41	55

A B C D E

3) 62 + 71 = 33 + ?

A	B	C	D	E
102	90	97	100	101

A B C D E

4) 43 – ? = 17

A	B	C	D	E
18	24	26	36	25

A B C D E

5) 89 – 32 = ?

A	B	C	D	E
59	42	47	55	57

A B C D E

6) 53 – 29 = 25 – ?

A	B	C	D	E
2	1	0	3	-1

A B C D E

7) 6 × 7 = ?

A	B	C	D	E
48	56	36	49	42

A B C D E

8) 3 × ? = 27

A	B	C	D	E
5	6	8	9	7

A B C D E

9) 19 × 4 = ?

A	B	C	D	E
76	66	74	78	67

A B C D E

10) 135 ÷ 27 = ?

A	B	C	D	E
3	5	4	6	7

A B C D E

11) 26 ÷ ? = 6.5

A	B	C	D	E
3.5	4	3	4.5	5

A B C D E

12) 250 ÷ 50 = ?

A	B	C	D	E
20	50	5	25	7

A B C D E

13) $\frac{1}{5} \times \frac{1}{2} = ?$

A	B	C	D	E
$\frac{3}{5}$	$\frac{2}{3}$	$\frac{1}{10}$	$\frac{2}{5}$	$\frac{1}{4}$

A B C D E

14) $\frac{3}{4} + ? = 1\frac{1}{2}$

A	B	C	D	E
$\frac{3}{4}$	$\frac{1}{5}$	$\frac{3}{5}$	$\frac{1}{2}$	$\frac{2}{3}$

A B C D E

15) $7\frac{3}{7} - 5\frac{6}{7} = ?$

A	B	C	D	E
$1\frac{2}{7}$	$1\frac{1}{5}$	$1\frac{3}{7}$	$1\frac{1}{2}$	$1\frac{4}{7}$

A B C D E

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Mechanical Aptitude Tests

Circle Answer

16) $7\frac{7}{8} - 2\frac{3}{4} = ?$

A	B	C	D	E
$6\frac{1}{8}$	$5\frac{2}{3}$	$5\frac{3}{8}$	$5\frac{1}{8}$	$6\frac{1}{4}$

A B C D E

17) 75% of 360 = ?

A	B	C	D	E
275	300	280	270	290

A B C D E

18) 60% of 400 = ?

A	B	C	D	E
210	240	230	250	245

A B C D E

19) 13% of 200 = ?

A	B	C	D	E
23	36	6.5	26	28

A B C D E

20) 35% of 500 = ?

A	B	C	D	E
180	170	165	177	175

A B C D E

21) 37 × 4.6 = ?

A	B	C	D	E
170.2	180.2	165.2	177.2	175.2

A B C D E

22) 28.9 – 7.3 = ?

A	B	C	D	E
2.6	20.6	21.6	21	26.1

A B C D E

23) 0.8 × 0.5 = ?

A	B	C	D	E
4.0	0.4	0.04	0.2	0.25

A B C D E

24) 1.6 × 1.5 = ?

A	B	C	D	E
3.1	2.4	2.6	4.2	2.3

A B C D E

25) 13.6 × 6.8 = ?

A	B	C	D	E
92	92.84	91.08	92.48	93.48

A B C D E

26) If 3 reams of paper costs $5.85 how much would 4 reams cost?

A	B	C	D	E
$7.95	$7.90	$7.75	$7.60	$7.80

A B C D E

27) If John starts work at 8:00 am and finishes at 5:15 pm. He has 60 minutes of breaks. How many hours does he work in 4 days?

A	B	C	D	E
32.5	34.5	24.5	33.5	33.0

A B C D E

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Circle Answer

28) A restaurant bill is made up of the following: $10.50 for starters, $26.55 for main courses and $7.95 for deserts, plus a 10% service charge. How much is the bill?

A	B	C	D	E
$48.50	$49.00	$50.00	$49.50	$50.50

A B C D E

29) A team of six lumberjacks cut an average of 18,000 cubic feet of timber in a week. How many cubic feet will four lumberjacks cut in three weeks?

A	B	C	D	E
36,000	60,000	68,000	7,200	6,000

A B C D E

30) A discount of 17% is offered on an item previously costing $2.00. What is the discounted price?

A	B	C	D	E
$1.69	$1.83	$1.66	$1.60	$1.65

A B C D E

End of Numerical Computation Test 2

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Mechanical Aptitude Tests

Numerical Computation Test 3: 30 Questions

Instructions: Answer as many questions as you can in 10 minutes. Circle the letter on the right which corresponds to the correct answer. Do not use a calculator.

1) 18 + 54 = 9 + ?

A	B	C	D	E
53	62	59	65	63

A B C D E

2) 43 + 15 = 27 + ?

A	B	C	D	E
33	31	29	30	21

A B C D E

3) 54 + 61 = 37 + ?

A	B	C	D	E
68	77	78	71	69

A B C D E

4) 39 – ? = 16

A	B	C	D	E
13	22	25	23	21

A B C D E

5) 85 – 29 = ?

A	B	C	D	E
56	54	66	65	55

A B C D E

6) 45 – 28 = 22 – ?

A	B	C	D	E
7	3	5	4	6

A B C D E

7) 7 × 7 = ?

A	B	C	D	E
56	49	42	47	59

A B C D E

8) 4 × ? = 28

A	B	C	D	E
4	8	6	7	5

A B C D E

9) 18 × 3 = ?

A	B	C	D	E
56	62	52	59	54

A B C D E

10) 174 ÷ 29 = ?

A	B	C	D	E
8	4	6	5	7

A B C D E

11) 35 ÷ ? = 7

A	B	C	D	E
6	5	4	8	3

A B C D E

12) 420 ÷ 7 = ?

A	B	C	D	E
70	80	65	60	75

A B C D E

13) $\frac{2}{5} \times \frac{1}{4} = ?$

A	B	C	D	E
$\frac{2}{7}$	$\frac{3}{9}$	$\frac{1}{10}$	$\frac{1}{5}$	$1\frac{1}{4}$

A B C D E

14) $\frac{7}{8} + ? = 1\frac{3}{4}$

A	B	C	D	E
$\frac{1}{4}$	$\frac{7}{8}$	$\frac{3}{5}$	$\frac{1}{2}$	$\frac{2}{3}$

A B C D E

15) $4\frac{3}{8} - 3\frac{1}{4} = ?$

A	B	C	D	E
$1\frac{1}{4}$	$\frac{7}{8}$	$1\frac{3}{5}$	$1\frac{1}{8}$	$1\frac{3}{8}$

A B C D E

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Mechanical Aptitude Tests

Circle Answer

16) $6\frac{3}{8} - 2\frac{1}{2} = ?$

A	B	C	D	E
$3\frac{3}{8}$	$3\frac{7}{8}$	$2\frac{3}{8}$	$3\frac{1}{8}$	$3\frac{1}{4}$

A B C D E

17) 25% of 180 = ?

A	B	C	D	E
35	42	40	30	45

A B C D E

18) 80% of 400 = ?

A	B	C	D	E
320	290	280	310	300

A B C D E

19) 28% of 200 = ?

A	B	C	D	E
54	57	56	55	52

A B C D E

20) 65% of 500 = ?

A	B	C	D	E
365	325	370	375	380

A B C D E

21) 27.6 + 8.3 = ?

A	B	C	D	E
39.9	34.9	36.9	35.9	35.8

A B C D E

22) 36.8 – 7.2 = ?

A	B	C	D	E
29.4	30.6	29.6	28.6	26.9

A B C D E

23) 0.8 × 0.25 = ?

A	B	C	D	E
1.8	2.0	0.22	0.2	0.15

A B C D E

24) 1.7 × 1.5 = ?

A	B	C	D	E
2.55	1.75	3.50	2.45	2.00

A B C D E

25) 14.4 ÷ 1.2 = ?

A	B	C	D	E
9.2	1.2	10.2	12.0	12.2

A B C D E

26) If 2 reams of paper costs $4.80 how much would 5 reams cost?

A	B	C	D	E
$12.80	$14.00	$12.00	$11.80	$24.00

A B C D E

27) If John starts work at 6:45 am and finishes at 2:15 pm. He has 60 minutes of breaks. How many hours does he work in 12 days?

A	B	C	D	E
68	77	76	80	78

A B C D E

Circle Answer

28) A restaurant bill is made up of the following: $22.50 for starters, $38.35 for main courses and $19.15 for deserts, plus a 15% service charge. How much is the bill?

A	B	C	D	E
$90.00	$92.00	$92.80	$94.80	$94.00

A B C D E

29) A team of three lumberjacks cut an average of 15,000 cubic feet of timber in a week. How many cubic feet will four lumberjacks cut in four weeks?

A	B	C	D	E
60,000	70,000	75,000	80,000	85,000

A B C D E

30) A discount of 15% is offered on an item previously costing $3.80. What is the discounted price?

A	B	C	D	E
3.23	3.25	3.22	3.20	3.18

A B C D E

End of Numerical Computation Test 3

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Numerical Computation Test 4: 30 Questions

Instructions: Answer as many questions as you can in 10 minutes. Circle the letter on the right which corresponds to the correct answer. Do not use a calculator.

1) 13 + 36 = 17 + ?

A	B	C	D	E
34	32	35	33	29

A B C D E

2) 47 + 17 = 36 + ?

A	B	C	D	E
25	29	28	38	27

A B C D E

3) 52 + 61 = 33 + ?

A	B	C	D	E
77	70	73	80	83

A B C D E

4) 42 – ? = 17

A	B	C	D	E
25	27	23	24	22

A B C D E

5) 89 – 22 = ?

A	B	C	D	E
71	69	77	76	67

A B C D E

6) 43 – 29 = 21 – ?

A	B	C	D	E
14	6	7	9	8

A B C D E

7) 8 × 9 = ?

A	B	C	D	E
88	63	79	72	64

A B C D E

8) 7 × ? = 84

A	B	C	D	E
13	12	11	14	9

A B C D E

9) 17 × 5 = ?

A	B	C	D	E
85	75	90	80	95

A B C D E

10) 108 ÷ 27 = ?

A	B	C	D	E
7	6	5	4	3

A B C D E

11) 36 ÷ ? = 6

A	B	C	D	E
6.5	5.8	5.7	5.5	6.0

A B C D E

12) 450 ÷ ? = 9

A	B	C	D	E
40	50	65	45	60

A B C D E

13) $\frac{3}{5} \times ? = 1\frac{1}{5}$

A	B	C	D	E
$1\frac{3}{5}$	2	$1\frac{2}{5}$	$2\frac{1}{5}$	$2\frac{1}{4}$

A B C D E

14) $1\frac{1}{4} + ? = 2\frac{3}{8}$

A	B	C	D	E
$1\frac{1}{4}$	$1\frac{3}{8}$	$1\frac{1}{8}$	$\frac{5}{8}$	$\frac{7}{8}$

A B C D E

15) $3\frac{3}{5} - 1\frac{4}{5} = ?$

A	B	C	D	E
$1\frac{1}{4}$	$1\frac{1}{5}$	$1\frac{3}{5}$	$1\frac{1}{2}$	$1\frac{4}{5}$

A B C D E

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Circle Answer

16) $7\frac{7}{8} - 3\frac{1}{4} = ?$

A	B	C	D	E
$4\frac{3}{8}$	$4\frac{2}{3}$	$3\frac{3}{8}$	$4\frac{5}{8}$	$4\frac{1}{4}$

A B C D E

17) 70% of 70 = ?

A	B	C	D	E
50	47	55	49	42

A B C D E

18) 80% of 400 = ?

A	B	C	D	E
300	320	340	325	280

A B C D E

19) 34% of 200 = ?

A	B	C	D	E
70	66	68	67	69

A B C D E

20) 75% of 500 = ?

A	B	C	D	E
370	365	360	355	375

A B C D E

21) 46.3 + 9.7 = ?

A	B	C	D	E
58	54	56	57	65

A B C D E

22) 38.6 – 17.2 = ?

A	B	C	D	E
21.4	18.4	19.4	20.6	20.4

A B C D E

23) 0.8 × 0.75 = ?

A	B	C	D	E
0.85	1.50	0.75	0.15	0.60

A B C D E

24) 1.4 × 1.5 = ?

A	B	C	D	E
0.70	2.80	0.75	2.15	2.10

A B C D E

25) 29.2 × ? = 7.3

A	B	C	D	E
1.25	0.25	1.20	0.20	0.33

A B C D E

26) If 4 reams of paper cost $10 how much would 7 reams cost?

A	B	C	D	E
$18.50	$16.50	$16.00	$17.50	$17.00

A B C D E

27) If John starts work at 8:30 am and finishes at 5:10 pm. He has 100 minutes of breaks. How many hours does he work in 5 days?

A	B	C	D	E
37	38	39	35	40

A B C D E

Circle Answer

28) A restaurant bill is made up of the following: $11.50 for starters, $27.55 for main courses and $10.95 for deserts, plus a 15% service charge. How much is the bill?

A	B	C	D	E
$57.50	$57.25	$57.00	$60.50	$55.50

A B C D E

29) A team of three lumberjacks cut an average of 45,000 cubic feet of timber in a week. How many thousand cubic feet will seven lumberjacks cut in two weeks?

A	B	C	D	E
21	105	225	210	22

A B C D E

30) A discount of 15% is offered on an item previously costing $5.00. What is the discounted price?

A	B	C	D	E
4.12	4.40	4.25	4.80	4.75

A B C D E

End of Numerical Computation Test 4

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Mechanical Aptitude Tests

Answers to Numerical Computation Tests 1-4

Question	Test 1	Test 2	Test 3	Test 4
1)	**B**	**D**	**E**	**B**
2)	**D**	**A**	**B**	**C**
3)	**A**	**D**	**C**	**D**
4)	**B**	**C**	**D**	**A**
5)	**E**	**E**	**A**	**E**
6)	**A**	**B**	**C**	**C**
7)	**B**	**E**	**B**	**D**
8)	**E**	**D**	**D**	**B**
9)	**C**	**A**	**E**	**A**
10)	**C**	**B**	**C**	**D**
11)	**C**	**B**	**B**	**E**
12)	**B**	**C**	**D**	**B**
13)	**B**	**C**	**C**	**B**
14)	**D**	**A**	**B**	**C**
15)	**E**	**E**	**D**	**E**
16)	**A**	**D**	**B**	**D**
17)	**D**	**D**	**E**	**D**
18)	**B**	**B**	**A**	**B**
19)	**B**	**D**	**C**	**C**
20)	**B**	**E**	**B**	**E**
21)	**A**	**A**	**B**	**C**
22)	**E**	**C**	**C**	**A**
23)	**B**	**B**	**D**	**E**
24)	**E**	**B**	**A**	**E**
25)	**B**	**D**	**D**	**B**
26)	**D**	**E**	**C**	**D**
27)	**C**	**E**	**E**	**D**
28)	**B**	**D**	**B**	**A**
29)	**A**	**A**	**D**	**D**
30)	**A**	**C**	**A**	**C**

Mechanical Aptitude Tests

Practicing Numerical Computation

The two biggest mistakes people make when faced with numerical computation tests are:

1. To fool themselves that they can remember their entire elementary maths curriculum even though they may not have used most of it for years.
2. To feel too embarrassed to sit down and relearn the parts that they may have forgotten.

Don't make these mistakes yourself. It's better to feel a little bit embarrassed now (in private) and quietly relearn how to do these things than to sit there in the test desperately trying to remember how to do them. Not only will taking action now improve your scores but it will avoid that sickening feeling after the test when you know that with a little more preparation you could have done a whole lot better.

Simple Arithmetic
You can achieve a big improvement in your scores by practicing your mental arithmetic until you are both quick and confident. Your score in the simple speed tests will be very much influenced by your ability to add, subtract, multiply and divide quickly and accurately.

In a typical test of 60 mixed questions, you may need to perform over two hundred separate multiplication and division operations. Knowing your multiplication tables (up to the ten times table) and being able to give an instant answer to any operation is essential. The key word here is instant. Most people can provide instant answers to about 80% of these operations and take a few seconds to think about the remainder.

These extra few seconds add up over the course of the test, where answering a single question can involve 3 or 4 of these simple operations. Shaving a few seconds off here and there gives you time to answer an extra four or five questions in a typical test and this will make a big difference to your final score.

Firstly, you need to forget any embarrassment you may feel about practicing this first-grade material and secondly you need to be totally objective. You are very unlikely to have any problems with the 1, 2, 5 and 10 times tables – so you can probably ignore them.

Mechanical Aptitude Tests

Take a sheet of A4 paper and list the numbers 3, 4, 6, 7, 8, 9 in a column down the left-side of the page and along the top margin as shown.

	3	4	6	7	8	9
3						
4						
6						
7						
8						
9						

Work from left to right and top to bottom through the grid, writing in each answer. If you pause for even a second to think about the answer then place a dash in the space and continue. This is very important – do NOT stop to think, either write the answer immediately or put a dash in the space. You MUST complete this exercise in less than 40 seconds.

The result is usually a sheet with a few dashes, where the correct answer did not come instantly to mind. You will need to spend a bit of time refreshing your memory on these operations – do this over a few days, spending a couple of minutes at a time to go over them until they are instantaneous. When you are completely happy, try the grid again – you should be able to complete it correctly in less than 40 seconds.

The rules for performing basic arithmetic operations with whole numbers (integers) should be familiar to you although you may find that there are some things you have forgotten. For example:

i) Multiplication or division of two integers with different signs gives a negative result; for example, 5 x -3 = -15.
ii) Multiplication or division of two negative integers gives a positive result; for example, -3 x -3 = 9.

There are some tips and tricks to help you with your mental arithmetic which are worth spending time to consider and practice. They will all make a small difference to your overall speed, but if you can master a few of them, and they are all straightforward, then the cumulative effect will enable you to answer a few more questions in the allotted time. As you already know, these few extra marks can make a big difference to how you are perceived as a job candidate.

Mechanical Aptitude Tests

Subtraction
Consider the sum: **a – b,** there are three approaches to this type of calculation.

1) Direct Calculation
When the digits of b are all smaller than the digits of a, the calculation can be done digit by digit. For example, evaluate 862 − 41 simply by subtracting 1 from 2 in the units place, and 4 from 6 in the tens place: 821.

2) Indirect Calculation
When the above situation does not apply, the problem can sometimes be modified:
If only one digit in b is larger than its corresponding digit in a, diminish the offending digit in b until it is equal to its corresponding digit in a. Then subtract further the amount b was diminished by from a. For example, to calculate 872 − 92, turn the problem into 872 − 72 = 800. Then subtract 20 from 800: 780.

If more than one digit in b is larger than its corresponding digit in a, it may be easier to find how much must be added to b to get a. For example, to calculate 8192 − 732, we can add 8 to 732 (resulting in 740), then add 60 (to get 800), then 200 (for 1000). Next, add 192 to arrive at 1192, and, finally, add 7000 to get 8192. Our final answer is 7460.

3) Look-ahead Borrow Method
This method can be used to subtract numbers left to right, and with a little practice it can dramatically speed up mental subtraction.

One place at a time is handled, left to right.
Example:

```
  4075
- 1844
------
```

Thousands: 4 – 1 = 3, look to right, 075 < 844, need to borrow.
3 – 1 = 2, say "Two thousand"

Hundreds: 0 – 8 = negative numbers not allowed here,
10 – 8 = 2, 75 > 44 so no need to borrow, say "two hundred"

Tens: 7 – 4 = 3, 5 > 4 so no need to borrow, say "thirty"

Ones: 5 – 4 = 1, say "one"

This gives 2231 as the answer.

Mechanical Aptitude Tests

Multiplication
Calculating products: a × b

Multiplying by 2
In this case, the product can be essentially calculated digit by digit. This is not exactly the case because it is possible to have a remainder, but if there is a remainder, it is always 1, which simplifies things greatly. Still, the product must be calculated from right to left: 2 × 167 is by 4 with a remainder, then a 2 (so 3) with another remainder, then a 2 (so 3). Thus, we get 334.

Multiplying by 5
To multiply a number by 5, first multiply that number by 10, and then divide it by 2. The following algorithm is a quick way to produce this result:

First, append a zero to right side of the desired number. Next, starting from the leftmost numeral, divide by 2 and append each result in the respective order to form a new number; fraction answers should be rounded down to the nearest whole number.

For example, if you intended to multiply 176 by 5, you would first append a zero to 176 to make 1760. Next, divide 1 by 2 to get .5, rounded down to zero. Divide 7 by 2 to get 3.5, rounded down to 3. Divide 6 by 2 to get 3. Zero divided by two is simply zero. The resulting number is 0330.

The final step involves adding 5 to the number that follows any single numeral in this new number that was odd before dividing by two; this is better understood through the example. In the original number, 176, the first place is 1, which is odd. Therefore, we add 5 to the numeral after the first place in our newly constructed number (0330), which is 3; 3+5=8. The numeral in the second place of 176, 7, is also odd. Therefore the number-place after the corresponding numeral in the constructed number (0830) is increased by 5 as well; 3+5=8. The numeral in the third place of 176, 6, is even, therefore the final number, zero, in our answer is not changed. That final answer is 0880. The leftmost zero can be omitted, leaving 880. So 176 times 5 equals 880.

Multiplying by 9
Since 9 = 10 − 1, to multiply by 9, multiply the number by 10 and then subtract the original number from this result. For example, 9 × 27 = 270 − 27 = 243.

Mechanical Aptitude Tests

Multiplying two 2 digit numbers between 11 and 19
To easily multiply 2 digit numbers together between 11 and 19 you can use this simple grid method.

Consider the calculation 1a x 1b.

xx = a + b
yy = a x b

This can be visualized as:

1	0	0
x	x	0
0	y	y

So, for example the calculation 17 x 16 can be visualized as:

1	0	0
1	3	0
0	4	2

Adding the columns in the grid gives the answer 272, like this…

1	0	0
1	3	0
0	4	2
2	**7**	**2**

Multiplying Any 2 digit Numbers Together
To easily multiply any 2 digit numbers together a simple algorithm is as follows:

ab * cd

100*(a*c) + 10*(b*c) + 10*(a*d) + b*d

For example, 23 * 47 can be reduced to

800 (which is 2 * 4 * 100)
120 (which is 3 * 4 * 10)
140 (which is 7 * 2 * 10)
21 (which is 7* 3)

Adding these gives 1081.

You will need to practice these methods a few times to become proficient, but they can save significant time answering numeric calculation and estimation questions.

Mechanical Aptitude Tests

Fractions

A *fraction* is a number of the form *a/b*, where *a* and *b* are integers.

The *a* is called the *numerator* of the fraction, and *b* is called the *denominator*.

For example, $\frac{3}{5}$ is a fraction that has 3 as its numerator and 5 as its denominator.

This can be thought of as 3 divided by 5.

If the numerator and denominator of the fraction are both multiplied by the same integer, the resulting fraction will be equivalent. If the numerator and the denominator of the above fraction are multiplied by 5 then this gives 15/25.

Therefore $\frac{3}{5} = \frac{15}{25}$

.

Adding, Subtracting and Comparing Fractions

To add two fractions with the same denominator, you simply add the numerators and keep the denominator the same.

$\frac{3}{5} + \frac{1}{5} = \frac{4}{5}$

If the denominators are *not* the same, you need to make them the same before doing the addition. To do this, you need to get the 'lowest common denominator'.

In mathematics, the lowest common denominator (abbreviated LCD) is the lowest common multiple of the denominators of a set of fractions. That is, it is the smallest (non-zero) number that is a multiple of the denominators.

For instance, the LCD of ½ and ¼ is 4 because the lowest common multiple of 2 and 4 is 4. Remember, the denominator says what the numerator is 'divided by'. Any number divided by 4 will be lower than if it were divided by 2. Therefore 4 is the lowest common denominator.

Likewise the LCD of ½ and $\frac{1}{3}$ is 6, because the smallest (non-zero) number that is a multiple of 2 and 3 is 6. Using the LCD (or any multiple of it) as a denominator, enables addition, subtraction or comparison of fractions, for example:

$$\frac{1}{2} + \frac{1}{4} = \frac{2}{4} + \frac{1}{4} = \frac{3}{4};$$

$$\frac{1}{2} - \frac{1}{3} = \frac{3}{6} - \frac{2}{6} = \frac{1}{6};$$

$$\frac{1}{3} < \frac{2}{5} \text{ since } \frac{5}{15} < \frac{6}{15}.$$

Subtracting Fractions
The process for subtracting fractions is, in essence, the same as that of adding them: find a common denominator, and change each fraction to an equivalent fraction with the chosen common denominator. The resulting fraction will have that denominator, and its numerator will be the result of subtracting the numerators of the original fractions. For instance,

$$\frac{2}{3} - \frac{1}{2} = \frac{4}{6} - \frac{3}{6} = \frac{1}{6}.$$

Multiplying Fractions
To multiply two fractions, multiply the two numerators and multiply the two denominators (the denominators need not be the same). For example:

$$\tfrac{3}{4} * \tfrac{1}{2} = \frac{3}{8}$$

Multiplying Fractions by Whole Numbers
If you have a quarter of a cake, and you multiply the amount by three, then you end up with three quarters. We can write this numerically as follows:

$$3 \times \frac{1}{4} = \frac{3}{4}$$

As another example, suppose that five people work for three hours out of a seven hour day (i.e. for three sevenths of the work day). In total, they will have worked for 15 hours (5 x 3 hours each), or 15 sevenths of a day. Since 7 sevenths of a day is a whole day, 14 sevenths is two days, then in total, they will have worked for 2 days and a seventh of a day. Numerically:

$$5 \times \frac{3}{7} = \frac{15}{7} = 2\frac{1}{7}$$

Multiplying Fractions by Fractions
Consider the cake example again; if you have a quarter of a cake, and you multiply the amount by a third, then you end up with a twelfth of the cake. In other words, a third of a quarter (or a third times a quarter), is a twelfth. This is because each quarter is split into three pieces, and four quarters times three makes 12 parts (or twelfths). We can write this numerically as follows:

$$\frac{1}{3} \times \frac{1}{4} = \frac{1}{12}$$

When fractions are multiplied by fractions, simply multiply the two numerators (the top numbers), and multiply the two denominators (the bottom numbers). For example:

$$\frac{5}{6} \times \frac{7}{8} = \frac{5 \times 7}{6 \times 8} = \frac{35}{48}$$

Dividing Fractions
To divide one fraction by another, first *invert* the fraction you are dividing by, and then proceed as in multiplication. For example:

¾ / ½ = ¾ * 2/1 = 6/4 = 1 ½

Mixed Numbers and Improper Fractions
In the example above, an expression such as 1 ½ is called a mixed number. It means 1 plus 1/2. For instance, you could have two entire cakes and three quarters of another cake. The whole and fractional parts of the number are written next to each other: 2 + ¾ = 2 ¾.

Whenever you are asked to perform calculations with mixed numbers, you may find it easier to convert the mixed number to an improper fraction.

An improper fraction can be thought of as another way to write a mixed number; in the "2 ¾" example above, imagine that the two entire cakes are each divided into quarters. Each entire cake contributes $\frac{4}{4}$ to the total, so $\frac{4}{4} + \frac{4}{4} + ¾ = \frac{11}{4}$ is another way of writing 2 ¾.

A mixed number can be converted to an improper fraction in three steps:

1. Multiply the whole part times the denominator of the fractional part.
2. Add the numerator of the fractional part to that product.
3. The resulting sum is the numerator of the new (improper) fraction, and the new denominator is the same as that of the mixed number.

Similarly, an improper fraction can be converted to a mixed number:

1. Divide the numerator by the denominator.
2. The quotient (without remainder) becomes the whole part and the remainder becomes the numerator of the fractional part.
3. The new denominator is the same as that of the original improper fraction.

If you are asked to solve the following for example: $7\frac{1}{8} - 2\,¾$

a. Convert $7\frac{1}{8}$ to an improper fraction $\frac{57}{8}$
b. Convert 2 ¾ to an improper fraction $\frac{11}{4}$
c. Determine the LCD – which is 8
d. Convert $\frac{11}{4}$ to $\frac{22}{8}$
e. Calculate $\frac{57}{8} - \frac{22}{8} = \frac{35}{8}$
f. Convert $\frac{35}{8}$ to $4\frac{3}{8}$
g. Therefore $7\frac{1}{8} - 2\,¾ = 4\frac{3}{8}$

Mechanical Aptitude Tests

Decimals
All numbers can be expressed in decimal form using base 10. A decimal point is used, and the place value for each digit corresponds to a power of 10, depending on its position relative to the decimal point. For example, the number 62.437 has 5 digits, where:

"6" is the "tens" digit; the place value for "6" is 10.
"2" is the "units" digit; the place value for "2" is 1.
"4" is the "tenths" digit; the place value for "4" is 1/10.
"3" is the "hundredths" digit; the place value for "3" is 1/100
"7" is the "thousandths" digit; the place value for "7" is 11000
.
Therefore, 82.537 is a short way of writing 60 + 2 + 0.4 + 0.03 + 0.007.

This numeration system has implications for the basic operations. For addition and subtraction, you must always remember to line up the decimal points:

For example: 126.5 + 68.231 = 194.731 can be written with the decimal points aligned as:

1	2	6	.	5		
	6	8	.	2	3	1
1	9	4	.	7	3	1

To multiply decimals, it is not necessary to align the decimal points. To determine the correct position for the decimal point in the product, you simply add the number of digits to the right of the decimal points in the decimals being multiplied. This sum is the number of decimal places required in the product.

For example: 15.381 * 0.14 = 2.15334

In this example the first number has 3 decimal places, the second number has 2 decimal places, and therefore the product must have 5 decimal places.

To divide a decimal by another, such as 62.744 ÷ 1.24, first move the decimal point in the divisor to the right until the divisor becomes an integer, then move the decimal point in the dividend the same number of places.

This gives 6274.4 ÷ 124

This procedure determines the correct position of the decimal point in the quotient (as shown). The division can then proceed as normal.

Conversion of Decimals and Fractions
Conversion from a given decimal to an equivalent fraction is straightforward. Since each place value is a power of ten, every decimal can be converted easily to an integer divided by a power of ten. For example,

84.1 = 841/10
9.17 = 917/100
0.612 = 612/1000

The last example can be reduced to lowest terms by dividing the numerator and denominator by 4, which is their *greatest common factor*.

The **greatest common factor** of two non-zero integers, is the largest positive integer that divides both numbers without remainder.

So, 612/4 = 153 and 1000/4 = 250. Therefore 0.612 = 153/250.

Any fraction can be converted to an equivalent decimal. Since the fraction *a/b* means *a* divided by *b*, we can divide the numerator of a fraction by its denominator to convert the fraction to a decimal.

Mechanical Aptitude Tests

Percentage
Percentage is a way of expressing numbers as fractions of 100 and is often denoted using the percent sign, "%". For example, "45.1%" (read as "forty five point one percent") is equal to 0.451.

Percentages are used to express how large one quantity is in terms of another quantity. The first quantity is then usually a part of or a change in the second quantity. For example, an increase of $ 0.15 on a price of $ 2.50 is an increase by a fraction of 0.15/2.50 = 0.06. Expressed as a percentage, this is therefore an increase by 6%.

Percentages May be Relative
Due to inconsistent usage, it is not always clear from the context what a percentage is relative to. When speaking of a "10% rise" or a "10% fall" in a quantity, the usual interpretation is that this is relative to the initial value of that quantity; for example, a 10% increase on an item initially priced at $200 is $20, giving a new price of $220; to many people, any other usage is incorrect.

In the case of interest rates, however, it is a common practice to use the percent change differently: suppose that an initial interest rate is given as a percentage like 10%. Suppose the interest rate rises to 15%. This could be described as a 50% increase, measuring the increase relative to the initial value of the interest rate.

However, many people say in practice "The interest rate has risen by 5%". To counter this confusion, the unit "percentage points" is sometimes used when referring to differences of percentages. So, in the previous example, "The interest rate has increased by 5 percentage points" would be an unambiguous expression that the rate is now 15%.

With changes, percentage can be of any positive value. For example, a 100% growth is synonymous with doubling; a growth of 100% starting from 200 units is 200 units, increasing the total to 400.

Percentages and Cancellations
A common error when using percentages is to imagine that a percentage increase is cancelled out when followed by the same percentage decrease. A 50% increase from 100 is 100 + 50, or 150. A 50% reduction from 150 is 150 – 75, or 75. The end result is smaller than the 100 we started out with. This phenomenon is due to the change in the "initial" value after the first calculation. In this example, the first initial value is 100, but the second is 150.

In general, the net effect is: $(1 + x)(1 – x) = 1 – x2$, that is a net decrease proportional to the square of the percentage change.

To use a specific example, stock brokers came to understand that even if a stock has sunk 99%, it can nevertheless still sink another 99%. Also, if a stock rises by a large percentage, the trader still loses all of the stock's value if the stock subsequently drops 100%, meaning it has a zero value.

Mechanical Aptitude Tests

Numerical Estimation Questions

Numerical estimation questions test your ability to make quick estimates of the answers to fairly straightforward numerical questions. To score well on these questions you will need to make quick approximations of the answer. You must avoid the trap of working out the answer exactly, which will take up too much time and prevent you from answering enough questions to get a good score.

Numerical Estimation	Operatives	Supervisory	Management
Craft & Technical	Y	Y	Y
Clerical & Administrative	Y	Y	Y
Police, Fire, Military etc.			
Management Trainee			
Graduate & Professional			

Numerical estimation is vital in many craft and technical jobs where the ability to quickly and accurately estimate material quantities is essential. The speed at which you can answer these questions is the critical measure, as most people could achieve a very high score given unlimited time in which to answer. You can therefore expect 25-35 questions in 10 minutes or so.

Even though numerical estimation questions appear straightforward, it can take some time to develop the optimum compromise between speed and accuracy. Before you attempt to answer each question, look at the range of answers available and ask yourself how accurate your estimate needs to be. For example, is an order of magnitude sufficient or does the answer need to be worked out to the nearest whole number?

If you are out of practice with arithmetic, then try re-learning the times tables up to 12 and practice rough and ready multiplication, division and percentage calculations. Practice can improve your test scores for all types of aptitude tests but numerical estimation is one area where it can really make a difference, so try as many examples as you can.

Mechanical Aptitude Tests

Numerical Estimation Test 1: 35 Questions

Instructions: You need to estimate the answers to these questions, as you do not have time to calculate them precisely. Answer as many questions as you can in 10 minutes. Circle the letter on the right which is nearest to the correct answer. Do not use a calculator.

1) 347 + 198 = ?

A	B	C	D	E
650	550	580	590	600

A B C D E

2) 3,509 + 3,492 = ?

A	B	C	D	E
7000	7200	7100	7250	6950

A B C D E

3) 989 + 413 + 498 = ?

A	B	C	D	E
2,600	900	1,100	1,900	3,200

A B C D E

4) 304 + 201 + 359 = ?

A	B	C	D	E
800	950	850	900	970

A B C D E

5) 89 + 21 + 48 + 32 = ?

A	B	C	D	E
190	170	290	230	220

A B C D E

6) 3,987 – 3,007 = ?

A	B	C	D	E
7,000	1,270	1,180	790	980

A B C D E

7) 9,601 – 481 = ?

A	B	C	D	E
9,200	9,100	8,100	9,000	8,050

A B C D E

8) 1,890 – 301 = ?

A	B	C	D	E
1,500	1,700	1,200	1,600	1,640

A B C D E

9) 7,814 – 3,010 = ?

A	B	C	D	E
4,400	3,900	4,800	4,200	5,800

A B C D E

10) 989 – 99 + 3,202 = ?

A	B	C	D	E
3,400	3,200	4,100	3,100	4,500

A B C D E

11) 69 × 70 = ?

A	B	C	D	E
490	4,650	5,000	4,800	4,600

A B C D E

12) 41 × 121 = ?

A	B	C	D	E
4,100	4,200	500	5,100	4,900

A B C D E

13) 5.8 × 6.1 × 9.8 = ?

A	B	C	D	E
480	250	560	350	260

A B C D E

14) 5.5 × 4 × 3.8 = ?

A	B	C	D	E
60	70	80	100	110

A B C D E

15) 313 × 2.03 = ?

A	B	C	D	E
710	660	530	690	630

A B C D E

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Circle Answer

16) 905 ÷ 49 = ?

A	B	C	D	E
18	14	13	15	12

A B C D E

17) 8,017 ÷ 390 = ?

A	B	C	D	E
200	150	50	15	20

A B C D E

18) 1,207 ÷ 72 = ?

A	B	C	D	E
20	16	140	160	14

A B C D E

19) 1,447 ÷ 12.3 = ?

A	B	C	D	E
100	150	120	140	14

A B C D E

20) 4,916 ÷ 711 = ?

A	B	C	D	E
11	5	14	7	4

A B C D E

21) 45% of 363 = ?

A	B	C	D	E
180	110	160	175	190

A B C D E

22) 62% of 987 = ?

A	B	C	D	E
610	670	560	640	680

A B C D E

23) 33% of 3,574

A	B	C	D	E
930	1180	900	1100	1400

A B C D E

24) 5% of 97,326

A	B	C	D	E
460	4400	5000	4500	4800

A B C D E

25) 11.5% of 78,754

A	B	C	D	E
7000	8000	9000	9500	8500

A B C D E

26) $1\frac{1}{4} + 1\frac{1}{5} \times 3\frac{1}{2} = ?$

A	B	C	D	E
8	5	7	4	6

A B C D E

27) $1\frac{3}{5} + 2\frac{4}{5} \times 7\frac{1}{2} = ?$

A	B	C	D	E
27	23	20	18	25

A B C D E

28) $3\frac{3}{16} \times 12\frac{7}{8} = ?$

A	B	C	D	E
30	40	50	60	45

A B C D E

29) $5\frac{3}{8} \times 3\frac{3}{16} = ?$

A	B	C	D	E
20	16	19	21	14

A B C D E

30) $3\frac{1}{16} \times 6\frac{1}{8} = ?$

A	B	C	D	E
25	24	23	19	22

A B C D E

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Circle Answer

31) If one ream of paper costs $3.95 how much would 12 reams cost?

A	B	C	D	E
$47.00	$37.00	$36.00	$44.00	$45.00

A B C D E

32) John starts work at 8:45 am and finishes at 5:15 pm. He has 90 minutes of breaks. How many hours does he work in 29 days?

A	B	C	D	E
180	200	220	240	260

A B C D E

33) A restaurant bill is made up as follows: $212.43 for starters, $128.52 for main courses and $78.96 for deserts, plus a 17% service charge. How much is the bill?

A	B	C	D	E
$400	$420	$440	$490	$460

A B C D E

34) A rectangular solid is 19 inches high, 19 inches wide and 19 inches long. What is its volume in cubic inches?

A	B	C	D	E
7,000	7,500	8,000	8,500	9,000

A B C D E

35) A cylindrical solid is 40 inches high, and has a diameter of 2 inches. What is its volume in cubic inches?

A	B	C	D	E
100	120	140	160	180

A B C D E

End of Numerical Estimation Test 1

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Numerical Estimation Test 2: 35 Questions

Instructions: You need to estimate the answers to these questions, as you do not have time to calculate them precisely. Answer as many questions as you can in 10 minutes. Circle the letter on the right which is nearest to the correct answer. Do not use a calculator.

1) 249 + 364 = ?

A	B	C	D	E
450	500	550	600	650

A B C D E

2) 3,905 + 4,204 = ?

A	B	C	D	E
8,700	9,000	7,000	8,100	8,500

A B C D E

3) 618 + 390 = ?

A	B	C	D	E
1100	1000	980	990	970

A B C D E

4) 106 + 317 + 339 = ?

A	B	C	D	E
700	740	760	790	720

A B C D E

5) 29 + 41 + 38 + 31 = ?

A	B	C	D	E
130	120	160	110	140

A B C D E

6) 4,998 – 2,005 = ?

A	B	C	D	E
2,900	2,950	3,000	2,850	3,150

A B C D E

7) 8,441 – 739 = ?

A	B	C	D	E
7,700	7,300	7,500	7,400	7,750

A B C D E

8) 1,689 – 402 = ?

A	B	C	D	E
1,250	1,290	1,180	1,165	1,300

A B C D E

9) 6,786 – 2,280 = ?

A	B	C	D	E
4,650	5,000	4,100	4,500	4,400

A B C D E

10) 714 – 88 + 7,459 = ?

A	B	C	D	E
8,250	7,750	7,900	7,950	8,100

A B C D E

11) 12 × 68 = ?

A	B	C	D	E
820	730	770	740	790

A B C D E

12) 39 × 119 = ?

A	B	C	D	E
3,600	3,700	4,500	4,000	3,200

A B C D E

13) 4.9 × 7.1 × 4.9 = ?

A	B	C	D	E
165	190	155	170	150

A B C D E

14) 5.5 × 8 × 2.9 = ?

A	B	C	D	E
130	110	155	160	90

A B C D E

15) 618 × 4.17 = ?

A	B	C	D	E
2,600	360	250	2,400	210

A B C D E

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Circle Answer

	A	B	C	D	E	
16) 448 ÷ 51 = ?	6	9	11	12	10	**A B C D E**
17) 4,442 ÷ 348 = ?	13	10	17	18	5	**A B C D E**
18) 1,408 ÷ 71 = ?	16	15	20	17	14	**A B C D E**
19) 1,201 ÷ 11.3 = ?	1,000	140	12	100	120	**A B C D E**
20) 5,907 ÷ 298 = ?	50	60	15	20	40	**A B C D E**
21) 45% of 721 = ?	375	350	325	300	270	**A B C D E**
22) 62% of 9,988 = ?	6,200	5,500	6,000	5,800	6,100	**A B C D E**
23) 33% of 8,980	3,300	3,000	2,700	2,600	3,900	**A B C D E**
24) 5% of 21,111 = ?	990	1,500	1,005	1,050	1,550	**A B C D E**
25) 11.5% of 81,854 = ?	9,410	8,900	9,000	9,800	9,900	**A B C D E**
26) $2\frac{7}{8} + 1\frac{3}{16} + 3\frac{1}{2} = ?$	$9\frac{1}{2}$	$6\frac{1}{2}$	$5\frac{1}{2}$	$8\frac{1}{2}$	$7\frac{1}{2}$	**A B C D E**
27) $2\frac{3}{5} + 1\frac{4}{5} + 7\frac{17}{20} = ?$	14	11	10	15	12	**A B C D E**
28) $1\frac{3}{16} + 9\frac{7}{8} = ?$	14	11	10	15	12	**A B C D E**
29) $14\frac{3}{8} - 8\frac{3}{16} = ?$	6	5	4	3	7	**A B C D E**
30) $7\frac{1}{16} \times 3\frac{1}{2} = ?$	$20\frac{1}{2}$	23	$24\frac{1}{2}$	$22\frac{1}{2}$	21	**A B C D E**

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Circle Answer

31) If one ream of paper costs $4.45 how much would 9 reams cost?

A	B	C	D	E
$39	$40	$44	$38	$37

A B C D E

32) John starts work at 7:45 am and finishes at 5:45 pm. He has 90 minutes of breaks. How many hours does he work in 10 days?

A	B	C	D	E
66	80	85	90	75

A B C D E

33) A restaurant bill is made up as follows: $192.88 for starters, $212.71 for main courses and $76.16 for deserts, plus a 5% service charge. How much is the bill?

A	B	C	D	E
$500	$550	$600	$650	$450

A B C D E

34) A rectangular solid is 12 inches high, 3 inches wide and 5 inches long. What is its volume in cubic inches?

A	B	C	D	E
150	200	250	180	230

A B C D E

35) A cylindrical solid is 8 inches high, and has a diameter of 4 inches. What is its volume in cubic inches?

A	B	C	D	E
40	100	60	70	80

A B C D E

End of Numerical Estimation Test 2

Mechanical Aptitude Tests

Numerical Estimation Test 3: 35 Questions

Instructions: You need to estimate the answers to these questions, as you do not have time to calculate them precisely. Answer as many questions as you can in 10 minutes. Circle the letter on the right which is nearest to the correct answer. Do not use a calculator.

1) 475 + 222 = ?

A	B	C	D	E
650	700	750	800	850

A B C D E

2) 4,115 + 3,918 = ?

A	B	C	D	E
7,750	6,750	7,250	8,000	8,250

A B C D E

3) 796 + 512 + 987 = ?

A	B	C	D	E
2,300	2,000	2,200	2,100	1,900

A B C D E

4) 708 + 497 + 111 = ?

A	B	C	D	E
1,550	1,500	1,700	1,400	1,300

A B C D E

5) 48 + 29 + 32 + 21 = ?

A	B	C	D	E
110	150	140	160	130

A B C D E

6) 7,879 – 1,790 = ?

A	B	C	D	E
8,200	6,000	9,600	9,000	8,600

A B C D E

7) 7,701 – 806 = ?

A	B	C	D	E
7,200	6,900	7,900	5,900	8,000

A B C D E

8) 2,924 – 318 = ?

A	B	C	D	E
2,350	2,550	2,600	2,400	2,650

A B C D E

9) 5,412 - 1,900 = ?

A	B	C	D	E
3,500	3,400	3,200	3,300	3,000

A B C D E

10) 618 – 86 + 3,089 = ?

A	B	C	D	E
3,200	3,100	3,400	3,500	3,600

A B C D E

11) 59 × 71 = ?

A	B	C	D	E
480	4,200	4,800	420	4,900

A B C D E

12) 68 × 122 = ?

A	B	C	D	E
8,000	820	8,050	8,300	800

A B C D E

13) 3.8 × 6.2 × 10.4 = ?

A	B	C	D	E
250	260	270	280	290

A B C D E

14) 5.5 × 7 × 3.9 = ?

A	B	C	D	E
190	180	170	160	150

A B C D E

15) 717 × 3.03 = ?

A	B	C	D	E
3,200	1,800	2,400	2,200	3,000

A B C D E

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Circle Answer

16) 361 ÷ 48 = ?

A	B	C	D	E
6	$7\frac{1}{2}$	$8\frac{1}{2}$	$8\frac{1}{4}$	9

A B C D E

17) 6,013 ÷ 197 = ?

A	B	C	D	E
50	45	60	30	40

A B C D E

18) 1,408 ÷ 34 = ?

A	B	C	D	E
80	40	60	20	15

A B C D E

19) 1,819 ÷ 8.9 = ?

A	B	C	D	E
370	400	200	300	350

A B C D E

20) 4,898 ÷ 124 = ?

A	B	C	D	E
20	25	30	35	40

A B C D E

21) 45% of 132 = ?

A	B	C	D	E
50	60	70	80	90

A B C D E

22) 62% of 722 = ?

A	B	C	D	E
350	400	550	450	300

A B C D E

23) 33% of 12,104

A	B	C	D	E
2,000	2,500	3,000	3,500	4,000

A B C D E

24) 5% of 40,019 = ?

A	B	C	D	E
1,000	2,000	3,000	4,000	5,000

A B C D E

25) 11.5% of 49,775 = ?

A	B	C	D	E
4,500	5,700	6,500	7,700	8,700

A B C D E

26) $2\frac{1}{16} + 7\frac{1}{5} + 3\frac{1}{2} = ?$

A	B	C	D	E
16	11	15	14	13

A B C D E

27) $2\frac{3}{5} + 4\frac{4}{5} + 1\frac{1}{2} = ?$

A	B	C	D	E
6	7	8	9	10

A B C D E

28) $6\frac{3}{16} + 9\frac{7}{8} = ?$

A	B	C	D	E
15	16	17	18	19

A B C D E

29) $15\frac{3}{8} - 4\frac{7}{16} = ?$

A	B	C	D	E
11	12	10	14	13

A B C D E

30) $6\frac{4}{5} \times 2\frac{1}{8} = ?$

A	B	C	D	E
12	15	19	17	21

A B C D E

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Circle Answer

31) If a ream of paper costs $7.47 how much would 11 reams cost?

A	B	C	D	E
$75	$70	$90	$60	$80

A B C D E

32) John starts work at 8:00 am and finishes at 4:45 pm. He has 90 minutes of breaks. How many hours does he work in 10 days?

A	B	C	D	E
72	62	78	60	70

A B C D E

33) A restaurant bill is made up as follows: $312.55 for starters, $512.52 for main courses and $278.96 for deserts, plus a 10% service charge. How much is the bill?

A	B	C	D	E
$1,250	$1,050	$1,200	$1,150	$1,100

A B C D E

34) A rectangular solid is 5.5 inches high, 9.5 inches wide and 10 inches long. What is its volume in cubic inches?

A	B	C	D	E
540	560	460	480	520

A B C D E

35) A cylindrical solid is 9 inches high, and has a diameter of 6 inches. What is its volume in cubic inches?

A	B	C	D	E
150	250	375	200	300

A B C D E

End of Numerical Estimation Test 3

Mechanical Aptitude Tests

Numerical Estimation Test 4: 35 Questions

Instructions: You need to estimate the answers to these questions, as you do not have time to calculate them precisely. Answer as many questions as you can in 10 minutes. Circle the letter on the right which is nearest to the correct answer. Do not use a calculator.

1) 3,506 + 198 = ?

A	B	C	D	E
3,800	3,700	3,650	3,750	3,770

A B C D E

2) 7,709 + 4,421 = ?

A	B	C	D	E
13,000	11,500	11,000	12,000	12,500

A B C D E

3) 588 + 608 + 981 = ?

A	B	C	D	E
2,600	2,500	2,200	2,400	2,100

A B C D E

4) 914 + 988 + 407 = ?

A	B	C	D	E
2,300	2,600	2,500	2,700	2,400

A B C D E

5) 57 + 51 + 28 + 19 = ?

A	B	C	D	E
160	120	170	180	150

A B C D E

6) 9,104 – 4,997 = ?

A	B	C	D	E
6,000	5,000	5,100	4,100	4,000

A B C D E

7) 6,695 – 579 = ?

A	B	C	D	E
5,900	5,800	6,100	5,100	5,500

A B C D E

8) 4,880 – 369 = ?

A	B	C	D	E
4,200	4,500	4,300	4,400	4,100

A B C D E

9) 8,120 – 5,611 = ?

A	B	C	D	E
3,600	2,800	2,700	2,500	2,600

A B C D E

10) 512 – 88 + 2,180 = ?

A	B	C	D	E
3,600	2,800	2,700	2,500	2,600

A B C D E

11) 48 × 51 = ?

A	B	C	D	E
3,600	2,800	2,700	2,500	2,600

A B C D E

12) 28 × 83 = ?

A	B	C	D	E
1,800	3,200	2,300	1,300	830

A B C D E

13) 6.9 × 4.9 × 7.1 = ?

A	B	C	D	E
2,000	160	200	220	190

A B C D E

14) 5.5 × 6 × 1.8 = ?

A	B	C	D	E
55	65	45	60	50

A B C D E

15) 508 × 5.91 = ?

A	B	C	D	E
1,800	3,200	2,300	1,300	3,000

A B C D E

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Circle Answer

16) 918 ÷ 94 = ?

A	B	C	D	E
15	22	10	25	20

A B C D E

17) 6,987 ÷ 129 = ?

A	B	C	D	E
20	50	40	30	35

A B C D E

18) 2,409 ÷ 31 = ?

A	B	C	D	E
80	44	31	55	120

A B C D E

19) 2,398 ÷ 11.8 = ?

A	B	C	D	E
110	300	60	120	200

A B C D E

20) 9,978 ÷ 509 = ?

A	B	C	D	E
10	20	30	40	50

A B C D E

21) 45% of 409 = ?

A	B	C	D	E
260	150	180	240	220

A B C D E

22) 62% of 308 = ?

A	B	C	D	E
200	160	250	110	150

A B C D E

23) 33% of 21,013 = ?

A	B	C	D	E
8,000	11,000	9,000	7,000	5,000

A B C D E

24) 5% of 79,899 = ?

A	B	C	D	E
3,000	2,000	4,000	2,500	3,500

A B C D E

25) 11.5% of 38,160 = ?

A	B	C	D	E
3,000	2,000	4,000	2,500	3,500

A B C D E

26) $5\frac{1}{40} + 1\frac{4}{5} + 3\frac{1}{2} = ?$

A	B	C	D	E
9	11	11.5	12.5	10.5

A B C D E

27) $1\frac{3}{5} + 2\frac{4}{5} + 9\frac{1}{2} = ?$

A	B	C	D	E
12	11	18	10	14

A B C D E

28) $12\frac{3}{16} + 14\frac{7}{8} = ?$

A	B	C	D	E
31	27	25	30	29

A B C D E

29) $15\frac{3}{8} - 6\frac{3}{16} = ?$

A	B	C	D	E
9	8	11	6	7

A B C D E

30) $7\frac{1}{16} \times 7\frac{1}{8} = ?$

A	B	C	D	E
65	70	50	40	60

A B C D E

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Circle Answer

31) If three reams of paper cost $12.66 how much would 20 reams cost?

A	B	C	D	E
$80	$85	$100	$120	$95

A B C D E

32) John starts work at 5:45 am and finishes at 3:15 pm. He has 90 minutes of breaks. How many hours does he work in 16 days?

A	B	C	D	E
210	160	100	130	75

A B C D E

33) A restaurant bill is made up as follows: $82.43 for starters, $128.52 for main courses and $48.96 for deserts, plus an 18% service charge. How much is the bill?

A	B	C	D	E
$400	$190	$300	$260	$240

A B C D E

34) A rectangular solid is 3.3 inches high, 3.3 inches wide and 24 inches long. What is its volume in cubic inches?

A	B	C	D	E
400	190	300	260	230

A B C D E

35) A cylindrical solid is 10 inches high, and has a diameter of 6 inches. What is its volume in cubic inches?

A	B	C	D	E
280	600	200	60	48

A B C D E

End of Numerical Estimation Test 4

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Answers to Numerical Estimation Tests 1-4

Question	Test 1	Test 2	Test 3	Test 4
1)	**B**	**D**	**B**	**B**
2)	**A**	**D**	**D**	**D**
3)	**D**	**B**	**A**	**C**
4)	**C**	**C**	**E**	**A**
5)	**A**	**E**	**E**	**E**
6)	**E**	**C**	**B**	**D**
7)	**B**	**A**	**B**	**C**
8)	**D**	**B**	**C**	**B**
9)	**C**	**D**	**A**	**D**
10)	**C**	**E**	**E**	**E**
11)	**D**	**A**	**B**	**D**
12)	**E**	**C**	**D**	**C**
13)	**D**	**D**	**A**	**D**
14)	**C**	**A**	**E**	**D**
15)	**E**	**A**	**D**	**E**
16)	**A**	**B**	**B**	**C**
17)	**E**	**A**	**D**	**B**
18)	**B**	**C**	**B**	**A**
19)	**C**	**D**	**C**	**E**
20)	**D**	**D**	**E**	**B**
21)	**C**	**C**	**B**	**C**
22)	**A**	**A**	**D**	**A**
23)	**B**	**B**	**E**	**D**
24)	**E**	**D**	**B**	**C**
25)	**C**	**A**	**B**	**C**
26)	**A**	**E**	**E**	**E**
27)	**A**	**E**	**D**	**E**
28)	**B**	**B**	**B**	**B**
29)	**B**	**A**	**A**	**A**
30)	**D**	**C**	**B**	**C**
31)	**A**	**B**	**E**	**B**
32)	**B**	**C**	**A**	**D**
33)	**D**	**A**	**C**	**C**
34)	**A**	**D**	**E**	**D**
35)	**B**	**B**	**B**	**A**

Mechanical Aptitude Tests

Numerical Reasoning Questions

Information is provided that requires you to interpret it and then apply the appropriate logic to answer the questions. In other words, you need to work out how to get the answer rather than what calculations to apply. Sometimes the questions are designed to approximate the type of reasoning required in the workplace.

Numerical Reasoning		Operatives	Supervisory	Management
Craft & Technical				Y
Clerical & Administrative				Y
Police, Fire, Military etc.	Y			
Management Trainee	Y			
Graduate & Professional	Y			

The questions will often use number series questions which represent the most popular type of numerical reasoning questions. Numerical reasoning questions are usually restricted to tests for more senior technical positions. You can usually expect 15-20 questions in 20-30 minutes.

Numerical reasoning is an increasingly popular way of assessing candidates during the job selection process. Many people who have been out of the education system for a while or who don't use maths on a day-to-day basis feel intimidated by these types of test. The important thing to remember is that you don't need to have studied mathematics to a high level to do well in these tests. They are primarily tests of reasoning ability and the math needed is invariably straightforward. Although you may need to get back up to speed with percentages, ratios, proportions, fractions and decimals.

You will usually be allowed to use a calculator for these types of question and investing in one which can handle fractions and percentages is a good idea. You should also try to work through a few numerical computation practice papers to get back into swing of these types of calculation.

Mechanical Aptitude Tests

Numerical Reasoning Test 1: 22 Questions
Instructions: Answer as many questions as you can in 20 minutes. Circle the letter on the right which corresponds to the correct answer. You can use a calculator.

Identify the missing number at the end of the series.

1) 3, 11, 19, 27, ?

A	B	C	D	E
33	35	37	39	41

A B C D E

2) 3, 6, 11, 18, ?

A	B	C	D	E
24	25	26	27	28

A B C D E

3) 516, 497, 478, 459, ?

A	B	C	D	E
436	440	438	452	442

A B C D E

4) 316, 323, 332, 343, ?

A	B	C	D	E
356	357	358	351	359

A B C D E

5) 662, 645, 624, 599, ?

A	B	C	D	E
587	566	589	575	570

A B C D E

Identify the missing number within the series.

6) 33, ?, 19, 12, 5

A	B	C	D	E
31	26	29	27	24

A B C D E

7) 11, 19, ?, 41, 55

A	B	C	D	E
31	29	26	39	34

A B C D E

8) 98, 94, ?, 70, 38

A	B	C	D	E
89	85	86	87	88

A B C D E

9) 86, ?, 79, 75, 72, 68

A	B	C	D	E
82	80	85	84	83

A B C D E

10) 20, 30, 25, 35, ?, 40

A	B	C	D	E
45	35	25	30	50

A B C D E

11) Identify the missing number.

7	4
5	6

49	16
25	?

A	B	C	D	E
41	36	35	18	37

A B C D E

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12) Identify the missing number.

4	14		11	31
35	26		73	?

A	B	C	D	E
51	56	45	55	52

A B C D E

13) Identify the missing number.

7	8		20	1
5	6		2	?

A	B	C	D	E
3	16	25	48	17

A B C D E

14) Identify the missing number.

41	44		72	78
36	66		62	?

A	B	C	D	E
120	122	130	132	98

A B C D E

15) Identify the missing number.

5	20	100	3	24
20	80	400	12	?

A	B	C	D	E
86	96	16	106	56

A B C D E

16) Identify the missing number.

8	?	6	9	7
5	7	3	6	4

A	B	C	D	E
16	14	11	10	9

A B C D E

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17) It costs a manufacturer *X* dollars per component to make the first 1,000 components. All subsequent components cost X÷3 each. When X = $1.50 How much will it cost to manufacture 4,000 components?

A	B	C	D	E
$3,500	$3,000	$4,000	$3,250	$4,500

A B C D E

18) A train travelling at 60 mph enters a tunnel that is 5 miles long. The train is one mile long. How many minutes does it take for the whole train to pass through the tunnel?

A	B	C	D	E
7	4	10	5	6

A B C D E

19) In the Shelbyville election, the Republican candidate received one and a half times as many votes as the Democrat candidate. The Democrat candidate received one third more votes than the Independent candidate. 900 votes were cast for the Independent candidate. How many votes were cast for the Republican candidate?

A	B	C	D	E
900	1,400	1,600	1,000	1,800

A B C D E

20) Anna and John both drive to their new home 400 miles away. Anna drives the family car at an average speed of 60 mph. John drives the removal truck at an average speed of 50 mph. During the journey, Anna stops for a total of 1 hour and 20 minutes, John stops for half as long. What is the difference in minutes between their arrival times?

A	B	C	D	E
60	55	40	90	80

A B C D E

21) A total of 800 copies of a CD were sold. 60% were sold at 50% discount, 20% were sold at 30% discount and the remainder were sold at the full price of $8.95. What was the approximate total revenue in dollars?

A	B	C	D	E
4,679	4,579	4,779	4,499	4,521

A B C D E

22) In a survey, 3/16 of people said that they preferred to use self-service gas stations. 5/8 said that they preferred not to pump their own gas. The remaining 75 respondents said that they had no clear preference. How many people preferred self service?

A	B	C	D	E
75	125	100	133	150

A B C D E

End of Numerical Reasoning Test 1

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Numerical Reasoning Test 2: 22 Questions
Instructions: Answer as many questions as you can in 20 minutes. Circle the letter on the right which corresponds to the correct answer. You can use a calculator.

Identify the missing number at the end of the series.

1) 3, 9, 15, 21, ?

A	B	C	D	E
30	27	25	28	29

A B C D E

2) 4, 13, 21, 28, ?

A	B	C	D	E
37	36	31	34	33

A B C D E

3) 798, 777, 756, 735, ?

A	B	C	D	E
712	711	720	710	714

A B C D E

4) 2, 5, 9, 14, 20, ?

A	B	C	D	E
23	29	27	25	28

A B C D E

5) 20, 16, 12.8, 10.24, ?

A	B	C	D	E
7.39	8.19	8.55	8.00	7.29

A B C D E

Identify the missing number or letter within the series.

6) 16, 18, 21, ?, 30

A	B	C	D	E
27	28	22	25	26

A B C D E

7) 97, 94, ?, 79, 67, 52

A	B	C	D	E
88	86	70	81	82

A B C D E

8) 21, ?, 8, 5, 3, 2

A	B	C	D	E
12	14	13	17	11

A B C D E

9) ?, 125, 64, 27, 8, 1

A	B	C	D	E
298	200	175	216	210

A B C D E

10) Q, S, ?, Z

A	B	C	D	E
T	U	V	W	X

A B C D E

11) Identify the missing number.

7	12
22	17

12	17
?	22

A	B	C	D	E
24	36	32	38	27

A B C D E

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12) Identify the missing number.

13	34		50	?
91	62		25	100

A	B	C	D	E
25	75	34	91	62

A B C D E

13) Identify the missing number.

6	12		14	20
1	7		9	?

A	B	C	D	E
14	21	12	18	15

A B C D E

14) Identify the missing number.

3	6		4	7
6	6		7	?

A	B	C	D	E
11	9	8	6	4

A B C D E

15) Identify the missing number.

14	28	74	98	154
7	14	37	49	?

A	B	C	D	E
76	59	77	80	88

A B C D E

16) Identify the missing number.

3	2	6	2	7
4	9	1	1	?

A	B	C	D	E
12	9	5	8	7

A B C D E

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17) Anna drives to her sisters at an average speed of 60 mph and then drives home again at an average speed of 30 mph, what is her average speed in mph for the whole trip?

A	B	C	D	E
50	40	45	48	37

A B C D E

18) A bank offers 4% per annum interest which is calculated and added at the end of the year. Another bank offers 10% per annum which is also calculated and added every year. What is the difference in dollars on a deposit of $1000 after two years?

A	B	C	D	E
128.40	12.80	126.10	26.10	124.80

A B C D E

19) Tom, Dick and Harry own stock in their company in the ratio 4:5:6. If the total value of the stock is $45,000, What value of stock would Harry need to give Tom for all three to own the stock equally?

A	B	C	D	E
$3,000	$4,000	$3,200	$3,500	$5,000

A B C D E

20) Anna and John both drive to their new home 600 miles away. Anna drives the family car at an average speed of 65 mph. John drives the removal truck at an average speed of 50 mph. During the trip, Anna stops for a total of 2 hours and 20 minutes, John stops for 90 minutes. What is the difference in minutes between their arrival times?

A	B	C	D	E
91	131	129	116	15

A B C D E

21) A total of 600 copies of a CD were sold. 55% were sold at 45% discount, 20% were sold at 25% discount and the remainder were sold at the full price of $9.95. What was the approximate total revenue in dollars?

A	B	C	D	E
$4,662	$3,660	$3,866	$3,992	$4,192

A B C D E

22) In a survey, 243 people were asked if they preferred butter, margarine or new Tastee low fat spread. The ratio of people who preferred Tastee to margarine was 7:2. Two thirds of the whole sample preferred butter. How many people preferred Tastee?

A	B	C	D	E
122	121	71	63	77

A B C D E

End of Numerical Reasoning Test 2

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Mechanical Aptitude Tests

Numerical Reasoning Test 3: 22 Questions
Instructions: Answer as many questions as you can in 20 minutes. Circle the letter on the right which corresponds to the correct answer. You can use a calculator.

Identify the missing number at the end of the series.

1) 7, 11, 15, 19, ?

A	B	C	D	E
22	23	24	25	26

A B C D E

2) 11, 16, 26, 41, ?

A	B	C	D	E
58	60	59	61	66

A B C D E

3) 28, 35, 42, 49, 56, ?

A	B	C	D	E
62	63	64	65	66

A B C D E

4) 97, 94, 88, 79, 67, ?

A	B	C	D	E
52	50	49	47	44

A B C D E

5) 72, 63, 54, 45, ?

A	B	C	D	E
32	33	36	39	35

A B C D E

Identify the missing number or letter within the series.

6) 195, ?, 180, 170, 165

A	B	C	D	E
192	188	185	190	182

A B C D E

7) 3, 5, 15, 17, 27, ?, 39

A	B	C	D	E
31	29	25	35	30

A B C D E

8) 54, ?, 28, 18, 10, 4

A	B	C	D	E
36	35	37	41	40

A B C D E

9) 1, 3, ?, 9, 27, 243

A	B	C	D	E
4	3	5	6	7

A B C D E

10) T, ? R, N, P

A	B	C	D	E
U	T	V	W	N

A B C D E

11) Identify the missing number.

3	9
81	3

4	8
?	4

A	B	C	D	E
124	128	64	48	28

A B C D E

Mechanical Aptitude Tests

12) Identify the missing number.

34	40		45	1
10	6		0	?

A	B	C	D	E
13	51	26	44	8

A B C D E

13) Identify the missing number.

3	6		2	6
1	8		1	?

A	B	C	D	E
2	4	6	8	10

A B C D E

14) Identify the missing number.

41	48		33	40
62	55		?	47

A	B	C	D	E
59	55	62	54	51

A B C D E

15) Identify the missing number.

3	17	6	14	10
5	5	25	?	7

A	B	C	D	E
13	12	25	17	3

A B C D E

16) Identify the missing number.

2	6	44	8	10
15	19	?	21	23

A	B	C	D	E
72	66	73	57	55

A B C D E

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17) It costs a manufacturer X dollars per component to make the first 500 components. All subsequent components cost X÷5 each. When X = $4.50 How much will it cost to manufacture 4,000 components?

A	B	C	D	E
$5,600	$4,600	$5,400	$5,200	$5,450

A B C D E

18) A bank offers 10% per annum interest which is calculated and added at the end of the year. Another bank offers 10% per annum which is calculated and added every six months. What is the difference on a deposit of $800 after one year?

A	B	C	D	E
$2.00	$2.60	$2.40	$2.20	$4.00

A B C D E

19) Components X,Y and Z are ordered in the ratio 1:5:4. How many Z components will be in an order for 8000 components?

A	B	C	D	E
3,200	1,600	6,400	4,600	1,800

A B C D E

20) Anna and John both receive stock as part of their remuneration. Anna receives $400 worth plus a bonus of 12%. John receives $300 worth plus a bonus of 20%. What is the difference between the values of the two bonuses?

A	B	C	D	E
$12.00	$10.00	$20.00	$14.00	$11.50

A B C D E

21) A total of 1600 copies of a CD were sold. 30% were sold at 55% discount, 10% were sold at 30% discount and the remainder were sold at the full price of $7.95. What was the approximate total revenue in dollars?

A	B	C	D	E
10,369	10,569	10,569	10,234	10,669

A B C D E

22) Anna bought $4,000 of company stock. She sold 75% of it when the value doubled, and the remainder at four times the purchase price. What was her total profit?

A	B	C	D	E
$4,000	$6,750	$6,000	$6,500	$5,000

A B C D E

End of Numerical Reasoning Test 3

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Mechanical Aptitude Tests

Numerical Reasoning Test 4: 22 Questions
Instructions: Answer as many questions as you can in 20 minutes. Circle the letter on the right which corresponds to the correct answer. You can use a calculator.

Identify the missing number at the end of the series.

1) 5, 12, 19, 26, ?

A	B	C	D	E
31	33	35	34	37

A B C D E

2) 11, 16, 26, 41, ?

A	B	C	D	E
51	56	61	66	46

A B C D E

3) 100, 96, 91, 85, ?

A	B	C	D	E
74	75	77	78	79

A B C D E

4) 5, 12, 26, 47, ?

A	B	C	D	E
66	65	60	70	75

A B C D E

5) 0, 4, 9, 13, 18, ?

A	B	C	D	E
22	20	24	21	25

A B C D E

Identify the missing number within the series.

6) ?, 14, 12, 11, 11, 12

A	B	C	D	E
14	17	18	15	16

A B C D E

7) 11, 30, ?, 68, 87, 106

A	B	C	D	E
50	52	40	49	47

A B C D E

8) 68, 72, 75, ?, 82, 86

A	B	C	D	E
80	78	77	81	79

A B C D E

9) ?, 30, 35, 25, 30, 20

A	B	C	D	E
40	45	25	20	30

A B C D E

10) 54, 40, 28, ?, 10, 4

A	B	C	D	E
24	16	18	14	15

A B C D E

11) Identify the missing number.

4	8
16	12

8	16
?	24

A	B	C	D	E
30	32	36	28	34

A B C D E

Mechanical Aptitude Tests

12) Identify the missing number.

22	22		44	1
13	3		6	?

A	B	C	D	E
14	44	13	9	22

A B C D E

13) Identify the missing number.

7	8		20	?
5	6		2	3

A	B	C	D	E
0	1	6	4	7

A B C D E

14) Identify the missing number.

41	48		33	40
62	55		54	?

A	B	C	D	E
55	62	42	49	47

A B C D E

15) Identify the missing number.

3	12	90	24	9
7	28	210	?	21

A	B	C	D	E
48	65	56	52	54

A B C D E

16) Identify the missing number.

1	1	1	3	?
2	1	0	2	2

A	B	C	D	E
0	1	2	3	4

A B C D E

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17) It costs a manufacturer X dollars per component to make the first 3,000 components. All subsequent components cost one quarter of X each. When X = $2.70 How much will it cost to manufacture 4,000 components?

A	B	C	D	E
$7,875	$8,770	$8,795	$8,705	$8,700

A B C D E

18) A bank offers 6% per annum interest which is calculated and added at the end of the year. Another bank offers 10% per annum which is calculated and added every six months. What is the difference on a deposit of $100 after one year?

A	B	C	D	E
$4.00	$4.05	$4.10	$4.15	$4.25

A B C D E

19) In the Shelbyville election, the Republican candidate received three times as many votes as the Democrat candidate. The Democrat candidate received one quarter less votes than the Independent candidate. 1200 votes were cast for the Independent candidate. How many votes were cast for the Republican candidate?

A	B	C	D	E
2,700	3,200	2,000	2,600	2,200

A B C D E

20) Anna and John both drive to their new home 280 miles away. Anna drives the family car at an average speed of 70 mph. John drives the removal truck at an average speed of 60 mph. During the journey, Anna stops for a total of 1 hour and 10 minutes, John stops for half as long. What is the difference in minutes between their arrival times?

A	B	C	D	E
5	10	12	15	25

A B C D E

21) A train travelling at 66 mph enters a tunnel that is 4.75 miles long. The train is three quarters of a mile long. How many minutes does it take for the whole train to pass through the tunnel?

A	B	C	D	E
8	5	14	16	10

A B C D E

22) Tom, Dick and Harry own shares in their company in the ratio 3:5:7. If the total value of the shares is $10,500. What value of shares would Harry need to give Tom for all three to own the shares equally?

A	B	C	D	E
$1,800	$700	$1,400	$1,200	$1,000

A B C D E

End of Numerical Reasoning Test 4

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Mechanical Aptitude Tests

Answers to Numerical Reasoning Tests 1-4

Question	Test 1	Test 2	Test 3	Test 4
1)	**B**	**B**	**B**	**B**
2)	**D**	**D**	**D**	**C**
3)	**B**	**E**	**B**	**D**
4)	**A**	**C**	**A**	**E**
5)	**E**	**B**	**C**	**A**
6)	**B**	**D**	**C**	**B**
7)	**B**	**A**	**B**	**D**
8)	**C**	**C**	**E**	**E**
9)	**A**	**D**	**B**	**A**
10)	**D**	**C**	**E**	**C**
11)	**B**	**E**	**B**	**B**
12)	**D**	**A**	**D**	**D**
13)	**A**	**E**	**A**	**B**
14)	**B**	**B**	**D**	**E**
15)	**B**	**C**	**A**	**C**
16)	**D**	**C**	**D**	**B**
17)	**B**	**B**	**C**	**B**
18)	**E**	**A**	**A**	**E**
19)	**E**	**A**	**A**	**A**
20)	**C**	**D**	**A**	**A**
21)	**B**	**E**	**D**	**B**
22)	**A**	**D**	**C**	**C**

Mechanical Aptitude Tests

Numerical Reasoning – Understanding Number Sequences
These questions require you to find the missing number in a sequence of numbers. This missing number may be at the beginning or middle but is usually at the end.

1. Find the next number in the series

4 8 16 32 --

A) 48 B) 64 C) 40 D) 46

2. Find the next number in the series

4 8 12 20 --

A) 32 B) 34 C) 36 D) 38

3. Find the missing number in the series

54 49 -- 39 34

A) 47 B) 44 C) 45 D) 46

4. Find the first number in the series

-- 19 23 29 31

A) 12 B) 15 C) 16 D) 17

These number sequences can be quite simple like the examples above. However, you will often see more complex questions where it is the interval between the numbers that is the key to the sequence.

5. Find the next number in the series

3 6 11 18 --

A) 30 B) 22 C) 27 D) 29

6. Find the next number in the series

48 46 42 38 --

A) 32 B) 30 C) 33 D) 34

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These simple number sequences usually consist of four visible numbers plus one missing number. This is because the test designer needs to produce a sequence into which only one number will fit. The need to avoid any ambiguity means that if the number sequence relies on a more complex pattern then there will need to be more visible numbers. For example;

7. Find the missing number in the series

4 3 5 9 12 17 --

A) 32 B) 30 C) 24 D) 26

8. Find the missing numbers in the series

5 6 7 8 10 11 14 -- --

A) 19 B) 17 C) 15 D) 16

9. Find the missing numbers in the series

1 -- 4 7 7 8 10 9 --

A) 6 B) 3 C) 11 D) 13

Answers
1. B – The numbers double each time
2. A – Each number is the sum of the previous two numbers
3. B – The numbers decrease by 5 each time
4. D – The numbers are primes (divisible only by 1 and themselves)
5. C – The interval, beginning with 3, increases by 2 each time
6. B – The interval, beginning with 2, increases by 2 and is subtracted each time
7. D – Each number is the sum of the previous and the number 3 places to the left
8. C A – There are 2 simple interleaved sequences 5,7,10,14,19 and 6,8,11,15
9. A D – There are 2 simple interleaved sequences 1,4,7,10,13 and 6,7,8,9

To solve these number sequence questions efficiently, you should first check the relationship between the numbers themselves looking for some simple arithmetic relationship. Then look at the intervals between the numbers and see if there is a relationship there. If not, and particularly if there are more than 4 numbers visible, then there may be two number sequences interleaved.

You will occasionally find multiplication, division, or powers used in these sequences, but test designers tend to avoid them as these operations soon lead to large numbers which are difficult to work out without a calculator.

Mechanical Aptitude Tests

Numerical Reasoning – Understanding Letters of the Alphabet as Numbers
Another type of sequence question which appears in these tests involves the substitution of letters of the alphabet for numbers. For example A=1, B=2 etc. It may seem strange to consider these as numerical reasoning questions but they actually work in the same way once you have changed them back into numbers.

1. Find the next letter in the series

B E H K --

i) L ii) M iii) N iv) O

2. Find the next letter in the series

A Z B Y --

i) C ii) X iii) D iv) Y

3. Find the next letter in the series

T V X Z --

i) Y ii) B iii) A iv) W

Answers
1. iii – There are two letters missing between each one, so N is next
2. i – There are 2 interleaved sequences A,B,C and Z,Y, so C is next
3. ii – Miss a letter each time and 'loop' back, so B is next

Because arithmetic operations cannot be performed on letters there is less room for ambiguity in these questions. This means that interleaved sequences can be used with fewer visible letters than in questions that use numbers. Question 17 for example can use 2 interleaved sequences even though only four letters are visible. This would be very difficult to achieve with numbers.

It is implicit in these 'alphabetic sequence' questions that the sequence 'loops' back around and starts again. See question 18. It is important to recognize this as it is not usually stated explicitly – you are just expected to know it.

If you see more than one of these questions in a test then it is almost certainly worth taking the time to write out the letters of the alphabet with their ordinal numbers underneath. You can then treat these questions in a similar way to number sequence questions. This can save a lot of time overall and avoids simple mistakes.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26

Mechanical Aptitude Tests

Explanations – Numerical Reasoning Test 1

1) The numbers in this series increase by 8 each time. therefore the next number is 35.

2) The difference between the numbers in this series increases by 2 each time – 3,5,7, etc. This will produce a difference of 9 between 18 and the next number in the series, which is therefore 27.

3) The numbers in this series decrease by 19 each time. Therefore the next number is 440.

4) The difference between the numbers in this series increases by 2 each time – 7,9,11, etc. This will produce a difference of 13 between 353 and the next number in the series, which is therefore 356.

5) The difference between the numbers in this series increases by -4 each time: -17,-21,-25, etc. This will produce a difference of -29 between 599 and the next number in the series, which is therefore 570.

6) The numbers in this series decrease by 7 each time. Therefore the second number is 26.

7) The difference between the numbers in this series increases by 2 each time – 8,10,12, 14. The third number in the series is therefore 29.

8) The difference between the numbers in this series increases by -4, -8, -16, -32. The third number in the series is therefore 86.

9) The difference between the numbers in this series alternates between -4 and -3. The second number in the series is therefore 82.

10) There are two number series here, the first is 20, 25, 30 and the second is 30, 35, 40. Signs that you need to consider two series are firstly six or more numbers and secondly, numbers in the series do not all increase or decrease.

11) The numbers in the box on the right are the squares of those in the corresponding box on the left. 7/49, 4/16, 5/25. Therefore the answer must be 36 which is the square of 6.

12) The numbers in the box on the right are $((n * 2) + 3)$ where n is the number in the corresponding box on the left. Therefore the answer must be $((26 * 2) + 3) = 55$.

13) The numbers in the boxes on the left adds up to 26. The answer must be 3 to make the numbers in the boxes on the right add up to 26.

14) The numbers in the box on the right are $((n * 2) - 10)$ where n is the number in the corresponding box on the left. Therefore the answer must be $((66 * 2) - 10) = 122$.

15) The numbers on the bottom row are four times those on the top row. Therefore the missing number is $4 * 24 = 96$.

16) The numbers on the bottom row are three less than those on the top row. Therefore the missing number is 7 + 3 = 10

17) The first 1000 components are $1.50 each which gives $1,500. The 3000 subsequent components cost $0.50 each which gives $1,500. Therefore the run of 4000 components will cost $3,000.

18) The train is travelling at a speed of one mile per minute, so the front of the train will exit the tunnel in 5 minutes. At this point there is still one mile of train in the tunnel since the train is one mile long. This will take a further one minute to clear the tunnel. Therefore the total time taken is six minutes.

19) If 900 votes were cast for the Independent candidate then the Democratic candidate must have received (900 + 300) 1200 votes. The Republican candidate must have received (1200 + 600) 1800 votes.

20) If Anna covers 400 miles in at an average speed of 60 mph then she must have driven for a total of 6 hours and 40 minutes. If she stops for 1 hour and 20 minutes then her total journey time is 8 hours. If John covers 400 miles in at an average speed of 50 mph then he must have driven for a total of 8 hours. If he stops for 40 minutes then his total journey time is 8 hours and 40 minutes. Therefore there is a 40 minute difference in their arrival times.

21) Of the 800 CDs sold:

20% or 160 were sold at $8.95 = $1432
20% or 160 were sold at $6.26 = $1001
60% or 480 were sold at $4.47 = $2145

Therefore the total revenue was $4578

22) The proportion of the sample who preferred not to pump their own gas was 5/8 which is the same as 10/16. Add to this 3/16 which is the proportion of the sample that prefers to pump their own gas. This means that 13/16 of the people who replied to the survey expressed a preference. This means that 3/16 (or 75 people) did not. The question states that 3/16 of the sample preferred to use self service and we know that 3/16 = 75 people, therefore 75 people preferred self-service.

Mechanical Aptitude Tests

Explanations - Numerical Reasoning Test 2

1) The numbers in this series increase by 6 each time. Therefore the next number is 27.

2) The difference between the numbers in this series decreases by 1 each time: 9, 8, 7, etc. This will produce a difference of 6 between 28 and the next number in the series, which is therefore 34.

3) The numbers in this series decrease by 21 each time. Therefore the next number is 714.

4) The difference between the numbers in this series increases by 1 each time: 3, 4, 5, 6, etc. This will produce a difference of 7 between 20 and the next number in the series, which is therefore 27.

5) The difference between the numbers in this series is (minus) the number divided by 5. For example 20/5 = 4. 20-4 = 16. Then 16/5 = 3.2. 16-3.2 = 12.8. To obtain the final number in the series 10.24/5 = 2.04. Therefore subtract 2.05 from 10.24 to give 8.19.

6) The difference between the numbers in this series increases by 1 each time: 2, 3, 4, 5, etc. This will produce a difference of 4 between 21 and the next number in the series, which is therefore 25.

7) The difference between the numbers in this series increases by -3 each time: -3, -6, -9, etc. Therefore the third number in the series is 88.

8) Subtract each number from the previous one to get the next in the series. For example, 8-5=3, 5-3=2, etc. The second number in the series is 13 since 21-13 = 8.

9) The numbers in this series are descending cubes of 6,5,4,3,2,1.

10) The interval between letters increases by 1 each time, so V is the third in the series.

11) The numbers in the box on the right are (n + 5) where n is the number in the corresponding box on the left. Therefore the answer must be (22 + 5) = 27.

12) The numbers in the boxes on the left add up to 200. The answer must be 75 to make the numbers in the boxes on the right add up to 200.

13) The numbers in the box on the right are (n + 8) where n is the number in the corresponding box on the left. Therefore the answer must be (7 + 8) = 15.

14) The numbers in the bottom left and top right of each group of 4 are multiplied and the answer is written in the top left and bottom right square. One the left 6 * 6 = 36 and on the right 7 * 7 = 49. Therefore the answer is 9. Note that even though logically 7 could be an answer, it is not one of the answer options – so you need to think a bit more laterally.

15) The number in the top row is divided by 2 to give the number in the bottom row. Therefore the answer is 77.

16) The top row and the bottom row must add up to 20. Therefore the answer is 5. Note that the presence of two adjacent 1's in the bottom row beneath different numbers offers a strong hint that the relationship can not be between corresponding numbers in the top and bottom rows.

17) It is not 45 mph. Imagine that Anna's sister lives 60 miles away. The outward journey will take 1 hour. The return journey will take 2 hours (60 miles at 30 mph). Anna has been travelling for 3 hours and has covered 120 miles. Her average speed is therefore (120/3 = 40 mph). Note that this is true however far she drives.

18) At the bank paying 4%, the amount will be $1040 ($1000 * 1.04) at the end of the first year and $1081.60 ($1040 * 1.04) at the end of the second year. At the bank paying 10% the amount will be $1100 ($1000 * 1.10) at the end of the first year and $1210 ($1100 * 1.10) at the end of the second year. Therefore the difference will be $128.40.

19) The stock is owned in the ratio 4:5:6 and the total value is $45,000.
To work out the value of their holdings add 4+5+6 = 15.
Divide $45,000 by 15 = $3,000. You can then calculate that
Tom owns $12,000 (4 * $3,000)
Dick owns $15,000 (5 * $3,000)
Harry owns $18,000 (6 * $3,000)

Therefore Harry would need to give Tom $3,000 worth of stock for all three to own $15,000 worth.

20) If Anna covers 600 miles in at an average speed of 65 mph then she must have driven for a total of 554 minutes. If she stops for 2 hour and 20 minutes then her total journey time is 694 minutes. If John covers 600 miles in at an average speed of 50 mph then he must have driven for a total of 720 hours. If he stops for 90 minutes then his total journey time is 810 minutes. Therefore there is a 116 minute difference in their arrival times.

21) Of the 600 CDs sold:

25% or 150 were sold at $9.95 = $1,492
20% or 120 were sold at $7.46 = $895
55% or 330 were sold at $5.47 = $1,805

Therefore the total revenue was $4,192

22) If two thirds of the sample of 243 preferred butter, then 81 (243/3) people preferred either margarine or Tastee. The Tastee and margarine are preferred in the ratio 7:2 and the total number preferring either is 81, then to work out the numbers of each add 7+2 = 9. Divide 81 by 9 = 9. You can then calculate that the number of people preferring Tastee will be (7 * 9 =) 63.

Mechanical Aptitude Tests

Explanations - Numerical Reasoning Test 3

1) The numbers in this series increase by 4 each time. The next number is 23.

2) The difference between the numbers in this series increases by 5 each time – 5, 10, 15, etc. This will produce a difference of 20 between 41 and the next number in the series, which is therefore 61.

3) The numbers in this series increase by 7 each time. The next number is 63.

4) The difference between the numbers in this series increases by 3 each time: -3, -6, -9, -12 etc. This will produce a difference of -15 between 67 and the next number in the series, which is therefore 52.

5) The numbers in this series increase by -9 each time. The next number is 36.

6) The difference between the numbers in this series alternates between -5 and -10. Therefore the second number in the series is 190.

7) The difference between the numbers in this series alternates between 2 and 10. Therefore the sixth number in the series is 29.

8) Each number in the series is a multiple of the two preceding numbers. The third number in the series will be 3 since 1 * 3 = 3. Whenever you see a large increase in the numbers in a series (for example, 9, 27, 243) this indicates that multiplication of the numbers themselves or of their differences defines the series.

9) The difference between the numbers in this series increases by 1 each time: -19, -20, -21, -22. The third number in the series is therefore 110.

10) The interval between the letters alternates between incrementing by 4 and decrementing by 2. Therefore the second letter is 'P'.

11) The number in the bottom left square is the product of the other three numbers.

12) The numbers in the boxes on the left add up to 90. The answer must be 44 to make the numbers in the boxes on the right add up to 90.

13) The numbers in the top boxes are multiplied to give the numbers in the bottom boxes – one digit per box. 3 * 6 = 18 (written as 1 and 8). Therefore the answer is 2 since 12 (written 1 and 2) is the product of 2 times 6.

14) Starting from the box on the top left, move clockwise adding 7 each time. The answer is 54 which is 47 +7.

15) The numbers in the top row add up to 50. Therefore the answer must be 13 to make the numbers in the bottom row add up to 50. Note that the presence of two adjacent 5's in the bottom row beneath different numbers offers a strong hint that the relationship can not be between corresponding numbers in the top and bottom rows.

16) The numbers on the bottom row are 13 more than those on the top row. Therefore the missing number is 44 + 13 = 57.

17) The first 500 components are $4.50 each which gives $2,250. The 3500 subsequent components cost $0.90 each which gives $3,150. Therefore the run of 4000 components will cost $5,400.

18) At the bank paying 10% interest calculated each year, the amount will be $880 ($800 * 1.10) at the end of the first year. At the bank paying 10% per annum added every 6 months, the amount will be $840 ($800 * 1.05) at the end of the first six months and $882 ($840 * 1.05) at the end of the year. Therefore the difference will be $2.00.

19) The components are ordered in the ratio 1:5:4 and the total order is for 8,000.
To work out the numbers of each add 1+5+4 = 10.
Divide $8,000 by 10 = 800. You can then calculate that the number of Z components will be (800 * 4 =) 3,200.

20) Anna receives a bonus of ($400 * 0.12 =) $48. John receives a bonus of ($300 * 0.20 =) $60. The difference between their bonuses is therefore $12.

21) Of the 1600 CDs sold:

60% or 960 were sold at $7.95 = $7632
10% or 160 were sold at $5.56 = $889
30% or 480 were sold at $3.57 = $1713

Therefore the total revenue was $10,234

22) Anna sold 75% of her stock when it was worth $8000. So she took $6000 cash, leaving her with $2000 worth of stock, which she had purchased for $1000. When this stock increased in value to $4000 she sold it and added this to the first $6000 giving her $10,000 in cash. Subtracting the initial $4000 coat of the stock, Anna has made $6,000.

Mechanical Aptitude Tests

Explanations - Numerical Reasoning Test 4

1) The numbers in this series increase by 7 each time. Therefore the next number is 33.

2) The difference between the numbers in this series increases by 5 each time – 5, 10, 15, etc. This will produce a difference of 25 between 41 and the next number in the series, which is therefore 66.

3) The difference between the numbers in this series increases by 1 each time: -4, -5, -6, etc. This will produce a difference of -7 between 85 and the next number in the series, which is therefore 78.

4) The difference between the numbers in this series increases by 7 each time: 7, 14, 21, etc. This will produce a difference of 28 between 47 and the next number in the series, which is therefore 75.

5) The difference between the numbers in this series alternates between 4 and 5. Therefore the last number in the series is 22.

6) The difference between the numbers in this series follow the sequence: -3, -2, -1, 0, +1, etc.

7) The numbers in this series increase by 19 each time. Therefore the third number is 49.

8) The difference between the numbers in this series alternates between 4 and 3. Therefore the fourth number in the series is 79.

9) The difference between the numbers in this series alternates between -5 and -10. Therefore the first number in the series is 40.

10) The difference between the numbers in this series decreases by 2 each time: -14, -12, -10, -8, etc. The fourth number in the series is therefore 18.

11) Starting from the box on the top left, move clockwise adding the number in the first box each time. The answer is 32 which is 24 + 8.

12) The numbers in the boxes on the left add up to 60. The answer must be 9 to make the numbers in the boxes on the right add up to 60.

13) The numbers in the boxes on the left add up to 26. The answer must be 1 to make the numbers in the boxes on the right add up to 26.

14) Starting from the box on the top left, move clockwise adding 7 each time. The answer is 47 which is 40 + 7.

15) The numbers which are one above the other are read as fractions then they are all equal. Therefore the correct answer is 56.

16) The numbers in the bottom row add up to 7. Therefore the answer must be 1 to make the numbers in the top row add up to 7. Note that the presence of three 1's in the top row beneath different numbers offers a strong hint that the relationship can not be between corresponding numbers in the top and bottom rows.

17) The first 3000 components are $2.70 each which gives $8,100. The 1000 subsequent components cost $0.67 each which gives $670. Therefore the run of 4000 components will cost $8,770.

18) At the bank paying 6%, the amount will be $106 ($100 * 1.06) at the end of the first year. At the bank paying 10% the amount will be $105 ($100 * 1.05) at the end of the first six months and $110.25 ($105 * 1.05) at the end of the year. Therefore the difference will be $4.25.

19) If 1200 votes were cast for the Independent candidate then the Democratic candidate must have received (1200 * 0.75) 900 votes. The Republican candidate must have received (900 * 3) 2700 votes.

20) If Anna covers 280 miles in at an average speed of 70 mph then she must have driven for a total of 4 hours. If she stops for 1 hour and 10 minutes then her total journey time is 310 minutes. If John covers 280 miles in at an average speed of 60 mph then he must have driven for a total of 4 hours and 40 minutes. If he stops for 35 minutes then his total journey time is 315 minutes. Therefore there is a 5 minute difference in their arrival times.

21) The simplest way to approach this question is to add the length of the train to the length of the tunnel as this gives the distance that the train needs to travel to clear the tunnel. This is 5.5 miles. Travelling at 66 mph the train will travel this distance in (5.5/66 = 0.08333 hours) which equals 5 minutes.

22) The stock is owned in the ratio 3:5:7 and the total value is $10,500.
To work out the value of their holdings add 3+5+7 = 15.
Divide $10,500 by 15 = $700. You can then calculate that
Tom owns $12,000 (3 * $2,100)
Dick owns $15,000 (5 * $3,500)
Harry owns $18,000 (7 * $4,900)

Therefore Harry would need to give Tom $1,400 worth of stock for all three to own $15,000 worth.

Abstract Reasoning

Author : Paul Newton
Version: 2.3

Mechanical Aptitude Tests

The Importance of Abstract Reasoning Tests

Abstract Reasoning assesses your ability to understand complex concepts and assimilate new information outside of your previous experience. The questions consist of items which require you to recognize patterns and similarities between shapes and figures. As a measure of reasoning, it is independent of educational and cultural background and can be used to provide an indication of intellectual potential.

These tests are of particular value when the job involves dealing with abstract ideas or concepts as many technical jobs do. However, as they also provide the best measure of your general intellectual ability they are very widely used and you will usually find some questions of this type whichever particular tests you are given.

These tests are particularly valued where the job you are applying for involves:

- A high degree of problem solving
- Dealing with complex concepts
- Performing non-routine tasks where initiative is required

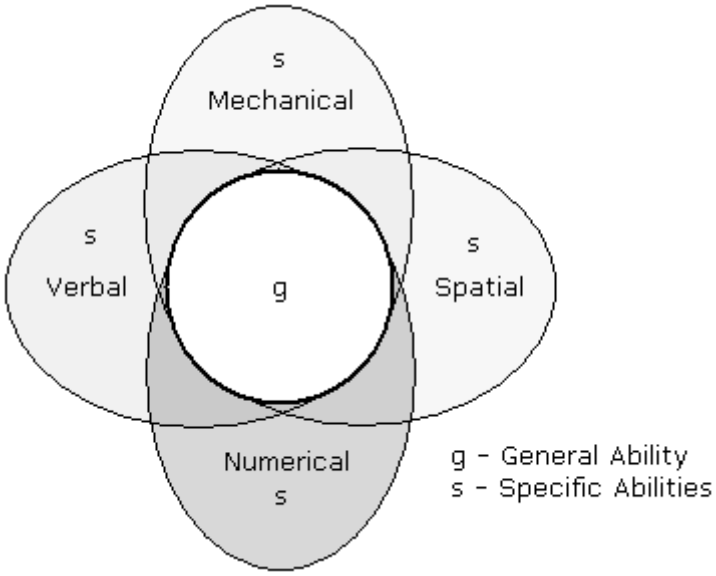
What do they Test?

The aptitudes and abilities measured by mechanical, spatial, verbal and numeric reasoning tests can easily be related to real world tasks and jobs, as many jobs require some degree of skill with words and numbers. Abstract reasoning tests on the other hand, seem to consist of questions which have little or no application in the real world. Yet these types of question appear in most aptitude tests. Why is this?

Abstract reasoning tests date back to research done by the psychologist Charles Spearman in the 1920's. Spearman used a statistical technique called factor analysis to examine relationships between people's scores on different tests of intelligence. He concluded that people who do well on some intelligence tests also do well on others (e.g. vocabulary, mathematics, spatial abilities). Conversely, if people do poorly on a particular intelligence test, they also tended to do poorly on other intellectual tests. This led him to believe that there are one or more factors that are common to all intellectual tasks.

Mechanical Aptitude Tests

As a result of this research Spearman developed a two-factor theory of intelligence.



As the diagram shows, Spearman said that intelligence is mainly made up of ‘g’, with bright people having a lot, and dull people having less. People may also vary according to their specific abilities, ‘s’, i.e. one person might be better at maths, while another would be very good verbally. However, Spearman placed much more importance on ‘g’ and believed that the most important information about someone’s intellectual ability is an estimate or measurement of ‘g’. Even though Spearman’s research was done many years ago, his theory of ‘g’ is still widely accepted by psychologists and a great deal of research has supported it.

Spearman defined ‘g’ as:

> “the innate ability to perceive relationships and educe co-relationships”

If we replace the word ‘educe’ with ‘work out’ then you can see why abstract reasoning questions are seen to be a good measure of general intelligence, as they test your ability to perceive relationships and then to work out any co-relationships without you requiring any knowledge of language or mathematics.

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Overview of Abstract Reasoning Tests

Abstract reasoning tests use diagrams, symbols or shapes instead of words or numbers. They involve identifying the underlying logic of a pattern and then determining the solution. Because they are visual questions and are independent of language and mathematical ability, they are considered to be an accurate indicator of your general intellectual ability as well as being 'culturally fair'.

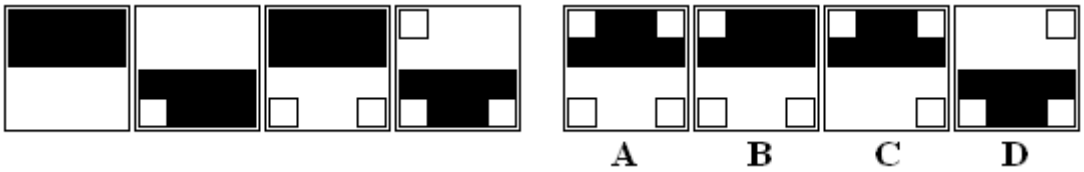
Questions tend to involve the repetition or change of the following:

- Shape
- Size
- Colour
- Pattern

These questions use symbols arranged in a straight line or in a pattern and you are required to identify the missing symbol or the next in the sequence. You can expect to be given slightly longer for these questions than for verbal and numeric ability questions. Thirty minutes to complete 20 questions would be typical.

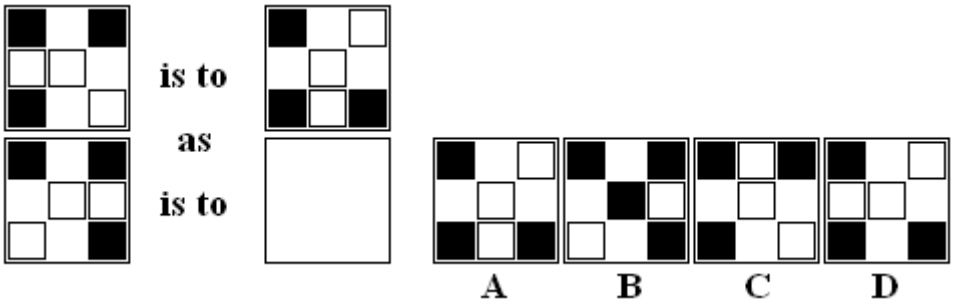
Sample Abstract Reasoning Questions

1) Which figure completes the series?



Hint: In this series the black rectangle is alternating from top to bottom and the number of white squares is increasing by one each time. Answer = A.

2) Which figure completes the statement?



Hint: Begin by comparing the top figures. Does each one contain the same number of elements? If so, does each contain the same elements? If so, the elements must have been moved in some way. This is usually done by reflection or rotation. Answer = C.

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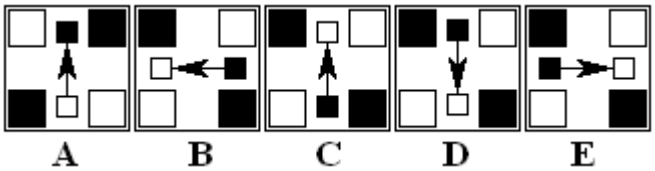
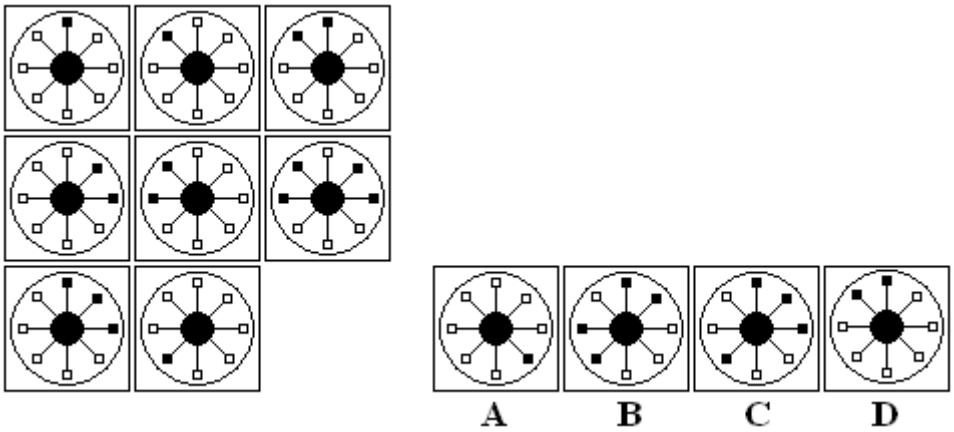
3) Which figure is the odd one out?



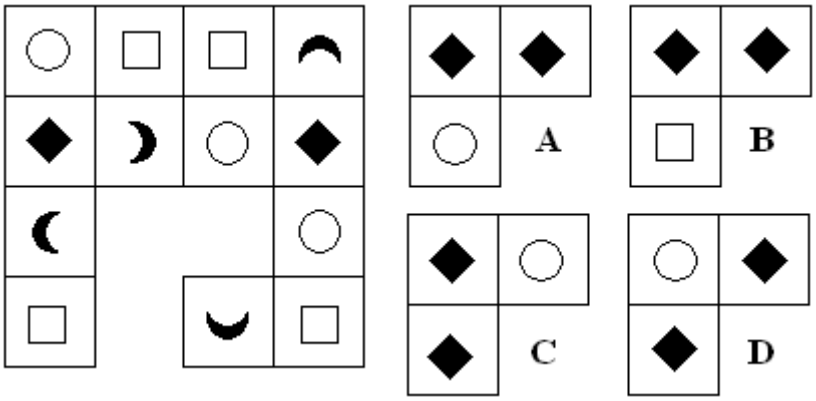
Hint: Begin by looking at the elements in each figure. Are there the same number in each? Are they the same? If so, then look at the configuration. Answer = A.

4) Which figure completes the series?



Hint: Begin by looking for a relationship between the figures in the top row. If you think you have found one, then check that the same relationship holds for the second row. Answer = C.

5) Which figure completes the grid?



Hint: Check to see if each row and column contains one, and only one, of each shape. If not, then divide the grid horizontally and vertically. Are they reflections? If not, are individual rows related in some way? What about individual columns? If not, divide the grid into four groups of four squares? Is there a relationship between these groups? Answer = A.

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Abstract Reasoning Questions

Abstract reasoning questions test your ability to identify patterns presented in diagrammatic form and are not dependent on your knowledge of English or maths. Because they are visual questions and are independent of language and mathematical ability, they are considered to be an accurate indicator of your general intellectual ability as well as being 'culturally fair'. Abstract reasoning ability is believed to be the best indicator of fluid intelligence and your ability to learn new things quickly.

These questions use symbols arranged in a straight line or in a pattern and you are required to identify the missing symbol or the next in the sequence. Abstract reasoning ability questions are invariably multiple-choice and strictly timed.

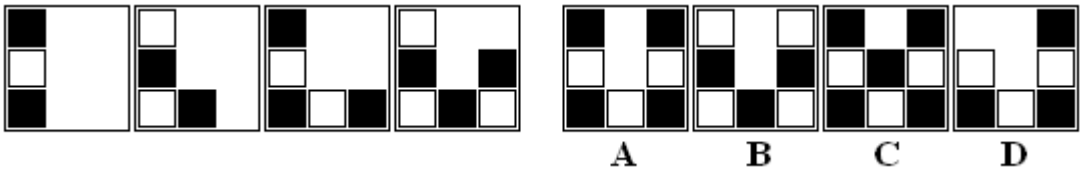
Abstract Reasoning		Operatives	Supervisory	Management
Craft & Technical			0	Y
Clerical & Administrative			0	Y
Police, Fire, Military etc.	0	0 - Occasionally used in selection.		
Management Trainee	Y			
Graduate & Professional	Y			

These sample question papers each contain 25 questions and have a suggested time limit of 20 minutes each. The questions are presented in Letter/A4 format for easy printing and self-marking.

Mechanical Aptitude Tests

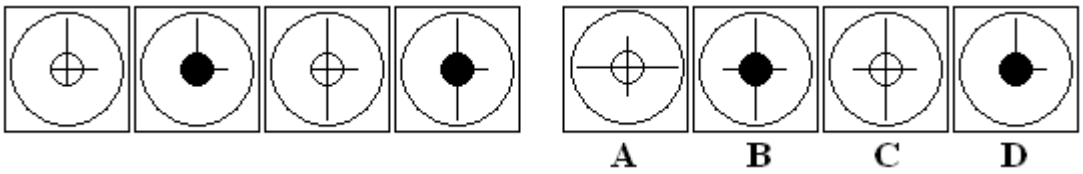
Test 1: 25 Questions
Answer as many questions as you can in 20 minutes. Circle the letter on the right which corresponds to the correct answer.

1) Which figure completes the series?



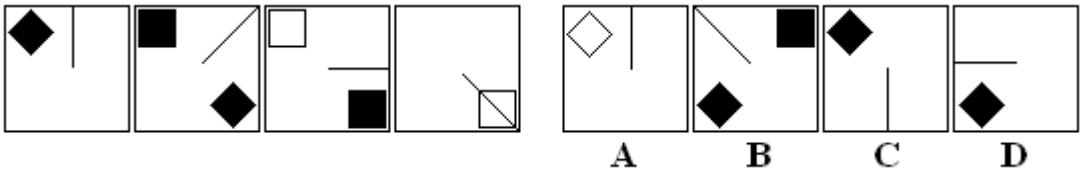
A B C D

2) Which figure completes the series?



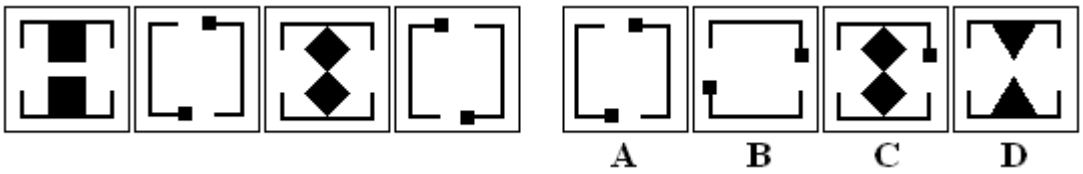
A B C D

3) Which figure completes the series?



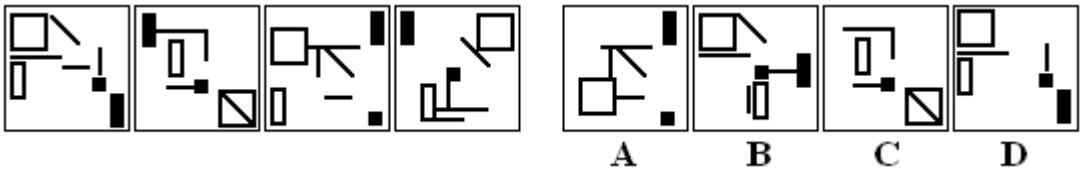
A B C D

4) Which figure completes the series?



A B C D

5) Which figure completes the series?



A B C D

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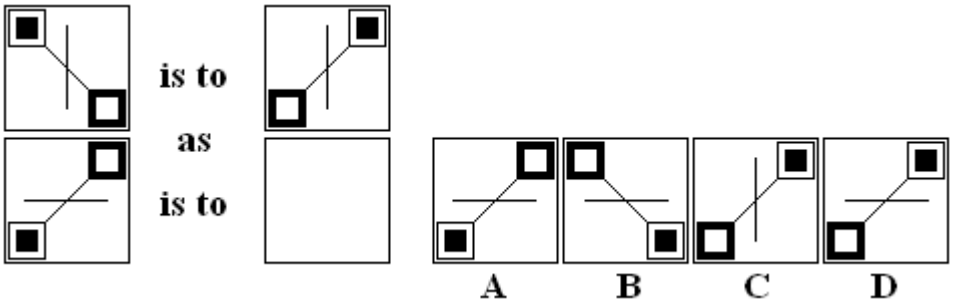
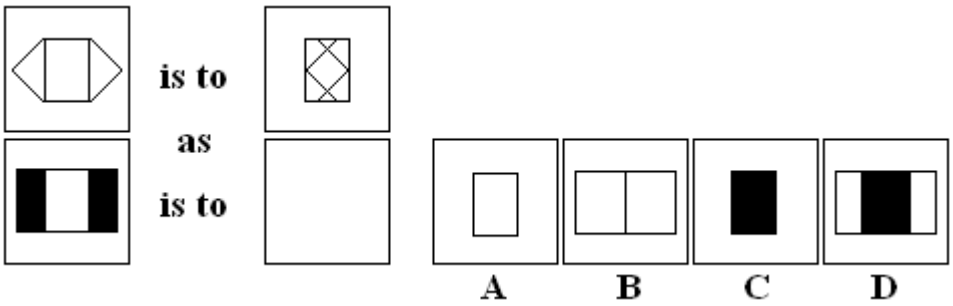
6) Which figure completes the statement?



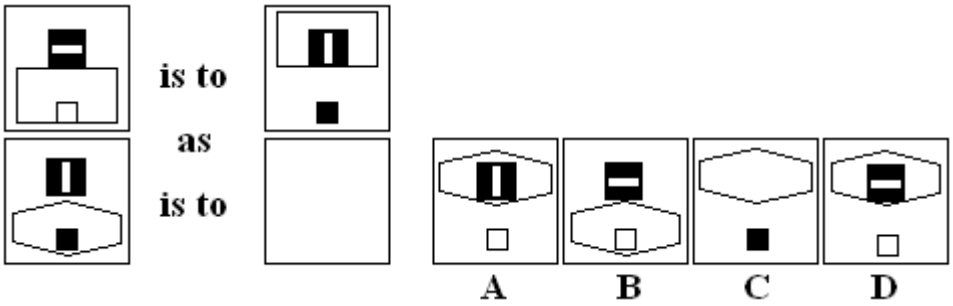
A B C D

7) Which figure completes the statement?



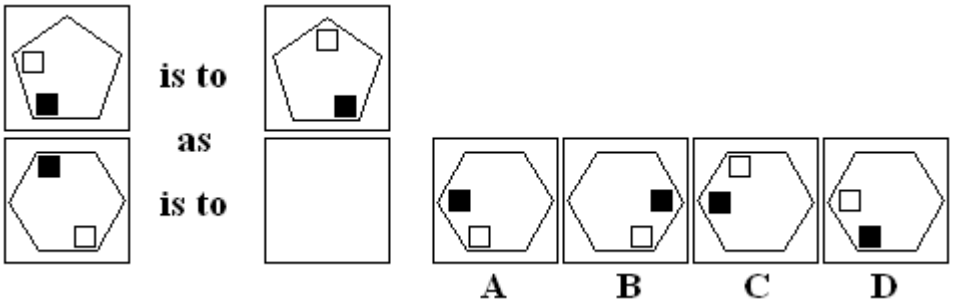
A B C D

8) Which figure completes the statement?



A B C D

9) Which figure completes the statement?



A B C D

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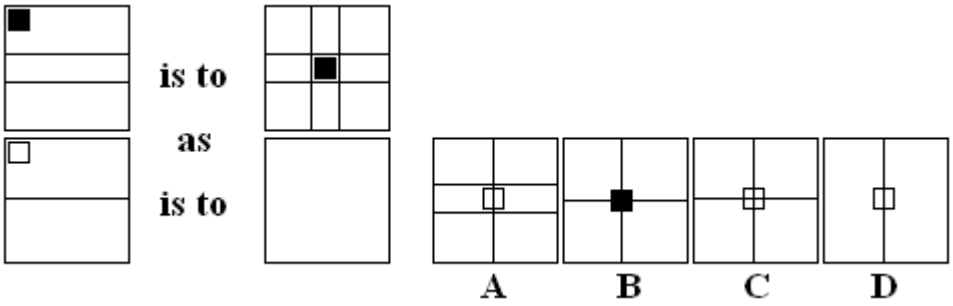
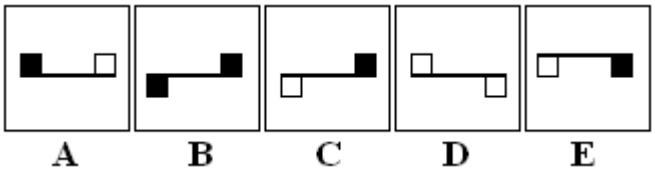
10) Which figure completes the statement?



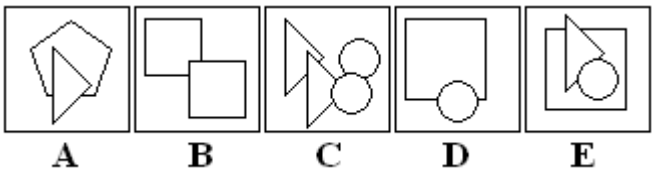
A B C D

11) Which figure is the odd one out?



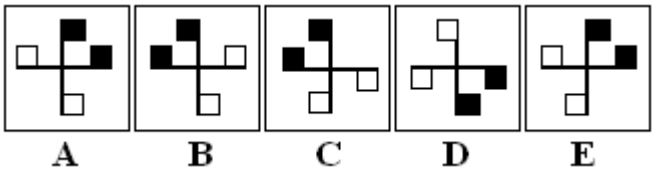
A B C D E

12) Which figure is the odd one out?



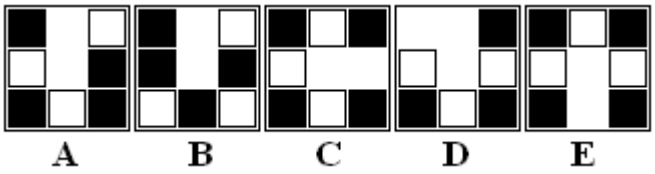
A B C D E

13) Which figure is the odd one out?



A B C D E

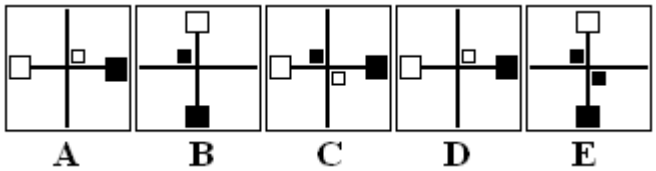
14) Which figure is the odd one out?



A B C D E

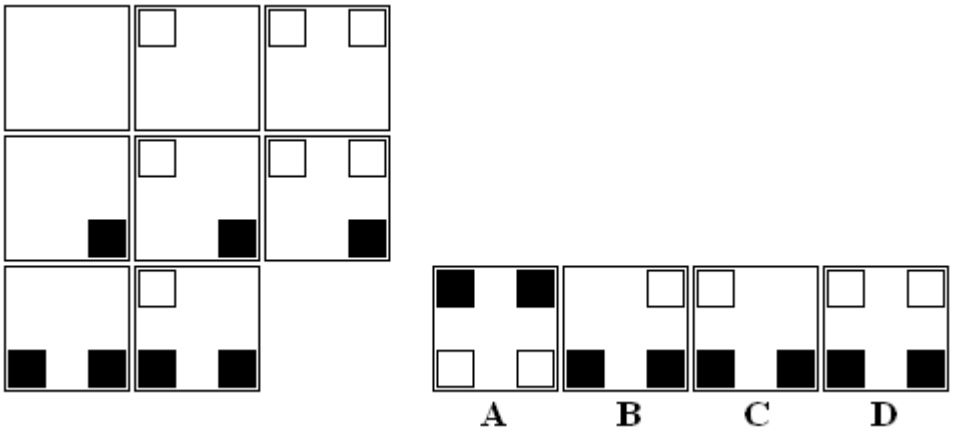
Mechanical Aptitude Tests

15) Which figure is the odd one out?



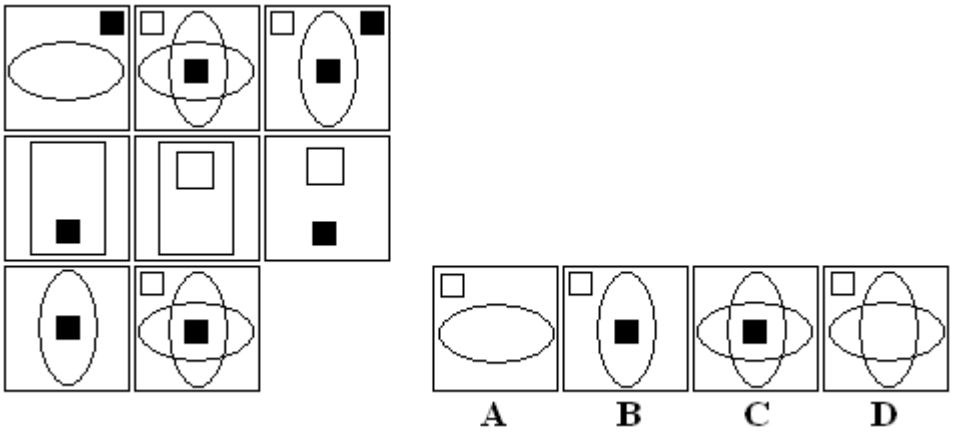
A B C D E

16) Which figure completes the series?



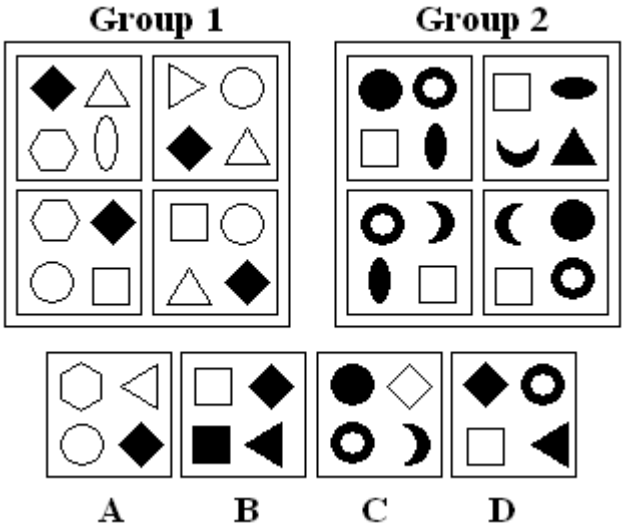
A B C D

17) Which figure completes the series?



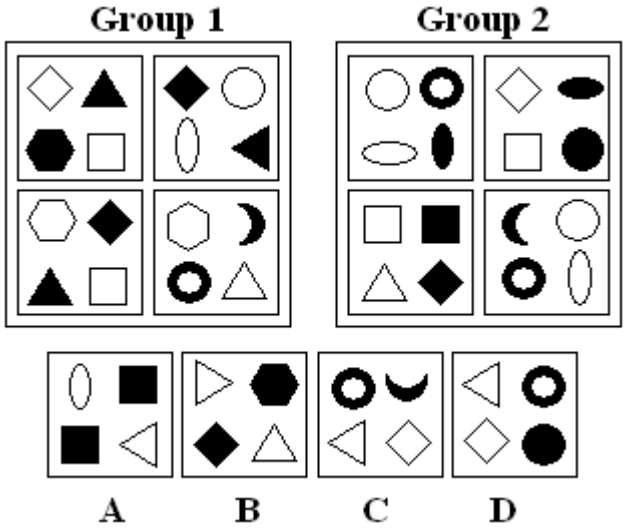
A B C D

18) Which figure belongs in neither group?



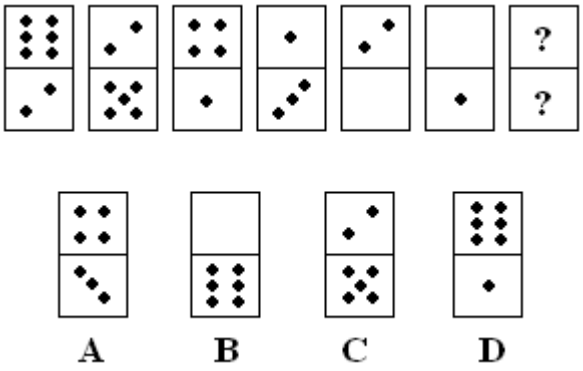
A B C D

19) Which figure belongs in neither group?



A B C D

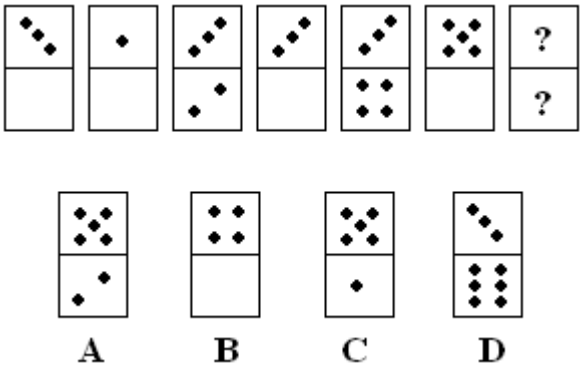
20) Which figure is next in the series?



A B C D

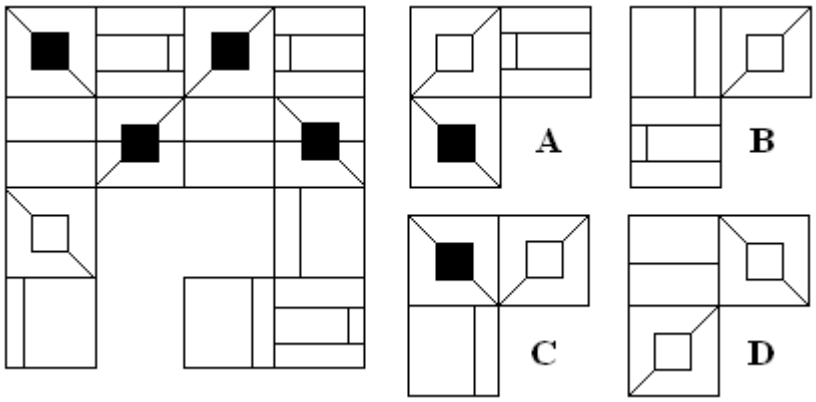
Mechanical Aptitude Tests

21) Which figure is next in the series?



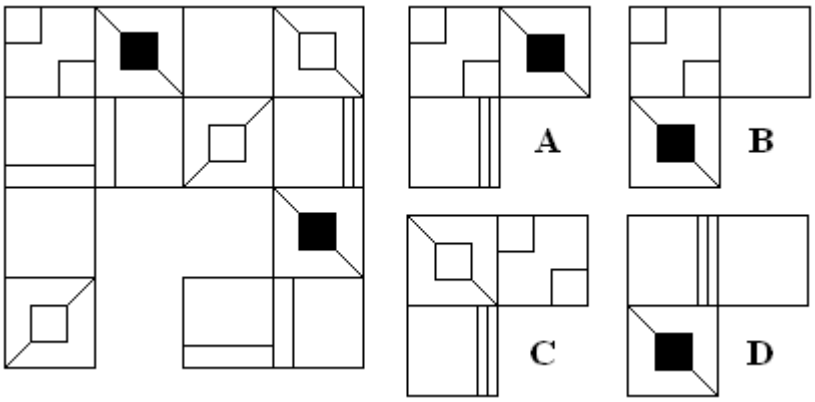
A B C D

22) Which figure completes the grid?



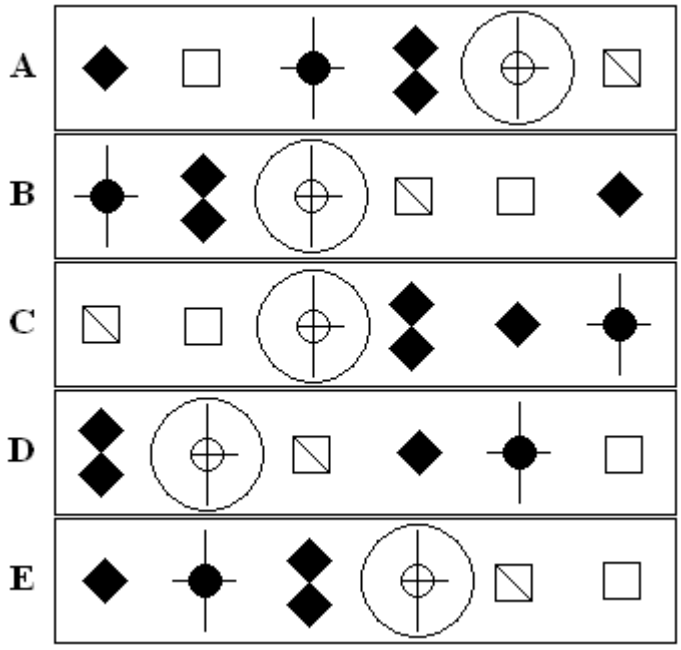
A B C D

23) Which figure completes the grid?



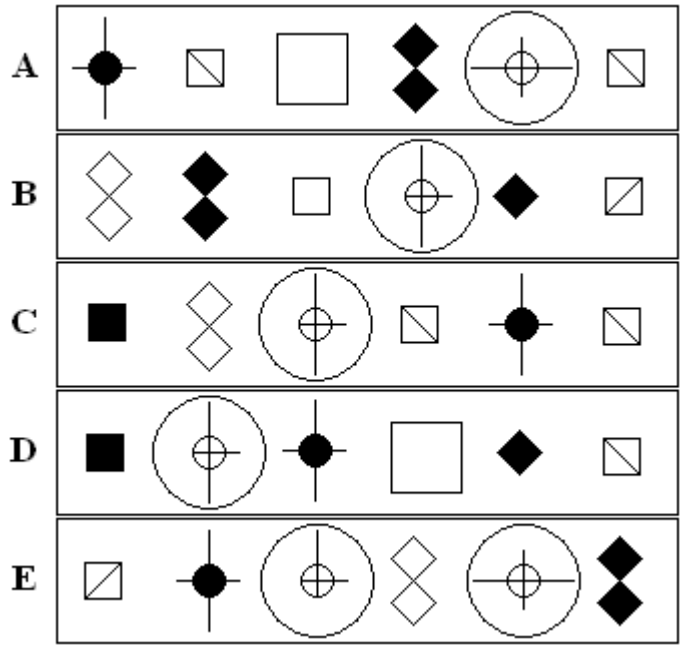
A B C D

24) Which figure is the odd one out?



A B C D E

25) Which figure is the odd one out?



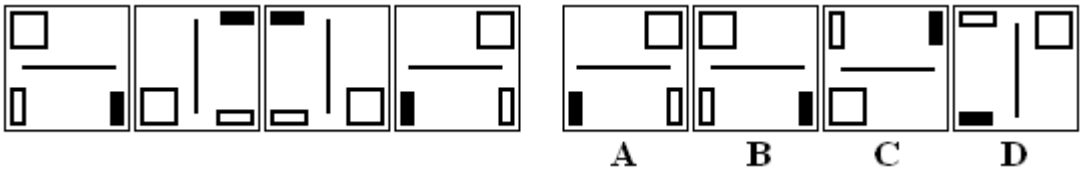
A B C D E

End of Abstract Reasoning - Test 1

Mechanical Aptitude Tests

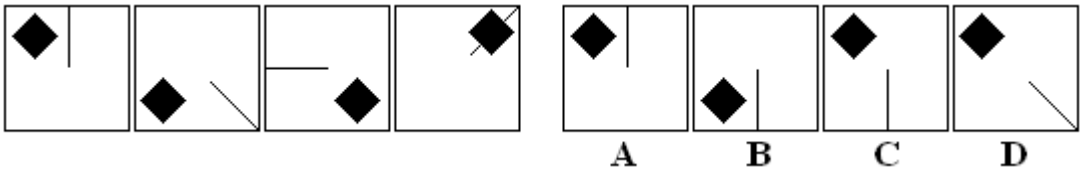
Test 2: 25 Questions
Answer as many questions as you can in 20 minutes. Circle the letter on the right which corresponds to the correct answer.

1) Which figure completes the series?



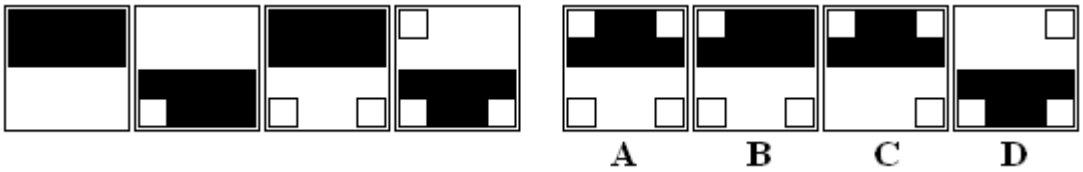
A B C D

2) Which figure completes the series?



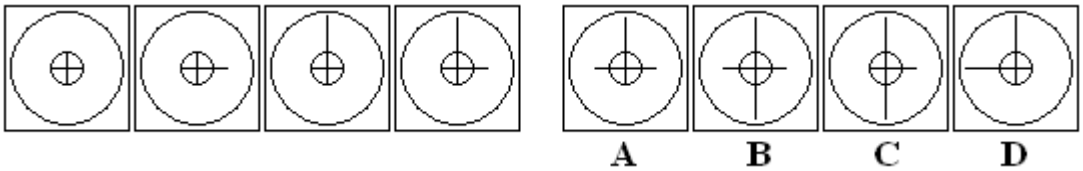
A B C D

3) Which figure completes the series?



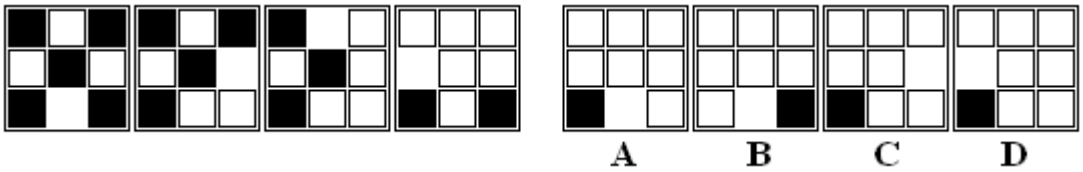
A B C D

4) Which figure completes the series?



A B C D

5) Which figure completes the series?



A B C D

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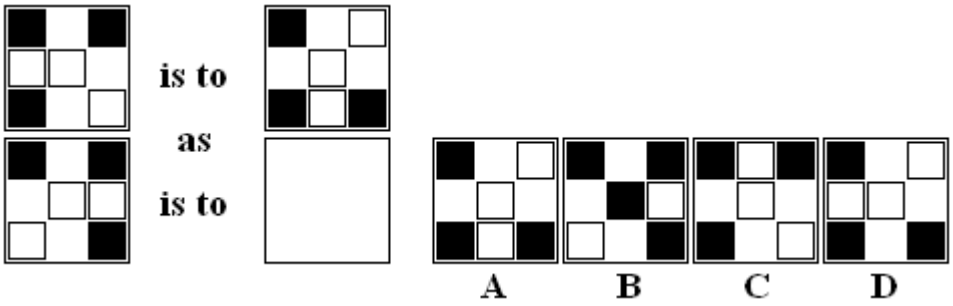
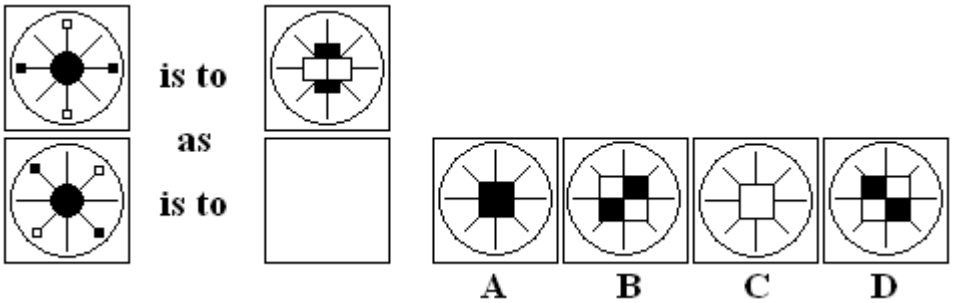
6) Which figure completes the statement?



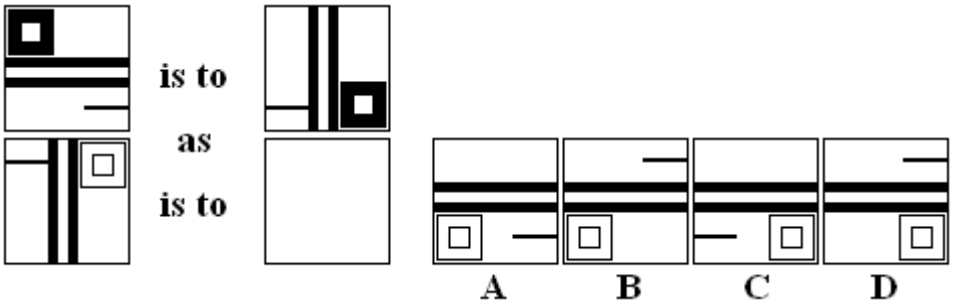
A B C D

7) Which figure completes the statement?



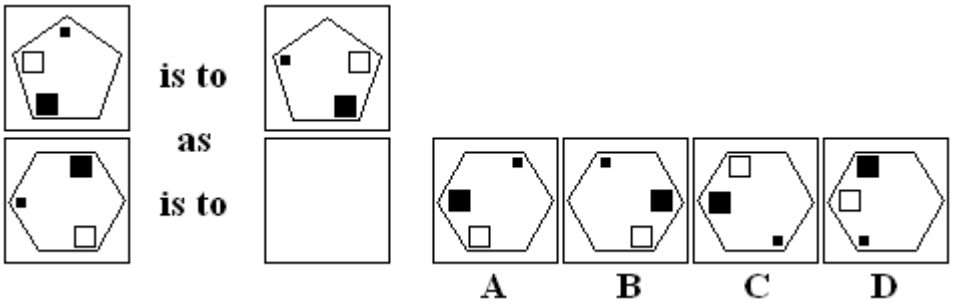
A B C D

8) Which figure completes the statement?



A B C D

9) Which figure completes the statement?



A B C D

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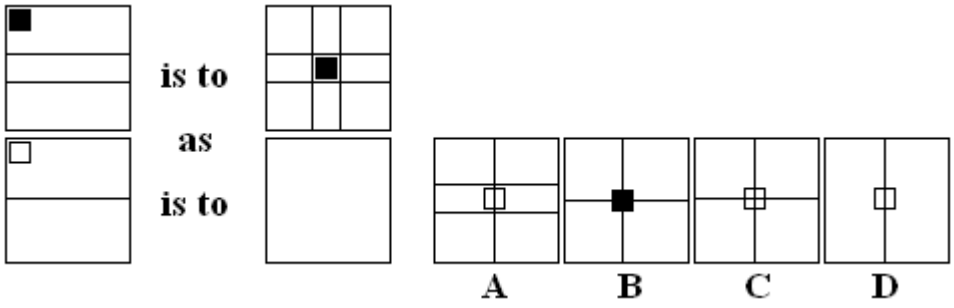
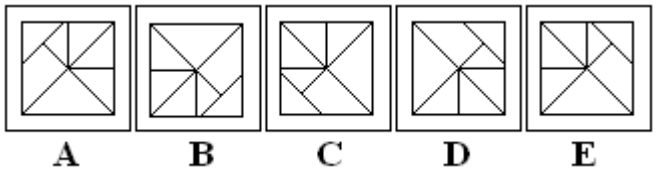
10) Which figure completes the statement?



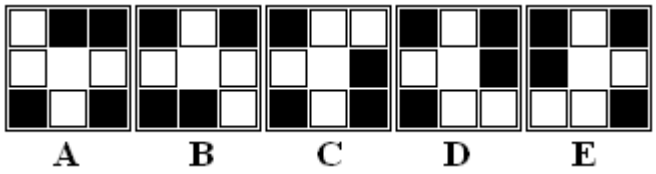
A B C D

11) Which figure is the odd one out?



A B C D E

12) Which figure is the odd one out?



A B C D E

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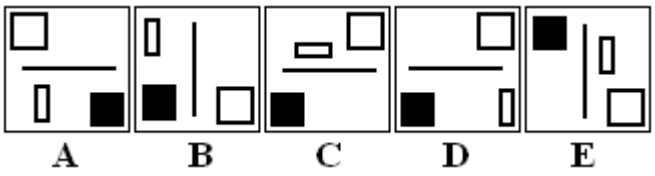
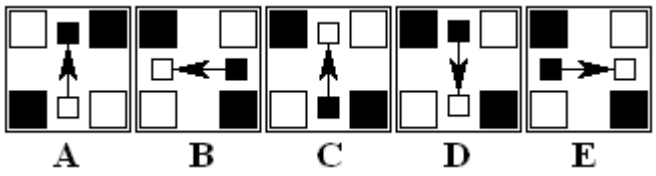


A B C D E

14) Which figure is the odd one out?



A B C D E

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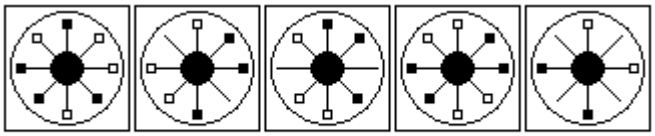
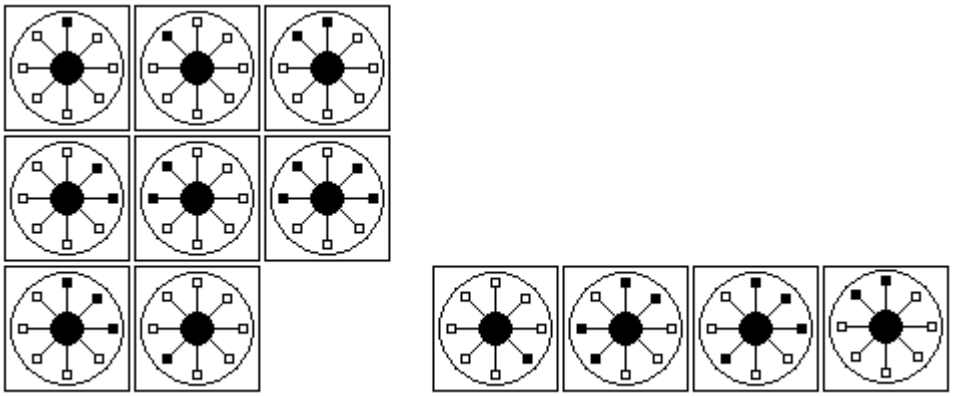
15) Which figure is the odd one out?



A B C D E

A B C D E

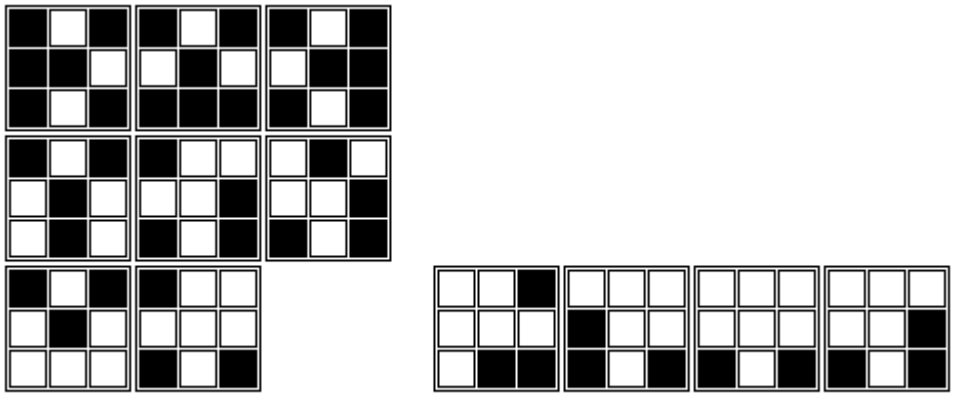
16) Which figure completes the series?



A B C D

A B C D

17) Which figure completes the series?

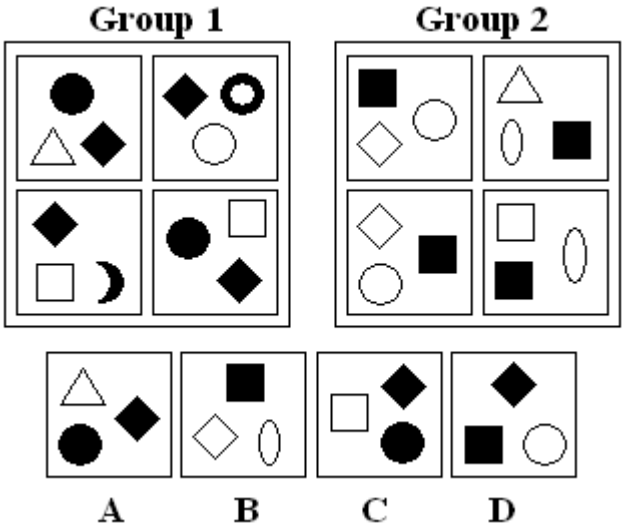


A B C D

A B C D

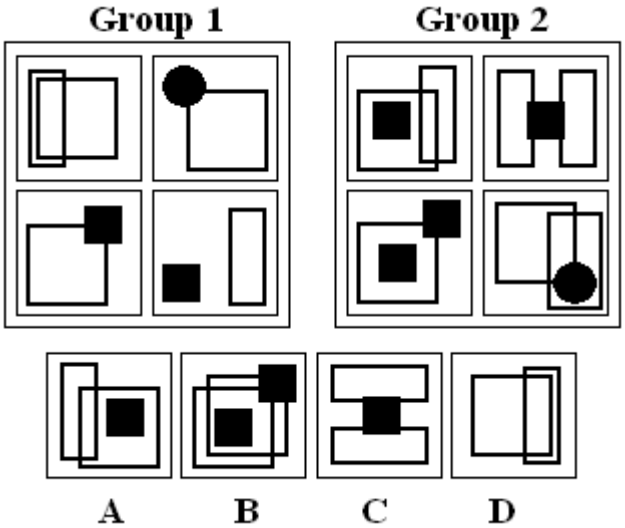
Mechanical Aptitude Tests

18) Which figure belongs in neither group?



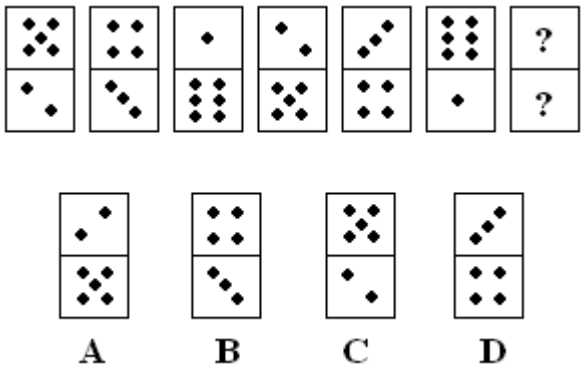
A B C D

19) Which figure belongs in neither group?



A B C D

20) Which figure is next in the series?



A B C D

Mechanical Aptitude Tests

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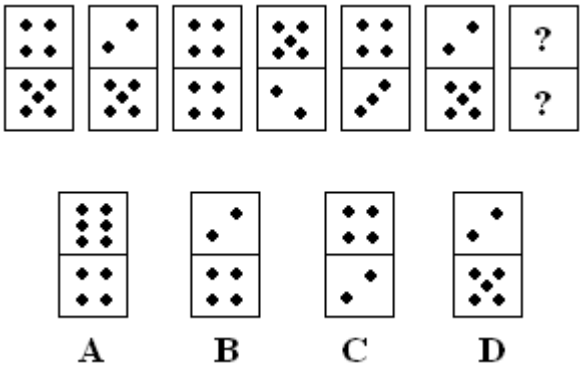
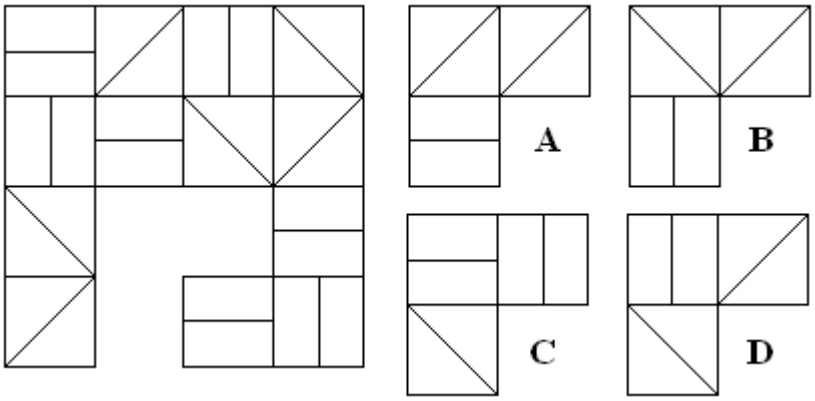


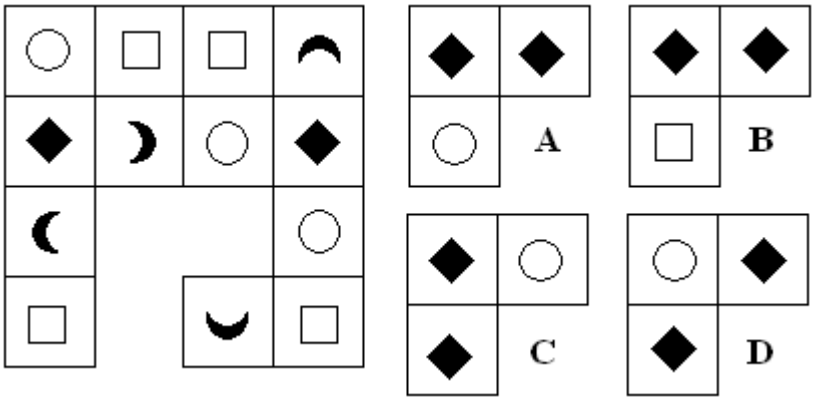
A B C D

22) Which figure completes the grid?



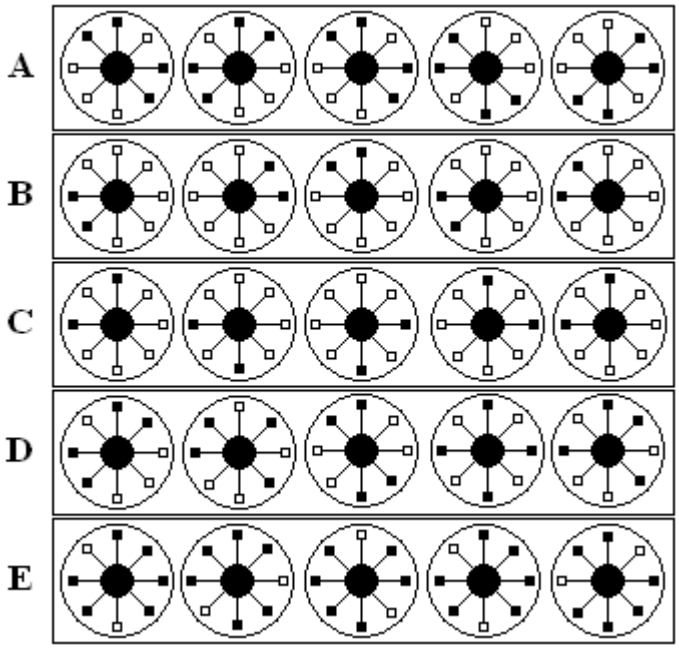
A B C D

23) Which figure completes the grid?



A B C D

24) Which figure is the odd one out?



A B C D E

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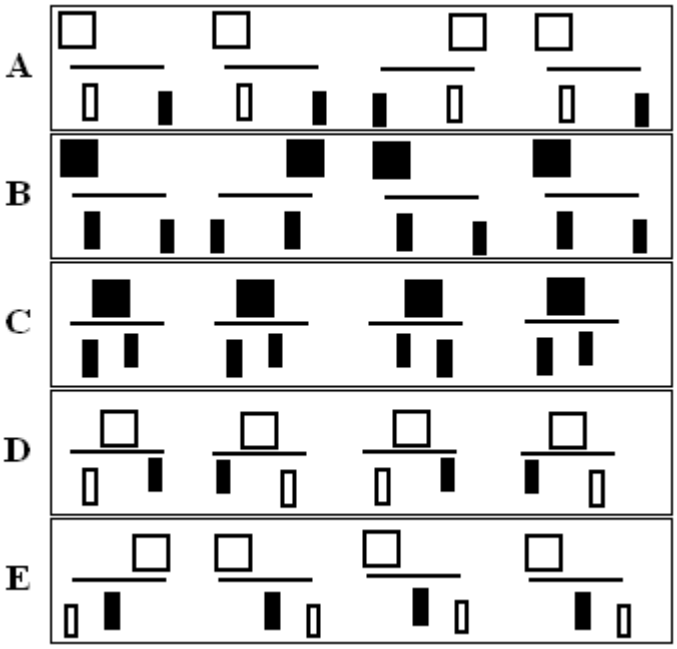


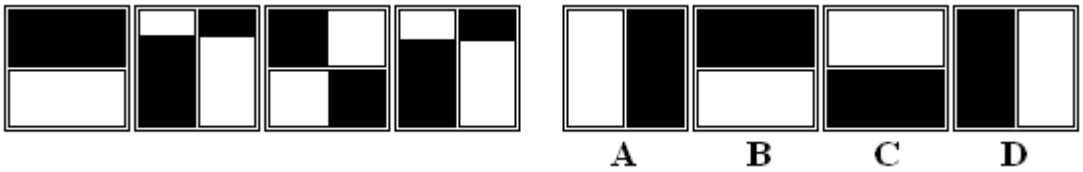
A B C D E

End of Abstract Reasoning - Test 2

Mechanical Aptitude Tests

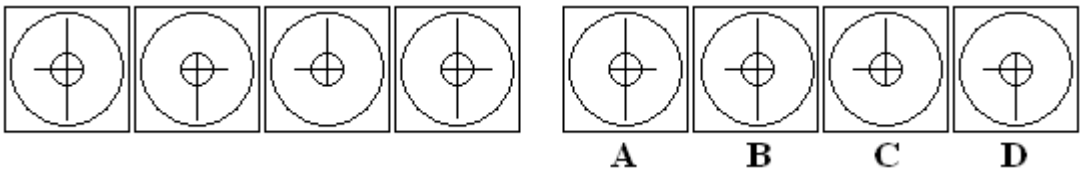
Test 3: 25 Questions
Answer as many questions as you can in 20 minutes. Circle the letter on the right which corresponds to the correct answer.

1) Which figure completes the series?



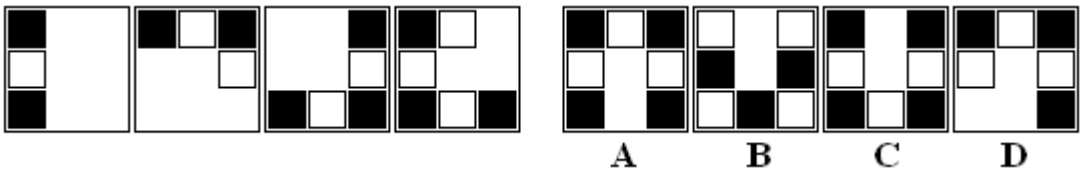
A B C D

2) Which figure completes the series?



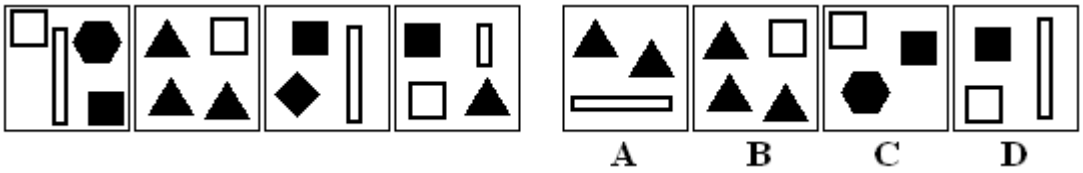
A B C D

3) Which figure completes the series?



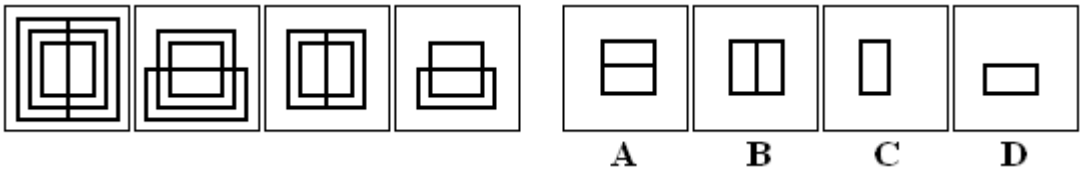
A B C D

4) Which figure completes the series?



A B C D

5) Which figure completes the series?



A B C D

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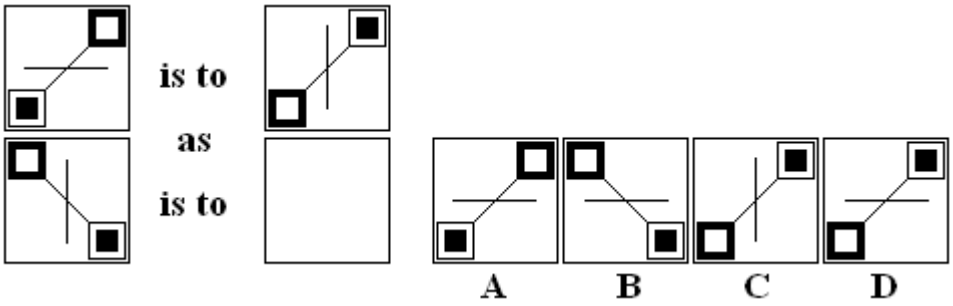
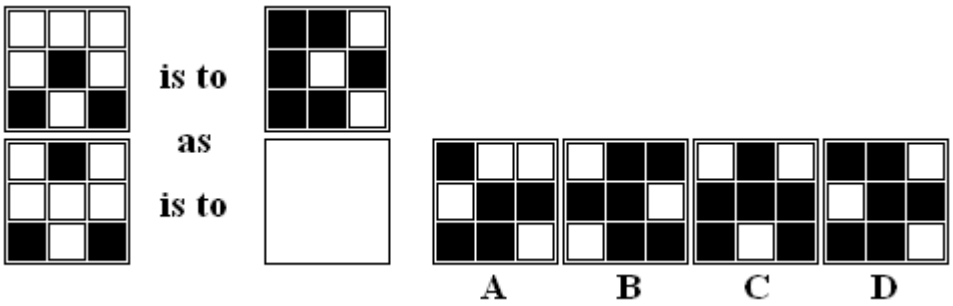
Mechanical Aptitude Tests

6) Which figure completes the statement?



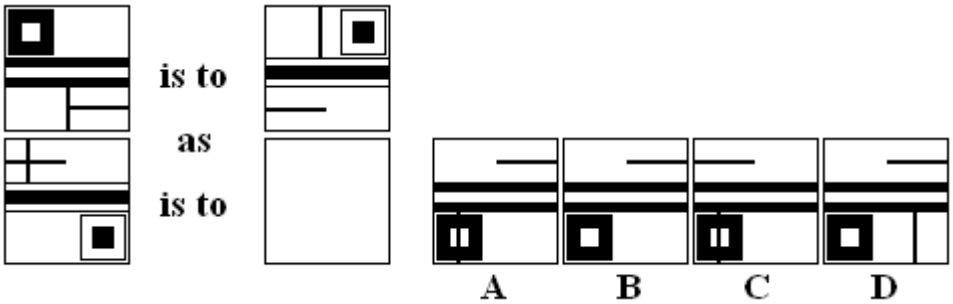
A B C D

7) Which figure completes the statement?



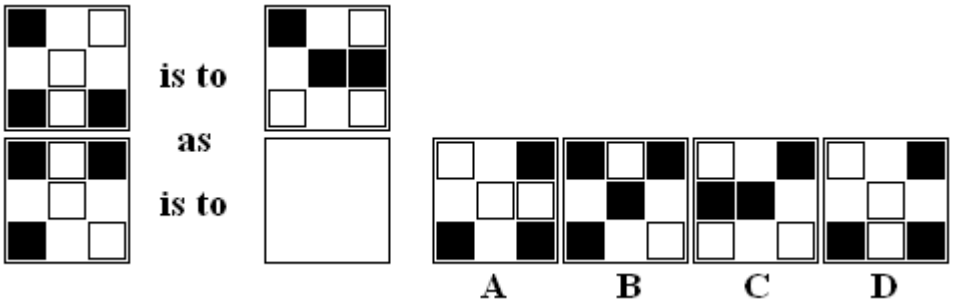
A B C D

8) Which figure completes the statement?



A B C D

9) Which figure completes the statement?



A B C D

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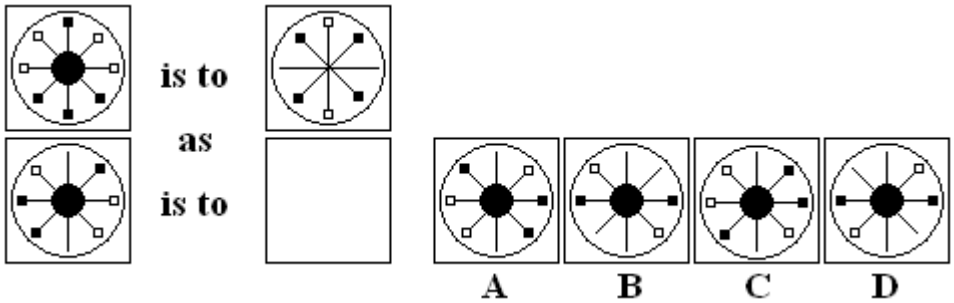
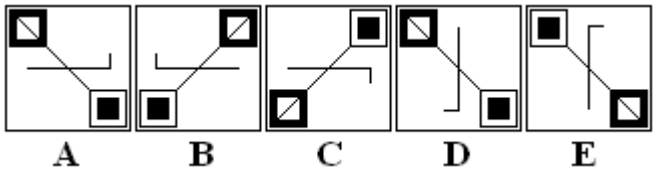
10) Which figure completes the statement?



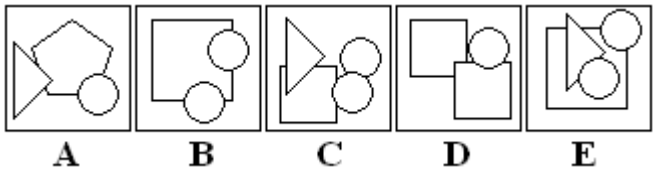
A B C D

11) Which figure is the odd one out?



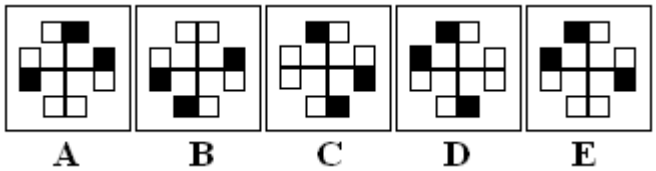
A B C D E

12) Which figure is the odd one out?



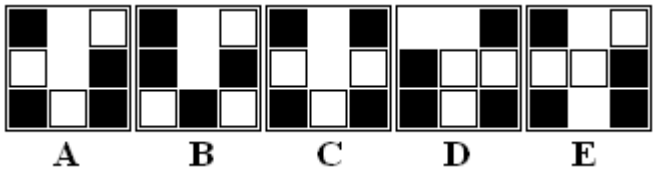
A B C D E

13) Which figure is the odd one out?



A B C D E

14) Which figure is the odd one out?



A B C D E

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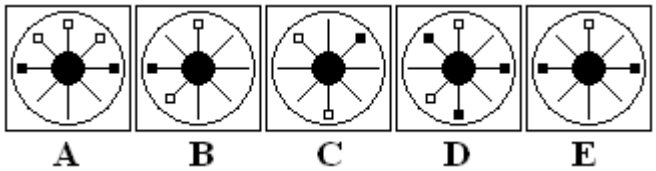
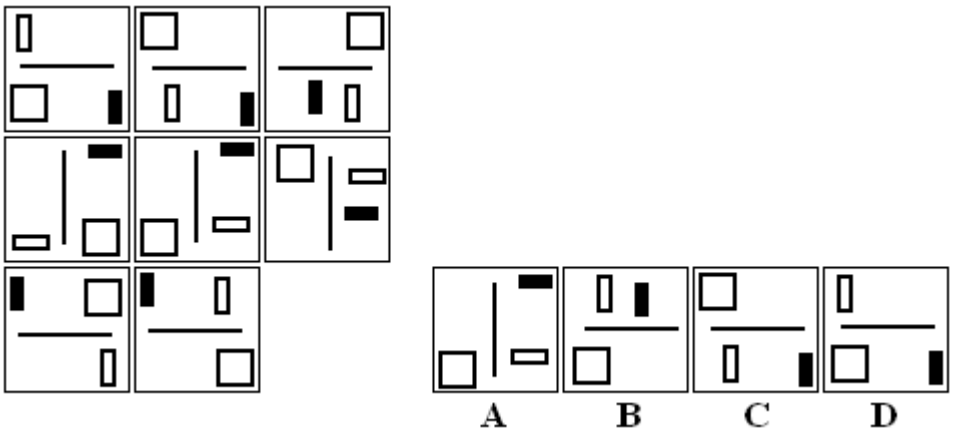
15) Which figure is the odd one out?



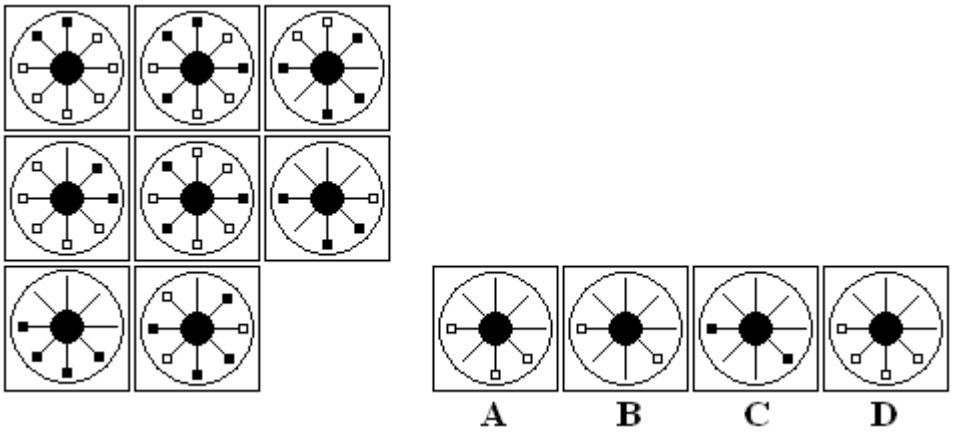
A B C D E

16) Which figure completes the series?



A B C D

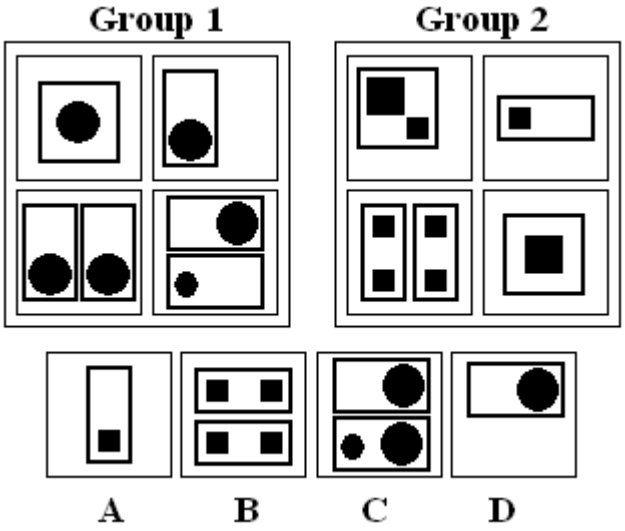
17) Which figure completes the series?



A B C D

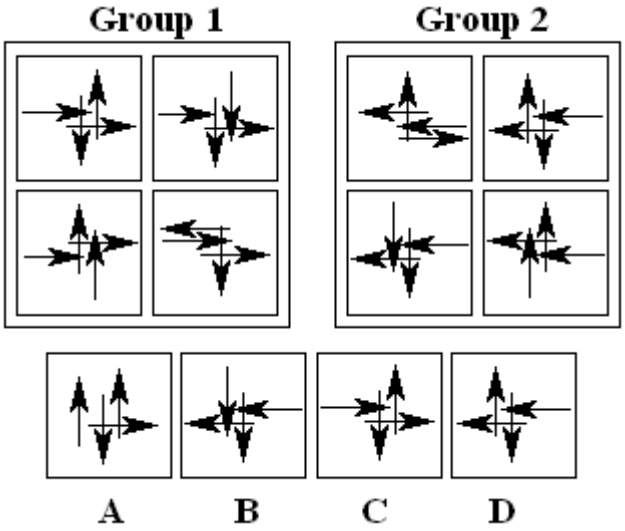
Mechanical Aptitude Tests

18) Which figure belongs in neither group?



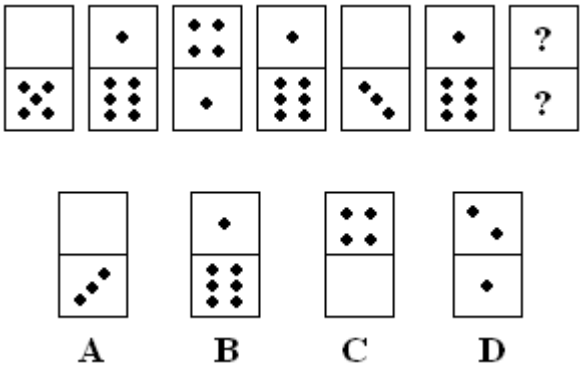
A B C D

19) Which figure belongs in neither group?



A B C D

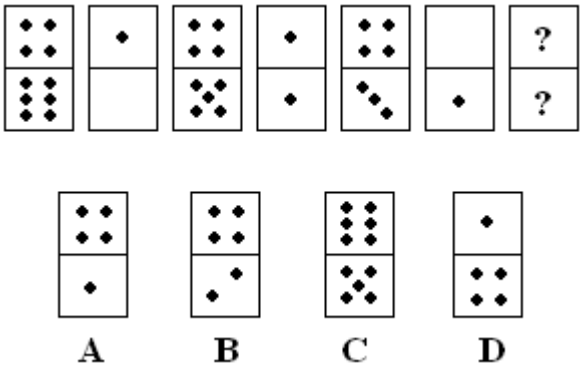
20) Which figure is next in the series?



A B C D

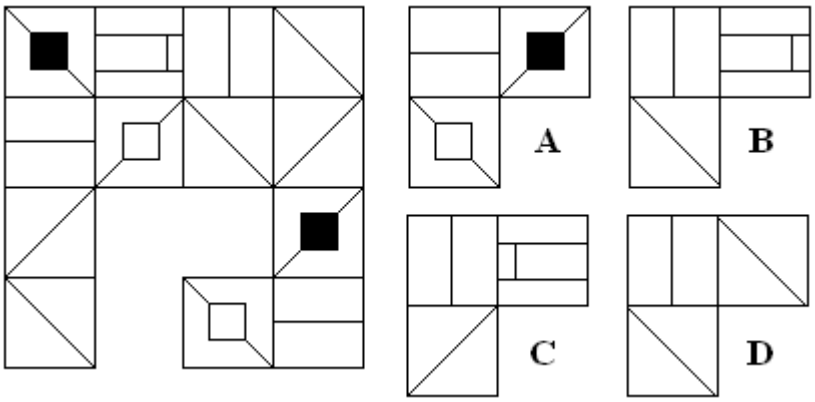
Mechanical Aptitude Tests

21) Which figure is next in the series?



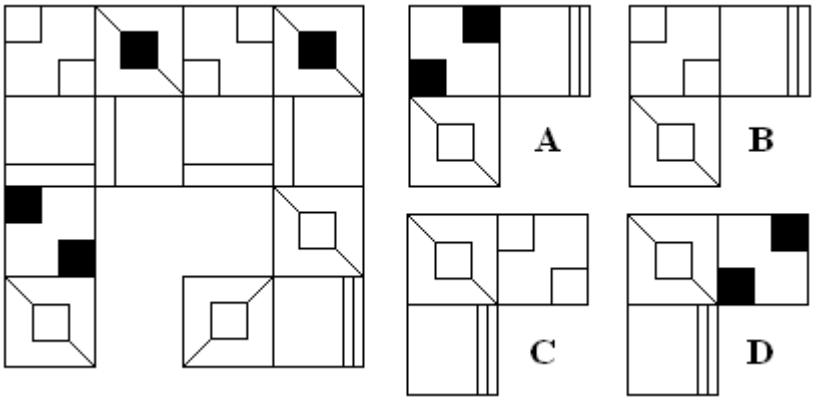
A B C D

22) Which figure completes the grid?



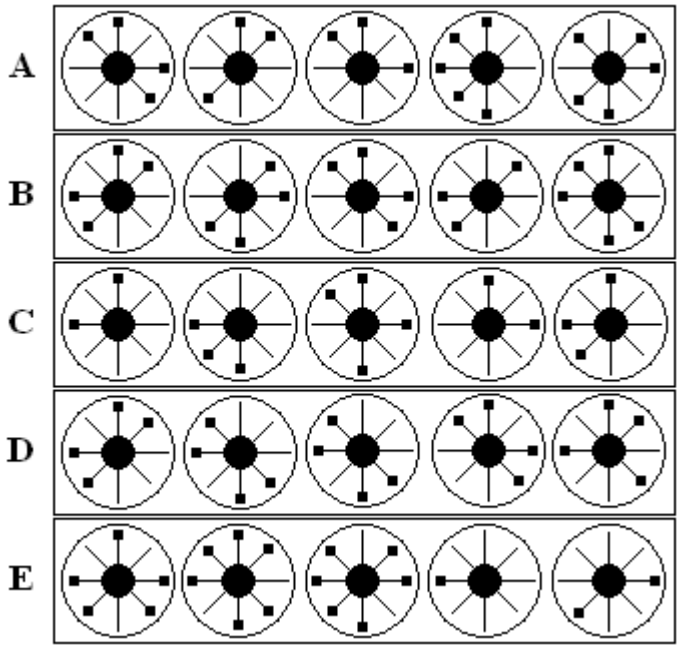
A B C D

23) Which figure completes the grid?



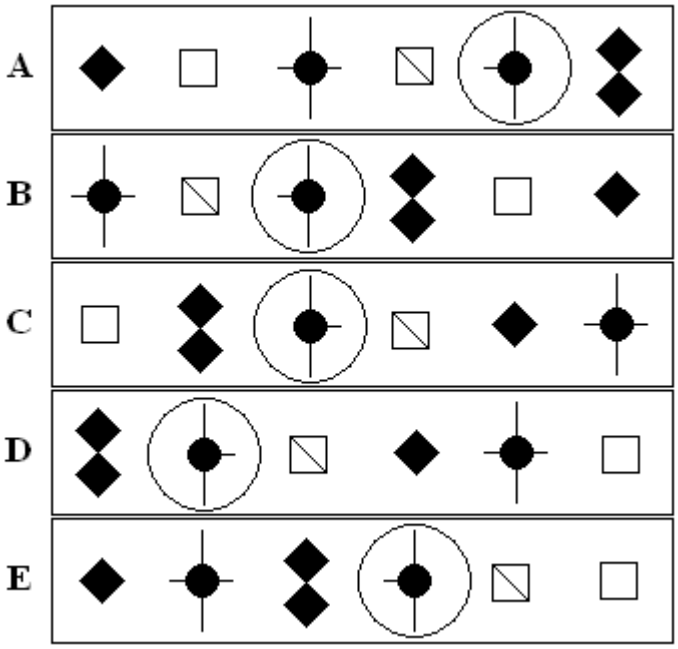
A B C D

24) Which figure is the odd one out?



A B C D E

25) Which figure is the odd one out?



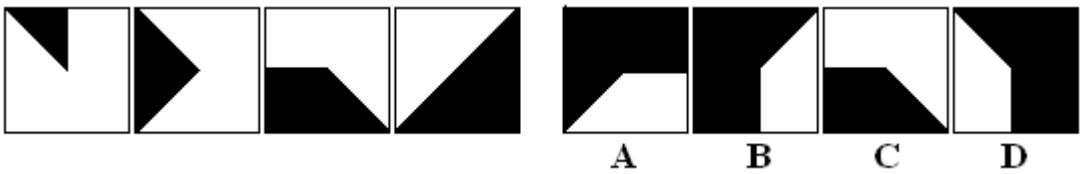
A B C D E

End of Abstract Reasoning - Test 3

Mechanical Aptitude Tests

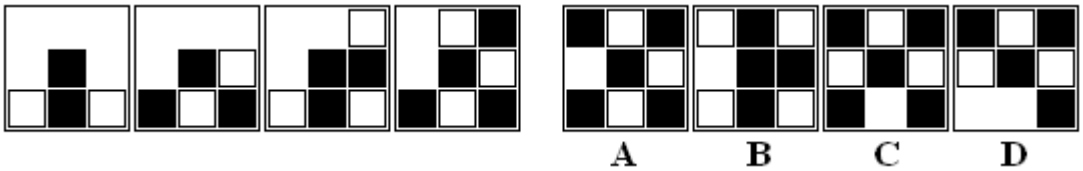
Test 4: 25 Questions
Answer as many questions as you can in 20 minutes. Circle the letter on the right which corresponds to the correct answer.

1) Which figure completes the series?



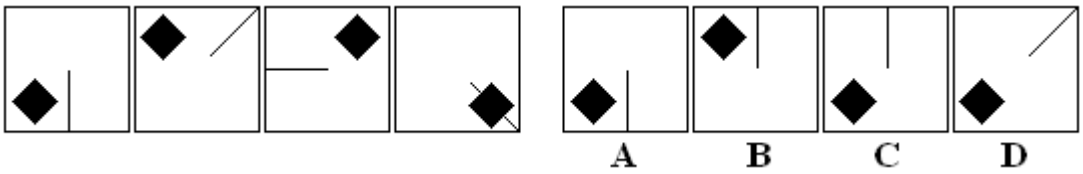
A B C D

2) Which figure completes the series?



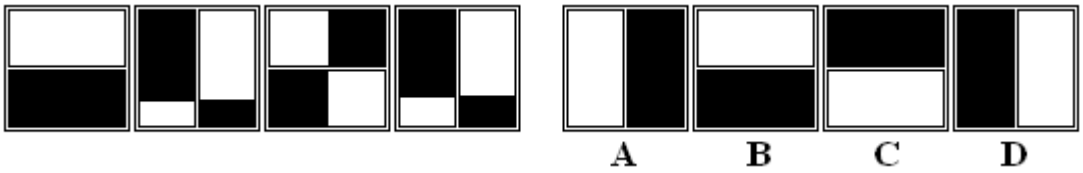
A B C D

3) Which figure completes the series?



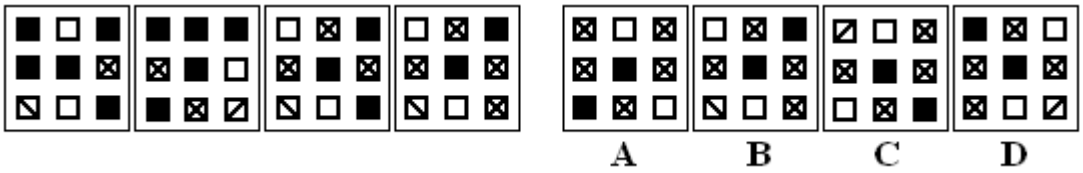
A B C D

4) Which figure completes the series?



A B C D

5) Which figure completes the series?



A B C D

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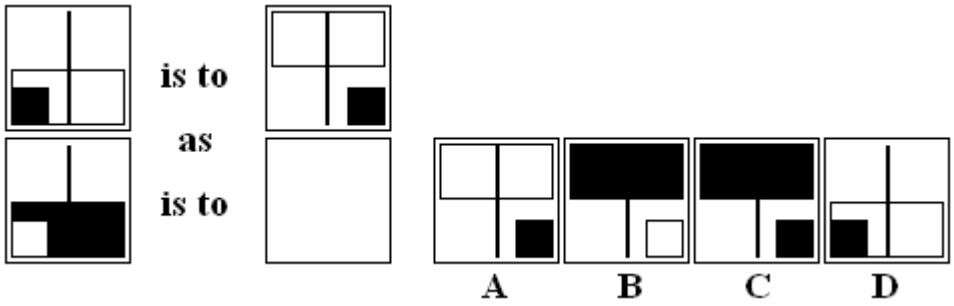
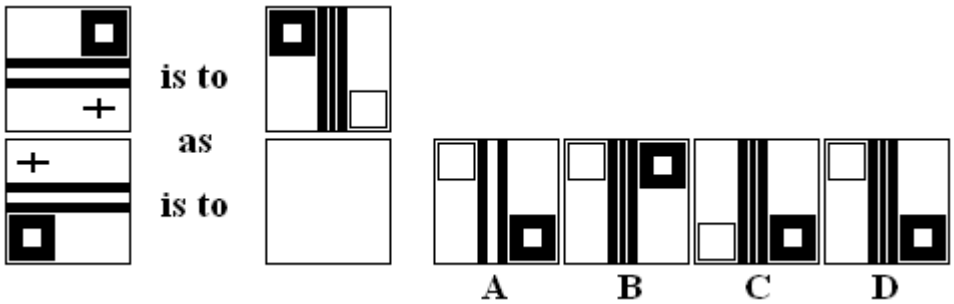
6) Which figure completes the statement?



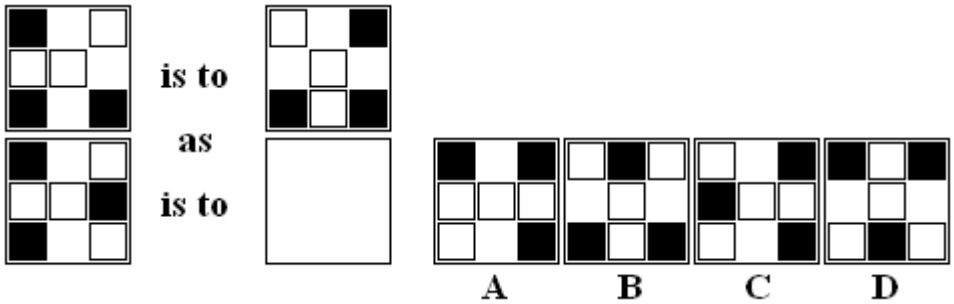
A B C D

7) Which figure completes the statement?



A B C D

8) Which figure completes the statement?



A B C D

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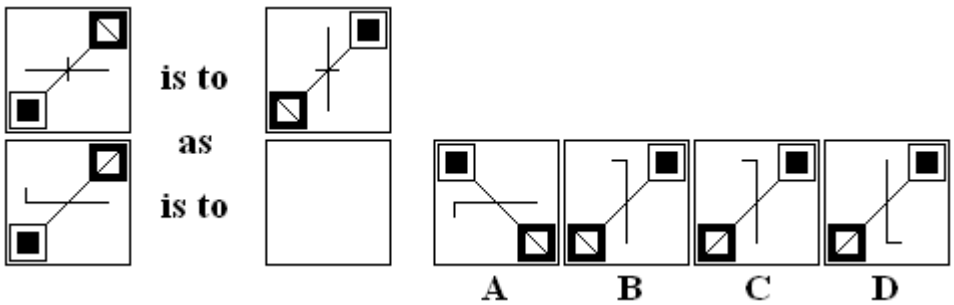

A B C D

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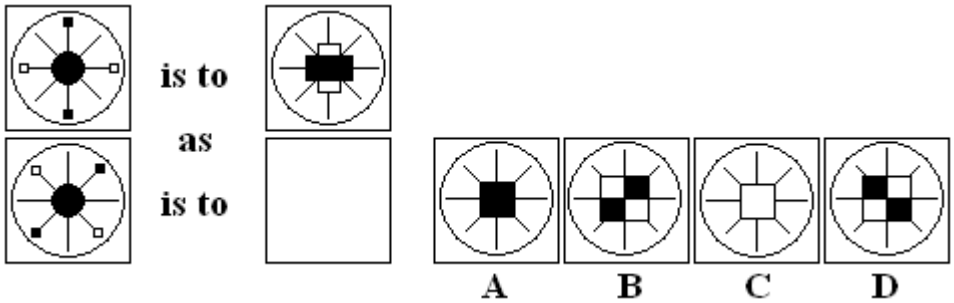
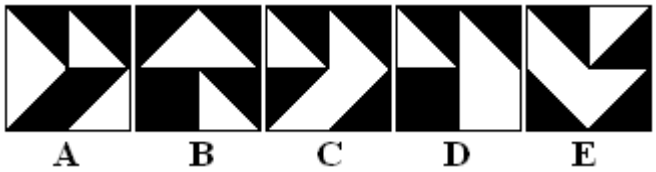
10) Which figure completes the statement?



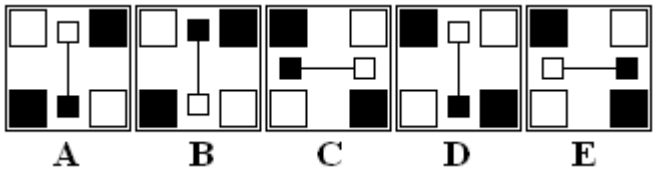
A B C D

11) Which figure is the odd one out?



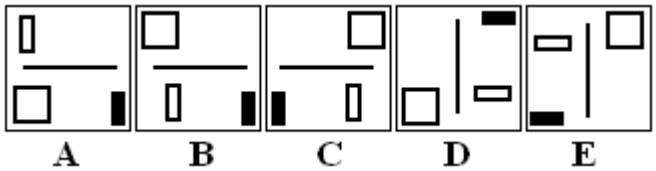
A B C D E

12) Which figure is the odd one out?



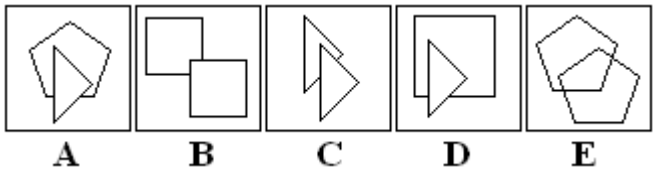
A B C D E

13) Which figure is the odd one out?



A B C D E

14) Which figure is the odd one out?



A B C D E

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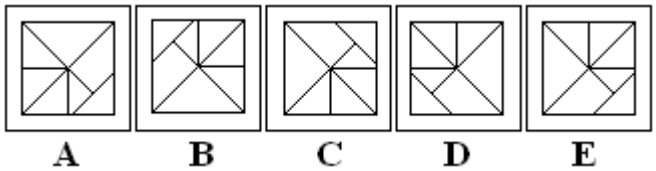
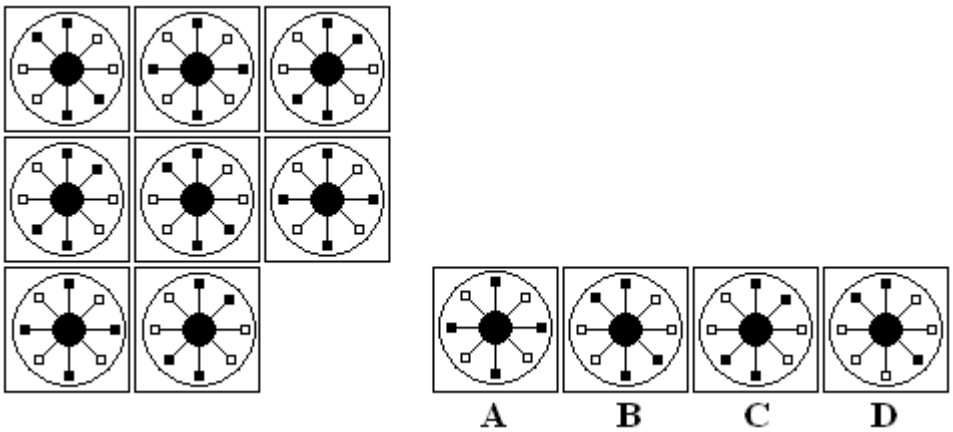
15) Which figure is the odd one out?



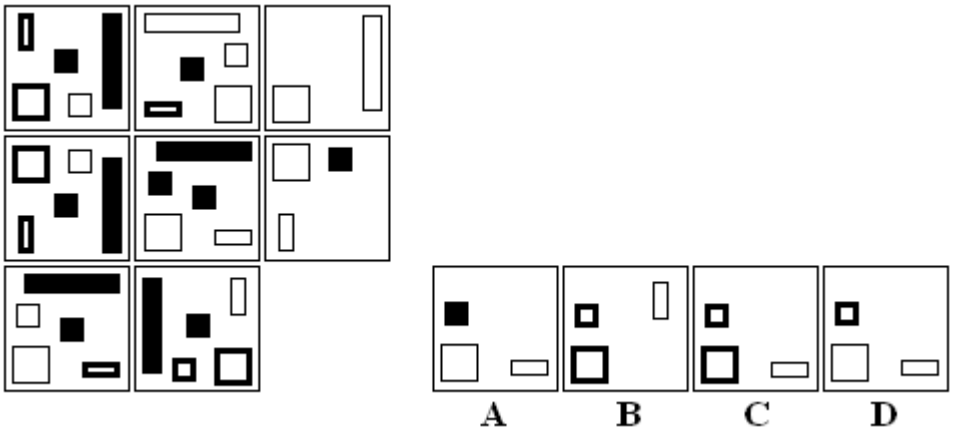
A B C D E

16) Which figure completes the series?



A B C D

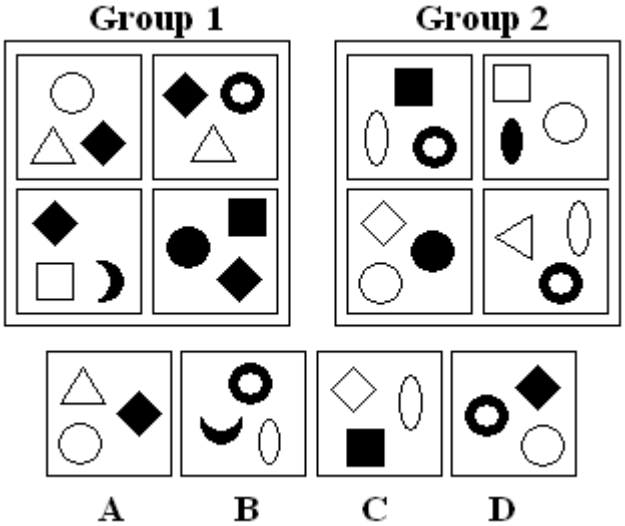
17) Which figure completes the series?



A B C D

Mechanical Aptitude Tests

18) Which figure belongs in neither group?



A B C D

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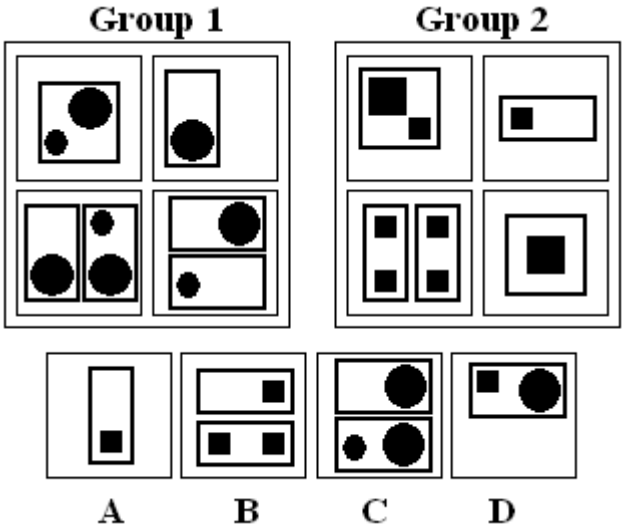
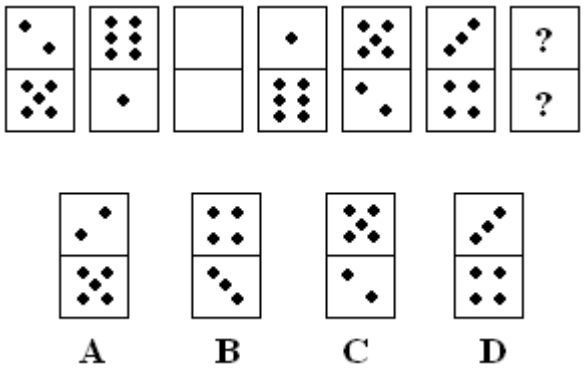


A B C D

20) Which figure is next in the series?



A B C D

Mechanical Aptitude Tests

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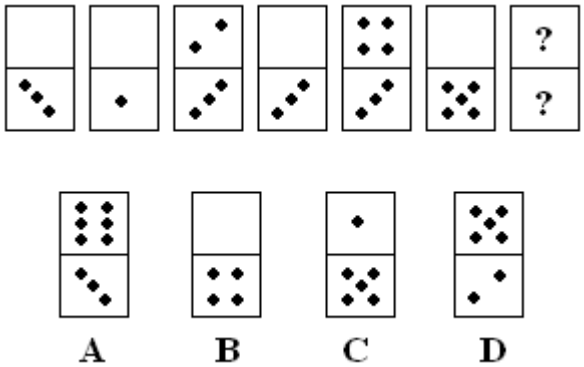
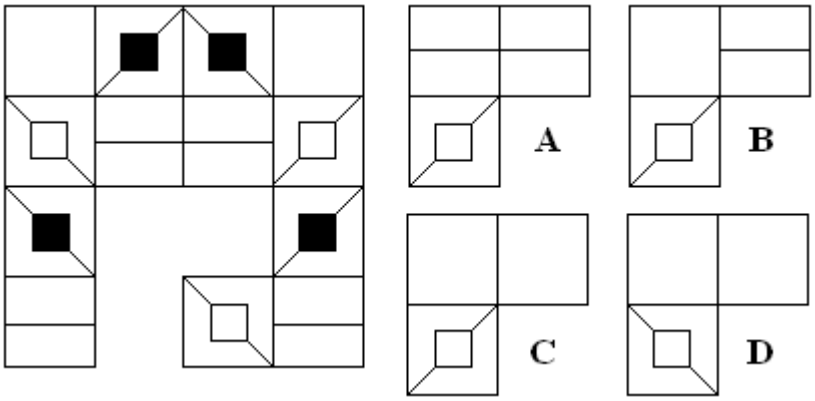


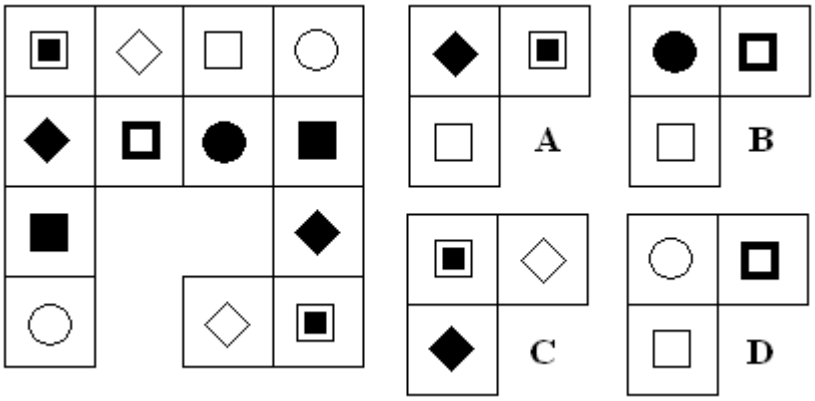
A B C D

22) Which figure completes the grid?



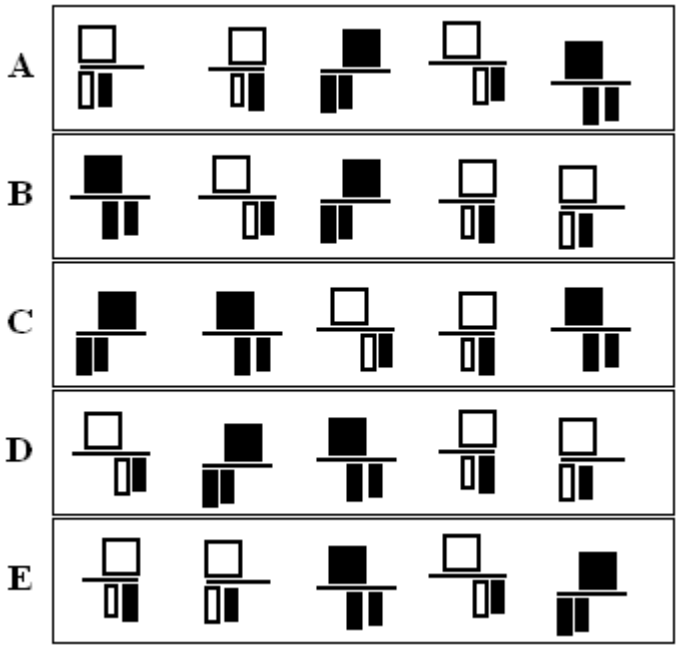
A B C D

23) Which figure completes the grid?



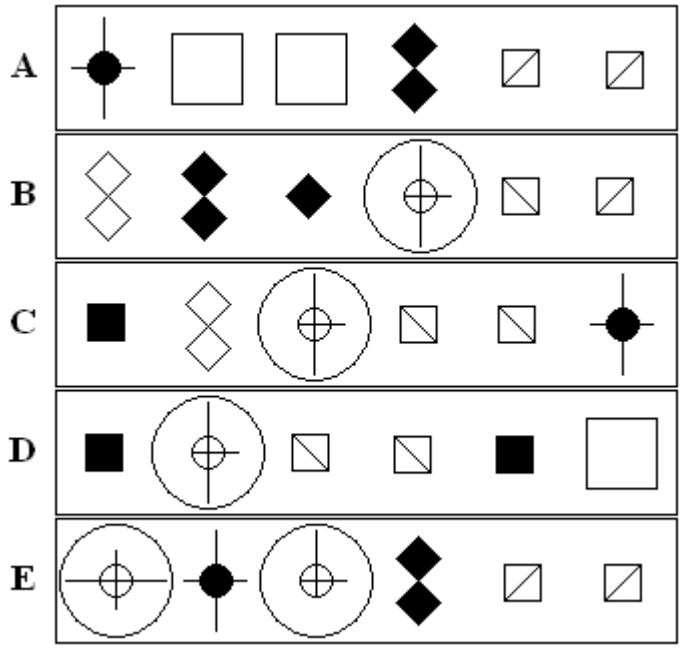
A B C D

24) Which figure is the odd one out?



A B C D E

25) Which figure is the odd one out?



A B C D E

End of Abstract Reasoning - Test 4

Mechanical Aptitude Tests

Answers to Abstract Reasoning Tests 1-4

Question	Test 1	Test 2	Test 3	Test 4
1)	**A**	**B**	**C**	**D**
2)	**C**	**C**	**D**	**B**
3)	**C**	**A**	**A**	**C**
4)	**D**	**D**	**A**	**C**
5)	**B**	**B**	**B**	**A**
6)	**B**	**C**	**B**	**B**
7)	**C**	**B**	**D**	**D**
8)	**D**	**B**	**A**	**B**
9)	**A**	**D**	**C**	**C**
10)	**C**	**C**	**D**	**D**
11)	**C**	**E**	**A**	**B**
12)	**D**	**D**	**B**	**D**
13)	**A**	**B**	**E**	**A**
14)	**D**	**A**	**C**	**D**
15)	**E**	**D**	**D**	**E**
16)	**D**	**C**	**B**	**B**
17)	**A**	**D**	**A**	**C**
18)	**C**	**D**	**C**	**B**
19)	**A**	**B**	**A**	**D**
20)	**B**	**C**	**D**	**D**
21)	**D**	**B**	**B**	**A**
22)	**B**	**D**	**C**	**C**
23)	**C**	**A**	**D**	**B**
24)	**C**	**D**	**C**	**C**
25)	**D**	**D**	**E**	**B**

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Mechanical Aptitude Tests

Explanations - Abstract Reasoning Test 1

1) A square is added each time and the colour of the squares is inverted each time. Option A is correct.

2) The centre circle is alternating between clear and solid. An extra line is added every second iteration. Option C is correct.

3) The short line moves 45% clockwise with each iteration. The shape which appears in the top left moves to the bottom right, replacing any shape which already there. Another shape may (or may not) appear in the top left. Option C is correct.

4) An alternating series in which the next figure requires the top and bottom elements plus two solid black shapes arranged vertically. Option D is correct.

5) The same eight elements are arranged randomly in each figure. Option B is the only option which has these same eight elements.

6) The operation involves a reflection in the vertical plane through the centre of the figure. Option B is correct.

7) There are three shapes in the figure. The two outer shapes are moved into the centre to cover the shape already there. Option C is correct.

8) The compound shape at the top rotates through 90 degrees. The large shape at the bottom moves to the top. The colour of the small shape at the bottom is inverted. Option D is correct.

9) The white square moves one place anticlockwise. The black square moves one place clockwise. Option A is correct.

10) The square in the top left corner moves to the centre. Any horizontal lines are duplicated and rotated through 90 degrees. Option C is correct.

11) Option C is the only figure in which opposite collared squares are on opposite sides of the line.

12) Adding the number of sides of the shapes in each figure gives eight, except for option D where it adds up to 5.

13) Option A is the odd one out. The others are all either reflections or rotations of the same figure.

14) Option D is the odd one out. In the other figures there is one more black square than there are white squares.

15) Option E is the odd one out. It is the only figure where there is a small black square adjacent to the large black square.

16) Option D completes the series. An extra white square is added for each column and an extra black square is added for each row.

17) Option A completes the series. Each figure in column three is a product of the figures in the preceding two columns, with the added rule that if the same shape appears in both columns then it does not appear in the third column.

18) The figures in Group1 contain a black diamond plus 3 white shapes.
The figures in Group 2 contain a white square plus 3 black shapes. Option C belongs in neither group.

19) Shapes are either curved or angular. The figures in Group1 black shapes at opposite corners. The figures in Group2 contain black shapes arranged vertically. Option C belongs in neither group.

20) The top halves of the dominoes are in descending sequence 6, 5, 4, 3, 2, 1, Blank. The bottom halves follow the descending sequence 2, 2, 1, 1, Blank, Blank, Six. Alternate dominoes are then inverted. Option B completes this sequence.

21) The top halves of the dominoes follow the sequence 3, Blank, 3, blank, etc. The bottom halves follow the ascending Blank, 1, 2, 3, etc. Alternate dominoes are then inverted. Option D completes this sequence.

22) The first and third columns are mirror images of each other, as are the second and fourth columns. Option B completes the grid.

23) The four squares which make the top left corner block are identical to the four squares which make the bottom right corner block. The four squares which make the top right corner block are identical to the four squares which make the bottom left corner block. Option C completes the grid.

24) Option C is the only figure which does not contain the sequence of the double-diamond, concentric circles and line and the square plus diagonal line.

25) Option D is the only figure which does not contain only two black shapes.

Mechanical Aptitude Tests

Explanations - Abstract Reasoning Test 2

1) The first figure is rotated through 90 degrees anticlockwise to produce the second figure. The second figure is then reflected in the vertical plane to produce the third figure. This sequence is repeated. Option B is correct as it is a reflection of the fourth figure in the vertical plane.

2) The line moves clockwise 135 degrees with each iteration, whilst the diamond moves anticlockwise from corner to corner. Option C is the next figure in the series.

3) The black rectangle alternates between the top and bottom position and a white square is added with each iteration. Option A is the next figure in the series.

4) The horizontal line which forms part of the centre cross alternates between short and long. Option D is the only figure where it is short.

5) The figure is rotated by 90 degrees anticlockwise and a black square is transformed to white with each iteration. Option B is the next figure in the series. Note that option A will not work because the black square is in the wrong position.

6) The figure is rotated through 90 degrees clockwise. Option c is correct.

7) The figure is rotated through 90 degrees and the black and white squares are enlarged and brought into the centre. Option B is correct.

8) The thick vertical lines are rotated through 90 degrees, the square-within-square is moved to the opposite corner and the line is reflected in the vertical plane. Option B is correct.

9) The black squares move one place anticlockwise. Option D is correct.

10) The square moves to the centre and the lines are duplicated and rotated through 90 degrees. Option C is correct.

11) Option E is the odd shape out as it is a reflection, not a rotation, of the others.

12) Option D is the odd shape out as it is not a rotation, of the others.

13) Option B is the odd one out. It is the only figure where the black and white square are not in opposite corners.

14) Option A is the odd one out. In the other figures the arrow points from a black to a white square.

15) Option D is the odd one out. In the other figures the black and white boxes are always opposite each other.

16) Option C completes the series. The black squares in columns one and two are combined in column three.

17) Option D completes the series. If a square is black in both row one and in row two then a black square appears in row three,.

18) The figures in Group1 contain a black diamond plus a black curved shape.
The figures in Group 2 contain a black square plus two other white shapes. Option D belongs in neither group.

19) The figures in Group1 contain two shapes. The figures in Group 2 contain three shapes. Option B belongs in neither group.

20) The top halves of the dominoes repeat the sequence 5, 3, 1. The bottom halves repeat the sequence 2, 4, 6. Alternate dominoes are then inverted. Option C completes this sequence.

21) The top halves of the dominoes repeat the sequence 4, 2, etc. The bottom halves repeat the sequence 5, 5, 4, 5, 3, 5, etc. Alternate pairs of dominoes are then inverted. Option B completes this sequence.

22) Option D is correct. Each row and each column has one line of each type.

23) The four squares which make up each corner block all rotations. Option A completes the grid.

24) Figure D is the odd one out. The other figures all contain five rotations of the same shape.

25) Figure D is the odd one out. The other figures contain three repetitions of the group plus one reflection. Figure D contains two repetitions plus two reflections.

Mechanical Aptitude Tests

Explanations - Abstract Reasoning Test 3

1) Subsequent figures are rotated through 90 degrees anticlockwise and an increasing amount of the original colour is inverted. Option C completes the sequence.

2) The horizontal line alternates between the left and right position. The vertical line alternates between 3 states top & bottom, bottom then top. Option D completes the series.

3) An extra square of alternating colour is added each time and the figure is rotated 90 degrees clockwise. Option A completes the series.

4) The number of sides belonging to the black shapes decreases each time. Option A completes the series.

5) Each figure is rotated through 90 degrees and one of the outermost shapes removed each time. Option B completes the series.

6) The figure is rotated through 90 degrees clockwise and then reflected in the vertical plane. Option B is correct.

7) The figure is rotated through 90 degrees anticlockwise and then the colours are inverted. Option D is correct.

8) The figure should be considered as four separate elements. The square-within-square is reflected in the vertical plane and the colours are inverted. The colours of the three thick lines are inverted. The horizontal line is reflected in the vertical plane and the vertical line is reflected in the horizontal plane. Option A is the correct answer.

9) The figure is rotated through 90 degrees anticlockwise and then the colours are inverted. Option C is correct.

10) This transformation follows three rules depending on the colour of the squares on each end of the three lines. If both squares are white, they are removed. If both squares are black, they become white. If there is a black and a white square, then both squares become black. Option D is correct.

11) Option A is the odd shape out as it is a reflection, not a rotation, of the others.

12) Option B is the odd one out. The sum of the sides of the shapes does not equal nine.

13) Option E is the odd one out as it is a reflection, not a rotation, of the others.

14) Option C is the odd one out as it contains no adjacent black squares.

15) Option D is the odd one out as the white square is opposite a black square.

16) Option B is correct. Each row is a 90 degree rotation of the row above.

17) Option A is correct. The squares in the third row are the sum of the squares in rows one and two, subject to these following rules. If the colour of the squares is the same then switch colours. If the colour of the squares is different then delete the squares.

18) The figures in Group1 contain circles which are always bounded by their own rectangle. The figures in Group 2 contain one or two black squares bounded by a rectangle. Option C belongs in neither group.

19) The figures in Group 1 all contain two arrows pointing to the right. The figures in Group 2 all contain two arrows pointing to the left. Option A belongs in neither group.

20) Every alternate domino is a 1 & 6, these can be ignored. The remainder follow a descending pattern of 5, 4, 3, 2 and a repeating pattern of Blank, 1, Blank, 1. Every other one of these is inverted. Option D continues this series.

21) Every other domino follows a descending pattern 4&6, 4&5, 4&3, 4&2. Option B continues this series.

22) The four squares which make up opposite corner blocks are mirror images. Option C completes the grid.

23) Columns two and four are identical. Column three is a mirror image of column one. Option D completes the grid.

24) Figure C is the odd one out. The black squares in each of the other figures add up to 20. The black squares in figure C add up to 14.

25) Figure E is the odd one out. In all of the other figures the horizontal line in the black-circle-within-white-circle is on the same side as the diagonal-line-within square.

Mechanical Aptitude Tests

Explanations - Abstract Reasoning Test 4

1) Subsequent figures are rotated through 90 degrees anticlockwise and a black triangle is added clockwise to the existing black shape. Option D completes the sequence.

2) Black squares and white squares are added alternately. Option B completes the sequence.

3) The line moves 135 degrees anticlockwise and the black diamond moves clockwise to the next corner. Option C completes the sequence.

4) Subsequent figures are rotated through 90 degrees clockwise and an increasing amount of the original colour is inverted. Option C completes the sequence.

5) Subsequent figures have an additional white-square-with-cross. Option A completes the sequence.

6) The figure is rotated through 90 degrees anticlockwise and the colours are inverted. Option C is correct.

7) This transformation follows three rules depending on the colour of the squares on each end of the three lines. If both squares are white, they are removed. If both squares are black, they become white. If there is a black and a white square, then both squares become black. Option D is correct.

8) The rectangle is reflected in the horizontal plane. The square is reflected in the vertical plane. Option B is the correct answer.

9) The square-within-square is reflected in the vertical plane. The cross becomes a white square. The black lines are rotated through 90 degrees and a black line is added between them. Option D is the correct answer.

10) The whole figure is rotated through 90 degrees anticlockwise. Option B is the correct answer.

11) Option B is the odd one out as it contains an additional black triangle.

12) Option D is the odd one out as it is a reflection, not a rotation, of the others.

13) Option A is the odd one out as it is neither a reflection nor a rotation of the others.

14) Option D is the odd one out as it is the only one in which the sum of the sides of the shapes is an odd number.

15) Option E is the odd one out as it is neither a reflection nor a rotation of the others.

16) Option B is correct. The squares in the third column are the sum of the squares in columns one and two, subject to the following rules. If the colour of the squares is the same then the result is black. If the colour of the squares is different then the result is white.

17) Option C is correct. The figures in the third column are the sum of the figures in columns one and two, subject to the following rules. If the same shape appears in column one and column 2 it is deleted. The remaining figure is rotated 90 degrees clockwise.

18) The figures in Group1 contain three shapes, one of which is curved. The figures in Group 2 contain three shapes, two of which are curved. Option B belongs in neither group.

19) The figures in Group1 circles within rectangles. The figures in Group 2 contain squares within rectangles. Option D belongs in neither group.

20) The top halves of the dominoes have the descending sequence 2, 1, Blank, six, five, etc. The bottom halves have the ascending sequence 5, 6, Blank, 1, 2, 3, etc. Alternate dominoes are then inverted. Option D completes this sequence.

21) The top halves of the dominoes have the ascending sequence Blank, 1, 2, 3, etc. The bottom halves haves repeat the sequence 3, Blank, 3, Blank, etc. Alternate dominoes are then inverted. Option A completes this sequence.

22) The halves of the grid are mirror images in the vertical plane. Option C completes the grid.

23) Diagonally opposite corners of the grid are colour inversions. Option B completes the grid.

24) Figure C is the odd one out. It is the only figure which does not contain one of each of the sets of shapes, the second and fifth are identical.

25) Figure B is the odd one out. It is the only one in which the diagonals in the diagonal-within-square are not the same.

Summary

Mechanical Aptitude Tests

Preparing Yourself for Selection Tests

Psychometric testing can take place at any stage in the recruitment process, but are usually used to screen candidates prior to the first interview. Some organisation prefer to use them later on, for example, prior to a second interview or short-listing, or at several times throughout the whole selection process.

Many organizations use verbal, numerical and abstract reasoning tests as a matter of routine irrespective of the precise demands of the job. Others apply psychometric testing in ways that are directly relevant to the job. For example, you may only have to take numerical reasoning tests if the job you're applying for requires good numerical skills.

After they have received candidates résumé's the organization will screen them against the job specification, discarding those where the qualifications or experience are judged to be insufficient. The remaining candidates will each be sent a letter telling them:

- Test date,
- Time,
- Place of the test,
- Format,
- Duration
- If there are breaks
- Types of tests
- Items that will be supplied
- Materials you need to bring
- Whether the test is paper based, PC-based or palm-top computer.

To ensure that everyone has the opportunity to prepare for the test and that nobody is going to be upset or surprised when they see the test paper, sample questions will be sent out 1-2 weeks before interviews. As part of the recruitment process, you should:

1. Be briefed about the purpose of the test before taking it
2. Have the results of the test provided to you in a private feedback session
3. Be informed of organizational policy about distribution and storage of the results.

When you receive this letter, if you have any special requirements you must notify the test centre immediately. This would include disabled access and any eyesight or hearing disability you may have. Large text versions of the test should be available for anyone who is visually impaired and provision for written instructions should be made for anyone with a hearing disability.

Mechanical Aptitude Tests

What You Can Expect on the Day

Tests usually take place in a dedicated room with desks are laid out in rows and there may be up to 25 other candidates. You will be provided with all of the materials you need including pencils and pocket calculators; you may even be inputting your test directly into a PC. The latter has advantages for the employer the results are available immediately, it is a cost-effective method to test and can be presented along with a computer generated analysis for feedback to the candidate.

Before the test begins you can expect the test administrator will you how the tests will be run to ensure it is as fair as possible for all of the candidates:

1. Provide a thorough explanation of what you will be required to do.
2. Timing of the tests and whether or not they will remind you of time left.
3. You will also be given the opportunity to ask any questions you have before the test begins.

During the tests if you mark your answers on the wrong answer sheet then you must inform the administrator so that this can be taken into account. It is extremely important to read your instructions and questions carefully.

Mechanical Aptitude Tests

When the Test Begins

There are a few tips which you can use to help you gain control of your nerves on the day of the test.

LISTEN	Listen carefully to the instructions you are given and ask for clarification if you need it.
CHECK	Check that your PC and all your equipment is in good working order.
READ	Read the questions carefully and underline key words to keep you focused.
IDENTIFY	Clearly identify those questions you want to go back to.
FOCUS	Keep your attention firmly on your test paper and don't be distracted by any other candidates.
PACE	Keep to your own unique pace developed during your practice sessions, maximising your accuracy.

If you feel that you cannot finish the test in the allotted time, don't panic, *some tests are designed to be impossible to finish.* If you deviate from your optimum pace you will only under-perform. The only change that you should make is to guess at questions that you know you find more difficult. This will give you more time to focus on those questions you are strong in.

If, on the other hand, you realize as the test progresses that you will finish with time to spare; do not deviate from your optimum pace. If you want to return to any questions clearly mark so you can quickly find them again. This will avoid you wasting time trying to find them, which would be better spent working out or guessing the answer.

Good Luck!